

On the Fundamental Difference between Dependent-up and Dependent-down Cases, and on their Manifestation in Turkic

Sabine Laszakovits, Ph.D.

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Dependent cases in Dependent Case Theory (DCT) fall into two directions: during case competition between two nominals, “dependent-down” cases are assigned to the lower nominal (such as accusative), and “dependent-up” cases are assigned to the higher nominal (such as ergative). It is a well-known, not often discussed, issue why accusative case can be assigned to multiple arguments, while ergative case can only ever appear on one argument, and must appear on the highest argument. In this thesis, I argue that this is not accidental but follows from computational considerations: a formal grammar checking whether a given dependent-down case is licensed is less complex (in a formal sense) than a formal grammar checking whether a given dependent-up case is licensed: For a dependent-down case, the possible licensors are the NP’s c-commanders, which is a single sequence, while for a dependent-up case, the possible licensors (the NP’s c-commandees) need to be looked for in every local subtree.

Chapter 2 gives an overview over the range of “versions” of DCT in the existing literature, and presents the assumptions made in this thesis. Chapter 3 reviews data from Turkic that illustrate that the same NP can serve as licensor for accusative on multiple lower arguments. Chapter 4 presents the argument that there is a complexity difference between dependent-up and dependent-down cases. Chapters 5 and 6 argue for a novel view on Turkic cases. Chapter 5 argues that in Turkic, dative is never a dependent-up case (against Baker & Vinokurova 2010), but always lexical/inherent: We don’t have evidence that it should be a dependent-up case, and (assuming chapter 4 is on the right track) because there is no Single Dative Generalization in Turkish. Chapter 6 argues that Turkic genitive is the dependent-up case in the nominal domain, assigned by case-competition against the possessum (Baker 2015a): If it were an unmarked case, we should observe GEN getting overwritten by movement into a different case assignment domain (like nominative), and its be-

ing an inherent/lexical case would violate Bobaljik's (2008) Revised Moravcsik Hierarchy for the accessibility of cases for φ -agreement.

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Sabine Laszakovits

Master of Arts, University of Connecticut, 2018

Magistra philosophiae, University of Vienna, Austria, 2013

Bachelor of Science, TU Wien, Austria, 2013

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Doctor of Philosophy Dissertation

**On the Fundamental Difference between Dependent-up and
Dependent-down Cases, and on their Manifestation in Turkic**

Presented by

Mag.phil. Sabine Laszakovits, B.Sc., M.A.

Approved by

Major Co-Advisor: Susanne Wurmbrand

Major Co-Advisor: Stefan Kaufmann

Associate Advisor: Jonathan Bobaljik

Associate Advisor: Andrea Calabrese

Associate Advisor: Thomas Graf

Associate Advisor: Whitney Tabor

University of Connecticut

2025

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Hello sweetie!

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Chapter 1

Overview and results of this thesis

1.1 Introduction

This thesis lives at the intersection of two fields within generative grammar: It contributes to theoretical aspects of Case Theory (§1.2), and it contributes to the analysis of cases in Turkic languages (§1.3). Regarding Case Theory, I use methodology from formal language theory to investigate the computational steps required for describing the distribution of dependent-up and dependent-down cases (§1.4). This mathematical formulation allows the results to remain independent from syntactic frameworks, and to be rigorously investigated to ensure that seemingly innocuous additions (such as feature percolation) do not lead to unintended consequences (such as vastly overgenerating the complexity class of predicted patterns). Regarding Turkic linguistics, I argue that dative (DAT) case is an inherent case assigned by Appl and Voice_{CAUSEE}, and that genitive (GEN) case is the dependent-up case in the nominal domain.

1.2 Results for Case Theory

I contribute to developing the formal mechanics of the framework known as Dependent Case Theory (DCT) (most notably Marantz 1991; Baker 2015b), which I introduce in detail in chapter 2.

The reader familiar with the literature on DCT may skip ahead to section 2.6, where I list the assumptions about the workings of DCT that I will make throughout this thesis. After discussing the application of the dependent-down case accusative (ACC) in Turkic in chapter 3, which includes a construction of theoretical interest where the same nominal serves as case-competitor for at least two ACC-markings (§3.4), I turn to the theoretical implications of this in §3.6. I show that recently proposed implementations of DCT, where case-competition is merely a specific instance of Agree, cannot account for the data observed for Turkic.

This leads me to a theoretical discussion of dependent-up case in chapter 4. There, I discuss ergative (ERG), which is not a case that occurs in Turkic languages, but is of great theoretical interest because empirically, we do not observe multiple ERG-markings on arguments in the same case-assignment domain, an observation that Nie (2017) has termed the “Single Ergative Generalization”. I discuss previous accounts for the Single Ergative Generalization as well as possible reasons for it. My hypothesis is that the Single Ergative Generalization is a fundamental property of all dependent-up cases. I provide arguments from formal language theory (§4.5) that dependent-up case assignment, like all licensing relationships where the licensee (the ERG-marking) c-commands the licensor (the case-competitor), are in a formal sense computationally *harder* to check than the opposite pattern, dependent-down case assignment, where the licensor c-commands the licensee. I argue that this is connected to two properties: the branching direction of trees, and the absence of restrictions on the search path for the licensor at branching nodes. I show that formal languages of trees in a commonly assumed sub-regular complexity class (TSL-2; Graf & Heinz 2015; Graf & Kostyszyn 2021; Heinz, Rawal & Tanner 2011) cannot capture the Single Ergative Generalization, and Sensing Tree Automata (Graf & De Santo 2019) cannot either, even with a look-ahead of 2.

If my hypothesis holds, this makes predictions for other dependent-up cases, namely that they should also be restricted to a single marking per case-assignment domain.

1.3 Results for Turkic linguistics

In chapter 5 I discuss the proposal by Baker (2015b) and Baker & Vinokurova (2010) that dative (DAT) in Turkic languages is a dependent-up case, which I argue against. My argument has two parts: first, we do not have evidence for any occurrence of DAT where it must be a dependent case — rather, all occurrences I have found can be explained by reference to lexical and inherent cases; and second, while difficult to process, we can observe multiple occurrences of DAT-markings per case-assignment domain.

In chapter 6, I argue that genitive (GEN) case in Turkic languages must be analyzed as the dependent-up case in the nominal case-assignment domain. I argue that Baker’s (2015a) proposal for the typologically frequent ERG-GEN syncretism can and should be extended to Turkic GEN. Dependent-up case is assigned to the possessor by case-competition of the possessor against an intermediate projection of the possessum. I show that analyzing GEN as dependent case rather than as inherent or unmarked case makes the correct predictions with respect to Bobaljik’s (2008) Revised Moravcsik (1974, 1978) Hierarchy. My hypothesis then predicts that there can only ever be a single GEN-marking per case-assignment domain, which is a well-known fact about Turkic GEN.

1.4 Results for computational linguistics and cognitive science

Overall, then, this thesis aims to contribute to the question of the computational complexity of dependencies in natural language syntax. This is of great interest to cognitive science in general. By assumption, the Language Faculty would make use of its resources optimally, thus any complex computations within its abilities should be observable in some form. Graf (2022a) formulates this as follows:

The central argument is that if a cognitive system must be TSL-2 to handle movement, then we should expect to see these TSL-2 resources be used in a variety of ways. For in-

stance, if the complexity of a system with movement and island constraints is not higher than that of the movement system without island constraints, additional explanations would be needed if no language ever exhibited island effects. Island constraints would inevitably be part of a linguistic ecosystem of free variation that is limited only by the available cognitive resources. Free variation limited to TSL-2 thus carves out a space within which we find some of the most surprising movement phenomena. (Graf 2022a: p. 184)

Investigating the computational complexity of dependencies attested in natural languages is thus of great interest to research on general cognition.

Computational linguistics, as the term will be used here, is the study of natural languages and their grammar through the eyes of computation theory and its subfield formal language theory. This informs theoretical linguistic theories from two perspectives: First, it can serve as meta-theory to theories of grammar, evaluating these proposals in terms of their computational feasibility and adding new considerations. For example, as Graf (2019a) discusses, enriching the representation with features can lead to a blow-up in the formalism's complexity that may not be desirable. A mathematical understanding of the properties of feature-based representations is thus imperative for theories of grammar.

Second, it brings new tools to the table that may help capture natural language phenomena to a greater degree and with more simplicity than existing tools employed by theoretical linguists. For example, monotonicity has been used to derive *ABA patterns for tense allomorphy (Moradi 2018) and for the PCC (Graf 2016, 2019b).

The immediate goal of computational linguistics is then to establish a hierarchy of complexity classes of grammars, and to identify which of these classes is the most suitable for natural language.

There are two factors that influence the complexity of natural language grammars: the data representation, and the shape of constraints on this data representation. These two factors are interconnected. For example, assuming a linear data structure for syntax, that is, without hierarchical structure, increases the workload of rules and constraints that regulate correct word orders to the

class of *mildly context-sensitive* grammars, while the usage of some hierarchical data structures can reduce the complexity of constraints at least as far down as to the class of *regular* grammars.

Considerations regarding complexity can draw independent motivation from the complexity in other modules, especially within the language module. It has been a motivating factor to choose data representations such that the complexity of phonological rules, morphological rules, and syntactic rules align (Graf & Heinz 2015 et seq.). Work in formal language theory since the 1950s has focused on defining some schematics for formal grammars and on identifying mathematical properties of these schematics. Famously, in his seminal work, Chomsky (1956, 1959) proposed four schematics for formal languages that stand in an entailment hierarchy, and discussed which of these could provide a complete string-based grammar for English:

- (1) regular languages (Type 3) $\subsetneq$ context-free languages (Type 2) $\subsetneq$ context-sensitive languages (Type 1) $\subsetneq$ recursively enumerable languages (Type 0)

Since then, research has identified the class of regular languages as an upper bound for natural language phonology (string-based, Kaplan & Kay 1994), morphology (string-based, Koskenniemi 1983), and syntax (tree-based, Kobele, Retoré & Salvati 2007). In recent decades, work has been undertaken to define and investigate some schematics that are less powerful than that of regular languages, collectively called the subregular languages (Chandlee 2014; Chandlee & Heinz 2017; Heinz 2011a,b). As the name suggests, subregular grammars can do a subset of the things that regular grammars can do. Since assuming the class of regular grammars for human language over-generates the actual grammars we see for individual natural languages, it is necessary to find a weaker complexity class. For a thorough introduction into the field of subregular linguistics and how it can inform theoretical linguistics, I refer the interested reader to Graf (2022b,c).

This thesis contributes to the research on subregular languages by investigating the suitability of different concrete proposals for subregular languages for the phenomenon of dependent-up case assignment/licensing (§4.5).

1.5 Summary

Overall, the results of the proposed work and the formal methodology it employs contribute to research in generative linguistics, Turkic linguistics, cognitive sciences, and computer sciences in at least the following ways:

1. Restricting the complexity class for the language module, and using computational considerations to model grammar, can derive typological patterns, including gaps in paradigms, that have not received explanations in less-computational theories of grammar.
2. The use of mathematics can facilitate the translation between different theories of grammar, thus enabling the exchange of results, questions, and deriving new insights from the translation procedure. It also allows results to remain accessible after the theoretical framework changes, as the language of mathematics and logic is unlikely to undergo a change as drastic as syntactic theories might.
3. If the identified narrowest complexity class for the language module is a general narrowest complexity class for cognition, identifying it will be of interest to research in other cognition sciences, such as neuro-linguistics where parsers depend on complexity considerations of the grammar, and may make predictions there. Among cognitive disciplines, grammar is uniquely well positioned for a detailed study of the complexity of constraints that the human brain can process because grammar can be formalized as symbolic and rule-based in the same way that mathematics and logic have been, which may be more challenging for other cognitive disciplines such as vision or memory.
4. Identifying the complexity class, and proposing a suitable data representation, for grammar may also have repercussions for computer linguistics and the automated language processing techniques employed there, as it may lead to a simpler design of algorithms and reduce their resource usage with respect to runtime and/or memory.

Chapter 2

Background on Dependent Case Theory (DCT)

2.1 Introduction

This chapter provides the theoretical background on case theory that I will assume in the remainder of this thesis, as well as a historical overview over the development of case theory in general and of Dependent Case Theory (DCT) in particular.

In section 2.2 I will begin by discussing the phenomenon of case in general. I present a common differentiation between different kinds of case: *structural* case, *inherent* case, and *lexical* case; providing some motivation and diagnostics for this differentiation and establishing the relevance of classifying cases into these categories for the present thesis.

In §2.3 I draw on Poole's (2024) presentation of the core assumptions of DCT and their contrast with the earlier forms of case theory as developed by Chomsky & Lasnik (1977) and Vergnaud (1977) et seq.. I do not go into details into how, why, and when DCT is the superior theory, as this has already been done at length in the literature, but I provide references for the interested reader.

Section 2.4 discusses some formalizations of DCT and their advantages and disadvantages. This provides historical background into the development of DCT. Importantly, subsection 2.4.5 details

the implementation of DCT in Minimalism by Preminger (2014) that I will assume in the remainder of this thesis (see also §2.6).

In §2.5 I discuss various current points of discussion within the details of formalizing DCT in Minimalism. Some of these questions require parametric answers, such as “Can a lexically case-marked nominal serve as case-competitor?” (“Yes” in Faroese, “No” in Icelandic; §2.5.2), while other questions are presumably principled, such as “What is the timing of dependent case assignment in the derivation?” (§2.5.6).

Finally, in §2.6 I summarize the assumptions about the workings of DCT that I will make for the remainder of this thesis, based on the discussion in previous sections. Section 2.7 summarizes this chapter.

2.2 Structural vs. inherent vs. lexical case

Case assignment has long been divided into predictable and exceptional case patterns. Predictable patterns are typically more frequent and can be acquired by a learner, resp. described by a grammarian, by stating a rule that applies to all but the exceptional cases. Such a pattern arises for instance as NOM-ACC patterns in Turkic transitives, (1).

- (1) a. *Ercan -Ø Aydoğan -ı gör -dü.*
Ercan -NOM Aydoğan -ACC see -PST
'Ercan saw Aydoğan.' *Turkish*
- b. *Ercan -Ø Aydoğan -ı bekle -di.*
Ercan -NOM Aydoğan -ACC wait -PST
'Ercan waited for Aydoğan.' *Turkish*
- c. *Ercan -Ø Aydoğan -ı öp -tü.*
Ercan -NOM Aydoğan -ACC kiss -PST
'Ercan kissed Aydoğan.' *Turkish*

Exceptional patterns arise in the presence of particular lexical items such as adpositions, (2), or verbs, (3). A learner needs to acquire, resp. a dictionary needs to state, the information which lexical items are exceptional in what way. Typically, such exceptional properties only depend on

one lexical item and remain constant in the presence of this lexical item, which is why they are taken to be a property of this lexical item itself. For example, the Turkish postposition *dolayı* ‘because of’ in (2a) consistently takes an ablative (ABL)-marked complement, no matter what the complement is or where in the clause this postposition and its complement appear. A case that is assigned in this manner is called *lexical case* or *idiosyncratic case*. (2b) gives an example of a postposition assigning lexical dative (DAT) to its complement, and the examples in (3) show verbs assigning lexical case to one of their arguments.

- (2) a. *Ercan -dan dolayı*
 Ercan -ABL because
 ‘because of Ercan’ *Turkish*
- b. *Aydoğan -a göre*
 Aydoğan -DAT according
 ‘according to Aydoğan’ *Turkish*
- (3) a. *Ercan -Ø Aydoğan -dan kork -uyor.*
 Ercan -NOM Aydoğan -ABL fear -PROG
 ‘Ercan is afraid of Aydoğan.’ *Turkish*
- b. *Ercan -Ø Aydoğan -a güven -iyor.*
 Ercan -NOM Aydoğan -DAT trust -PROG
 ‘Ercan trusts Aydoğan.’ *Turkish*

Sometimes it seems that cases express meaning themselves. For example, in some languages the case instrumental (INS) is only ever used to express (as its name indicates) that something is used as an instrument or tool. Similarly, Turkic locative (LOC) indicates a physical or temporal location, and in a few verbs is used to express a metaphorical location — but it never expresses something like instruments or accompaniment. These cases are called *semantic cases*, and I will follow many others in assuming that the stable semantics originates from a lexical item such as a silent adposition that assigns this case as lexical case:

[...] lexically governed case[s] are always contained within PPs, overt [...] or covert [...] (see, among others, Bittner & Hale (1996), Řezač (2008), Caha (2009), Pesetsky (2013), and Baker (2015b) that dative, genitive, instrumental, ablative arguments are

hidden PPs).

(Anagnostopoulou & Sevdali 2020: p. 1040 f.)

Predictable cases can be split into two sub-types (Woolford 2006): *inherent cases* that are “inherently” associated with a particular thematic role, and *structural cases* that are independent of thematic roles and are assigned only based on syntactic structure. This thesis will investigate the distinction between these two types in detail, arguing in chapter 5 that Turkic datives are not structural but inherent cases, and in chapter 6 that Turkic genitives are not inherent, but structural cases. As teaser, consider the fact that in many languages in ditransitives the GOAL-argument is consistently DAT-marked. This might be seen either (A) as a property of the lexical items (the ditransitive verbs), analyzing this DAT-marking as lexical case, or (B) as a rule about arguments bearing the θ -role GOAL in general such as “all nominals bearing the θ -role GOAL receive DAT”, analyzing this DAT as inherent case, or (C) as a rule about the structural configuration between GOALS and the other internal argument of ditransitives, cashing out on the generalization that in ditransitives with GOAL-arguments, the DAT-marked argument seems to always c-command the other (often ACC-marked) argument (but see §5.3.4.2 on distinctions between GOALS that are higher or lower than the THEME), and analyzing this DAT as a structural case.

Since Baker’s (1988) seminal proposal known as the UTAH (cf. also Baker 1997), thematic roles are taken to correspond essentially to structural positions. For instance, an argument is interpreted as the GOAL if and only if it is first-Merged in the specifier position of an applicative phrase (Pylkkänen 2008). Then, inherent DAT case may be assigned to the GOAL by the applicative head Appl. Inherent cases are thus cases assigned by functional heads that in Government and Binding theory used to also assign thematic roles, and nowadays, given the UTAH, *define* thematic roles. This is illustrated in Fig. 2.1, where assignment is indicated by an arrow from the assigner to the recipient.

Recent works in Distributed Morphology (DM) have argued that what used to be lexical items in syntactic derivations (such as the terminal node V indicating the insertion of a verb) should not be analyzed as simple heads but instead should be analyzed as split into roots and categorizing heads (such as the verbalizer *v*). Roots can in principle be turned into any category, and are

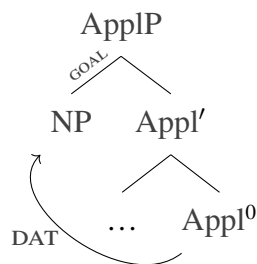


Figure 2.1: Inherent case assignment

thus typically not assumed to carry any syntactic features, including those that would specify lexical case-assignments by the root. Preminger (2024) and Sigurðsson (2012) note that, given these assumptions, it must be the categorizing head (such as v) that is responsible for “lexical” case assignment. Categorizing heads are functional heads, and thus, the theoretical distinction between lexical cases and inherent cases has essentially disappeared. Preminger (2024) proposes to subsume lexical and inherent case assignment under the term “head case”, i.e., case that is assigned by a syntactic head. In this thesis, I will continue to use the traditional terminological distinction between “lexical case” and “inherent case” as described above and as teased apart by Woolford (2006), but I do not assume a categorical ontological distinction between them.

Theories of case differ mostly in their treatment of structural cases. Dependent Case Theory (DCT) is a theory of case that sub-divides structural cases into *dependent cases* and *unmarked cases*: Dependent case is assigned to a nominal depending on whether it stands in a particular structural configuration with another nominal, and unmarked case is the elsewhere-case assigned if no other case, structural or other, can be assigned. In the next section, I will present the principles of Dependent Case Theory (DCT) and contrast it with other common assumptions about case in modern generative syntax.

2.3 Case Theories

Case theory in Generative Grammar emerged in the 1970s with Chomsky & Lasnik (1977), Fillmore (1968), Vergnaud (1977), and Chomsky (1980, 1981) et seq. (cf. the overview articles by

Bobaljik & Wurmbrand 2009; Pesetsky & Torrego 2011). It was characterized by three independent properties, as summarized succinctly in Poole (2024):

1. Case has the “purpose” to license the presence of overt¹ nominals in the clause. Nominals need to receive case during the derivation, otherwise the derivation fails. This condition is called the “Case Filter” as it serves to “filter out” malformed derivations.
2. Case is assigned to a nominal as a reflex of φ -agreement happening with that nominal. For instance, the subject is assigned nominative (NOM) case if and when the finite verb φ -agrees with the subject.
3. The process that assigns case to a nominal is initiated by a functional head, such as by finite T for NOM-case marking on subjects.

Because of property 3, this case theory is often referred to as functional-head case theory (FHCT). However, it is the Case Filter (property 1) that is the most prominent part of FHCT and the most debated part as well.² Famously, Marantz’s (1991) seminal paper on DCT has taken issue with the Case Filter and by association with properties 2 and 3 as well. DCT, as Marantz (1991) defined it and as it is usually assumed, dispenses with all three properties, proposing instead the following opposites:

- 1’. Case is independent of nominal licensing (if there is such a thing as nominal licensing at all).
I.e., if a nominal does not receive case, it does not follow that the derivation will fail.
- 2’. Case is independent of φ -agreement. I.e., it is possible that φ -agreement targets a nominal N_1 and yet case-marking appears on another nominal N_2 .

¹ This contrasts with “covert”, i.e., unpronounced, nominals such as PRO, and serves to explain why infinitives in English and many other languages cannot have overt subjects.

² I will not be discussing FHCT and its comparison to DCT here, as this has been done quite extensively in the literature already. For evidence and theory-internal arguments from languages across several language families, such as Chukchi (Chukotko-Kamchatkan), Icelandic and Greek (Indo-European) Finnish (Uralic), Kalaallisut (Inuit-Yupik-Unangan), Korean (Koreanic), Sakha (Turkic), Shipibo (Panoan), Yimas (Ramu-Lower Sepik), I refer the interested readers to the following papers, among many others: Anagnostopoulou & Sevdali 2020; Baker 2014b, 2015b; Baker & Bobaljik 2017; Baker & Vinokurova 2010; Bobaljik 2008; Bobaljik & Branigan 2006; Levin 2017; Levin & Preminger 2015; Maling 1993a, 2009; Maling & Kim 1992; Marantz 1991; McFadden 2004; Poole 2022; Preminger 2011, 2014; Yip, Maling & Jackendoff 1987; Zaenen, Maling & Thráinsson 1985. Here, I will only discuss aspects of case theory to the extent that they are relevant for the present thesis.

- 3'. Case is independent of functional heads. Functional heads are not involved in structural case assignment.

Often, evidence for 2' and 3' is cited to argue for 1'. For instance, Gribanova (2017) presents evidence from Uzbek and argues that *GEN*-assignment cannot be done through Agree because the verb copies the φ -features of NP_1 but *GEN* appears on NP_2 (to be returned to in 6.5.3). Similarly, Baker & Vinokurova (2010) argue that *ACC*-assignment in Sakha cannot be done by the functional head Voice because *ACC* appears even when we have reason to believe that there is no Voice head present in the structure (to be returned to in 3.4.2). Not following this same conflation, Poole (2024) has recently argued that we can (and should) dispense with the Case Filter while keeping the other two properties, i.e. he advocates for a combination of 1'+2+3. I will discuss his proposal in more detail in §2.4.5 and argue against it in §3.6, where I show that, in addition to the Case Filter, we need to dispense with property 3 and thus with property 2 as well, thus returning to Marantz's (1991) original 1'+2'+3'.

2.4 A brief history of Dependent Case Theory

2.4.1 Introduction

In this section, I present some early and modern implementations of the core insights of Dependent Case Theory (DCT). I start with the proposal by Marantz (1991), who introduced the term “Dependent Case Theory”, and whose formulation is often still referred to in modern papers. Besides Marantz's implementation, subsection 2.4.2 also presents Bittner & Hale's (1996) mechanism, which is equally couched in the contemporary framework Government and Binding theory (GB) (Chomsky 1980, 1981, 1986). As alternatives to GB, §2.4.3 presents Zaenen, Maling & Thráinsón's (1985) implementation of DCT in Lexical-functional grammar (LFG), and §2.4.4 presents Yip, Maling & Jackendoff's (1987) approach using autosegmental tiers for case-assignment domains. In §2.4.5 I turn to more modern implementations in Minimalism (Chomsky 1993, 1995, 2000, 2001) and I present and contrast Preminger's (2014) proposal with Poole's (2024) counter-

proposal for dependent-case assignment via Agree. §2.4.6 summarizes this discussion.

2.4.2 Government and Binding theory

Marantz (1991) has separated the notion of structural cases into what he calls “dependent cases” and “unmarked cases”. Dependent cases are assigned to an NP depending on the presence of another NP with particular properties (to be defined). For example, accusative is a dependent case that is only assigned in the presence of a structurally higher NP that receives nominative (to be revised). Similarly, in some ergative languages, ergative is assigned in dependence on the presence of a structurally lower NP receiving absolutive.

Unmarked cases are (somewhat unfortunately named) structural cases that are not dependent on another NP. The term “unmarked” does not refer to the actual morphological marking of these cases (Marantz presents data from Georgian and Japanese, where nominative is realized by an overt affix), but rather refers to the fact that these cases are somewhat uninteresting compared to dependent cases.^{3,4} For example, the case on the sole argument of an intransitive verb cannot be a dependent case, because there is no other NP that this case could depend on.⁵ Instead of a dependent case, this NP receives an unmarked case. Examples include nominative, absolutive, and in some languages arguably genitive inside the DP (see e.g. Baker 2015b: §4.4.1 for Japanese, but cf. my §6 on arguments that genitive (GEN) in Turkic is a dependent case).

In addition to dependent and unmarked cases, Marantz proposes “default cases” (one per language). Default cases serve as a backup-plan in the assignment of case to ensure that *every* NP will receive *some* case. Marantz (1991) proposes that case assignment happens after the derivation has completed checking its validity, and so the derivation cannot become invalidated for failure to

³ In his discussion of agreement, Marantz (1991: p. 28) uses the term “unmarked subjects” to not refer to “subjects with unmarked cases”, but to subject NPs whose properties will trigger this or that rule for agreement: Subjects of transitive clauses, carrying nominative or ergative, are “unmarked” for the purposes of agreement, even though ergative is not an unmarked case, while subjects of intransitives, still carrying nominative or ergative, as well as subjects with lexical case marking, are “marked”.

⁴ Baker (2015b) draws a parallel between “unmarked cases” and empty morphological marking. He suggests to distinguish between “unmarked nominative case” in some languages and “marked nominative case” in other languages where the nominative case marking is morphologically visible, *mutatis mutandis* for absolutive (ABS).

⁵ Except if one assumes the reality of Baker’s (2015b) “negative dependent cases”, which he proposes are assigned to an NP depending on the absence of another NP in the same case-assignment domain.

assign case — an idea prominently taken up by the subsequent literature, most notably Preminger (2014).

There is a crucial distinction between unmarked cases, which make up for the lack of dependent cases, and default cases, which make up for the lack of any case. It is that unmarked cases apply under certain conditions, whereas there are no conditions at all for default cases. The condition for the assignment of unmarked nominative is that it can only happen within a clausal case-assignment domain. In contrast, the condition for the assignment of unmarked GEN is that it can only happen in the nominal case-assignment domain. Marantz (1991) employs case-assignment domains only for unmarked cases, while later proposals also adopt them for dependent cases (e.g. Baker & Vinokurova 2010; Baker 2015b; criticized by den Dikken 2023).

Marantz’s (1991) theory of case assignment proceeds along the disjunctive hierarchy given in (4). “Hierarchy” means that the options for case assignment are considered in order (here: top-down), and “disjunctive” means that as soon as one form of case assignment succeeds, the following options are no longer considered.

- (4) Case realization disjunctive hierarchy (Marantz 1991: p. 24, ex. (29))
- a. lexical case
 - b. dependent case
 - c. unmarked case
 - d. default case

Cases, once assigned, cannot be overwritten.⁶ An NP that has been assigned lexical dative will bear this case marking in any position it moves to, such as to a subject position during passivization or raising in Icelandic. Similarly, an NP that carries dependent accusative will bear this case even if it scrambles past the subject.

A formal account of essentially Marantz’s (1991) case theory within GB was proposed by Bittner & Hale (1996). In this highly technical paper the authors argue that structural case is not assigned

⁶ But see, e.g., Richards (2013), Baker (2015b: §6.3.4.), and Levin (2017) on case stacking, i.e., an NP undergoing movement and receiving different cases in different positions, more than one of which are morphologically realized.

by functional heads in the sense of FHCT; instead, while mediated by (functional or lexical) heads, structural case crucially arises from the presence or absence of other NPs. Their rule for dependent case assignment is very similar to Marantz's (1991): An NP is assigned dependent case if and only if there is another case-less NP present in the same case-assignment domain. The role of the mediating head is to limit the case-assignment domain and to ensure a particular configuration between the two NPs.⁷ In the spirit of Burzio's Generalization (Burzio 1986), the rule for accusative assignment states that the mediating head must be V and that there must be a higher argumental NP present inside the VP-complex. Similarly, ergative is mediated by T. Regarding structural dative, they give a rule for Inuit stating that dative is assigned to an NP through mediation by V if the NP does not c-command V. This is a round-about way of saying that dative can only be assigned inside the VP, parallel to what Marantz (1991) proposed. The only unmarked case in their system is nominative, which, in accordance with Marantz (1991), occurs on NPs that do not qualify for a lexical or dependent case.

Within the intricacies of late GB, Bittner & Hale's (1996) treatment of lexical, dependent, and unmarked cases is quite elegant. Following Fillmore (1968), they argue that all NPs are selected by a head K that realizes the NP's case morphology. For NPs with lexical case, K and its case feature are selected for by a lexical item. For dependent cases, K is empty and must thus obey the Empty Category Principle (Chomsky 1986), which for Bittner & Hale can be fulfilled by the mediating head. At Spell-out, the identity of the mediating head (and optionally additional conditions) determine which dependent case will be realized by K. NPs realized with unmarked case are "bare", i.e., not selected by any K. Bittner & Hale (1996) argue that nominative is identical to not having any case at all; an idea going back to Jakobson (1936) that has been picked up again by Kornfilt & Preminger (2015) and Preminger (2014, 2024) and will be discussed in §6.3.2. In my understanding, the absence of K makes any overt realization of unmarked case morphology impossible, and is thus only eligible if NOM is morphologically unmarked as well.

⁷ Specifically, the head needs to locally c-command the case receiver and govern the case-competitor (Bittner & Hale 1996: p. 12, ex. (22)).

2.4.3 Lexical-functional grammar

Zaenen, Maling & Thráinsson (1985) show that in Icelandic, subjects as well as objects can have any of the four cases *NOM*, *GEN*, *DAT*, and *accusative (ACC)*. They distinguish between semantic cases, lexical cases, and structural cases. Structural cases are assigned on the basis of grammatical functions (GFs) and will change with them, e.g. during passivization or raising. Zaenen, Maling & Thráinsson 1985 provide an analysis of the Icelandic data in the framework Lexical-functional grammar (LFG) by mapping thematic roles (θ -roles) to GFs and GFs to case. For example, in a transitive clause with two θ -roles *AGENT* and *THEME*, a rule states that the highest θ -role (here: *AGENT*) is mapped to the GF subject, and next-highest θ -role (here: *THEME*) is mapped to object (Zaenen, Maling & Thráinsson 1985: p. 466). Then the rule for case assignment applies, mapping the highest GF that has not yet been mapped to a case (here: the subject) to *NOM*, and the second-highest (here: the object) to *ACC* (Zaenen, Maling & Thráinsson 1985: p. 466). If the highest θ -role is assigned lexical case by V, *NOM* is assigned to the *THEME*, still with the GF direct object. An example of such a construction is given in (5), and the mappings are illustrated in Fig. 2.2.

- (5) *Henni hefur alltaf þótt Ólafur leiðinlegur.*
 her.DAT has always thought Olaf.NOM boring.NOM
 ‘She has always considered Olaf boring.’

*Icelandic*⁸

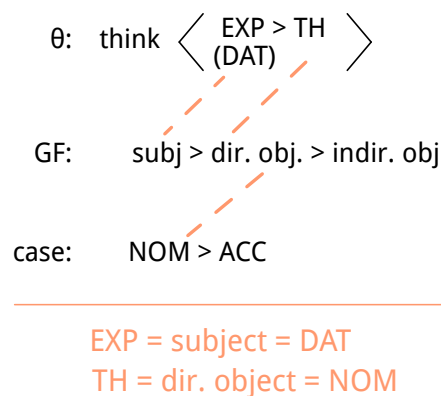


Figure 2.2: The mapping from thematic roles to grammatical functions to case in (5)⁹

⁸ Zaenen, Maling & Thráinsson 1985: p. 447, ex. (13)

⁹ The illustration in Fig. 2.2 does not follow the proposal in Zaenen, Maling & Thráinsson (1985) exactly: Zaenen, Maling & Thráinsson (1985) do not assume a hierarchy for θ -roles. Instead, they state separate rules such as “*AGENT*,

In an intransitive clause, no matter the single argument's θ -role, this θ -role will be mapped to the GF subject, and, as long as it does not receive lexical case, subsequently be assigned NOM. Zaenen, Maling & Thráinsson's (1985) approach thus ensures that ACC is only assigned if NOM is also assigned, and — by definition of GFs — that NOM is assigned to the structurally higher of two NPs with structural case.

2.4.4 Autosegmental tiers

Yip, Maling & Jackendoff (1987: p. 100) argue that structural case is assigned to an NP depending on which other NPs in the case domain receive structural case. They treat the sequence of structural cases as floating elements on a tier — inspired by autosegmental tiers in phonology (Goldsmith 1976) — that are associated to NPs on the NP-tier.

Lexical cases arrive on a separate tier and are assigned before structural cases are assigned. In Fig. 2.3, lexical **DAT** is assigned to the subject NP. The object receives structural case, which is **NOM**, because it is the left-most unassociated case on the structural case tier. Structural **ACC** remains “floating” on the structural case tier. Similarly, when it is the direct object that is lexically case

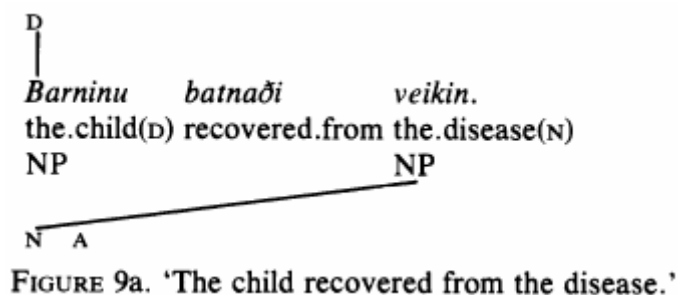


Figure 2.3: A transitive verb with a lexically case marked subject (from Yip, Maling & Jackendoff 1987: p. 223).

marked, the subject will receive `NOM`, and `ACC` will remain floating.

In addition to the tier with lexical cases (here: $\langle \text{DAT} \rangle$) and the tier with the structural cases $\langle \text{NOM}, \text{ACC} \rangle$, Yip, Maling & Jackendoff (1987: p. 233) propose that there is a third tier consisting of the

if available, is mapped to subject". The mapping from GFs to cases does proceed along a hierarchy, but without them calling it such: they state rules such as "the highest GF becomes *NOM*, the next highest *ACC*".

single case $\langle \text{GEN} \rangle$. This tier is provided by an NP (i.e., in the nominal CAD), and the $\langle \text{NOM}, \text{ACC} \rangle$ tier is provided by TP (in the clausal CAD). In this way, Yip, Maling & Jackendoff (1987) differentiate case-assignment domains not just for unmarked cases, as Marantz (1991) did, but also for dependent cases: cases that are dependent on the previous assignment of an unmarked case.

2.4.5 Minimalism

In Minimalism, case is encoded as feature on the highest nominal projection (DP resp. NP, or optionally an extra, dominating KP). Preminger (2014: pp. 204–208, §9.3.2) proposes an implementation along the following lines: Upon Merge of a nominal into the syntactic structure, the head that selected the nominal may also assign a case value if this head is idiosyncratically specified to do so in the lexicon. This encodes lexical case assignment or, if this head is understood to contribute a thematic interpretation, inherent case. Subsequently the grammar employs an operation that assigns the value [case: DEP] (with DEP for ‘dependent’) if applicable. This operation is called Case Competition and, Preminger (2014) proposes, exists in addition to the standard Minimalist operations Merge, Move, and Agree. Depending on whether the grammar is specified for a dependent-up or dependent-down case (or both) for the given CAD, Case Competition checks for configurations where a nominal N_1 with an unvalued case feature [case: ____] c-commands another nominal N_2 with an unvalued case feature [case: ____] and assigns to either the c-commanding nominal (N_1) or to the c-commanded nominal (N_2) the case value [case: DEP], which upon Spell-out will map to a morphological case paradigm. Finally, any nominal that has not received any case value during the derivation will remain case-value-less [case: ____] and upon Spell-out be mapped to the unmarked case of its CAD. In this way Preminger (2014) has not only implemented Marantz’s (1991) proposal in Minimalism but has also derived Marantz’s (1991) case assignment hierarchy (4): lexical cases are assigned before dependent cases precisely because valuation upon selection/Merge happens earlier than valuation during the derivation (dependent cases), which in turn happens earlier than no valuation at all (unmarked cases).

Recently, Poole (2024) has argued that the fourth operation that Preminger (2014) proposes (Case

Competition) may not be necessary and can be done through Agree instead. The dependency between the case marking [case: DEP] on a nominal N_1 and the presence of another nominal N_2 with unvalued case [case:] can be modeled by having a mediating head equipped with stacked probes. This is illustrated in Fig. 2.4 for dependent-down case assignment (ACC) and in Fig. 2.5 for dependent-up case assignment (ergative (ERG)). Importantly, as a feature of Poole’s (2024) proposal, the probe stacks in both examples are identical. Poole’s (2024) proposed derivation proceeds

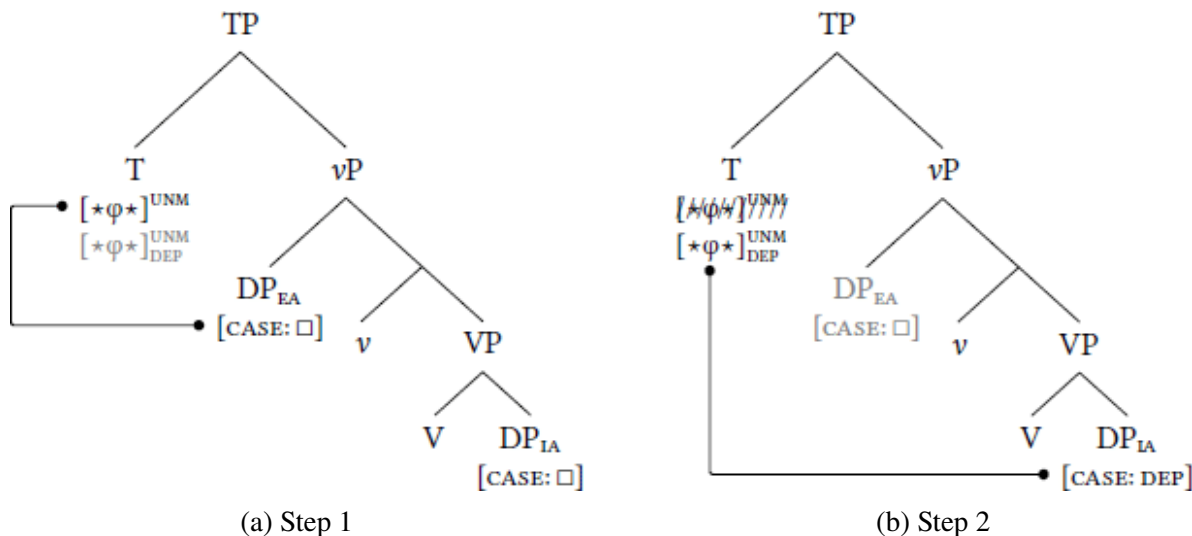


Figure 2.4: Assignment of dependent-down case (here: ACC) via Agree (figures from Poole 2024: exs. (10a, b))

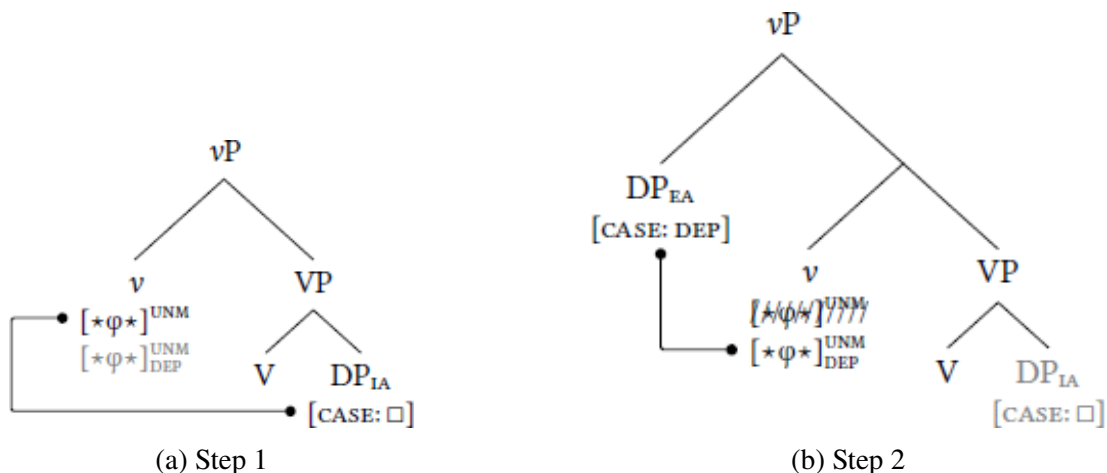


Figure 2.5: Assignment of dependent-up case (here: ERG) via Agree (figures from Poole 2024: exs. (11a, b))

as follows: Upon Merge of the mediating head and its complement, and again upon Merge of the

projection of the head and the specifier, the active Probe will try to value its φ -features by looking for a goal while case-discriminating for only those goals whose case feature is currently unvalued [case: ____]. Upon Agree, the Probe is specified to leave the case value of its goal unvalued. This is illustrated in Step 1 of dependent-down (Fig. 2.4a) and dependent-up (Fig. 2.5a) case assignment. In both examples, the first Probe finds the licensing nominal: for dependent-down case assignment, a NOM-marked subject, and for dependent-up case assignment, an ABS-marked object. Before becoming inactive, the first Probe renders its goal invisible to subsequent Probes on the same head.

The second Probe, on its turn, again looks for a goal with φ -features, discriminating for goals with unvalued case, and ignoring the now-invisible goal of the first Probe. It looks first in its complement (Fig. 2.4b), and if it cannot find a goal there, in its specifier (Fig. 2.5b). It copies the goal's φ -features and as a reflex of this Agree process, values the goal's case feature to [case: DEP].

If either the first or the second Probe cannot find a goal, the probing head will remain unvalued and realize default agreement.

However, Preminger (2019) argues that on certain assumptions about how agreement and clitic doubling work, this approach could predict that if there were such a thing as “abstract” φ -agreement, i.e., an agreement Probe on a syntactic head whose exponents are systematically $\emptyset$ across the entire paradigm, then the presence of such a Probe would have to be postulated by every learner of every language, which would lead to every language obeying the PCC, contrary to fact. This is an argument that Poole (2024) does not address.

Apart from this theoretical argument, there may be an empirical way to distinguish Preminger's (2014; 2024) proposal from Poole's (2024) — but to my knowledge it has not been attested to-date: If we were to find evidence that dependent cases can be assigned outside of CADs, i.e. in the context where Marantz (1991) and Schütze (2001), a.o., postulate default cases, this would be an argument for the independence of dependent case assignment of mediating heads and thus for Preminger's (2014; 2024) mediation-less approach over Poole's (2024) Agree-based proposal.

2.4.6 Summary

In this section I have presented different implementations of the core insight of DCT: that the assignment of some cases is dependent on the presence of other nominals with particular cases (Marantz 1991; Poole 2024; Preminger 2014) resp. on previous assignment of other cases (Yip, Maling & Jackendoff 1987; Zaenen, Maling & Thráinsson 1985). The particular formulations of this insight are as different as the syntactic frameworks they have been formulated in. In §2.4.2 I have briefly introduced formulations in Government and Binding theory, in §2.4.3 one proposal in Lexical-functional grammar, in §2.4.4 an outlier formulated using autosegmental tiers, which are known from phonology, and finally in §2.4.5 I have presented proposals in the current framework of Minimalism.

I will now proceed to discussing some points of variation between languages with dependent cases (§2.5) and then turn to the definition that I will be using in this thesis (§2.6).

2.5 Points of variation

2.5.1 Introduction

In this section, I will discuss some points of variation between different proposals and assumptions about DCT regarding under precisely what conditions the operation Case Competition can apply and value an NP's [case: ____] feature. Sometimes there are principal (pun intended) differences in the formalization of DCT between papers, especially if they discuss the same languages and data points. Then we expect that one formalization will turn out to be more correct than the others. Sometimes, however, the differences arise across different languages. Then, there may be a range of possibilities for concrete parameter settings regarding the operations in DCT. Both questions are still open areas of investigation. My aim here is merely to illustrate that the assumptions about DCT as discussed in the sections above and below are open to debate regarding details in the formalization, but the underlying principle of Case Competition remains the defining feature of all flavors of DCT: that it applies when two nominals with particular properties stand in a particular configuration.

Baker (2015b) formulates this as in (6):

- (6) “If XP bears c-command relationship R to ZP in local domain WP , then assign case V to XP .” (Baker 2015b: p. 229, ex. (1))

I will discuss below that X and Z also need to have particular properties that may be principled or parameterized, including their previous case-marking, their nominal-ness, and their status as argument or adjunct (Baker 2015b), and whether the $R(X, Z)$ or $R(Z, X)$ has triggered Case Competition in this derivation before (Baker 2014b: p. 355, 2015b; cf. §4).

2.5.2 Previous case-marking on the undergoing and competing NPs

Most theories of DCT that I am aware of assume that for Case Competition to apply, both NPs must be unvalued for case. Here are some quotations from the literature stating and/or assuming this prerequisite:

If there are two distinct argumental NPs in the same phase such that NP1 c-commands NP2, then value the case feature of NP2 as accusative unless NP1 has already been marked for case. (Baker & Vinokurova 2010: p. 595, ex. (4b))

If there are two distinct argumental NPs in the same phase such that NP₁ c-commands NP₂, then value the case feature of NP₁ [NP₂] as accusative [ergative] unless NP₁ [NP₂] has already been marked for case. (collapsing Baker 2014b: p. 343, exs. (4) and (5))

[...] dependent case can only arise if neither of the noun phrases in question has been previously assigned case by other means [means other than Case Competition] (e.g. [lexical/oblique case]). (Kornfilt & Preminger 2015: p. 115)

The hallmark of a dependent case is its reliance on the presence of a local caseless nominal. (Abramovitz 2020: p. 14)

Earliness: Upon (re)merging α into the structure, if α c-commands β and both α and β have unvalued case features, establish a dependent-case relationship between α and β . [...] The formulation of Earliness [...] also reiterates the restriction that two DPs can enter into a dependent-case relationship only if they both presently have unvalued case features ([CASE: ____]) [...]. (Poole 2022: p. 355)

[...] the hypothesis (originally due to [Marantz 1991]) that case is not assigned by designated heads in the syntactic structure but distributed dependent on the number of case-needy noun phrases in a particular local domain, on the basis of their hierarchical relations. (den Dikken 2023: p. 482)

However, it seems that in some languages, the receiving NP can have been valued for case before, resulting in case-stacking, such as has been reported for Quechua (Lefebvre & Muysken 1988, cited after Baker 2015b: §6.3.4, pp. 281–285), Korean (Baker 2015b: §6.3.4, pp. 281–285; Levin 2013, 2015, 2017; Schütze 2001), and Lardil (Richards 2007, 2013). As Baker (2015b: p. 284) discusses, if we accept that this requires UG to allow multiple case values per nominal in some form, we are confronted with two options: Either all languages assign each nominal one case value *per CAD*, except in most languages only one value gets realized; or there is parametric variation such that most languages allow only one case value per nominal, while a select few allow more than one. As far as I can tell, the data are limited and not yet fully understood, so I will leave this possibility aside and have only mentioned it for completeness.

As for the competitor's prior case marking, we might differentiate between prior structural and prior lexical case marking. I have not found any papers making this distinction. Most papers assume full-out that no prior case marking on the competitor is possible (cf. the above quotes). For instance, Baker (2015b: p. 219) discusses the appearance of dependent ERG on adverbs in Warlpiri and Diyari and does not consider the idea that this ERG might be assigned by Case Competition against the already-ERG-marked subject (to be returned to in §4). I have not come across any data that would make it necessary to let an NP valued as [case: DEP] participate as competitor in further applications

of Case Competition. In tripartite languages, where in a transitive clause the subject is dependently *ERG*-marked *and* the object is dependently *ACC*-marked, both cases are assumed to be assigned simultaneously with a single application of Case Competition.¹⁰ In §6.3.2.3 I will discuss proposals by Baker (2015b), Baker & Vinokurova (2010), and Kornfilt & Preminger (2015) about object-topicalization constructions in Sakha, where the *ACC*-marked object has topicalized to a position higher than the *NOM*-marked subject: the subject remains *NOM* and cannot receive *ACC* by Case Competition against the higher *ACC*-marked nominal.¹¹

However, Baker (2015b: §5.1, pp. 184–197) discusses in detail that it should be a parameter setting whether *lexically* case-marked NPs can count as competitors.¹² Famously, Icelandic and Faroese contrast in this property: while in Icelandic lexically *DAT*-marked subjects cannot trigger dependent *ACC* on their objects (Marantz 1991; Yip, Maling & Jackendoff 1987; Zaenen, Maling & Thráinsson 1985), this is possible in Faroese (Asarina 2011; Barnes 1986; Eythórsson & Barðdal 2005; Eythórsson & Jónsson 2003; Jónsson & Eythórsson 2005; Petersen 2002). Baker (2015b: §5.1.3–4, pp. 188–197) argues carefully for the same contrast in Tamil vs. Kannada. Like Faroese, Tamil has predicates with *DAT*-subjects and *ACC*-objects. Baker shows that these *DAT*-subjects are real subjects because they (i) are required arguments, (ii) can be controlled when embedded under ‘want’, (iii) can bind anaphors, and (iv) can be clefted, as can arguments but not adjuncts.¹³ This

¹⁰ Abramovitz (2020) discusses data from Koryak where an *ABS*-marked *wh*-phrase can be observed to move successively and trigger dependent *ERG*-marking from every intermediate landing site. This is not an instance of the competitor having prior case-marking, but if similar data could be found in a language that is tripartite where the moving element is already *ACC*-marked and still triggers *ERG* wherever it lands, this would be an instance of the competitor (the moving element) having prior dependent case-marking.

¹¹ This is not the only way to rule out dependent case on the subject in OSV topicalizations in Baker’s (2015b) system. I have counted four reasons: 1. The competitor (the topicalized object) is already valued as [case: *DEP*] and thus can’t compete; 2. The competitor has become an adjunct, and adjuncts can’t compete against arguments (Baker 2015b: §6.2.2–6.3.2, pp. 256–272); 3. The undergoer (the subject) has already been valued for unmarked case [case: *UNM*] and might not be able to be revalued (this is unclear to me, see my discussion in §2.5.3 below); 4. The derivation remembers the prior application of Case Competition when the subject competed against the object leading to [case: *DEP*] valuation of the object, and the derivation doesn’t allow re-application of Case Competition on the same two NPs, even in reversed roles (Baker 2015b: §6.1, pp. 230–252, and my §4).

¹² It might be fruitful to investigate whether there could be an implication here, such as: If a nominal with lexical/inherent case can serve as competitor in a language *L*, a nominal with dependent case can also serve as competitor in *L*. I am not aware of Baker (2015b) discussing this. Crucially, when stating the parameter in question, he uses the phrasing “[NP Case:OBL] is a case competitor” vs. “only [NP Case:____] is a case competitor”, omitting further possibilities (Baker 2015b: 196, ex. (22)). (Baker uses the term “oblique case” to refer to “lexical/inherent case”; (p. 184).)

¹³ Baker (2015b: §5.1.3, pp. 188–194) makes the point that Tamil also has a minimally different *DAT*-‘subject’ construc-

contrasts with DAT-subjects in Kannada, which are like Icelandic DAT-subjects in that they are real subjects (argumental NPs) with lexical DAT-marking, but do not trigger dependent ACC-case on their objects, yielding transitive predicates with a DAT-NOM case pattern — by parameter setting, Baker (2015b: §5.1.4, pp. 194–197) proposes.

Baker (2015b), who assumes contra Preminger (2011, 2014) that NPs bearing an unmarked case are *valued* as such, e.g. as [case: NOM], allows the NP receiving dependent case to change its case value from [case: UNM] to [case: DEP] under certain circumstances. Likewise, an NP already valued as [case: UNM] can sometimes be a competitor in Case Competition. This model has some advantages and some disadvantages compared to Preminger’s (2014) model where unmarked case is unvalued case [case: ____], that I will discuss in §2.5.3 below.

2.5.3 The nature of unmarked case

Baker (2015b) proposes that unmarked cases correspond to a feature value, e.g. [case: NOM] or [case: ABS], which I generalize to [case: UNM], and that this value is assigned to a caseless NP as a “last-resort, if-all-else-fails” (p. 150) option.

In principle, this is not so different from Preminger’s (2014) proposal where NPs that do not receive a lexical, inherent, or dependent case value during the derivation remain caseless, which is then realized morphologically as the corresponding unmarked case.

There is a disadvantage to Baker’s (2015b) proposal. Sometimes an NP does not receive a case value in its original CAD, which should result in the valuation [case: UNM], but when this NP moves into a different CAD, it may receive dependent case there. Baker (2015b: §6.3.3, especially pp. 278–281) discusses that in this case, this NP may re-value its case feature from [case: UNM] to [case: DEP]. He writes, “there is some empirical reason to think that unmarked case assigned to a lower copy in a chain is not automatically inherited by a higher copy in the same chain” (p. 278), and

tion where the DAT-marked nominals are not real subjects but rather embedded inside an adjoined PP headed by a silent P that is responsible for the DAT-assignment. In these constructions, the object does not receive dependent ACC, but appears with unmarked NOM, and the DAT-marked nominals (i) are optional like adjuncts, (ii) cannot be controlled when embedded under ‘want’, rather it is the NOM-object that can be controlled, (iii) cannot bind anaphors, and (iv) cannot be clefted, like adjuncts but unlike arguments.

“the lower member of a multi-membered chain does not receive unmarked case because there is a chance of a higher member of the chain getting a more distinctive case” (p. 279), while in contrast, “[d]ependent case features that are added to one copy in a movement chain automatically belong to all copies in the chain” (p. 272, ex. (65)). He shows that in Finnish, unmarked partitive can be overwritten by dependent *ACC*¹⁴, in Cuzco Quechua¹⁵ and Sakha¹⁶, unmarked *NOM* can be overwritten by dependent *ACC*, and in Japanese, unmarked *GEN* can be overwritten by dependent “marked *NOM*”¹⁷. This contrasts with dependent *ACC* in Sakha¹⁸, dependent *DAT* in Sakha¹⁹, dependent *ERG* in Tongan²⁰, and dependent *GEN* in Cuzco Quechua²¹, which do not get overwritten after movement into a different CAD, Baker (2015b) argues. For Baker, there is thus an inherent stipulation in that the feature value [case: *UNM*] is categorically different from the value [case: *DEP*] in that [case: *UNM*] can be re-valued, but [case: *DEP*] cannot be re-valued. This stipulation is completely unnecessary in Preminger’s (2014) proposal: For Preminger, a nominal that remains case-value-less [case: ____] in its original CAD may at any point in the derivation receive a case-value later whenever such an opportunity arises. This is an immediate and strong advantage to Preminger’s (2014) approach to unmarked case.

However, Baker (2015b) presents data that allegedly support his treatment of unmarked case as valued case precisely because an NP with unmarked case does not re-value its case with dependent case when we might expect it to. He argues that by making modifications to his stipulation of when [case: *UNM*] can be revalued, he can account for these data. Under Preminger’s (2014) proposal, a novel mechanism that would prohibit this late valuation seems necessary. I will first present Baker’s (2015b) findings and then show that there might be additional factors at play such that Preminger’s (2014) theory remains feasible even for these data.

The relevant data come from bare-NP adverbs in Cuzco Quechua and Diyari, which can par-

¹⁴ Baker 2015b: p. 278, ex. (71)

¹⁵ Baker 2015b: p. 279, ex. (73)

¹⁶ Baker & Vinokurova 2010, see also §3.4.

¹⁷ Baker 2015b: p. 280, ex. (74)

¹⁸ Baker 2015b: p. 272, ex. (64)

¹⁹ Baker 2015b: p. 273, ex. (66a), but see my chapter 5 on arguments that *DAT* in Sakha is not a dependent case.

²⁰ Baker 2015b: p. 274, ex. (67)

²¹ Baker 2015b: p. 281, ex. (76)

ticipate in Case Competition. (7) illustrates dependent ACC-marking on the adverb ‘tomorrow’, assigned by Case Competition against the subject ‘Juan’ (Baker 2015b: p. 260, ex. (44)). Baker (2015b: pp. 215–217) illustrates in detail that this ACC is in fact dependent case. (7) can be accounted for under two theories: that ‘Juan’ might be unvalued for case; or that ‘Juan’ might be valued as [case: UNM] and allowed to act as competitor with this value.

- (7) a. *Xwancha paqarin -ta Lima -man ri -nqa.*
 Juan.NOM tomorrow -ACC Lima -to go -3SG.FUT
 ‘Tomorrow Juan will go to Lima.’ *Cuzco Quechua*²²
- b. ??*Xwancha paqarin Lima -man ri -nqa.*
 Juan.NOM tomorrow Lima -to go -3SG.FUT
 ‘Tomorrow Juan will go to Lima.’ *Cuzco Quechua*²³

In (8a–b), the adverb ‘tomorrow’ precedes the subject. We might expect that the adverb might now be in a position c-commanding the subject and might thus compete against it, valuing or re-valuing ‘Juan’ as [case: ACC]. However, this does not happen, (8b); instead the subject must appear with unmarked case, (8a), as it would in the absence of the adverb.

- (8) a. *Paqarin Xwancha Lima -man ri -nqa.*
 tomorrow.NOM Juan.NOM Lima -to go -3SG.FUT
 ‘Tomorrow Juan will go to Lima.’ *Cuzco Quechua*²⁴
- b. **Paqarin Xwancha -ta Lima -man ri -nqa.*
 tomorrow.NOM Juan -ACC Lima -to go -3SG.FUT
*Cuzco Quechua*²⁵

Baker (2015b: p. 260, ex. (43)) proposes that this happens because adverbs, and more generally adjuncts, are evaluated by the case algorithm in a later cycle than arguments. Thus, the argument-only cycle will fixate ‘Juan’ as [case: NOM], and when the argument-and-adjunct cycle tries to apply Case Competition between ‘tomorrow’ and ‘Juan’ in (8), it cannot re-value ‘Juan’ from [case: NOM] to [case: ACC].²⁶ A similar construction is found in the tripartite language Diyari, Baker (2015b:

^{22,23} Lefebvre & Muysken 1988: p. 50, cited after Baker 2015b: p. 215, ex. (59b), p. 256, ex. (37b).

²⁴ Lefebvre & Muysken 1988: p. 50, cited after Baker 2015b: p. 215, ex. (59a), p. 256, ex. (37a).

²⁵ Lefebvre & Muysken 1988: p. 50, cited after Baker 2015b: p. 256, ex. (37a).

²⁶ In the early parts of Baker (2015b), he speculates whether unmarked case is special in that it can be overwritten by

p. 257, ex. (39), p. 260, ex. (45)) proposes. In a regular transitive clause, (9), the subject ‘old man’ is marked ERG and the object ‘boomerang’ is marked ACC.

- (9) *Pinada -li kira anka -yi malka -li .*
 old.man -ERG boomerang.ACC make -PRS mulga -ERG
 ‘The old man is making a boomerang from mulga.’ *Diyari* ²⁷

When the predicate contains a cognate object, however, Austin (1981) argues and Baker (2015b) employs, such as ‘speak (a language, a word)’ in (10), the cognate object ‘language’ is still ACC-marked, but the subject is no longer ERG-marked, but appears with NOM-marking instead.

- (10) *ḡaṇi [ṇiṇa -ya yawada] yaṭa -yi .*
 me.NOM 3^{SG.NF}.ACC -near language speak -PRS
 ‘I speak this language.’ *Diyari* ²⁸

Baker (2015b) proposes that, as has been claimed for other languages, cognate objects relate to manner adverbs and thus behave like adjuncts. This then allows Baker (2015b) to employ a single account for Cuzco Quechua and Diyari: the subject is fixated as [case: UNM] at the end of the intransitive arguments-only cycle and cannot be re-valued by Case Competition against an adjunct.²⁹ In contrast with this account, for Preminger’s (2014) account of unmarked case deriving the ungrammaticality of (8b) seems hard, if not impossible. What might prohibit valuation of ‘Juan’ as [case: ACC] and of ‘old man’ as [case: ERG] in competition against the adverb resp. cognate object?

I would like to propose that these data should not be considered an argument for Baker’s (2015b)

dependent case, or whether unmarked case is special in that it does not yet get assigned when the nominal may still undergo further competitions or movements into a different CAD. At this point, however, it seems he has decided that overwriting of unmarked case with dependent case is not an option.

²⁷ Austin 1981: p. 120, cited after Baker 2015b: p. 147, ex. (56).

²⁸ Baker 2015b: p. 257, ex. (39), presumably after Austin (1981: p. 116). See also Baker 2015b: p. 257, fn. 25 on a correction to Austin’s (1981) glosses.

²⁹

[...] Spell Out is structured as follows: first heads, specifiers, and complement are ordered, then morphological merger under adjacency takes place, then adverbs are interleaved into the string at the proper places. [...] Now suppose that dependent case assignment also happens first at roughly the same point in the structured spell out process that morphological merger happens: when heads, complements, and specifiers are present, but before adjuncts are. [...] the case of the arguments will generally not change, because it has already been assigned and cannot be reassigned [...]. On this view, the asymmetry between arguments and adjuncts in terms of case marking reduces to a difference in the timing of when they are incorporated into the representation, an idea that is also relevant to other kinds of argument-adjunct asymmetries. (Baker 2015b: p. 259)

proposal over Preminger’s (2014) for two reasons: First, the data from Cuzco Quechua are on closer inspection not actually more problematic for Preminger (2014) than for Baker (2015b) (I have nothing to say about cognate objects at this point); and second, data from Koryak reported by Abramovitz (2018, 2020) go against Baker’s (2015b) proposal that adjuncts are evaluated on a later cycle than arguments. I discuss these two arguments in turn.

Regarding Cuzco Quechua, I suspect that it is the details of this construction that may hold the answer to how Preminger’s (2014) proposal could account for these data. Let us assume that the different word orders between (7a) and (8) come with (perhaps subtle) further differences, such as topicalization. There might be intermediate scrambling-like movement steps that create a c-command configuration where the subject c-commands the adverb even when the surface order is adverb-subject as in (8). Then, the subject would value the adverb as [case: DEP] before the adverb undergoes further movement above the subject, and, as discussed above, this value then prohibits the adverb from acting as case-competitor against the subject. Evidence that this might be the correct direction to go comes from examples like (11), which Baker (2015b) mentions but does not discuss further:

- (11) *Paqarin -ta Xwancha Lima -man ri -nqa.*
 tomorrow -ACC Juan.NOM Lima -to go -3SG.FUT
 ‘Tomorrow Juan will go to Lima.’ *Cuzco Quechua*³⁰

If the ACC-marking on the adverb in (11) is structural (dependent) and assigned by Case Competition against the subject, the adverb must have been in a position c-commanded by the subject at the time of Case Competition, and must have undergone its movement to a position higher than the subject only after receiving its ACC-value. If this is correct and Case Competition applies before the adverb moves to sentence-initial position in (11), then it is not far-fetched to say that this ordering of operations also happens in (8b), accounting for the absence of ACC-marking on the subject. The astute reader will notice that there seems to be optionality in whether the adverb ‘tomorrow’ preceding the subject ‘Juan’ carries ACC, (11), or NOM, (8a). I tentatively suggest that the optionality

³⁰ Lefebvre & Muysken 1988: p. 50, cited after Baker 2015b: p. 215, ex. (59a).

lies in the adverb's first-Merge position: while in (11), as discussed above, 'tomorrow' is has a landing site below the subject, in (8a) it may be base-generated so high that it is outside the clausal CAD altogether, bleeding the application of Case Competition.

As Abramovitz (2018, 2020) has shown in detail, Koryak patterns differently from Cuzco Quechua and Diyari. For Baker's (2015b) assumptions about the timing of unmarked case assignment to work, he would need to assume either that it is subject to parametric variation, differentiating between Cuzco Quechua and Diyari above on the one hand (where adjuncts in the sense of adverbs cannot act as competitors against arguments), and Koryak on the other hand (where adjuncts in the sense of intermediate landing sites can); or that there is a categorical difference between adverbs (Cuzco Quechua) and cognate objects (Diyari) on the one hand, and intermediate landing sites of successive-cyclic movement (Koryak) on the other hand. I now present the Koryak data in question.

In the Chukotko-Kamchatkan ergative language Koryak, a successive-cyclically moving ABS-marked wh-object can trigger dependent-up case-marking (ERG and DAT) in all its intermediate landing sites (Spec,CP and Spec,vP). An example of this is illustrated in (12) below: In the declarative sentence, (12a), the matrix subject 'I' of the (intransitive) matrix predicate 'hear' is marked ABS. In the interrogative version, (12b), with the wh-element moving to sentence-initial position, the subject of the same intransitive predicate 'hear' must appear with ERG-marking instead of ABS. Abramovitz's (2020) proposed analysis for this Case Competition is illustrated in Fig. 2.6.³¹

- (12) a. γammo $t\text{-}\partial$ valom $\text{-}\partial\text{-}k$, ano $\text{?ew}\eta\text{ato}$ -na $\text{-}k$ $\emptyset$ -
 me._[ABS] 1SG.S/A-EP- hear -EP-1SG.S that Hewngyto -OBL.SG -ERG 2/3.S/A.IND-
 $j\text{-}\partial$ $t\hat{c}im$ -aw -nin $koj\eta$ $\text{-}\partial\text{-}t$.
 CAUS-EP- break -VBLZ -3SG.A>3.O cup -EP-ABS.DU
 'I heard that Hewngyto broke two cups.' Koryak³²
- b. jej $\text{-}i$ { $^{\text{ok}}\gamma\partial$ -nan / $*\gamma\partial t\hat{c}i$ } $\emptyset$ - valom -na $\text{-}t$, ano
 what -ABS.DU { $^{\text{ok}}\text{you}$ -[ERG] / $*\text{you}$ [ABS] } 2/3.S/A.IND- hear -3.O -3DU that
 $\text{?ew}\eta\text{ato}$ -na $\text{-}k$ $\emptyset$ - $j\text{-}\partial$ $t\hat{c}im$ -aw -nin
 Hewngyto -OBL.SG -ERG 2/3.S/A.IND- CAUS-EP- break -VBLZ -3SG.A>3.O

³¹ Jonathan Bobaljik (p.c.) points out to me that for this argument to go through, an alternative declarative construction with ERG-marking on the subject due to prolepsis resp. LDA would need to be shown to be unacceptable. I do not currently have that data available.

³² Abramovitz 2020: p. 17. ex. (27b)

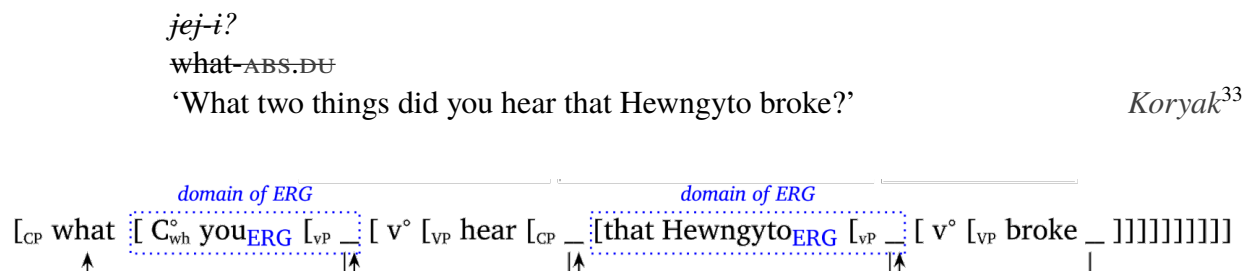


Figure 2.6: The analysis of (12b) by Abramovitz (2020: p. 23, ex. (44e))

I conclude from this that Baker’s (2015b) proposal that unmarked case corresponds to a case value [case: UNM] is not well-founded, and will assume Preminger’s (2014) proposal that unmarked case is unvalued case [case: ____] instead.

2.5.4 Restrictions on the nominalness of the competing NP

Another property of nominals that can be relevant for whether they can act as case-competitors is their nominal-ness. Baker (2015b: §5.2) discusses the (non)-availability of nominalized clauses to enter Case Competition as recipients or competitors, and Baker (2015b: §5.3) argues that phonetically null NPs such as *pro* or *PRO* can be ordered on a hierarchy according to their nominal-ness, and the same hierarchy is used in parameter setting to specify the cut-off point for what degree of nominal-ness is required resp. sufficient to enter Case Competition. This is illustrated in Fig. 2.7. Further languages that might be added to this diagram are Ukrainian allowing the implicit *AGENT* of passives to act as case-competitor against the *THEME* (Lavine 2013; Sobin 1985), and Icelandic (Maling & Sigurjónsdóttir 1997, 2002) and German (Laszakovits 2017b) where the implicit *AGENT* of passives does not suffice, but uncontrolled *PRO* does.

2.5.5 Case-assignment domains

Baker (2015b: §4) proposes that case-assignment domains (CADs) correspond to Spell-out domains, i.e., the complements of phase heads, which, following Chomsky (2000, 2001), he assumes are *C*, *v*, *D*, and some *Ps*, among possibly others, such as *Poss* and *Pred* (Baker 2015a, 2015b:

³³ Abramovitz 2020: p. 17, ex. (27a)

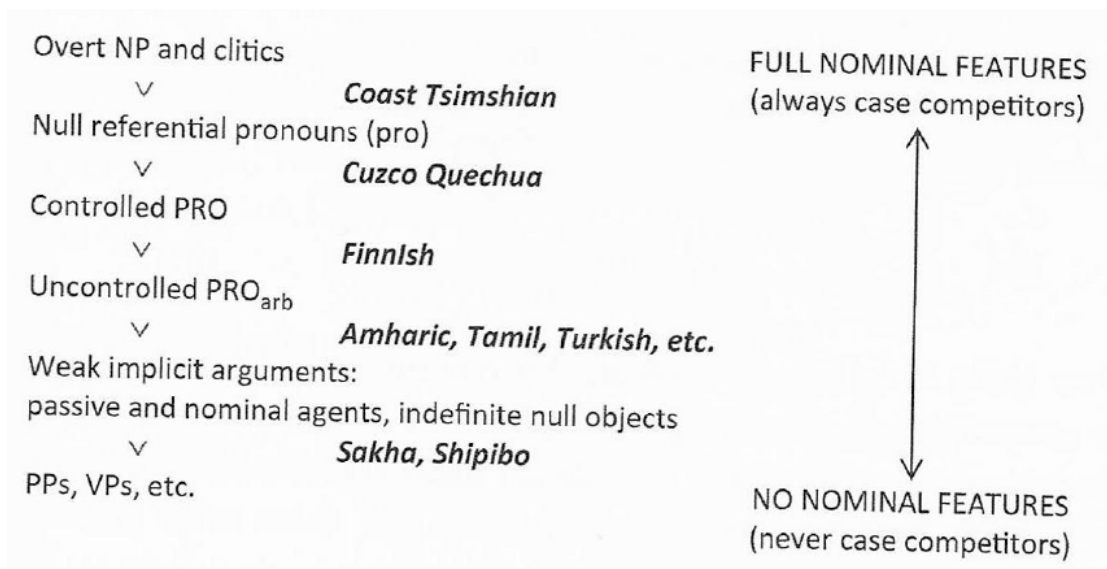


Figure 2.7: Baker's (2015b: p. 211, ex. (52)) hierarchy of features of the competing NP

p. 177–180). He argues that the evidence that CADs and Spell-out domains align is found because variations in case marking correspond with the presence and absence of phase heads as diagnosed by independent criteria such as restructuring.

In Cuzco Quechua infinitives, the embedded object is marked with ACC or unmarked for case, depending on the embedding predicate. Baker (2015b) discusses the proposal by Lefebvre & Muysken (1988), who state that this is an effect of restructuring as evidenced by the clustering of the following properties: clitic climbing, attachment of a validation clitic, NP scrambling, wh-movement, quantifier float, and negation. Following Wurmbrand's (2003) theory of restructuring as truncation, whereby restructuring predicates embed a non-phasal verbal phrase, and non-restructuring predicates embed (up to) a phasal CP, Baker (2015b) understands this as an argument that CADs are sensitive to phase heads in the sense that they correspond to Spell-out domains.

Likewise, in Shipibo infinitives selected by non-restructuring predicates (such as *shinan*- 'think'), the matrix subject is always marked ERG, irrespective of the embedded predicate's argument structure, but with restructuring matrix predicates (such as *atipan*- 'can', 'be able'), the subject's case-marking is sensitive to the embedded predicates transitivity. This suggests, Baker (2014b, 2015b) argues, that non-restructuring predicates introduce a separate CAD, while with restructuring pred-

icates, the matrix and embedded CADs merge, which corresponds to the observed phasehoods, as diagnosed for Shipibo by the distribution of switch-reference markers, (13), desiderative constructions, and progressive clauses.

- (13) a. # [*Jose -kan* [*Rosa oin-ti*] *shinan -a*]-*ra* , *e-a kena-ke* .
 José -ERG Rosa.ABS see-INF think -O=S -PRT me-ABS call -PRF
 [#O] O=S S
 intended: ‘When José thought to see Rosa, she (Rosa) called me.’ *Shipibo*³⁴
- b. [*Jose -kan Rosa kena-ti atipan -a*]-*ra* , *bene -ke* .
 José -ERG Rosa.ABS call -INF can -O=S -PRT be happy-PRF
 O O=S S
 ‘Because José could call Rosa, she (Rosa) was happy.’ *Shipibo*³⁵

That CADs should correspond to Spell-out domains is a good first hypothesis under Chomsky’s (1995) Minimalist considerations. However, Poole (2022) contests it, offering evidence from Finnish where he argues that CADs and Spell-out domains do not align: In Finnish *-ma* infinitives, the matrix subject can enter Case Competition against the embedded object inside the infinitive, as (14a,b) illustrates. The ACC-marking on the embedded object in (14a) is dependent on the presence of a matrix subject because in the imperative version (14b), without a matrix subject, the case marking on the embedded object changes from dependent (ACC) to unmarked (NOM). The same case change can be observed when removing the matrix subject through passivization, or by replacing the matrix predicate with a predicate that assigns a lexical (quirky) case to its external argument, Poole (2022: p. 365) writes.

- (14) a. *Hän kävi* [*avaa -ma -ssa ove -n*].
 s/he.NOM went open -INF -INE door -ACC
 ‘S/he went to open the door.’ *Finnish*³⁶
- b. *Käy* [*avaa -ma -ssa ovi*].
 go.IMP open -INF -INE door. NOM
 ‘Go open the door!’ *Finnish*³⁷

With a ditransitive matrix predicate, selecting a GOAL in addition to the infinitive, it is still the matrix subject that licenses the dependent case on the embedded object, and it cannot be the matrix

³⁴ Baker 2015b: p. 123, ex. (20a)

³⁵ Baker 2015b: p. 123, ex. (21a)

³⁶ Poole 2022: p. 365, ex. (35a)

³⁷ Poole 2022: p. 365, ex. (35b)

object that does so, as the following contrast shows. If the matrix object ‘Jukka’ were a potential competitor against the embedded object ‘book’, (15b) should allow ACC marking on ‘book’, which it doesn’t.

- (15) a. *Maija pyysi Jukka -n [luke-ma-an kirja -n]*.
 Maija.NOM asked Jukka -ACC read-INF-ILL book -ACC
 ‘Maija asked Jukka to read the book.’ Finnish³⁸
- b. *Pyydä { Jukka / *Jukka -n } [luke-ma-an { kirja / *kirja -n }]!*
 ask.IMP Jukka.NOM Jukka -ACC read-INF-ILL book.NOM book -ACC
 ‘Ask Jukka to read the book!’ Finnish³⁹

Under the theory that CADs are Spell-out domains, two possible reasons for this come to mind, both of which are wrong, Poole argues. First, it might be that the matrix object cannot serve as competitor in the first place, for an as-of-yet unknown reason. However, famously (Kiparsky 2001; Maling 1993a), in Finnish, some adverbs can receive dependent ACC case by competition against the object. This is still possible here: (16) shows that the matrix object ‘Jukka’ can compete against lower matrix adverb ‘for a third time’, assigning the adverb dependent ACC, and disproving the hypothesis that the matrix object cannot serve as competitor at all.

- (16) *Pyydä Jukka [kolmanne-n kerra-n] [luke-ma-an kirja]*.
 ask.IMP Jukka.NOM third -ACC time -ACC read-INF-ILL book.NOM
 ‘Ask Jukka for the third time to read the book!’ Finnish⁴⁰

Second, it might be the case that the matrix object simply does not c-command the infinitive, voiding the application of Case Competition between them. However, this is not so either, Poole argues, because the matrix object can bind into the infinitive, and binding requires c-command. In (17), the embedded object ‘record’ is marked by a 3SG possessive suffix *-nsä* that is subject to Condition A, Poole claims citing (18) as evidence. From the fact that in (17), the suffix can reference the matrix object (index 2), it must follow that the matrix object ‘Pekka’ c-commands the embedded object ‘record’ inside the infinitive, he claims.

- (17) *Maija₁ pyysi Peka₂ -n [tuo -ma -an levy -nsä_{1/2/*3}]*.
 Maija.NOM asked Pekka -ACC bring -INF -ILL record.ACC -POSS.3SG

³⁸ Poole 2022: p. 366, ex. (36b); Vainikka 1989: p. 267

³⁹ Poole 2022: p. 366, ex. (39b); Vainikka 1989: p. 268

⁴⁰ Poole 2022: p. 368, ex. (42a); Maling 1993b: p. 69

- ok ‘Maija₁ asked Pekka₂ to bring Maija₁’s record.’ *Finnish*⁴¹
 ok ‘Maija₁ asked Pekka₂ to bring Pekka₂’s record.’
 # ‘Maija₁ asked Pekka₂ to bring Liisa₃’s record.’ (where Liisa is some third person given in the discourse)
- (18) *Poika₁ myi marsu -nsa₁/*₂ .*
 boy.NOM sold guinea.pig.ACC -POSS.3SG
 ok ‘The boy₁ sold his₁ own guinea pig.’ *Finnish*⁴²
 # ‘The boy₁ sold Liisa₂’s guinea pig.’ (where again, Liisa is some third person given in the discourse)

The problem with the theory that CADs are Spell-out domains, Poole (2022) argues, is this: the accessibility of Spell-out domains, managed by the PIC, is assumed to be “binary” in the sense that the content of the Spell-out domain is accessible either to all operations in the higher phase, or to none. Here, this doesn’t work: the infinitive must remain accessible to Case Competition against the matrix subject but not to Case Competition against the matrix object, both of which c-command it and qualify as competitors. Poole (2022) proposes that instead of phases, operations in narrow syntax, which includes Case Competition but does not include pure-LF-operations like binding, should be subject to the Williams Cycle (van Riemsdijk & Williams 1981; Williams 1974, 2003, 2013): “Within the current XP, a syntactic operation may not target an element across YP, where Y is higher than X in the functional sequence.” (Poole 2022: p. 383, ex. (77), p. 388, ex. (87))

Notwithstanding Poole’s (2022) argument, I will assume that case-assignment domains correspond to Spell-out domains, and that there may be an alternative account of the Finnish binding examples (17)–(18) that does not include c-command of the matrix object into the infinitive, which I leave as an open problem. As things stand at this point in my investigation, I am not convinced that appealing to the Williams Cycle instead of Spell-out domains is really necessary, and I will continue to work with Spell-out domains as CADs in the way that Baker & Vinokurova (2010) and Baker (2014b, 2015a,b) propose.

I will close this section by saying that den Dikken (2023) has proposed to use Spell-out domains as CADs in a different matter from Baker: den Dikken (2023: p. 492, ex. (15)) proposes that the Spell-

⁴¹ Poole 2022: p. 367, ex. (40b); Vainikka 1989: p. 270

⁴² Poole 2022: p. 367, ex. (40a); Nelson 1998: p. 187

out domain (and thus the CAD) encompasses the entire phase, including the phase projection's specifier and adjuncts (i.e., not just the phase head's complement), and that any material in non-terminal positions at the phase edge, as well as the left-most spelled out element of the phase, remain accessible to the next Spell-out domain and CAD.⁴³

2.5.6 The timing of case assignment

There is variation among implementations of DCT with respect to when exactly during the derivation the operation Case Competition applies. The earliest possibility is at the time when the second (structurally higher) nominal gets Merged into the structure. This is what Baker & Vinokurova (2010), Preminger (2014), and Poole (2022) propose. Baker & Vinokurova (2010) write, “[w]e assume that the rules [of DCT] apply immediately when the configurations they describe are first created” (p. 604). They use this assumption to account for a problem raised by Legate (2008: p. 89): objects scrambling above subjects to a position where the scrambled object will c-command the subject, yielding OSV order, crucially does not affect the dependent case marking on neither object nor subject. Baker & Vinokurova (2010) account for these constructions by saying that dependent case assignment (the operation Case Competition) applies before this movement happens, namely when the subject is Merged at which point the subject c-commands the object, and not vice-versa. While for Baker & Vinokurova (2010) this proposal is a means to capture the data under discussion (and is revised in Baker 2015b, see below), Poole (2022) makes an argument that Case Competition must necessarily apply “as early as possible, a principle that I called Earliness” (p. 387). He motivates this through other aspects of his proposal: The Williams Cycle applies in syntax — on the one hand because it regulates movement, which is syntactic, and on the other hand because it refers to syntactic Labels. By simplicity, Poole argues, all other operations that the Williams Cycle regulates should also be assumed to apply in the syntax, and his paper argues that the application of Case Competition is regulated by the Williams Cycle. Similarly, the proposal by Preminger (2014),

⁴³ Den Dikken's (2023) purpose in developing his theory is to account for the case patterns of applicatives of unaccusatives in Shipibo and Nez Perce in a unified manner, and without making recourse to stipulating a silent PP, as Baker (2014b, 2015b) has proposed, that encompasses the applied argument and thereby bleeds a c-command relationship and also Case Competition between the applied argument and the *THEME*.

which argues carefully that φ -agreement is sensitive to case and that φ -agreement can feed syntactic operations, draws from this the conclusion that because Case Competition feeds φ -agreement, and φ -agreement is part of syntax proper, Case Competition, too, must be part of syntax proper. As far as I can tell, however, Preminger (2014) does not take a stance on exactly when during syntax Case Competition must apply.

An alternative and frequently seen proposal is that Case Competition applies during Transfer from syntax to PF (Chomsky 2004: p. 107), known as Spell-out, which leads to the prediction that case-assignment domains correspond to Spell-out domains. The exact predictions that this proposal makes depend, of course, on the particulars of one's theory of phases and Spell-out. Baker (2015b), following Chomsky (2000, 2001), assumes that Spell-out domains are the complements of phase heads. Borrowing Chomsky's terminology, Baker (2015b) proposes with Legate (2003) that the phase head v always triggers the Spell-out of its complement, but can differ parametrically between languages between being a "strong" or a "weak" phase head, which Baker (2015b) understands to mean whether the Phase Impenetrability Condition (PIC) applies to its Spell-out domain (strong), or not (weak). Baker uses this mechanism to distinguish languages with DOM from languages without DOM; which I will return to in §3.3. Den Dikken (2023) also takes Case Competition to apply at Spell-out, but his theory of phases differs from Baker's (2015b) significantly. As mentioned in the previous subsection, for den Dikken, upon completion of the projection of a phase head, the entire phase (including its specifier(s) and adjoined edge positions) is sent to PF, where first, linearization happens (the decision which copies of a movement chain get realized, following Fox & Pesetsky 2005) and then Case Competition applies and considers only elements with a morpho-phonological realization. The elements at the phase's edge down to the first spelled out element remain accessible to the next application of Spell-out. Den Dikken (2023) shows that this set of assumptions, together with constraints on antilocality and remnant movement, can account for the case patterns in applicatives of unaccusatives in Shipibo (Baker 2014b) and Nimiipuutímt (also known as Nez Perce) (Deal 2019) in a unified way and without appealing to Baker's (2015b) "strong" and "weak" phase heads. However, from my understanding of the mechanisms in den Dikken's (2023) paper,

while it derives the case patterns of applicatives of unaccusatives, I fail to see how it could account for something like the case preservation during object-topicalization past the subject: If only the copies in final resp. criterial positions are realized, and realization feeds Case Competition, at the time of Case Competition the object c-commands the subject, and thus it should be the subject that receives ACC-marking resp. the object that receives ERG-marking, contrary to fact.

A third option is that Case Competition applies even later, during Vocabulary Insertion of the respective NP. This was the original view taken by Marantz (1991) and supported by McFadden (2004) and Bobaljik (2008) based on lack of evidence to the contrary and because at the time, a major point was to contrast DCT with “abstract Case theory” as advanced by Vergnaud (1977) and much work since, as discussed in §2.3 above.

2.5.7 Summary

In this section, I have discussed principled and parametric variation in some recent implementations of DCT and the data it derives in different languages.

Subsection 2.5.2 has discussed the common assumption that the competing nominal must not yet have been case-marked with structural case, but can in some languages have been marked with lexical case.

This raises the question of whether unmarked case is its own case value, as Baker (2015b) assumes, or merely the realization of the absence of a value, as in Preminger’s (2014) proposal, and I have discussed the advantages and problems of both proposals in §2.5.3.

In §2.5.4 I briefly mention parametric variation in the necessary nominalness of the competitor.

Subsection 2.5.5 has dealt with the nature of CADs, which are often assumed to correspond to Spell-out domains, but Finnish might be a counter-example, pending further data.

§2.5.6 has discussed three major proposals regarding when in the derivation the operation Case Competition applies: as soon as the configuration is met, at Spell-out, or at Vocabulary Insertion.

2.6 The flavor of DCT assumed in this work

In this thesis, I will assume a formalism of DCT that captures (i) the analysis by Baker (2015b) and Baker & Vinokurova (2010) for Turkic ACC, and (ii) the analysis by Kornfilt & Preminger (2015) and Levin & Preminger (2015) for Turkic NOM, together with (iii) the syntactic implementation of DCT in Minimalism proposed by Preminger (2014, 2024). This section spells out exactly how this plays together.

Nominals (the highest projection in the nominal Phase, whichever projection this turns out to be) carry a syntactic feature [case: ____] that enters the derivation unvalued. This feature may receive a value in the course of the derivation, but does not have to. A value can be assigned by a syntactic head under local c-command; this subsumes what has been called “lexical”, “idiosyncratic”, “quirky”, and “semantic” case, where the assigning head is a lexical element, such as Icelandic DAT-marked subjects (Zaenen, Maling & Thráinsson 1985) and German DAT-marked objects in simple transitives, and it also includes “inherent” case, where the assigning head is an element that provides a thematic interpretation, such as the functional head Appl assigning DAT case to the applicative argument (Pylkkänen 2008, see also §5.3.2). The assigning head can be the same head that first-Merges the nominal, or a different head. For instance, Preminger (2024: §4.3.1.2, pp. 84 f.) discusses two possible analyses of English nominative (i.e. the distribution of the words *I* (except after *and*), *she*, *he*, *we*, and *they*) that involve assignment of a case value by a head that is not the Merging head. His first proposal (Preminger 2024: pp. 80 f.) is that in English, NOM is the unmarked case, and ACC is both a dependent case and also an idiosyncratic case assigned by prepositional complementizers in ECM constructions. Thus, in the example in (19), the prepositional complementizer *for* assigns idiosyncratic ACC to *them* inside its complement under local c-command without for being responsible for first-Merging *them*.

- (19) [*For* [*them to win*]] *would be impossible.* *English*

Preminger’s alternative proposal for English NOM (§4.3.1.2, pp. 84: f.) is the opposite, namely that

ACC is the unmarked case, and NOM is an idiosyncratic case assigned by finite T to its specifier⁴⁴, where, again, T is not the head that first-Merges *them*. Thus, Preminger (2024) proposes to call this case “head case” to tease out the commonality, but I will continue to use the established (although somewhat vague, see Woolford 2006) terms “lexical case” and “inherent case”.

Nominals that have not received a case value in or shortly after they were first-Merged (but see §2.5.2 above) may receive a value through the application of the operation Case Competition in the course of the derivation.⁴⁵ I follow Baker (2015b), Marantz (1991), and Preminger (2014), i.e., in assuming that the mechanism by which the value [case: DEP] is assigned is a mechanism independent of the standard Minimalist triple Merge, Move, and Agree, contrary to the proposals by Bittner & Hale (1996) and Poole (2024) who attempt to formalize dependent case with existing operations, which I will return to in §3.6. Rather, dependent case is assigned by the mechanism that Marantz’s (1991) has described as (20) and that is often referred to as “case competition”⁴⁶

- (20) Dependent case is assigned by V+I [after Merge of the phase head defining the clausal case-assignment domain, i.e., C] to a position [nominal] governed by V+I [inside this domain] when a distinct position governed by V+I is:
- a. not “marked” (not part of a chain governed by a lexical case determiner) [not yet valued for (lexical) case; see §2.5.2]
 - b. distinct from the chain being assigned dependent case

⁴⁴ Technically, this is not c-command, but m-command (Aoun & Sportiche 1983).

⁴⁵ Preminger (2014) proposes that this value is [case: DEP] (for “dependent”) rather than a case-specific value such as [case: ACC]. The advantage of using [case: DEP] is that if a language uses the same case paradigm for a lexical case and a dependent case, syntax may still need to distinguish the nature of this case marking for constructions such as passivization, which may be possible only for dependently case-marked nominals, and not for lexically case-marked nominals.

For tripartite languages with a dependent-up and a dependent-down case, syntax would obviously need to distinguish between these values. Similarly, under Podobryaev’s (2013) system of DAT as “intermediate” dependent case, i.e., a dependent case that is assigned to the middle one of three nominals, more than one value for dependent cases will be necessary as well.

Taking the value assigned by Case Competition as [case: DEP] makes a prediction: Take a language with different dependent cases defined in different case-assignment domains (CADs) (Baker 2015b). Let *N* be a nominal that receives the value [case: DEP] in some CAD *D*₁ and subsequently moves to a different CAD *D*₂ where it is spelled out. Then, if *N* is valued as [case: DEP], we expect *N* to carry the morphology associated with the dependent case defined for *D*₂, even though *N* Case Competition applied in *D*₁. I am unaware of any such data, thus I consider this prediction untested.

⁴⁶ To my knowledge, the first usage of this term was by Bittner & Hale (1996).

Dependent case assigned up to subject: ergative

Dependent case assigned down to object: accusative

(Marantz 1991: p. 25, ex. (30))

I will understand case-competition as developed by Preminger (2014) in the contemporary Minimalist framework and as outlined in §2.4.5 above: During the derivation, the operation Case Competition may apply in a configuration where two as-yet-caseless nominals stand in a local c-command configuration (i.e., in the same case-assignment domain). I will tentatively assume that Case Competition applies upon Transfer, i.e. when a Phase is closed off and its Spell-out domain is shipped to the interfaces. Case Competition may value the higher nominal's case feature as “dependent-up” case (typically ERG) and/or the lower nominal's as “dependent-down” case (typically ACC) — see Deal (2015) on tripartite languages and Baker (2014a) on languages without dependent cases. At PF, upon Vocabulary Insertion, the case value will be realized by the corresponding case paradigm.

Finally, if at the end of the derivation a nominal has not had its case feature valued [case: ____], it will be realized morphologically with the case inflection of the unmarked case specified for the case-assignment domain it is in. There is no kind of penalty for a derivation containing nominals with unvalued [case: ____].

2.7 Conclusions

This chapter has served to set the theoretical background for the arguments that will be presented in this thesis. I have motivated the interest in case in general (section 2.2), presented the place of Dependent Case Theory in the spectrum of case theories (§2.3), and given an overview over the historical development of DCT (§2.4). Then, I have proceeded to discussing the State of the Art of research on the details of DCT. While §2.4.5 has given an introduction into a common formalization (Preminger 2014), section 2.5 has discussed in detail several open questions about how to formalize the principles and parameters of DCT. Building on these considerations, I have then in §2.6 summarized the assumptions about DCT on which I will build my arguments in this

thesis, i.e., the present “flavor” of DCT.

Chapter 3

The relevance of dependent-down case assignment for syntactic theory

3.1 Introduction

In this chapter I discuss accusative (ACC) case in Turkic languages as an example of dependent-down case assignment, i.e., dependent case assignment to the *lower* one of two NPs in a c-command configuration.

In section 3.2 I show that ACC in Turkic is a structural case, as defined in §2.2, i.e., either a dependent or an unmarked case.

In sections 3.3 and 3.4 I review two constructions that illustrate the dependency of ACC case on the presence of a c-commanding NP in the relevant case-assignment domain (CAD): both NPs must be in the Spell-out domain of a phasal C head. In Differential Object Marking (DOM) constructions (§3.3), ACC is assigned only if the direct object is a topic and moves out of the Spell-out domain in the complement of Voice into the Spell-out domain in the complement of C, where it is c-commanded by the subject NP.

In §3.4 I discuss ‘raising to ACC’ constructions, where subjects of embedded clauses receive ACC marking in the matrix CAD — iff there is a c-commanding NP in the matrix domain. Again, this

movement to the edge of the embedded clause is topicalization movement, and the position at the edge makes the low subject accessible to the matrix clause for case assignment, but also for binding, etc. I review data by Baker & Vinokurova (2010) that illustrate the assignment of two instances of ACC that cannot be explained other than that the assigner is the c-commanding subject.

In §3.5 I present a novel proposal about the assignment of ACC to the complement of some nominals, arguing that these nominals are nominalizations of verbs and that ACC is assigned iff the nominalization contains a c-commanding NP: PRO.

In §3.6 I turn to the implications of the assignment of multiple instances of ACC case by the same licensing/assigning NP. I argue that it differentiates the two proposals from §2.4.5 about how to implement DCT in Minimalism, in favor of adding the operation Case Competition to the list of core Minimalist operations.

§3.7 concludes this chapter.

3.2 Turkic accusative is a structural case

3.2.1 Introduction

We start by establishing that accusative (ACC) in Turkish and Sakha, but presumably in all Turkic languages, is a structural, and not a lexical case. As introduced in §2.2, this means that ACC appears in particular syntactic configurations, and not in dependency of particular lexical or functional items. In other words, the rules of ACC assignment are rules in the syntax, and not rules that refer to information stored in the lexicon.

This section presents four diagnostics of the structural case properties of accusative: four configurations, where ACC is assigned independently of lexical information. This is to show that in these configurations, ACC is a structural case. However, we also need to make sure that there are not additional usages of this ‘ACC’ morphology, i.e., two (or more) cases that are homophonous — chapter 5 will show that this is what happens with Turkic dative (DAT). To show that in addition to its appearance as structural case, ACC morphology doesn’t also have a usage as, e.g., lexical case,

we can only provide the absence of evidence of a lexical case assignment rule, which I am unaware of in Turkic.^{1,2,3}

In §3.2.2 I show that ACC disappears under passivization, a syntactic operation. Section 3.2.3 introduces the appearance and disappearance of ACC in Differential Object Marking (DOM), which will be discussed in more detail in §3.3. Similarly, section 3.2.4 briefly introduces the appearance of ACC on subjects of complement clauses, which I will come back to in detail in §3.4. Section 3.2.5 shows the appearance of ACC on undergoers of causatives (causees) depending on the presence of other arguments. Section 3.2.6 concludes.

3.2.2 Accusative is lost in passivization

Like in many languages across the world, passive in Turkish is both a subject-demoting and an object-promoting operation: In the corresponding active clause, the nominative-marked subject

¹ Cf. Göksel & Kerslake 2005: §13.1.2 (pp. 141–144), §14.3.3.2 (p. 175 f.), §15.5 (p. 201)

² In some Indo-European languages such as German, we see both structural and lexical properties of ACC: structural ACC, as discussed in the following, disappears on passives a.o., while another kind of ACC is used on adverbs of durations a.o.:

- (i) *Ich habe [den ganzen Tag] gearbeitet.*
 me have.1SG the.ACC whole.ACC day worked.PPP
 ‘I have worked the whole day long.’

German (Butt 2006: p. 7)

This ACC does not disappear under passivization — unlike in Finnish (Kiparsky 2001; Maling 1993a, 2009; McFadden 2004; Poole 2015). In this thesis, I will use passivization as a diagnostic for whether a case is structural. For instance, in §5 I will argue that DAT case on GOALS and causees in Turkic is an inherent case and does not disappear under passivization.

However, note that Anagnostopoulou & Sevdali (2015, 2020) have claimed that DAT and GEN in Ancient Greek are lexical/inherent cases and yet *do* alternate with NOM in passives. See Anagnostopoulou & Sevdali (2020: pp. 1041–1045) for two tentative proposals on how this might work.

³ In Ottoman Turkish, there were two adjectives/adverbs taking an ACC-marked object that might be considered instances of lexical ACC case:

- (i) *nitek -ler -i haiz adam*
 qualification -PL -ACC possessing man
 ‘the man possessing the qualifications’

Ottoman Turkish (Solak 2021: p. 2, ex. (2))

- (ii) a. [*Namaz -ı müteakip*] *defn -ed -il -ecek.*
 prayer -ACC following burial -do -PASS -FUT
 ‘S/he will be buried following the prayer.’
 b. [*Namaz -ı müteakiben*] *defn -ed -il -ecek.*
 prayer -ACC following burial -do -PASS -FUT
 ‘S/he will be buried following the prayer.’

Ottoman Turkish (Solak 2021: p. 2, ex. (5))

Ottoman Turkish (Solak 2021: p. 2, ex. (5))

For my take on these adjectives and on similar ACC-assigning nominalizations, see §3.5.

disappears and may appear as *tarafından* ‘by’-phrase; the ACC-marked direct object of the active clause becomes the NOM-marked subject; the verb takes a passivization morpheme, and φ -agrees with this new subject. This is illustrated in (1).

- (1) a. *Sabine bu tez -i yaz -ıyor.*
 Sabine.NOM this thesis -ACC write -PROG
 ‘Sabine is writing this thesis.’ Turkish
- b. *Bu tez (Sabine tarafından) yaz -ıl -ıyor.*
 this thesis.NOM Sabine by write -PASS -PROG
 ‘This thesis is being written (by Sabine).’ Turkish

In contrast to accusative, lexical cases do not disappear under passivization and cannot undergo promotion to subject. Turkish has “impersonal passives” that are subject-demoting non-promoting impersonals that also use the passive morpheme on the verb (Legate et al. 2020). (2a) shows an active clause with an ablative-marked THEME, which is an instance of lexical case. (2b) is the impersonal version of (2a). (2c) is the ungrammatical object-promoting passive of (2a).

- (2) a. *Herkes köpek -ten kork -ar.*
 everyone.NOM dog -ABL fear -AOR
 ‘Everybody is afraid of the dog/of dogs.’ Turkish
- b. *Köpek -ten kork -ul -ur.*
 dog -ABL fear -PASS -AOR
 ‘People are afraid of the dog/of dogs.’ Turkish
- c. **Köpek kork -ul -ur.*
 dog.NOM fear -PASS -AOR
 intended: ‘The dog is feared.’ Turkish

If the predicate is ditransitive, taking for instance both an ACC-marked and an ABL-marked object, promotion of the ACC-marked object remains possible, as might be expected. This is exemplified in (3).

- (3) a. *Ali Bahar -dan bu kitab -ı dilen -di.*
 Ali.NOM Bahar -ABL this book -ACC beg -PST
 ‘Ali begged Bahar for this book.’ Turkish

- b. *Bu kitap (Ali tarafından) Bahar -dan dilen -il -di.*
 this book.NOM Ali by Bahar -ABL beg -PASS -PST
 ‘This book was begged from Bahar (by Ali).’ *Turkish*
- c. **Bahar bu kitap -ı dilen -il -di.*
 Bahar.NOM this book -ACC beg -PASS -PST
 intended: ‘Bahar was begged for this book.’ *Turkish*

3.2.3 Accusative is gained or lost via DOM

Under the theory of Differential Object Marking (DOM) that I, among many others, adopt for Turkic, we can see the instance of accusative that appears/disappears depending on the direct object’s specificity/definiteness/topic-hood, behaving like a structural case. Following Baker (2015b), Baker & Vinokurova (2010), and Diesing (1992) among many others, I take the difference in interpretation and case-marking to arise because of different structural positions — a consequence of syntax. I will discuss this approach and its consequences in detail in §3.3.

3.2.4 Accusative is gained on subjects of complement clauses

In Turkish, Sakha, and Uyghur, subjects of finite complement clauses may alternate between *NOM*- and *ACC*-marking. This section aims to briefly argue that this assignment of accusative is structural. I will come back to this construction in more detail in §3.4.

Şener (2008, 2011) has shown for Turkish that accusative case is possible only when the embedded subject is topicalized, and not when it is focalized. Assuming that topic-hood corresponds to a structural position (Spec,TopP), this indicates that this accusative is assigned in this position. But how can we be sure that it is not the Top head assigning inherent accusative to its specifier? Baker & Vinokurova (2010) have shown for Sakha that this accusative is still subject to the structural accusative assignment rule: it is dependent on the presence of a higher NP etc., typically the matrix subject. In (4), both the matrix predicate ‘become sad’ and the embedded predicate ‘come’ are intransitive, thus neither of them can be the source of the *ACC*-marking on the embedded subject ‘Aisen’.

- (4) *Keskil [Aisen -y kel -be -t dien] sonun -tan xomoj -do.*
 Keskil Aisen -ACC **come** -NEG -AOR that news -ABL **become.sad** -PST
 ‘Keskil became sad that Aisen is not coming.’

*Sakha*⁴

This is also the case in (5) with the intransitive predicates ‘become certain’ and ‘go’. However, in (5), contrary to (4), there is no matrix subject present, and this correlates with the impossibility of ACC-marking on the embedded subject.

- (5) *Bügün munnjax -xa [Masha -(*ny) ehiil Moskva -qa bar -ya dien]*
 today meeting -DAT Masha -(*ACC) next.year Moscow -DAT **go** -FUT that
cuolkajdan -na.
become.certain -PST
 ‘It became clear today at the meeting that Masha will go to Moscow next year.’

*Sakha*⁵

Whenever the matrix predicate does not have a (separate) subject, accusative case is not possible on the embedded subject, even if it is topicalized. Given this dependency on properties of the matrix clause, we can state that accusative case in these constructions is not an inherent case assigned by Top.

3.2.5 Accusative is gained in causatives of intransitives

Another construction where we can see structural accusative is in causatives of verbs that do not take an ACC-marked argument in their base form. Examples of such verbs include intransitives and verbs taking one or more lexically case-marked arguments: Under causativization, the causee will be assigned accusative case.

- (6) Unaccusative verb:

- a. *Ali uyan -dı.*
 Ali.NOM wake.up -PST
 ‘Ali woke up.’

Turkish

- b. *Bahar Ali -yi uyan -dır -dı.*
 Bahar.NOM Ali -ACC wake.up -CAUS -PST
 ‘Bahar woke Ali up.’

Turkish

⁴ Vinokurova 2005: p. 366, fn. 15, ex. (2b)

⁵ Baker & Vinokurova 2010: p. 619, ex. (47a)

(7) EXPERIENCER verb with a lexically ABL-marked argument:

- a. *Aygül köpek -ten kork -ar.*
Aygül.NOM dog -ABL fear- AOR
‘Aygül is afraid of dogs.’ Turkish
- b. *Bolgün Aygül -ü köpek -ten kork -ut -tu.*
Bolgün.NOM Aygül -ACC dog -ABL fear -CAUS -PST
‘Bolgün made Aygül fear dogs.’ Turkish

(8) Unergative verb with two lexically case-marked arguments:

- a. *Ayşe İstanbul -dan Kuşadası -a taşın -dı.*
Ayşe.NOM İstanbul -ABL Kuşadası -DAT move -PST
‘Ayşe moved from İstanbul to Kuşadası.’ Turkish
- b. *Bergül Ayşe -yi İstanbul -dan Kuşadası -a taşın -dır -dı.*
Bergül.NOM Ayşe -ACC İstanbul -ABL Kuşadası -DAT move -CAUS -PST
‘Bergül made Ayşe move from İstanbul to Kuşadası.’ Turkish

One might entertain the hypothesis that this accusative on the causee could be an inherent case assigned by the causative head to its specifier — similar to what has been claimed for ergative by Legate (2008) a.m.o., namely that it is assigned to Spec,VoiceP by Voice; cf. a discussion of this approach in Baker & Bobaljik (2017). However, in Turkish and Sakha, not all causees receive ACC-marking. Causatives of predicates that already contain a structurally case-marked argument will appear with dative-marked causees. We can see this with ACC-objects (9), bare-DOM objects (10), and even with double causatives of intransitives (11).

(9) Transitive verb with an ACC-marked argument:

- a. *Aydost kitab -ı oku -du.*
Aydost.NOM book -ACC read -PST
‘Aydost read the book.’ Turkish
- b. *Öğretmen Aydost -a kitab -ı oku -t -tu.*
teacher.NOM Aydost -DAT book -ACC read -CAUS -PST
‘The teacher made Aydost read the book.’ Turkish

(10) Transitive verb with a bare argument:

- a. *Aydost kitap oku -du.*
 Aydost.NOM book read -PST
 ‘Aydost read books.’

Turkish

- b. *Öğretmen Aydost -a kitap oku -t tu.*
 teacher.NOM Aydost -DAT book read -CAUS -PST
 ‘The teacher made Aydost read books.’

Turkish

(11) Causativized intransitive, whose lower causee is ACC-marked:

- a. *Bahar Ali -yi uyan -dır -dı.*
 Bahar.NOM Ali -ACC wake.up- CAUS -PST
 ‘Bahar woke Ali up.’

Turkish

- b. *Cem Bahar -a Ali -yi uyan -dır -t -tı.*
 Cem.NOM Bahar -DAT Ali -ACC wake.up -CAUS -CAUS -PST
 ‘Cem made Bahar wake up Ali.’

Turkish

In particular, note that the high causee in (11) (‘Bahar’) is DAT-marked like the causee of the transitives in (9)–(10) (‘Aydost’), and not ACC-marked like the causees of verbs with a lexically case-marked argument in (7)–(8) (‘Aygöl’, ‘Ayşe’). This patterning is another argument for the structuralness of the accusative case on the low causee in intransitives, (6) resp. (11) (‘Ali’): If the ACC-marking in (6) and (11) were lexical ACC, we would expect the high causee to bear structural ACC, as it does in the lexically case-marking predicates in (7)–(8).

Another argument that the ACC-marking on causees is structural case comes from the fact that when a causative is passivized, an ACC-marked causee can be promoted to NOM, (12), while a DAT-marked causee cannot, (13).

- (12) a. *Ayşe Ali -yi resm -e bak -tır -dı.*
 Ayşe.NOM Ali -ACC picture -DAT look -CAUS -PST
 ‘Ayşe made Ali look at the picture.’

*Turkish*⁶

- b. *Ali resm -e bak -tır -ıl -dı.*
 Ali.NOM picture -DAT look -CAUS -PASS -PST
 ‘Ali was made to look at the picture.’

*Turkish*⁷

⁶ Öztürk 2007: p. 310, ex. (19b)

⁷ Öztürk 2007: p. 310, ex. (19c)

- (13) a. *Ali Ayşe -ye bu kitab -ı oku -t -tu.*
 Ali.NOM Ayşe -DAT this book -ACC read -CAUS -PST
 ‘Ali made Ayşe read this book.’ *Turkish*⁸
- b. **Ayşe bu kitab -ı oku -t -ul -du.*
 Ayşe.NOM this book -ACC read -CAUS -PASS -PST
 intended: ‘Ayşe was made to read this book.’ *Turkish*⁹

In section 5.4 I will come back to causatives of verbs with a structurally case-marked argument in more detail.

3.2.6 Summary

In this section, I have shown that accusative in Turkish is a structural case. I have not shown that it is always a structural case, as there might be constructions where it is a lexical/inherent case, having, accidentally, two assignment rules. The absence of a lexical assignment rule for ACC can only be shown through negative evidence: I am unaware of any ACC-assigning lexical or functional items in modern Turkish.

I have given positive evidence that accusative is a structural case in some constructions. Whenever accusative case can disappear, e.g., turn into nominative under passivization, this accusative must have been a structural case: assigned to a grammatical function, rather than to a thematic role (§3.2.2).

Whenever accusative case can be dropped from a direct object under the rules of Differential Object Marking (that I will come back to in §3.3), this accusative case must be structural, because DOM is a structural phenomenon.

Whenever accusative appears on the subjects of complement clauses, this accusative case must be a structural case because we can observe the dependency of this accusative on the presence of the matrix subject, a structural condition (§3.2.4).

Whenever accusative case appears on the causee, this accusative must be a structural case. This is so for the following two reasons. First, it can alternate with dative case, depending on the presence

⁸ Öztürk 2007: p. 309, ex. (17b)

⁹ Öztürk 2007: p. 309, ex. (18a)

of other structurally-case-marked arguments, in accordance with the structural case assignment rules for accusative case, and thus not depending on the causee's thematic role, which is constant among all causatives. Second, ACC-marked causees pattern with other structural cases in triggering dative on the high causee in causatives of causatives, and not with lexically case-marked arguments, which would trigger accusative on the causee.

Having established that we may treat all instances of accusative as structural case, I will now turn to three constructions where we can observe the rule of accusative assignment in detail (§3.3, §3.4, §3.5). Section 3.6 discusses the relevance of these assignment rules to this thesis. §3.7 concludes the discussion of accusative case.

3.3 Differential Object Marking (DOM)

3.3.1 Introduction

In this section I will illustrate and discuss the phenomenon of Differential Object Marking (DOM) as it appears in Turkic languages: the presence or absence of accusative (ACC) on the direct object (DO) of a clause. I will show that the distribution of ACC follows the rules introduced in §2.6 given a particular syntactic analysis of the verbal domain and topicalization-induced object movement that I will show is independently motivated.

Cross-linguistically, in languages with DOM, DOs are partitioned into (at least) two classes (Bossong 1985, 1991). Typically, one class receives a marker (such as ACC or an adposition), and the other class receives no marker ("bare object"). In some languages (such as Finnish), both classes receive a case marker (Finnish: accusative vs. partitive case). The partitioning factor between the classes varies between languages, but typically follows a notion of prominence on the semantic scales of animacy and/or definiteness/specificity (Aissen 2003). For Turkic languages, only the latter scale has been claimed to be of relevance.

The most simple case of DOM is illustrated for Sakha below in (14)–(15):

- (14) a. *Min saharxaj sibekki ürgee -ti -m.*
 me yellow flower pick -PST -1SG
 ‘I picked a yellow flower.’ or ‘I picked yellow flowers.’ *Sakha*¹⁰
- b. *Min saharxaj sibekki -ni ürgee -ti -m.*
 me yellow flower -ACC pick -PST -1SG
 ‘I picked the yellow flower.’ or ‘I picked a specific yellow flower.’ *Sakha*¹¹
- (15) a. *Masha salamaat sie -te.*
 Masha porridge eat -PST.3SG
 ‘Masha ate porridge.’ *Sakha*¹²
- b. *Masha salamaat -y sie -te.*
 Masha porridge -ACC eat -PST.3SG
 ‘Masha ate the porridge.’ *Sakha*¹³

Whether the connection between ACC and the DO’s interpretation is a direct or an indirect connection has been the source of much debate. Most proposals of DOM in Turkish that I am aware of tackle the problem from the semantic side: What is the difference in interpretation when a DO is marked with ACC or not? Different solutions abound.¹⁴ Here, I will assume the purely syntactic proposals by Baker (2015b), Baker & Vinokurova (2010), and Kelepir (2001).¹⁵ Whether or not a DO receives ACC-marking depends on the object’s structural position (whether it is in the VP-Phase or CP-Phase), as the following subsections will illustrate. This structural position determines the DO’s interpretation: Diesing (1992) has shown that NPs receive an indefinite/non-specific interpretation when they remain inside the VP, and a definite/specific interpretation when they move

¹⁰ Vinokurova 2005: p. 322, ex. (7a)

¹¹ Vinokurova 2005: p. 322, ex. (7b)

¹² Baker 2015b: p. 125, ex. (22b)

¹³ Baker 2015b: p. 125, ex. (22a)

¹⁴ Classification of previous proposals for DOM in Turkic, taken from the literature review in Özge (2011) and Bolgün (2005): **Discourse-linking** (Enç 1991; Nilsson 1985; Zidani-Eroğlu 1997); **Definiteness** (Ergin 1958; Erguvanlı 1984; Sebütekin 1971); **Referentiality** (Dede 1986; Kornfilt 2008); **Specificity** (Enç 1991; Erguvanlı 1984; von Heusinger 2002; von Heusinger & Kornfilt 2005; Kornfilt 1997; Swift 1963); **Presuppositionality** (Kelepir 2001; Kennelly 1997); **Individuation/particularization** (Bolgün 2005; Erguvanlı Taylan & Zimmer 1994; Kılıçaslan 2006; Nilsson 1985); **Totality/delimitedness** (Nakipoğlu 2009; Nilsson 1985). **Generalized Skolem term** (Özge 2011).

¹⁵ Kalin & Weisser (2019) argue that in languages with asymmetric DOM, i.e., when in a conjunction of DOs, it is possible for only one of the conjuncts to carry the respective marking, DOM cannot arise by movement of the respective DO because this would constitute a violation to the Coordinate Structure Constraint. While most of the languages in their sample allow asymmetric DOM, Turkish does not, however, the closely related Urum does (Böhm 2015). It is thus possible that the nature of DOM within the Turkic family is not uniform. Here, I will focus on the Turkish pattern and assume a movement analysis of DOM, leaving a detailed analysis of DOM in other Turkic languages for future research.

out of the VP. Similarly, Şener (2010) has shown for Turkish that movement of the DO out of its base position is always topicality-induced, and we know that topics tend to have a definite/specific interpretation.

The rest of this section is structured as follows: In §3.3.2, I will show that the assignment of ACC is due to its structural position: The DO must have moved out of the Spell-out domain of the verbal phase head in order to receive ACC in the CP-domain. In §3.3.3, I show that this movement of the DO is due to topicality, and only topicality, but not due to specificity or definiteness, as has been claimed. I will give examples where topics are neither definite nor specific to show the difference. In section §3.3.4, I summarize how the presented examples fit into DCT as formalized in §2.6 by extending the proposals of Baker (2015a,b) and Baker & Vinokurova (2010).

3.3.2 Accusative is assigned outside of the VP domain only

The examples in (14) and (15) made it seem like the only difference between the sentence-pairs was the presence or absence of ACC-marking. This subsection will show that there is more going on in the syntax than merely the different cases on the direct object (DO). Rather, the DOs with and without ACC-marking are in different structural positions: outside and inside the VP. We can make this visible by adding low adverbs that mark the VP-boundary. In (16), the DO appears to the left of the adverb (i.e., outside the VP) and must appear with ACC-marking. In (17), the DO appears between the adverb and the verb (i.e., inside the VP) and cannot appear with ACC-marking in an unmarked reading.

- (16) a. *Masha salatmaat -y türgennik sie -te.*
Masha porridge -ACC quickly eat -PST.3SG
'Masha ate the porridge quickly.'

*Sakha*¹⁶

- b. **Masha salatmaat türgennik sie -te.*
Masha porridge quickly eat -PST.3SG

*Sakha*¹⁷

^{16,17} Baker 2015b: p. 126, ex. (23a)

- (17) a. #*Masha türgennik salamaat -y sie -te.*
 Masha quickly porridge -ACC eat -PST.3SG
Sakha^{18,19}
- b. *Masha türgennik salamaat sie -te.*
 Masha quickly porridge eat -PST.3SG
 ‘Masha ate porridge quickly.’
Sakha²⁰

Apart from low adverbs, we can also use indirect objects (IOs) to diagnose whether a DO is in its base position or has moved. Like in (16), the examples in (18) show that a DO that appears to the left of an IO must be marked with ACC, and, similarly, we see in (19) that ACC-marking is not possible between the IO and the verb.²¹

- (18) a. *Min kinige -ni Masha -qa bier -di -m.*
 I book -ACC Masha -DAT give -PST -1SG
 ‘I gave the book to Masha.’
Sakha²²
- b. **Min kinige Masha -qa bier -di -m.*
 I book Masha -DAT give -PST -1SG
Sakha²³
- (19) a. #*Min Masha -qa kinige -ni bier -di -m.*
 I Masha -DAT book -ACC give -PST -1SG
Sakha²⁴
- b. *Min Masha -qa kinige bier -di -m.*
 I Masha -DAT book give -PST -1SG
 ‘I gave Masha a book.’
Sakha²⁵

We have now seen that ACC can only appear on DOs that have moved out of the VP. This is schematized in Fig. 3.1. In the next subsection, we will see that ACC appears on all DOs outside the VP, even when they are associated with an indefinite/non-specific interpretation.

¹⁸ Regarding this example’s felicity, Baker & Vinokurova (2010: p. 602) note: “[A]ccusative marking on the post-adverbial object in [(17a)] is not strictly impossible, but it is given a special interpretation, as having contrastive focus on the accusative-marked object. We tentatively assume that this is not a syntactically simple structure; rather it is derived by a series of movements into the left periphery of the clause, driven by considerations of focus and topic. See Vinokurova (2005: p. 211) for some possible derivations.” I will return to this shortly in section 3.3.3.

^{19,20} Baker & Vinokurova 2010: p. 602, ex. (12b)

²¹ In my reading of Baker & Vinokurova (2010), the acceptability of (19a) parallels that of (17a) noted in footnote 18.

^{22,23} Baker & Vinokurova 2010: p. 602, ex. (13b)

^{24,25} Baker & Vinokurova 2010: p. 602, ex. (13a)

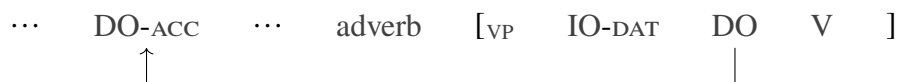


Figure 3.1: The movement of the direct object past adverbs of manner feeds ACC case assignment

3.3.3 The movement of the direct object out of the VP is topicalization

In this subsection, we see that assignment of ACC depends only on the structural configuration of arguments, and does not directly depend on the DO's definiteness or specificity. In the examples below, an indefinite/non-specific DO receives ACC-marking solely because it appears in a topic position.²⁶ Topics tend to be definite/specific, but do not have to be, as the examples show.

The first possibility of having an indefinite/non-specific DO with ACC-marking arises because of a property of Turkish information structure: Foci are located in the linearly immediately preverbal field²⁷, while topics must move to the left. By putting an adverbial (20) or another argument (21) into the focus position, we force the DO to move to the left, outside of the verbal domain, into an intermediary topic position.

In (20) and (21), the adverbial *hızlı* 'fast' appears in focus position, and the ACC-marked DO can receive an indefinite interpretation. The focus on the adverbial becomes especially clear in (21).

- (20) a. *Ben kitab -ı hızlı oku -r -um.*
 I book -ACC fast read -AOR -1SG
 (i) 'It is fast that I will read the book.' (a definite book) *Turkish*²⁸
 (ii) 'It is fast that I read books.' (non-specific books)
- b. **Ben kitap hızlı oku -r -um.*
 I book fast read -AOR -1SG *Turkish*

²⁶ Aydemir (2004) and Kamali (2015) argue that there is a difference between Turkish (i) indefinite NPs, which appear with a numeral or indefinite expression in the same NP, and are the canonical means to introduce new discourse referents; and (ii) "bare" NPs, which merely indicate a category (Göksel & Kerslake 2005), are number-neutral, and cannot typically introduce new discourse referents. This distinction goes outside the scope of this project, and so I will treat these classes alike. I refer the interested reader to Sağ-Parvardeh (2019) for a thorough analysis of the semantics of kinds in Turkish and English.

²⁷ Cf. Erguvanlı 1984; Erkö 1983; Göksel 1998; Göksel & Kerslake 2005; İşsever 2003; Kennelly 1997; Kornfilt 1997; Şener 2010; a.o.

²⁸ Aygen 1999: p. 1

- (21) A: *Sen fazla yavaş -ça kitap oku -r -muş -sun.*
 you very slow -ly book read -AOR -COP.EVID -2SG
 ‘(I notice that) You read books really slowly.’ *Turkish*
- B: *Yoo, ben kitap -ı hızlı oku -r -um.*
 No I book -ACC fast read -AOR -1SG
 ‘No way, I read books really fast!’ (non-specific books) *Turkish*

In (22), the subject NP *Oya* appears in focus position, and the DO can receive an indefinite interpretation (22a) and must have ACC-marking (22b).

- (22) a. *Dün kitap -ı Oya al -dı.*
 yesterday book -ACC Oya buy -PST
 (i) ‘It was Oya who bought the book yesterday.’ (a definite book) *Turkish* ²⁹
 (ii) ‘It was Oya who bought books yesterday.’ (non-specific books)
- b. **Dün kitap Oya al -dı.*
 yesterday book Oya buy -PST *Turkish* ³⁰

The following two examples illustrate this with contexts. Example (23) consists of two sentences. (23a) initializes the utterance, introducing the VP *balık tut-* ‘to fish’, lit. ‘to catch fish’. (23b) and (23c) are two options of continuing (23a): (23b) with ACC-marking, and (23c) without. Since the VP ‘catch fish’ is given information, and the new information comes from the adverbial *yapayalnız* ‘completely alone’, the DO, despite referring to an indefinite number and non-specific identities of fish, must carry ACC.

- (23) a. *Keyf -im yalnız balık tut -mak -tır.*
 pleasure -POSS.1SG alone fish catch -INF -COP
 ‘My pleasure is to go fishing on my own.’ *Turkish* ³¹
- b. *Balığ -ı yapayalnız tut -mak -tır.*
 fish -ACC completely alone catch -INF -COP
 ‘That is, to go fishing completely alone.’ (non-specific fish) *Turkish* ³²
- c. **Balık yapayalnız tut -mak -tır.*
 fish completely alone catch -INF -COP *Turkish* ³³

^{29,30} Laszakovits 2014: p. 172, ex. (11)

^{31,32,33} Nilsson 1991: p. 100

Example (24) constructs a dialogue that puts focus on V by making all other arguments and adverbials salient, including the DO. (24a) is uttered by the first discourse participant, introducing everything except V. (24b) and (24c) are two possible answers. (24b), despite its ACC-marking, allows an indefinite/non-specific interpretation of the DO. I suggest that this is the case because the DO is in a topic position. (24c), without ACC-marking, is not a felicitous answer to (24a) — presumably because the lack of ACC-marking makes the DO be part of the focused phrase, which doesn't fit the context.

- (24) a. *Oya dün kitap iş -in -i ne yap -tı?*
 Oya yesterday book matter -CMPD -ACC what do -PST
 'What did Oya do regarding a book/books yesterday?' *Turkish*³⁴
- b. *Oya dün kitab -ı AL -dı.*
 Oya yesterday book -ACC buy -PST
 'Oya BOUGHT a book/books yesterday.' (non-specific books) *Turkish*³⁵
- c. *#Oya dün kitap AL -dı.*
 Oya yesterday book buy -PST *Turkish*³⁶

We have now seen that the assignment of accusative does not depend on the DO's definiteness or specificity, but on its structural position: ACC cannot be assigned inside the VP, and in transitive clauses, ACC will be assigned any time the DO is outside the VP. This is consistent with the rule formulation of dependent-down case above in section 2.6.

3.3.4 Summary

The phenomenon of DOM in Turkic is quite complex. From my understanding of the literature, no one seems to have a complete account of the ACC-marking/word order/interpretation patterns. For the purposes of this subsection, I will adopt the proposal developed in Baker (2015b) and Baker & Vinokurova (2010), which I outline here, proposing a slight modification to avoid some edge cases that are not solved there convincingly. Finding a full account of DOM in Turkic remains

^{34,35,36} Laszakovits 2014: p. 172, ex. (12a, b)

an open problem, and solving it is not in the scope of this thesis.

The following quote from Baker (2015b: p. 126) sums up his proposal of ACC-assignment in DOM:

Following the idea of Baker & Vinokurova (2010) (but changing the implementation somewhat), we can analyze this sort of DOM in terms of [Voice] being a phase head, causing VP to be a spell out domain. If the object remains in VP, then it is properly inside this spell out domain, whereas the subject is not. There is one, and only one, NP in VP when VP is spelled out, so that NP does not get dependent case. However, the object can move out of VP into [VoiceP], a familiar kind of object shift. This affects its word order with respect to adverbs, and also its interpretation, assuming in the tradition of Diesing (1992) that (roughly) NPs inside VP get one sort of interpretation and NPs outside it get another. In Sakha [and other Turkic languages], this object shift also affects the morphological case. If NP moves out of vVP, then it is not spelled out (only) with VP. Rather, it is spelled out with TP when the next phase head (C) is reached. The subject is also part of the TP domain, so dependent case marking does apply, and the object gets accusative.

In Turkic, subjects can remain in their base-position in Spec, VoiceP (Öztürk 2005, a.m.o.). And crucially, DOs can receive ACC-marking even when the subject stays in Spec, VoiceP. Having shown that DCT's rule for ACC-marking applies in the CP-domain, this must mean that there must be a position for DOs in the CP-domain where they are c-commanded by low subjects. For Baker (2015b), assuming a structure like in Fig. 3.2, this leaves only one possibility: a moving DO must land as adjunct to the verbal Spell-out domain, i.e., to VP.

This is where the waters get murky because we would expect the escape hatch to be an adjunct site to the phasal phrase (VoiceP), not to the Spell-out domain. In §3.5, I will split V into a root and a verbalizer, and I assume that the complement of V is introduced as the complement of the root. This then gives us the extra landing site of the DO that we need; Fig. 3.3.

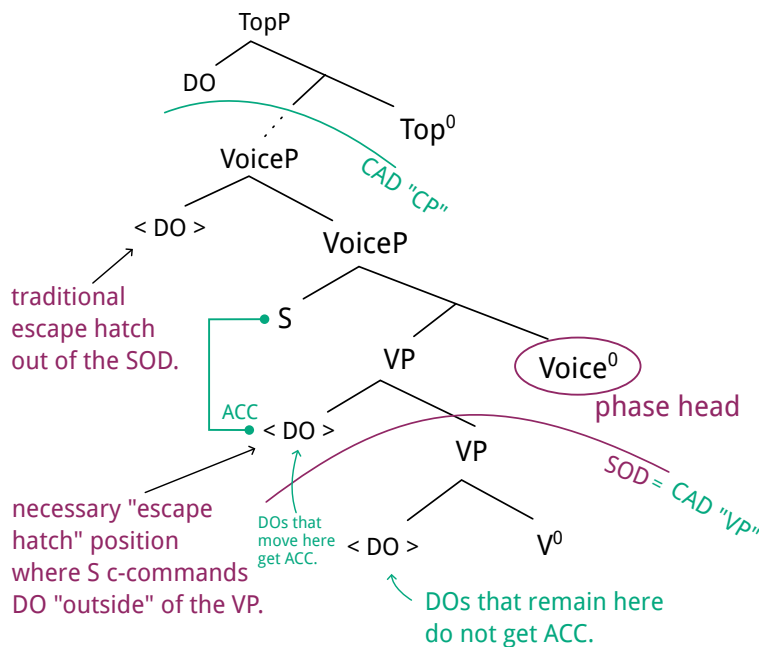


Figure 3.2: The assignment of ACC in DOM according to Baker (2015b)

3.4 Cross-clausal A/A'-dependencies (CC $\mathcal{A}$ s)

3.4.1 Introduction

In this section I present an accusative phenomenon that, first, has been offered as evidence for the necessity for DCT over FHCT (Baker & Vinokurova 2010; Kornfilt & Preminger 2015) — FHCT cannot account for the data while DCT can —, and second, and more importantly for this thesis, this phenomenon illustrates very clearly the $1 : n$ ratio of ACC assignment: We will see that 1 licenser (a c-commanding NP) licenses accusative on more than one, and arguably arbitrarily many (n , for any $n \in \mathbb{N}$), licensees. The ramifications for this pattern on the complexity of grammar will be taken up in §3.6.

By cross-clausal A/A'-dependency (CC $\mathcal{A}$) (terminology from Lohninger, Kovač & Wurmbrand 2022) we understand a construction where an argument from an embedded clause (finite or nominalized) enters into an A-dependency with an element from the matrix clause. Typically, for this A-dependency to be established, the argument in question needs to first move to an A'-position inside the embedded clause, thus leading to an apparent “improper movement” violation (Chomsky

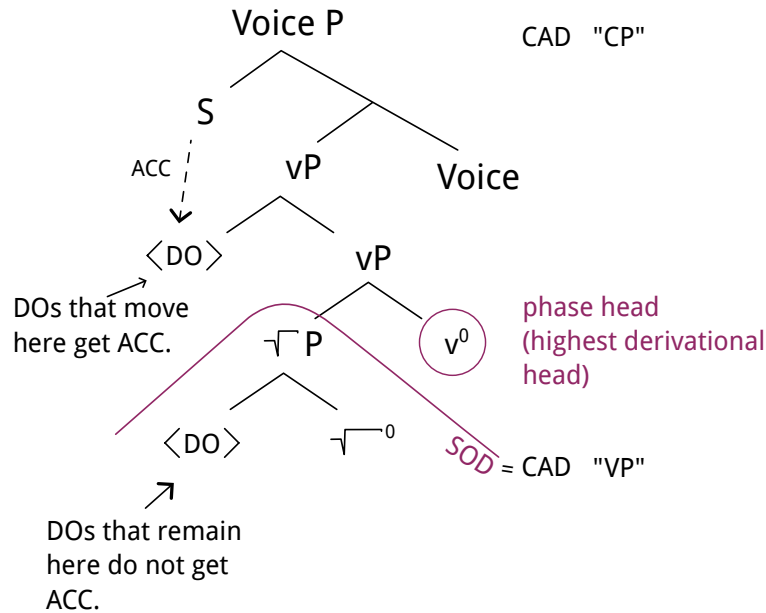


Figure 3.3: Creating an extra landing site for DOM by splitting V

1973, 1981; May 1979). Lohninger, Kovač & Wurmbrand (2022) have investigated CCAs cross-linguistically and argue for a five-way typology based on the syntactic interactions of the cross-clausal argument with its environment. One type is prolepsis; the other four types are subsumed in Fig. 3.4.³⁷ The matrix predicate selects for a C.R phrase, which combines the A'-properties of C with the A-properties of a relator R.³⁸ In this way, the specifier of CP.R (labeled DP.2 in Fig. 3.4) is both in an A-position and an A'-position and can enter into case competition and other A-relations with elements in the matrix clause (indicated in Fig. 3.4 by the solid arrow).

With this understanding of CCA, I now turn to Turkic. The rest of this section is structured as follows: §3.4.2 will introduce the phenomenon of CCAs in Sakha, based on Baker & Vinokurova

³⁷ The four constructions subsumed in Fig. 3.4 differ further with respect to whether DP.2 is base-generated in Spec,CP.R or not, and if it isn't, how the two parts of the A- and A'-probe on C.R interact with each other to move it there: (i) conjunctive satisfaction [A+A'] (the goal needs to satisfy the probe's A- and A'-properties); (ii) separate dependent satisfaction [A/A'] (the A- and A'-probes probe separately, potentially finding different goals, which will lead to the derivation crashing, no agreement being established, or other kinds of blocking effects — this is Lohninger, Kovač & Wurmbrand's (2022) analysis for Turkish, which I will come back to in §3.4.3 and which will be depicted in Fig. 3.6 on page 97); and (iii) separate independent satisfaction [A][A'] (the A- and A'-probes probe separately, and there are no further consequences whether they find the same goal or different goals). See Lohninger, Kovač & Wurmbrand (2022) for more details and on how to derive different kinds of hyperraising and LDA constructions in a variety of languages in this way.

³⁸ Lohninger, Kovač & Wurmbrand (2022) follow den Dikken (2006, 2017) here in that the R head relates its specifier and complement via predication (similarly to argument-introducing heads).

³⁹ From Lohninger, Kovač & Wurmbrand 2022: p. 13, ex. (29b)

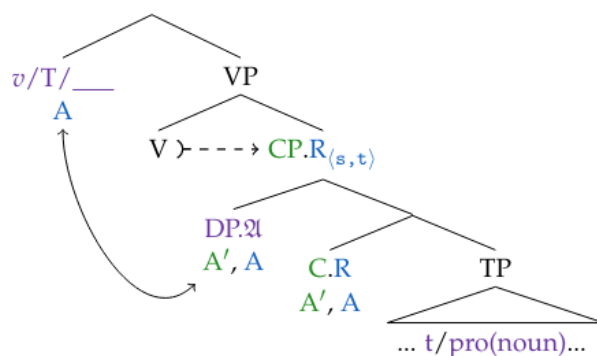


Figure 3.4: Lohninger, Kovač & Wurmbrand's (2022) analysis of CCᵐ-dependencies cross-linguistically³⁹

(2010). §3.4.3 presents evidence for the position of DP.ᵐ in Spec,TopP (Spec,CP.R) from Turkish (Şener 2008, 2011), and §3.4.4 presents evidence from indexical shifting in Uyghur (Major 2022; Shklovsky & Sudo 2014). §3.4.5 concludes the discussion of CCᵐs before I turn to the importance of the 1 : *n* pattern in §3.6.

3.4.2 CCᵐs in Sakha: dependency on matrix subject

In Sakha CCᵐ-constructions, the subject of an embedded clause (finite (25) or nominalized (26)) can optionally receive accusative marking:

- (25) a. *Min [ehigi bügün kyaj -yax -xyt dien] erem -mit -im.*
 me you today win -FUT -2SG that hope -PTCP -1SG
 'I hoped that you would win today.' Sakha⁴⁰
- b. *Min [ehigi -ni bügün kyaj -yax -xyt dien] erem -mit -im.*
 me you -ACC today win -FUT -2SG that hope -PTCP -1SG
 'I hoped that you would win today.' Sakha⁴¹
- (26) a. *Min [ehigi bügün kyaj -byk -kyt -yn] ihit -ti -m.*
 me you today win -PTCP -2SG.POSS -ACC hear -PST -1SG
 'I heard you won today.' Sakha^{42,43}

^{40,41} Baker & Vinokurova 2010: p. 615, ex. (39a)

⁴² Baker & Vinokurova 2010: p. 615, ex. (39b)

⁴³ I have corrected *ihit-ti-im* to *ihit-ti-m* in analogy to Baker & Vinokurova's (2010) examples (40), (41a), and (41b), and in accordance with the personal suffixes for the past tense given in Landmann (2016: p. 69).

- b. *Min [ehigi -ni bügün kyaj -byk -kyt -yn] ihit -ti -m.*
 me you -ACC today win -PTCP -2SG.POSS -ACC hear -PST -1SG
 'I heard you won today.'

Sakha⁴⁴

As Baker & Vinokurova (2010: §3.5, pp. 615–620) show, the availability of this ACC marking is dependent on the presence of a matrix subject, and is not dependent on other sources of ACC marking (such as functional heads) in either the embedded clause or the matrix clause. They present two main arguments for this. First, they argue that the ACC-marking on the embedded subject cannot stem from a Voice-head because ACC-subjects happen when neither the embedded predicate nor the matrix predicate is transitive. This is illustrated in (27) and (28). FHCT cannot derive the grammaticality of ACC in these examples. Recall that FHCT states that ACC is assigned by Voice whenever Voice introduces a DP (AGENT) in its specifier and selects a transitive VP.

- (27) *Keskil [Aisen -y kel -be -t dien] sonun -tan xomoj -do.*
 Keskil Aisen -ACC **come** -NEG -AOR that news -ABL **become.sad** -PST
 'Keskil became sad that Aisen is not coming.'

Sakha⁴⁵

- (28) *Masha [Misha -ny yaldj -ya dien] tönün -ne.*
 Masha Misha -ACC **fall.sick** -FUT.3SG that **return** -PST
 'Masha returned (for fear) that Misha would fall sick.' [sic]

Sakha⁴⁶

- (29) *Sargy [kim -i daqany tönn -üm- üö dien] erenner -ilin -ne.*
 Sargy who -ACC PRT **return** -NEG -FUT.3SG that **promise** -PASS -PST
 'Sargy was promised that nobody would return.'

Sakha⁴⁷

As expected under both DCT and FHCT, when there is no subject NP in the matrix clause, the embedded subject cannot receive ACC-marking. This is illustrated in (30) and (31) with impersonal matrix predicates.

- (30) *Bügün munnjax -xa [Masha -(*)ny ehiil Moskva -qa bar -ya dien]*
 today meeting -DAT Masha -(*)ACC next.year Moscow -DAT go -FUT that

⁴⁴ Baker & Vinokurova 2010: p. 615, ex. (39b)

⁴⁵ Vinokurova 2005: p. 366, fn. 15, ex. (2b)

⁴⁶ Baker & Vinokurova 2010: p. 618, ex. (44b)

⁴⁷ Baker & Vinokurova 2010: p. 618, ex. (45)

cuolkajdan -na.

become.certain -PST

‘It became clear today at the meeting that Masha will go to Moscow next year.’ *Sakha*⁴⁸

- (31) [*Aisen* -(*)y *massyyna atyylah* -ar -a] *naada buol* -la.
Aisen -(*)ACC car buy -AOR -3SG **need become** -PST
 ‘It became necessary for Aisen to buy a car.’ *Sakha*⁴⁹

Baker & Vinokurova’s (2010) second argument against FHCT comes from data with a single transitive active Voice head, and yet more than one ACC-marked NP. In (32), the embedded subject *Misha* carries ACC-marking despite the fact that the embedded predicate *kel*- ‘come’ is not transitive, and the matrix predicate *xomuj*- ‘tidy up’ has already assigned its ACC-case to its internal argument *djie* ‘house’. The ACC-marking on *Misha* can thus not stem from either Voice-head, and is argued by Baker & Vinokurova (2010) to stem from the matrix subject *Masha*.

- (32) *Masha* [*Misha* -ny *kel* -ie *dien*] *djie* -ni *xomuj* -da.
Masha *Misha* -ACC come -FUT that house -ACC tidy -PST
 ‘Masha tidied up the house (thinking) that Misha would come.’ *Sakha*⁵⁰

The same holds for (26b) above, repeated below as (33): the DO of ‘hear’ is the intransitive nominalization *kyaj-byk* ‘having won’, which carries ACC. There is no other transitive VoiceP that could license the second ACC that we see on the embedded subject *ehigi-ni* ‘you-ACC’.

- (33) *Min ehigi* -ni *bügün kyaj* -byk -kyt -yn *ihit* -ti -m.
 me you -ACC today win -PTCP -2SG.POSS -ACC hear -PST -1SG
 ‘I heard you won today.’ *Sakha*⁵¹

3.4.3 CCAs in Turkish are topics

Like Sakha, Turkish allows optional ACC-marking on subjects of finite complement clauses:

⁴⁸ Baker & Vinokurova 2010: p. 619, ex. (47a)

⁴⁹ Baker & Vinokurova 2010: p. 619, ex. (47b)

⁵⁰ Vinokurova 2005: p. 368

⁵¹ Baker & Vinokurova 2010: p. 615, ex. (39b)

- (34) a. *Pelin* [*ben* *Timbuktu* -*ya* *git* -*ti* -*m*] *san* -*iyor*.
 Pelin me Timbuktu -DAT go -PST -1SG believe -PROG
 ‘Pelin believes that I went to Timbuktu.’ *Turkish*⁵²
- b. *Pelin* [*ben* -*i* *Timbuktu* -*ya* *git* -*ti* -*m*] *san* -*iyor*.
 Pelin me -ACC Timbuktu -DAT go -PST -1SG believe -PROG
 ‘Pelin believes that I went to Timbuktu.’ *Turkish*⁵³

Şener (2008, 2011) presents evidence that the ACC-marked subject must be located (or have been located) at the phase edge of the finite embedded clause, in a position he analyzes as Spec,TopP. Under his analysis, when a subject-NP is interpreted as a topic, it moves to Spec,TopP, and in this position, sitting on the phase edge of the embedded clause, it is able to receive ACC-case from matrix Voice under FHCT. Due to its position on the phase edge, the now-ACC-marked subject may also move into the matrix clause if its topic-hood scopes over elements of the matrix clause. In particular, Şener presents evidence

- (i) that ACC-subjects must be the left-most element inside the embedded clause save for adverbs,
- (ii) that ACC-subjects can be contrastive topics, but never bear focus (neither contrastive nor presentational focus),
- (iii) that ACC-subjects do not reconstruct, which is typical for topicalization in Turkish (evidence comes from idioms, variable binding, and weak cross-over), and
- (iv) that ACC-subjects cannot receive ACC-marking if they are not at the edge: When a projection higher than TopP is merged, the only possibility being WhP, Spec,WhP will be the edge position, which is occupied by a wh-operator.

The following examples from Şener (2008) illustrate (ii). In (35) we see that an ACC-subject of a finite complement clause can be a contrastive topic.

- (35) A: What about John? Did Pelin tell you what he ate at the party?
 B: Well, he didn't know about John, but ...

⁵² Şener 2008: p. 1, ex. (2a)

⁵³ Şener 2008: p. 2, ex. (3)

- a. *Pelin* [*Mete*_{CTOP} *istakoz -dan*_{CFOC} *ye -di*] *san -ıyor -muş.*
 Pelin Mete lobster -ABL eat -PST believe -PROG -EVID

Turkish 54

- b. *Pelin* [*Mete -yi*_{CTOP} *istakoz -dan*_{CFOC} *ye -di*] *san -ıyor -muş.*
 Pelin Mete -ACC lobster -ABL eat -PST believe -PROG -EVID
 ‘Pelin believes that Mete ate from the lobster (at the party).’

Turkish 55

In contrast, example (36) shows that ACC-subjects cannot have presentational focus, and example (37) shows that ACC-subjects cannot have contrastive focus.

(36) A: Do you know who showed up at Mert’s party?

B: I haven’t asked Mert himself about it, but ...

- a. *Pelin* [*Sinan*_{PFOC} *git -ti diye*] *duy -muş.*
 Pelin Sinan go -PST that hear -EVID.PST

Turkish 56

- b. #*Pelin* [*Sinan -ı*_{PFOC} *git -ti diye*] *duy -muş.*
 Pelin Sinan -ACC go -PST that hear -EVID.PST
 ‘Pelin heard that Sinan went (to the party).’

Turkish 57

(37) A: Do you know if everyone (he invited) went to Mert’s party?

B: I haven’t talked to Mert but ...

- a. *Pelin* [*yalnızca Sinan*_{CFOC} *git -ti diye*] *duy -muş.*
 Pelin only Sinan go -PST that hear -EVID.PST

Turkish 58

- b. #*Pelin* [*yalnızca Sinan -ı*_{CFOC} *git -ti diye*] *duy -muş.*
 Pelin only Sinan -ACC go -PST that hear -EVID.PST
 ‘Pelin heard that only Sinan went (to the party).’

Turkish 59

Şener’s (2008) analysis of the projections of a finite complement clause in Turkish is illustrated in Fig. 3.5, which I have extracted from the discussions in his paper. If the embedded subject is a contrastive topic, it moves to Spec,TopP of the embedded clause. Şener shows that an ACC-marked embedded subject must be the left-most element in the embedded clause, save for some adverbs.

^{54,55} Şener 2008: p. 22, ex. (49)

^{56,57} Şener 2008: p. 21, ex. (47)

^{58,59} Şener 2008: p. 21, ex. (48)

TopP, being the highest phrase in the embedded clause, extends the embedded clause's CP phase with Top becoming the new phase head, making Spec,TopP an escape hatch for Top's Spell-out domain. The NP in Spec,TopP will be spelled out only at the next phase (matrix Voice), and can until then interact with elements in the matrix clause and receive ACC under FHCT.

Şener (2008, 2011) takes case-assignment to happen via assignment by a functional head (Voice, T). For DCT on the other hand, the ACC on the embedded subject would need to be licensed by a c-commanding NP in the matrix clause. Note Şener's (2008) insight that it is the subject's position at the phase edge of the embedded clause that allows an element of the matrix clause (whether it be Voice or an NP) to enter an A-dependency with it that results in ACC-marking. This insight applies in a DCT-framework equally since DCT's case-assignment domains correspond to Phases.

In Lohninger, Kovač & Wurmbrand's (2022) framework, Turkish CCAs are of "type 4": The joint complementizer-relator head C.R contains both an A-probe (case) and an A'-probe (topicalization). The two probes probe separately but are dependent on each other's goal in that they must both select the same DP for feature checking to be successful. This is illustrated in Fig. 3.6.

Parallel to Sakha in §3.4.2 above, this raising-to-ACC in Turkish finite complement clauses is only possible if the matrix predicate is transitive — for Şener (2008, 2011): if Voice can assign ACC; for DCT: if a matrix subject is present. (38) illustrates the base-line pattern that in passives, the internal argument appears with NOM, (38a), and cannot appear with ACC, (38b).

- (38) a. *Makarna ye -n -di.*
 pasta.NOM eat -PASS -PST
 ‘Pasta was eaten.’ Turkish ⁶¹
- b. **Makarna -yı ye -n -di.*
 pasta -ACC eat -PASS -PST
 intended: ‘Pasta was eaten.’ Turkish ⁶²

The following example shows that the availability of raising-to-ACC is *not* affected by passivization of the embedded predicate.

⁶⁰ From Lohninger, Kovač & Wurmbrand 2022: p. 28, ex. (48)

^{61,62} Şener 2011: ex. (5a)

- (39) *John* [*makarna -yı ye -n -di diye*] *duy -du.*
 John.NOM pasta -ACC eat -PASS -PST that hear -PST
 ‘John heard that pasta was eaten.’ *Turkish*⁶³

In (40), we see that raising-to-ACC *is* affected by passivization of the *matrix* predicate: If the matrix predicate is passivized — for Şener (2008, 2011): if Voice cannot assign ACC; for DCT: if the subject NP has been demoted — raising-to-ACC is not possible anymore, (40b), while the version with NOM-subject is acceptable, (40a), as expected.

- (40) a. [*Pelin Timbuktu -ya git -ti diye*] *bil -in -iyor.*
 Pelin Timbuktu -DAT go -PST that know -PASS -PROG
 ‘Pelin is known to have gone to Timbuktu.’ *Turkish*⁶⁴
- b. *[*Pelin -i Timbuktu -ya git -ti diye*] *bil -in -iyor.*
 Pelin -ACC Timbuktu -DAT go -PST that know -PASS -PROG
 intended: ‘Pelin is known to have gone to Timbuktu.’ *Turkish*⁶⁵

In the following, I will assume that the differences between Turkish and Sakha are as minimal as the data allow; i.e., that the evidence on the source of ACC given for Sakha in §3.4.2 also holds for Turkish, and that the evidence for the position of the ACC-marked subject given for Turkish in this section (§3.4.3) also holds for Sakha.^{66,67,68}

⁶³ Şener 2011: ex. (5b)

⁶⁴ Şener 2011: ex. (6a)

⁶⁵ Şener 2011: ex. (6b)

⁶⁶ We have seen that raising-to-ACC in Turkish is sensitive to matrix passivization, like raising-to-ACC in Sakha. In Sakha, raising-to-ACC is also impossible with inherently intransitive matrix predicates such as ‘it became certain that’. The question arises whether this also replicates to Turkish raising-to-ACC constructions, to the extent that Turkish allows finite complement clauses (with or without the complementizer *diye*) to these predicates. However, fieldwork is outside the scope of this thesis, and so I will leave these questions for future research.

⁶⁷ Another fieldwork question that arises is this: Şener (2008, 2011) only discusses finite *complement* clauses. In Sakha (§3.4.2) raising-to-ACC can also apply in finite *adjunct* clauses (cf. ex. (32) above). Does this extend to Turkish as well?

⁶⁸ We know that the pattern from Sakha nominalized complement clauses does not extend to Turkish: in Turkish nominalized complement clauses, the subject always carries GEN, and cannot raise to receive ACC. This is expected because the GEN-marking on the embedded subject is precisely what is different between Turkish and Sakha: in Sakha, the subjects of nominalized complement clauses carry NOM before they raise to ACC. The overwritability of NOM by ACC, but not of GEN by ACC, is something that I will discuss in detail in §6.3.

However, Turkish nominalized adjunct clauses famously carry NOM-marking on their subjects. (For discussion of this fact see §6.6.3.) So we might expect that these subjects *can* raise to ACC in Turkish, like subjects of finite clauses. This remains to be verified.

3.4.4 CCAs in Uyghur: matrix binding domain

3.4.4.1 Introduction

Like Sakha (§3.4.2) and Turkish (§3.4.3), Uyghur allows ACC-marking on subjects of finite embedded clauses. Interestingly, the alternation between NOM and ACC on the subject NP indicates whether a subject indexical is interpreted relative to the embedded (NOM) or matrix (ACC) context. An example is given in (41).

- (41) a. *Ahmet* [*men* *ket* *-ti* *-m*] *di* *-di*.
 Ahmet me leave -PST -1SG say -PST
 (i) #‘Ahmet said that I_{speaker} left.’ (non-shifted) Uyghur⁶⁹
 (ii) ✓‘Ahmet said that he_{Ahmet} left.’ (shifted)
- b. *Ahmet* [*men -i* *ket* *-ti*] *di* *-di*.
 Ahmet me -ACC leave -PST say -PST
 (i) ✓‘Ahmet said that I_{speaker} left.’ (non-shifted) Uyghur⁷⁰
 (ii) #‘Ahmet said that he_{Ahmet} left.’ (shifted)

I will not be concerned here with the peculiar properties and possible analyses of indexical shifting in Uyghur. Rather, I will focus on the properties of ACC-case assignment and show that CCAs in Uyghur behave like those in Sakha and Turkish.

Shklovsky & Sudo (2014) have argued at length that Uyghur NOM- and ACC-subjects occupy different structural positions: below and above the context-shifting/monster operator, respectively. Their proposal is illustrated in Fig. 3.7. Major (2022) argues against this by showing that ACC-subjects and the monster operator are mutually exclusive. His proposal is illustrated in Fig. 3.8. In the following, I will summarize the arguments from both papers showing that ACC-subjects are structurally higher than NOM-subjects.

⁶⁹ Shklovsky & Sudo 2014: p. 383, ex. (4b)

⁷⁰ Shklovsky & Sudo 2014: p. 386, ex. (12b)

⁷¹ From Shklovsky & Sudo 2014: p. 393, ex. (33)

⁷² From Major 2022: p. 782, exx. (3) and (4)

3.4.4.2 Binding Theory: Principle A

We can show that for the purposes of Principle A (and also Principle B, §3.4.4.3), ACC-subjects are part of the matrix binding domain, and NOM-subjects are part of the embedded binding domain. We will use the reciprocal anaphor *birbiri* ‘each other’, which needs to be bound by a local plural antecedent. (42) establishes the base-line data: *birbiri* can be bound in the same clause (42a), but not from the matrix clause into the embedded clause (42b).

- (42) a. *Tursun bilen Ali birbiri -ni kör -d -i.*
 Tursun with Ali each.other -ACC see -PST -3SG
 ‘Tursund and Ali saw each other.’ Uyghur⁷³
- b. **Tursun bilen Ali [Mahinur -(ni) birbiri -ni kör -d -i] dé -d -i.*
 Tursun with Ali Mahinur -(ACC) each.other -ACC see -PST -3SG say -PST -3SG
 intended: ‘Tursun and Ali said that Mahinur saw each other.’ Uyghur⁷⁴

In (43a), *birbiri* with NOM case cannot be bound by the matrix subject, while in (43b) *birbiri-ni* with ACC case *can* be bound by the matrix subject. This must mean that the ACC-subject, but not the NOM-subject, is in the matrix binding domain.

- (43) a. **Tursun bilen Ali [birbiri ut -i -du] dé -d -i.*
 Tursun with Ali each.other win -NPST -3SG say -PST -3SG Uyghur⁷⁵
- b. *Tursun bilen Ali [birbiri -ni ut -i -du] dé -d -i.*
 Tursun with Ali each.other -ACC win -NPST -3SG say -PST -3SG
 literally: ‘Tursun and Ali said that each other will win.’ Uyghur⁷⁶

3.4.4.3 Binding Theory: Principle B

Using the pronoun *men* ‘I’, which obeys Principle B, i.e., must not be bound in its binding domain, we can see evidence that shows again that ACC-subjects are in the matrix binding domain, and NOM-subjects aren’t. The acceptability of (44a) shows that the NOM-marked embedded subject can be in a different binding domain than the coreferential matrix subject, while the unacceptability of (44b)

⁷³ Major 2022: p. 786, ex. (13a)

⁷⁴ Major 2022: p. 786, ex. (13b)

^{75,76} Major 2022: p. 786, ex. (13c)

shows that the ACC-marked subject *must* be in the same binding domain as the coreferential matrix subject.

- (44) a. *Men* [*peqet men* *-la nan ye -men*] *di -dim*.
 I only I -only bread eat -IPFV.1SG say -PST.1SG
 ‘I said that only I eat bread.’ Uyghur⁷⁷
- b. **Men* [*peqet men -i* *-la nan ye -men*] *di -dim*.
 I only I -ACC -only bread eat -IPFV.1SG say -PST.1SG
Uyghur^{78,79}

In §3.4.4.2 above, we saw that ACC-subjects *can* be bound in the matrix binding domain, otherwise (43b) would be as unacceptable as (42b). From the data in (44), we know that the ACC-subject *must* be bound in the matrix binding domain, and cannot be bound in the embedded binding domain, because otherwise (44b) would have a successful derivation, i.e., be judged acceptable.

3.4.4.4 Passive

Much like we have seen for Sakha above, for at least some speakers⁸⁰, ACC-subjects in Uyghur become impossible when the matrix verb is passivized, i.e., the subject NP is demoted and thus there is no longer a case-competitor for the ACC-subject. This is illustrated in (45).

- (45) a. *Manga* [*Aygül* *ket -t -i*] *dé -yil -d -i*.
 me.DAT Aygül leave -PST -3SG say -PASS -PST -3SG
 ‘It was told to me that Aygül left.’ Uyghur^{81,82}

⁷⁷ Shklovsky & Sudo 2014: p. 391, ex. (27a)

⁷⁸ Shklovsky & Sudo 2014: p. 391, ex. (27b)

⁷⁹ I wonder whether one factor for ungrammaticality in this example might be the presence of agreement, which we don’t usually see with ACC-subjects in Uyghur. However, the examples that I have seen discuss agreement use a different tense, so perhaps there is a specialty to IPFV/NPST. Note, though, that Shklovsky & Sudo (2014: ex. (27c)) give a grammatical version of (44b), which uses a not-coreferent subject so that *men-i* can be unbound in the matrix binding domain, and in this example they give the verb agreement as *ye-du* ‘eat-IPFV.3SG’. Let us assume, however, that they checked for this (Major 2022 doesn’t contradict them either) and that (44b) with *ye-du* instead of *ye-men* would also be unacceptable.

⁸⁰ I am not sure how uncontroversial these data are. Shklovsky & Sudo (2014: p. 390, ex. (23)) report that (45a) and (45b) are both acceptable, while Major (2022: p. 785, fn. 6) writes, “Contrary to what [Shklovsky & Sudo 2014] found, the consultants I have worked with universally reject accusative subjects under passivization.” I take it to be relevant that Major consultants rejected the ACC-subjects, while I must remain agnostic about the I-language of Shklovsky & Sudo consultants.

⁸¹ Shklovsky & Sudo 2014: p. 390, ex. (23)

⁸² Major 2022: p. 786, ex. (12)

- b. %Manga [Aygül -ni ket -t -i] dé -yil -d -i.
 me.DAT Aygül -ACC leave -PST -3SG say -PASS -PST -3SG
 intended: ‘It was told to me that Aygül left.’

Uyghur^{83,84}

(46) shows that passivization of the embedded verb does not influence the possibility of ACC-marking on the embedded subject, as we expected under DCT, parallel to Turkish above.

- (46) Ahmet [istikan -ni buz -ul -di] di -di.
 Ahmet cup -ACC break -PASS -PST.3SG say -PST.3SG
 ‘Ahmet said the cup broke.’

Uyghur⁸⁵

3.4.4.5 Scrambling

Shklovsky & Sudo (2014) show that ACC-subjects can scramble within the matrix clause, while this is not possible for NOM-subjects. As Major (2022) correctly notes, this leaves inconclusive whether the lowest position of ACC-subjects is inside or outside the embedded clause.

- (47) a. Istikan -ni Ahmet [istikan -ni buz -ul -di] di -di.
 cup -ACC Ahmet break -PASS -PST.3SG say -PST.3SG
 ‘Ahmet said the cup broke.’ Uyghur⁸⁶
- b. Mahinur Tursun -ni Alim -gha [Tursun -ni ket -t -i] dé -d -i.
 Mahinur Tursun -ACC Alim -DAT leave -PST -3SG say -PST -3SG
 ‘Mahinur said to Alim that Tursun left.’ Uyghur⁸⁷
- c. Mahinur Tursun -ni tünügün [Tursun -ni két -i -du] dé -d -i.
 Mahinur Tursun -ACC yesterday leave -NPST -3SG] say -PST -3SG
 ‘Mahinur said yesterday that Tursun will leave.’ Uyghur⁸⁸

3.4.4.6 Prosodic phrasing

Major (2022) and Major & Mayer (2019) present preliminary evidence that the ACC-marked subject forms an Intonational Phrase (IP) with the matrix subject to the exclusion of the rest of the

⁸³ Shklovsky & Sudo 2014: p. 390, ex. (23)

⁸⁴ Major 2022: p. 786, ex. (12)

⁸⁵ Shklovsky & Sudo 2014: p. 392, ex. (28a)

⁸⁶ Shklovsky & Sudo 2014: p. 392, ex. (28b)

⁸⁷ Major 2022: p. 787, ex. (15a)

⁸⁸ Major 2022: p. 787, ex. (15b)

embedded clause. Example (48) contains an ACC-marked non-shifted subject, with its spectrogram given in Fig. 3.9. Major (2022) and Major & Mayer (2019) propose to analyze this construction as in Fig. 3.10: “the only low target on the right edge of any prosodic constituent in [Uyghur] is at the right edge of Intonational Phrases (labeled L% [and highlighted in yellow] in figure [3.9]), which roughly correlate with clauses” (Major 2022: p. 788)

- (48) *Mahinur men -i [~~men~~ böljürgen -ni tünügün mijiwet -t -i] dé -d -i.*
 Mahinur me -ACC strawberry -ACC yesterday squeeze -PST -3SG say -PST -3SG
 ‘Mahinur said that I_{speaker} squeezed the strawberry yesterday.’ *Uyghur*⁸⁹

Ideally, one would consider a minimal pair with a NOM-marked shifted subject to see that this Intonational Phrase boundary is not present there. Major & Mayer (2019) analyze the indexically shifted sentence in (49), where the embedded subject is pro-dropped, providing its spectrogram, Fig. 3.11, and a corresponding schema, Fig. 3.12.

- (49) *Mahinur [böljürgen -ni tünügün mijiwet -t -im] dé -d -i.*
 Mahinur strawberry -ACC yesterday squeeze -PST -1SG say -PST -3SG
 ‘Mahinur said that he_{Mahinur} squeezed the strawberry yesterday.’ *Uyghur*⁹⁰

In contrast to the non-shifted construction in (48), (49) contains only one Intonational Phrase boundary at the end of the sentence.

At this time it is unclear to me why (48) contains an Intonational Phrase boundary between the matrix constituents (matrix subject and ACC-marked embedded subject) and the embedded clause, while (49) does not contain an Intonational Phrase boundary between the matrix subject and the embedded clause. Major (2022) and Major & Mayer (2019) draw the conclusion that the presence of the Intonational Phrase in (48) must mean that the ACC-subject has moved not only to the edge of the embedded clause, but into the matrix clause proper. Having replicated their argument, I leave the discussion at that until the data are understood in more detail.

⁸⁹ Major 2022: p. 787, Fig. 1

⁹⁰ Major & Mayer 2019: p. 251, ex. (11)

3.4.4.7 Summary

Taken together, Shklovsky & Sudo (2014) and Major (2022) agree that in Uyghur, too, ACC-subjects are in a higher structural position than NOM-subjects, and that ACC-subjects are in the matrix clause for purposes of binding (§§3.4.4.2–3.4.4.3), scrambling (§3.4.4.5), and prosody (§3.4.4.6); and, DCT claims, also for the purposes of case assignment (see §3.4.4.4).

3.4.5 Summary

In this section, I have presented data from Sakha, Turkish, and Uyghur cross-clausal A/A'-dependencies with ACC-marked subjects of finite complement clauses. These subjects originate inside the embedded clause and A'-move to a position in the left periphery of the embedded clause (Spec,CP.R). In this position, or perhaps in subsequent positions within the matrix clause, they can enter into A-dependencies such as case-marking and binding with elements of the matrix clause.

The data from Sakha in §3.4.2 in particular show that the ACC-dependency is between the embedded subject and a c-commanding NP in the matrix clause, such as the matrix subject. We have seen that the matrix subject can license more than one ACC-marking in this way, and we will pick up the theoretical relevance of this right away in §3.6.

3.5 Verbal-noun constructions in Turkish

3.5.1 Introduction

Turkish has constructions, here referred to as “verbal-noun constructions (VNCs)” (after Keskin 2009), that look like a noun with verbal properties (a verbal noun, VN; essentially a nominalization) that can assign accusative (ACC) to the internal argument (IA) of the nominalized predicate. The first interesting part about VNs is that unlike typical gerund-type nominalizations, they are morphologically simplex within Turkish, having been loaned as nominalizations into Turkish from Arabic, Persian, and French, mostly. The second interesting part about VNs is that they also appear

as the left part in light-verb constructions (LVCs). Keskin (2009: p. 82, ftn. 6, p. 84, ftn. 8) writes that he counted 202 VNs in Turkish, 111 of which co-occur with ACC-marked objects, and all of them also appear in LVCs.

(50a) and (51a) show examples of a light-verb construction, and (50b) and (51b) show the corresponding verbal-noun constructions.⁹¹

- (50) a. *Timur Anadolu -yu mahv -et -ti.*
Timur Anatolia -ACC devastation -do -PST
'Timur devastated Anatolia.' Turkish
- b. [*Timur -un Anadolu -yu mahv -ı*] *Moğol -lar -ı tatmin et*
Timur -GEN Anatolia -ACC devastation -3SG.POSS Mongol -PL -ACC satisfy do
-me -di.
-NEG-PST
'Timur's devastation of Anatolia did not satisfy the Mongols.' or
'Timur's devastating Anatolia ...' Turkish⁹²
- (51) a. *Başkan siz -i aff -et -ti.*
president you.PL -ACC pardon -do -PST
'The president pardoned you.' Turkish
- b. [*Başkan -ın siz -i aff -ı*] *biz -i mutlu et -ti.*
president -GEN you.PL -ACC pardon -3SG.POSS us -ACC happy do -PST
'The president's pardon of you made us happy.' or
'The president's pardoning you ...' Turkish⁹³

The class of VNCs is rather diverse, and no definite set of properties unites them, apart from the defining availability of ACC and the existence of related LVCs. The properties in which VNCs vary include, but are not limited to: 1. whether consultants 'feel' a silent LV *et-*; 2. whether the VN can be modified by adjectives or adverbs or both⁹⁴; 3. whether the VN is actually a bare object

⁹¹ I follow Modern Turkish orthography with respect to which VNs are spelled together with the following light verb, indicated by morpheme-separating dashes, and which aren't.

⁹² Keskin 2009: p. 78, ex. (9d)

⁹³ Keskin 2009: p. 158, ex. (182b)

⁹⁴ For discussion cf. Keskin 2009: §3.2.1.1.2, §3.2.1.2.1, §3.2.2.2.2, §3.3.2.4.

of the full verb *et-* ‘do’⁹⁵; 4. whether the ACC-marked internal argument can undergo DOM⁹⁶; and 5. whether the VN is in the origin language verbal or a nominalization itself;

I leave a detailed taxonomy to further research, and will try to argue here for another unifying property: that ACC is available only when the VNC contains a higher, structurally-case-marked NP: the external argument of the LV-predicate, which I will refer to here as the “subject” of the corresponding VN.

In (50b), and likewise in (51b), the subject of *mahv* ‘devastation’ is the GEN-marked DP that looks like a regular possessor and agrees with the possessive agreement suffix on *mahv*. Since Turkish is a pro-drop language, the subject NP can be replaced with *pro*, but the agreement will continue to show the presence of *pro*, (52).

- (52) [(*On -un*) *Anadolu -yu mahv -i*] *Moğolları tatmin etmedi.*
 s/he -GEN Anatolia -ACC devastation -3SG.POSS ...
 ‘His/her devastating Anatolia ...’ Turkish

In contrast, when the agreement suffix is dropped, ACC case becomes impossible, (53).

- (53) *[*Anadolu -yu mahv*] *Moğolları tatmin etmedi.*
 Anatolia -ACC devastation ...
 literally: ‘*The devastating Anatolia ...’ Turkish⁹⁷

Keskin (2009: p. 116) notes that “when subject agreement morphology (i.e. φ -features instantiating D^0 , [...]) is not present in the VNC, accusative assignment to the object is barred.” (see also Keskin 2009: §3.2.4.1, pp. 116–118) I will argue that this observation is exactly right, but will draw different conclusions from Keskin, who goes on to say that “[t]his suggests that accusative assignment depends on [the subject agreement] φ -features.” (p. 116)

In my 2023 presentation, I have argued that the ACC-assignment depends not on the subject agree-

⁹⁵ I suspect that this may be the case for *dua et-* ‘to pray’ and *telefon et-* ‘to call (on the phone)’, which behave differently from other LVs in that the “VN” can (i) be relativized out of its LVCs, (ii) be modified by demonstratives and accordingly undergo DOM, and (iii) be promoted to subject under passivization. Cf. Keskin (2009: p. 89 f.) for discussion. Key & Tat (2015) call this type of LVC “Type IV” and also categorize *dua et-* as this type.

⁹⁶ Cf. Keskin 2009: p. 79 f., p. 118 f. for ‘no’, and Akkuş 2015: pp. 137–140 for ‘yes’.

⁹⁷ Keskin 2009: p. 116, ex. (84a)

ment, but on the presence of the subject NP, which may be pro-dropped, but whose presence we can track through its agreement morphology. I.e., when the subject is present but silent, φ -agreement will show up as in (52); but when the subject is absent altogether as in (53), φ -agreement is impossible.⁹⁸ In this property, I will show, VNCs are completely parallel to Turkish *-mA* nominalizations (“short infinitive”; “action nominal” in Kornfilt 1997).

The rest of this section is structured as follows: In §3.5.2, I introduce *-mA* nominalizations. In §3.5.3 and §3.5.4 I discuss two diagnostics to show that *-mA* nominalizations and VNs both do not contain an AGENT. In §3.5.5 I present my analysis of VNs. Section 3.5.6 concludes.

3.5.2 Similarity to Turkish ‘-mA’ nominalizations

-mA nominalizations can appear with or without possessive agreement. When they carry agreement, (54), the possessor/subject can appear with GEN-marking, (54a), or can undergo pro-drop, (54b). In both cases, ACC-marking on the internal argument of the nominalized verb is possible, (55). As nominals, these nominalizations bear case-marking according to their context in the superordinate clause.⁹⁹

- (54) a. (Ben) [*Ahmed -in öl -me -sin*]-den kork -uyor -du -m.
 me Ahmet -GEN die -mA -POSS.3SG -ABL fear -PROG -PST -1SG
 ‘I was afraid that Ahmet would die.’

lit. ‘of Ahmet’s dying’

*Turkish*¹⁰⁰

- b. *Doktor* [*pro sigara iç -me -sin*]-i yasak et -ti.
 doctor cigarette drink -mA -POSS.3SG -ACC forbidden do -PST
 ‘The doctor forbade him/her to smoke.’

lit. ‘forbade his/her smoking’

*Turkish*¹⁰¹

⁹⁸ Note that the 3SG.POSS agreement marker *-(s)I(n)* also appears in compounds, where it does not indicate agreement with a 3SG possessor. For clarity, I gloss the agreement morpheme as ‘3SG’ and the compound marker as ‘CMPD’, even though they are presumably the same morpheme. Cf. Öztürk & Taylan (2016) for a recent analysis of the function of this morpheme, and my discussion in §6, especially §6.5.3.

⁹⁹ I write the case marking outside of the phrase-delimiting bracket, but I do not mean for this to denote that the case suffix attaches to the entire phrase. Since nothing can ever intervene between *öl-me-si(n)* and *-den*, the case marking is clearly on the nominalization itself.

¹⁰⁰ Kornfilt 1997: p. 51, ex. (222)

¹⁰¹ Ersen-Rasch 2012: p. 269, translation from German to English by me.

- (55) *Önemli ol -an [siz -in bu ev -i beğen -me -niz]-Ø -Ø.*
 important be -REL you.PL -GEN this house -ACC be.pleased -mA -POSS.2PL -NOM -COP
 ‘The important thing is that you (should) like this house.’

lit. ‘your being pleased with this house’

*Turkish*¹⁰²

On the other hand, when possessive agreement is absent on a -mA nominalization, ACC-case is not possible: a bare object is ok, (56a), but ACC-marking is not, (56b).

- (56) a. *[Resim boz -ma] işlem -i için aşağıdaki alet -ler gerekli -dir.*
 painting ruin -mA process -CMPD for following tool -PL required -COP
 ‘For the process of ruining paintings, the following tools are required.’

- b. **[Resim -ler -im -i boz -ma] işlem -i için ...*
 painting -PL -1SG.POSS -ACC ruin -mA process -CMPD for ...
 intended: ‘For the process of ruining my paintings, ...’

Turkish

Nor is there an “understood” subject in a -mA nominalization. The examples in (57) to (61) illustrate the difference in interpretation between a -mA nominalization, (57a), (58a), (59)–(61), and a “full” infinitive with a PRO-subject (cf. the next two sections) ending in -mAG, (57b), (58b).

- (57) a. *[Vergi kaçır -ma] Elif -i ürküt -ür.*
 tax smuggle -mA Elif -ACC frighten -AOR
 ‘[The concept of] Tax evasion frightens Elif.’

*Turkish*¹⁰³

- b. *[Vergi kaçır -mak] Elif -i ürküt -ür.*
 tax smuggle -INF Elif -ACC frighten -AOR
 ‘[The idea] To evade taxes [herself] frightens Elif.’

*Turkish*¹⁰⁴

- (58) a. *[Hayvan -lar -a eziyet et -me] günah -tır.*
 animal -PL -DAT torture do -mA sin -COP
 ‘Animal cruelty is a sin.’

*Turkish*¹⁰⁵

- b. *[Hayvan -lar -a eziyet et -mek] günah -tır.*
 animal -PL -DAT torture do -INF sin -COP
 ‘It is a sin to torture animals.’

*Turkish*¹⁰⁶

¹⁰² Göksel & Kerslake 2005: p. 420, ex. (75)

^{103, 104, 105, 106} Ersen-Rasch 2012: p. 235, translations from German to English by me.

- (59) *iç -me suy -u*
 drink -mA water -CMPD
 ‘potable water’ Turkish
- (60) *oku -ma kitab -ı*
 read -mA book -CMPD
 ‘reader, primer, storybook’ Turkish
- (61) *ara -ma motor -u*
 search -mA engine -CMPD
 ‘search engine’ Turkish

In the next two sections, I will apply two formal diagnostics for the presence or absence of PRO:

1. agent-oriented adverbs like *bile bile* ‘knowingly’, and 2. the subject-oriented anaphor *kendi* ‘by myself/...’. to VNCs, -mA nominalizations, and for comparison also to -mAG infinitives¹⁰⁷ containing PRO.

3.5.3 Diagnostic: Agent-oriented adverbs

The idea is that agent-oriented adverbs (AOAs) are possible only when there is an AGENT present that they can refer to. (Passives are complicated, cf. Laszakovits 2017b a.m.o., but we will not use passives here.) In non-passives, the AGENT is typically the subject, so we can use AOAs to test for

¹⁰⁷ Due to morphological particularities, I will exclude forms in -mAyA (DAT), -mAyI (ACC), and -mAnIn (GEN). As Ersen-Rasch (2012: p. 235 f.) writes, before an orthography reform in 1977, there were morphological forms clearly corresponding to declined infinitives: -mAğ-A (-mAG-DAT) and -mAğ-I (-mAG-ACC). The usage of these forms instead of declined -mA above corresponded exactly to the usages of -mAG and -mA as it is still today. This can be seen on the following example:

- (i) *Ayşe öl -meğ -e karar ver -miş -ti. Öl -me -ye karar ver -mek, öl -dür -meğ -e karar*
Ayşe die -mAG -DAT decision give -PRF -PST die -mA -DAT decision give -INF die -CAUS -INF -DAT decision
ver -mek -ten güç -tür.
 give -INF -ABL difficult -COP
 ‘Ayşe had decided **to die**. It is harder to decide **for dying** than it is to decide **to kill**.’ Turkish
 (Hikmet 1965: p. 61, cited after Ersen-Rasch 2012: p. 235, translations from German to English by me)

The orthography reform abolished the forms -mAğA and -mAğI in favor of -mAyA and -mAyI, thus collapsing the distinction between declined -mAG and -mA. In a few elicitation attempts with middle-aged native speakers, I have noticed that together with the spelling, the introspective awareness and also the pronunciation difference between ⟨y⟩ [j] and ⟨ğ⟩, pronounced as [j] with high vowels and as [ɣ] with dark vowels, seems to have been lost. Perhaps native speakers who have been schooled in orthography well before 1977 would be more aware of this distinction. In Modern Turkish, it seems to be a topic of discussion whether -mAyA, -mAyI, and -mAnIn count as declinations of -mA or -mAG, or possibly as either (Ersen-Rasch 2012: p. 235; Göksel & Kerslake 2005: p. 418).

the syntactic presence of a subject. (62) lists some AOAs for Turkish:

- (62) a. *bile bile, bilerek* ‘knowingly’
 b. *kasıtlı olarak, kasten, mahsus* ‘on purpose’
 c. *amacıyla, niyetiyle, maksadıyla* ‘with intent’
 d. *ikazıma rağmen* ‘despite my warning’
 e. *inatla* ‘stubbornly’

How do we tell whether the adverbs in (62) are actually AGENT-oriented? We try to insert them into contexts that do not have a (human/sentient) AGENT and expect them to be nonsensical. This is illustrated in (63).

- (63) a. #*Rüzgar bile bile es -iyor.*
 wind knowingly blow -PROG
 intended: ‘The wind is blowing on purpose.’ Turkish

With the unnaturalness of (63), we have established that the adverb *bile bile* in (62) is indeed an AOA. Now let us test whether we can insert it in *-mAG* infinitives, (64), *-mA* nominalizations without agreement, (65), and *-mA* nominalizations with agreement, (66).

- (64) [*Resim -ler -im -i bile bile boz -mak*] *salaklık -tır.*
 painting -PL -POSS.1SG -ACC knowingly ruin -INF stupidity -COP
 ‘To knowingly ruin my paintings is a stupidity.’ Turkish

- (65) a. [*Tablo değiştir -me*] *işlem -i için aşağıdaki alet -ler gerekli -dir.*
 painting alter -mA process -CMPD for following tool -PL required -COP
 ‘For the process of altering paintings, the following tools are required.’ Turkish
 b. *[*Bile bile tablo değiştir -me*] *işlem -i için ...*
 knowingly painting alter -mA process -CMPD for ...
 ‘For the process of knowingly altering paintings, ...’ Turkish

- (66) [(Senin) *resim -ler -im -i bile bile boz -ma -n*] *hoş değil.*
 (your) painting -PL -POSS.1SG -ACC knowingly ruin -mA -POSS.2SG nice is.not
 ‘Your ruining my paintings on purpose is not nice.’ Turkish

We find that *-mA* nominalizations without agreement, which were not compatible with ACC in (56b),

are also incompatible with AOAs, (65), while *-mAG* infinitives allow *ACC* and AOAs, (64), and *-mA* nominalizations with subject and agreement also allow *ACC* and AOAs, (66). Ideally we would also apply this test to VNCs with and without agreement, but my consultants didn't like VNCs to begin with.¹⁰⁸

3.5.4 Diagnostic: Subject-oriented anaphora

Wurmbrand (2002) has used anaphor binding as a diagnostic for the presence or absence of PRO in (non)-restructuring infinitives, arguing that in the absence of a matrix subject, if an anaphor inside an infinitive can be bound, it must be getting bound by PRO. Let us apply the same test to Turkish.

Turkish *kendi* resp. *kendi+POSS* has multiple uses and properties (Göksel & Kerslake 2005; Kornfilt 1997). Here we are interested in the usage of the adverbial anaphor *kendi-* with possessive agreement with the subject that translates to '(I did this) by myself.' Göksel & Kerslake (2005: p. 267) describe this usage as subject-oriented. We can now apply the test whether *kendi+POSS* can be inserted as adverbial to the three constructions under consideration. (67) shows that the subject-oriented anaphor (SOA) is acceptable in *-mAG* infinitives, as expected.

- (67) [*Kendin -e araba al -abil -mek*] *bir statü sembol -ü -dür.*
 self -DAT car buy -can -INF one status symbol -CMPD -COP
 'Being able to buy a car for oneself is a status symbol.' Turkish

The examples in (68) show that the SOA is not acceptable in *-mA* nominalizations without agreement, as predicted.

- (68) a. [*Araba al -ma*] *prosedür -ü karmaşık ol -abil -ir.*
 car buy -mA procedure -CMPD complicated be -can -AOR
 'The procedure of buying a car can be complicated.' Turkish
- b. *[*Kendin -e araba al -ma*] *prosedür -ü karmaşık ol -abil -ir.*
 self -DAT car buy -mA procedure -CMPD complicated be -can -AOR
 'The procedure of buying a car for oneself can be complicated.' Turkish

¹⁰⁸ Claudia Römer has speculated (p.c.) that VNCs may be relics from Ottoman Turkish that survived the Turkish language reform.

In (69), we see that *-mA* nominalizations with agreement allow the SOA, again as predicted.

- (69) Öğretmen, [(benim) ödev -i kendi -m yap -ma -m]-ı
 teacher.NOM me.GEN assignment -ACC self -1SG.POSS make -mA -1SG.POSS -ACC
 söyle -di.
 say -PST
 ‘The teacher said that I should do the assignment (by) myself.’ Turkish

Keskin has provided the following tests of SOAs in VNCs. As predicted, he finds that VNCs behave like *-mA* nominalizations: without agreement on the VN, an SOA is not possible, (70), while with agreement it is, (71).

- (70) ??[Kendin -e hizmet] insan -ı yozlaştır -ır.
 self -DAT service human.being -ACC corrupt -AOR
 ‘Self-servitude corrupts people.’ Turkish¹⁰⁹
- (71) [İnsan -ın kendin -e hizmet -i] de, aile -sin -e, toplum
 human.being -GEN self -DAT service -3SG.POSS too family -3SG.POSS -DAT society
 -un -a hizmet -i de kamu hayrı için -dir.
 -3SG.POSS -DAT service -3SG.POSS too public good for -COP
 ‘Both one’s self-servitude and one’s service to one’s family and society is for the public good.’ Turkish¹¹⁰

In addition to (70) and (71) above, Keskin (2009: p. 112 f.) shows that (70) becomes acceptable if the agreement-less VNC *kendin-e hizmet* is not in subject position, but in object position, which makes it possible for the subject DP to bind the anaphor inside the VNC, thus showing that the problem with (70) is indeed the lack of an antecedent.

3.5.5 Analysis of VNCs

The reader may have noticed that all the VNs discussed here are translated as nominalizations in our object language: *mahv* ‘devastation’, *istila* ‘invasion’, etc. The same holds true for all the other VNs I have found discussed in the literature (Akkuş 2015; Keskin 2009; Sezer 1991; Solak 2021).

¹⁰⁹ Keskin 2009: p. 112, ex. (76b)

¹¹⁰ Keskin 2009: p. 112, ex. (76d)

This raises the question whether VNs simply *are* nominalizations: lexicalized ones (like English *destruction*) or productive gerund-like ones (like English *destroying*; Abney 1987; Chomsky 1970). The latter options seems particularly interesting since English gerunds can also take ACC-marked objects, and this may derive from gerunds' verbal core. Additionally, the derivation of gerunds from verbs would account nicely for the identical case patterns between VNCs and LVCs “without one single exception” in Keskin's (2009: p. 82, ftn. 6, p. 84, ftn. 8, pp. 90, 99) collection of 202 VNs.

I will propose here that VNs are *not* derived from LVCs, but share the same core: they spell out only the root, which can be combined with a verbalizer *v* (the light verb) or a nominalizing categorizer *n*. What differentiates VNs from regular (non-verbal) nouns is that VNs, like verbs, can take a complement projected by the root, which is not possible for nouns otherwise (see, i.a., Öztürk & Taylan 2016: p. 101, and the references therein: Coon 2013; Gallego 2009; Grimshaw 1990; Hale & Keyser 2002; Kayne 2008; Marantz 2000; Mateu i Fontanals 2002).

We can see evidence for this in Akkuş's (2015) account of lexical blocking. Akkuş discusses LVC-VNC pairs where the VNs differ¹¹¹, although they remain derivationally related in the donor language. Borrowed mostly from French, some of these pairs are illustrated in (72).

- (72) a. *restore et-* ‘to restore’, *restorasyon* ‘restoration’
 b. *finanse et-* ‘to finance’, *finansman* ‘financing’
 c. *sabote et-* ‘to sabotage’, *sabotaj* ‘sabotage’

They all follow the pattern in (73) whereby the original infinitive (*restore*, from French *restaurer*) appears only with *et-* in an LVC, and the original nominalization (*restorasyon*, from French *restauration*) appears only in a VNC.

- (73) a. *Belediye bina -yı restore et -ti.*
 municipality building -ACC restore do -PST
 ‘The municipality restored the building.’

*Turkish*¹¹²

¹¹¹ Keskin (2009) must not have considered these.

¹¹² Akkuş 2015: p. 140, ex. (16a)

- b. **Belediye bina -yı restorasyon et -ti.*
municipality building -ACC restoration do -PST
*Turkish*¹¹³
- c. *belediye -nin bina -yı restorasyon -u*
municipality -GEN building -ACC restoration -POSS.3SG
'the municipality's restoration of the building'
*Turkish*¹¹⁴
- d. **belediye -nin bina -yı restore -si*
municipality -GEN building -ACC restore -POSS.3SG
*Turkish*¹¹⁵

Akkuş analyzes this as an instance of lexical blocking: the availability of *restorasyon* in the Turkish lexicon blocks insertion of *restore* as VN. For VNs like *mahv* and *istila*, only a single lexical item is available, which is sufficiently underspecified to appear in both environments.

Key & Tat (2015) and Solak (2021) argue that a VN spells out multiple syntactic heads, depending on which heads would be necessary in the non-borrowed Turkish version. I refer the reader to the respective papers for details about their proposals, and will suffice it here to say that I believe that these papers are on the right track: For the nominalization-like VNs discussed earlier, such as *mahv*, *istila*, *rezervasyon*, etc., I propose that they contain (almost) the same functional projections as *-mA* nominalizations, which, as I have shown above, do not contain an AGENT/PRO. Let us assume the structure of a *-mA* nominalization and of a VN as sketched in Fig. 3.13^{116,117,118}. Neither structure contains an AGENT or PRO. ACC case on the IA is only possible if (i) a higher c-commanding NP is merged in the possessor position of the nominal (not depicted in Fig. 3.13), and (ii) the IA leaves

¹¹³ Akkuş 2015: p. 141, ex. (17a)

¹¹⁴ Akkuş 2015: p. 141, ex. (17b)

¹¹⁵ Akkuş 2015: p. 140, ex. (16b)

¹¹⁶ The present proposal does not clash with Baker's (2003, 2015a) generalization that gerund-type nominalizations (such as English *breaking*), which he also calls verbal nouns because they have properties of both verbs and nouns, must always be bimorphemic (*break-ing*): Baker's (2003) Reference Predication Constraint states that a syntactic node can either introduce a referential index (as nouns do), or can project a specifier (as predicates do), but no syntactic node can do both. If a VN would realize a single syntactic node, this would clash with the Reference Predication Constraint. Thus, Baker (2015a, 2003) concludes, the two properties need to be split onto two syntactic nodes: one predicate (*break*) that introduces the specifier, and one nominalizer (*-ing*) that introduces a referential index. Unlike English *break-ing*, VNs like *mahv* do not display more than one morpheme within Turkish, but if there is indeed more than one derivational head involved, they are compatible with the Reference Predication Constraint nonetheless.

¹¹⁷ I have assumed here that in Fig. 3.13a, there is essentially active Voice present in a *-mA* nominalization because the exponents of passive, causative, etc. Voice can appear, even though clearly there is no PRO being projected.

¹¹⁸ I remain agnostic as to whether Fig. 3.13a also contains head movement like Fig. 3.13b or not.

its base position to topicalize closer to the possessor following the properties of DOM as described in §3.3.¹¹⁹ The following exponence rules account for the distribution of *restore* and *restorasyon*, spelling out the same root.

- (74) a. $\sqrt{22} \rightarrow \text{restore}$
 b. $\sqrt{22.v.Voice.n} \rightarrow \text{restorasyon}$

Rule (74b) being more specific than rule (74a) is inserted into the more specific context in Fig. 3.13b, while by the Subset Principle, (74b) cannot apply to the terminal node $\sqrt{22}$ in Fig. 3.13a.

The Arabic VNs that do not alternate between VNCs and LVCs can be specified with a single exponent rule:

- (75) $\sqrt{37} \rightarrow \text{mahv}$

If Fig. 3.13 were minimally different in that the roots were given as $\sqrt{37}$, exponent rule (75) would apply to both configurations, yielding the nominalized LVC *mahv et-me* on the left and the VNC *mahv* on the right. This is the desired result.

Having argued that VNs are not simple nouns but nominalizations with a verbal core, it follows

¹¹⁹ The DOM possibilities in VNCs are illustrated in (i) with the VN *transfer* from the LVC *transfer et-* ‘to transfer’. In (ii), we see that adverb placement in this construction mirrors that of DOM in matrix clauses: an ACC-marked (moved) DO precedes low adverbs, while a bare (unmoved) DO follows low adverbs.

- (i) a. *takım-in futbolcu transfer -i*
 team-GEN football.player transfer -POSS.3SG
 ‘the team’s football player transfer’ *Turkish (Akkuş 2015: p. 137, ex. (11a))*
 b. *takım-in futbolcu -yu transfer -i*
 team-GEN football.player -ACC transfer -POSS.3SG
 ‘the team’s transfer of the football player’ *Turkish (Akkuş 2015: p. 137, ex. (10a))*
- (ii) a. *takım-in futbolcu -yu beklenmeyen transfer -i*
 team-GEN football.player -ACC unexpected transfer -POSS.3SG
 ‘the team’s unexpected transfer of the football player’ *Turkish (Akkuş 2015: p. 139, ex. (12))*
 b. **takım-in futbolcu beklenmeyen transfer -i*
 team-GEN football.player unexpected transfer -POSS.3SG
Turkish (Akkuş 2015: p. 139, ex. (14a))
 c. *takım-in beklenmeyen futbolcu transfer -i*
 team-GEN unexpected football.player transfer -POSS.3SG
 ‘the team’s unexpected transfer of football players’ *Turkish (Akkuş 2015: p. 139, ex. (13))*
 d. *#takım-in beklenmeyen futbolcu -yu transfer -i*
 team-GEN unexpected football.player -ACC transfer -POSS.3SG
 ✓ ‘the team’s transfer of the unexpected football player’
 # ‘the team’s unexpected transfer of the football player’ *Turkish (Akkuş 2015: p. 140, ex. (14b))*

directly why their internal arguments can receive ACC-marking and undergo DOM alternations: the internal argument is projected as complement to the VN (i.e., to the root, cf. Bobaljik & Harley 2017; Harley 2013, 2014a,b; Harley, Tubino Blanco & Haugen 2017; Kratzer 1996; Marantz 1997 i.a.), and may move by topicalization into the higher CAD, where it is assigned ACC by case-competition against the (later GEN-marked) c-commanding NP in Spec,PossP. In the absence of the PossP projection (such as in bare VNCs and *-mA* nominalizations), such case-competition is not possible and ACC case is not available, as illustrated above.

This is how VNs differ from lexicalized nominalizations such as *yıkım* ‘destruction’ (from the root *yık-* ‘destroy’), which, unlike verbs, cannot project their internal argument in their complement.

- (76) a. **bina -yı yıkım*
 building -ACC destruction
 intended: ‘the destruction *(of) the building’ *Turkish*
- b. **bina yıkım*
 building destruction
 intended: ‘the destruction *(of) a building’ *Turkish*

Rather, I follow Öztürk & Taylan (2016) in that it is a functional head which is required to introduce a noun’s arguments¹²⁰, whether the noun is inherently relational (like *parent*) or not. I will come back to this in detail in chapter 6.

3.5.6 Summary

In summary, what I hope to have shown in this section is that the puzzling phenomenon of verbal nouns (VNs), where seemingly a noun can assign ACC to its complement, can be captured by acknowledging that (i) these nouns are actually nominalizations, and (ii) ACC is available exactly when there is a competing NP present in the structure, which in VNCs appears as GEN-marked possessor and triggers possessive agreement on the VN. VNs realize roots only (except for lexical blocking in some French borrowings) and (yet) select their internal argument (Harley 2013, 2014a). Selecting

¹²⁰ Perhaps similarly to Kratzer’s (1996) Voice, which introduces the AGENT argument via Event Identification.

their arguments ensures that VNs select the same arguments with the same case patterns in both light-verb constructions and verbal-noun constructions. The lexical underspecification allows the VN to appear both as the left part in an LVC as well as in a VNC. We have seen that there are exceptions to this when the root plus the syntactic heads in a VNC have their own lexical entry, which will block insertion of the underspecified entry. The fact that there are DOM-alternations on the internal argument in VNCs (Akkus 2015) indicates that we are dealing with structural accusative case here.

3.6 The relevance of multiple assignments of Acc for syntactic theory

3.6.1 Introduction

As discussed above in §2.4.5, there are two competing proposals about how to implement the rules of DCT in Minimalism. On the one hand, Baker (2015b), Marantz (1991), McFadden (2004), Preminger (2014), Yip, Maling & Jackendoff (1987), and Zaenen, Maling & Thráinsson (1985) and others propose that the assignment/licensing of dependent case is an operation in syntax or morphosyntax that has not previously been attested in phenomena other than case, and is thus best characterized by a novel operation, which, following Marantz's (1991) original proposal, is called Case Competition. Minimalism thus contains not three but four core operations: Merge, Move (resp. internal Merge), Agree, and Case Competition.

Under some sets of assumptions, there is a question about how to express DCT in a way that is compatible with Minimalism without postulating the novel operation Case Competition. Clem & Deal (2024) and Poole (2024) argue that this fourth operation is not necessary because the assignment/licensing of structural case to NP₁ depending on properties of the competing NP₂ can actually be modeled by Agree: a functional head containing ordered probes first probes for the competing NP₂ and only if it finds this goal will the second probe search for NP₁ and value its [case: ____]

feature as dependent case as reflex of this Agree operation. Thus, in Preminger’s (2024) terminology, this formulation of dependent case is actually parallel to inherent case in that it is assigned by a syntactic head, which Preminger calls “head case”.

In this section I will argue that the fact that the same competitor can feed ACC licensing/assignment on multiple nominals in Turkic (which I call “1 : n licensing” (and Poole 2022: p. 366 calls “covariance”), see §3.4 and Fig. 3.14) mediates between these two proposals in favor of the extra operation Case Competition (§3.6.3). This argument will be based on Laszakovits (2018), where I have shown how to model DCT in Minimalist Grammars, a rigorous formalism for Minimalist syntax (Stabler 1997 et seq.). In §3.6.4 I discuss the implication of extending Rooryck & vanden Wyngaerd’s (2011) proposal from binding to case: nominals with unvalued case contain a probe that searches for a case-competitor. §3.6.5 concludes.

3.6.2 Case Competition

Let us first look at how Baker (2015b) and Preminger (2014) deal with 1 : n licensing of ACC. In Fig. 3.15¹²¹, phasal heads (Voice, C) are circled and colored, and their complements, Spell-out domains, are boxed in colored, dashed lines. Case assignment is assumed to happen at Spell-out. Consider the highest phasal head C. In Baker’s (2015b) algorithm, upon Merge of this C, its complement is considered in the following way: First, ignore all adjuncts (the embedded CP). Then compute all new pairs of c-commanding NPs. This is the pair $\langle NP_2, NP_3 \rangle$, that has not been considered previously because at the Spell-out of VP, NP_2 had not yet been Merged. Assign ACC to the second NP in each pair; this is NP_3 . Then, include adjuncts in the consideration. Compute all new pairs of c-commanding NPs. This gives the pair $\langle NP_2, NP_1 \rangle$; the pair $\langle NP_2, NP_3 \rangle$ is ignored because it is no longer ‘new’. Assign ACC to the second NP in each pair; this is NP_1 . Baker (2015b)

¹²¹ Technically, in Fig. 3.15, the landing sites of NP_1 and NP_3 are implemented as adjuncts, which matters for Baker’s (2015b) algorithm. I have argued in §3.3 that the movement of NP_3 , and in §3.4.3 that the movement of NP_1 , is topicalization movement. Thus, we could assume that there are intermediate topic projections present, and that the topicalized NPs are actually in specifiers of those TopPs, allowing us to consider NP_3 on the first cycle of Baker’s (2015b) assignment of dependent cases. In the present example, if NP_3 actually *is* an adjunct to VP and only assigned ACC in the adjunct-considering cycle of Baker’s (2015b) algorithm, no differences in case assignment would arise. This is not so for all instances of DOM-triggered ACC-assignment, though.

closes his algorithm with the step: assign unmarked case (NOM) to all other NPs: NP₂. Under Preminger’s (2014) mapping of morphological cases to syntactic feature values, unmarked cases are represented by empty feature values, so NP₂ would simply remain unvalued, which will spell out as NOM. Thus, in this implementation of dependent case assignment with Case Competition, it makes absolutely no difference whether NP₂ serves as licensor/competitor for one or more other dependent-case-receiving NPs.

3.6.3 Dependent-down case as reflex of Agree in φ -features

Now consider the alternative formulation by Clem & Deal (2024) and Poole (2024). As introduced in §2.4.5, Poole (2024) writes that for dependent-down case assignment, on an assumed feature sequence $fseq = \langle C \succ T \succ v \succ V \rangle$, “some head above both DP_{EA} and DP_{IA}” (i.e., T or C; let’s assume C) carries the relevant probes: the first probe to become active searches for an NP with unvalued [case: ____], finds the nominal labeled NP₂ in Figs. 3.14 and 3.15, and copies its φ -features, which may or may not be realized morphologically as agreement with the NOM- or ABS-marked argument. If this Agree operation has been successful, the second probe will become active, search for a distinct NP₁ with unvalued [case: ____], copy NP₁’s φ -features (which, again, may or may not be realized morphologically^{122,123}), and value the NP’s case to [case: DEP] (in Turkic: [case: ACC]). I leave open whether by Poole’s (2024) algorithm, the second Agree operation finds NP₁ or NP₃ in Fig. 3.14, but crucially it finds only either one.

In order to model a 1 : n pattern of ACC assignment, Poole (2024) would need to make one of three additional assumptions. His first option is to let the ACC-assigning head carry multiple ACC-assigning probes. Such a head would behave in constructions with only one ACC-marked NP in the same way that his proposed probe stack works in intransitives: the lower probes do not find a goal and Agree simply fails. The problem with this option is that there is no definite number of ACC-probes that we can stipulate that this head carries: Is it 2? 3? 4? 5? Processing will not let the

¹²² By Bobaljik’s (2008) prediction, if an agreement probe in a language tracks goals valued with a dependent case, this probe must also track goals with an unmarked case (empty value).

¹²³ Jonathan Bobaljik (p.c.) has mentioned to me that in Gilligan’s (1987) survey of 100 languages, languages with subject and object agreement outnumber languages with just subject agreement by 2 : 1.

number go very high, but a model of competence (Marr’s (1982) “computational” level of analysis) should not care about hardware-limitations (Marr’s “physical” level). Thus this option would need to leave the number of probes unbounded. I.e., the grammar will contain an unbounded number of C-heads: one for each number of ACC-assigning probes. As Savitch (1993) points out, assuming unboundedness usually allows for more compact grammars — it is boundedness that needs to be motivated, not unboundedness.¹²⁴ Choosing this option to salvage Poole’s (2024) proposal would push the formal complexity of syntax beyond the class of regular (i.e., Chomsky’s (1959) Type 3) tree languages — a result that is very much undesirable (see §1 and §4.5).

Poole’s (2024) second option to model 1 : n licensing of ACC is to let the single ACC-assigning probe find multiple goals. Simultaneous agreement with multiple goals has been proposed by Hiraiwa (2001) and termed “Multiple Agree”. It is not clear to me which goal’s φ -features are copied onto the probe; perhaps there is a prediction to be made about the distinct goals’ φ -features needing to match in a certain way.¹²⁵ Contrary to Poole’s (2024) other assumptions about the properties of Agree, Multiple Agree for ACC seems an ad-hoc stipulation, and will most likely yield some wrong predictions.

The third option is to leave the probing head as Poole (2024) proposed, but to allow multiple such heads in the derivation. A 1 : 2 pattern could be captured by allowing both T and C to carry one ACC-assigning probe. For more than two instances of ACC-marking, however, more heads would be necessary. Recursion comes to mind, as well as cartography. In either case, such a stipulation also seems unmotivated at this point.

3.6.4 Dependent-down case via Agree between nominals

In order to model dependent-down case via Agree, one would need to find a way to match the number of probes to the number of case-receiving nominals. As I have shown in the preceding

¹²⁴ An example of that is Joshi, Becker & Rambow (2000), who argue that German scrambling obeys principled bounds that coincide with the expressivity limits of Tree-Adjoining Grammars (TAGs).

¹²⁵ Such as: where multiple probing of objects by a single Probe is advocated for in the literature, there’s widespread evidence that it yields PCC effects (some kind of limitation on feature combinations in a single Probe; e.g. Stegovec 2019, 2020). I thank Jonathan Bobaljik for pointing me in this direction.

subsection 3.6.3, this cannot be achieved by placing all probes onto the same syntactic head. Could we model this in a way such that the n -many probes are distributed across the n -many ACC-marked nominals, with each nominal containing its own probe?

Such an approach has been proposed by Rooryck & vanden Wyngaerd (2011) for binding: an anaphor, requiring an antecedent, contains a probe for φ -features, which searches for a suitable goal and enters into an Agree relationship with the licensing antecedent. As long as the antecedent can serve as goal more than once (i.e., the Activity Condition does not apply) unboundedly many anaphors will be able to be licensed.¹²⁶

Can Rooryck & vanden Wyngaerd's (2011) approach, then, extend to the licensing of dependent-down case? Let us see what this would entail. Any nominal N_P with as-of-yet unvalued [case: ____] would need to contain a probe P searching for a nominal N_C that qualifies as case-competitor. For simplicity, let us assume that case-competitors must have unvalued [case: ____] and no other requirements, but see §2.5.4 and §2.5.5 for discussion. Upon matching with the goal N_C , the probe P would then get valued as [case: ACC]. Where does the value 'ACC' come from? It might be copied over from the goal N_C . But then, the value 'ACC' must be independent of N_C 's actual case value, which is empty at the time of Agree, and may remain empty or may be valued at a later point in the derivation. This means that N_C would need to have two case values, currently unvalued [case_{self}: ____] and pre-valued [case_{other}: ACC], and that, crucially, N_P 's probing feature [case_{self}: ____] needs to match with N_C 's other feature [case_{other}: ACC] in order to copy its value, and not with N_C 's identically labeled feature [case_{self}: ____].

I have investigated this form of $1 : n$ licensing from a mathematical point of view in Laszakovits (2018), and I have shown that dependent-down case licensing can be modeled in a Minimalist Grammar (Stabler 1997, 2011).

The picture becomes even more complicated in languages with more than one dependent case, such as tripartite languages, where a nominal qualifying as case-competitor can provide the values

¹²⁶ Besides binding, such a $1 : n$ licensing configuration also appears in the licensing of negative polarity items (NPIs), where a single operator creating a downward-entailing context suffices to license unboundedly many NPIs; as well as in other constructions.

‘ACC’ and ‘ERG’, or in languages where the dependent-down case differs depending on the case-assignment domain, as Baker (2015b) has proposed. Then, nominals may need to carry more than one [case_{other}] feature, and the matching-procedure would be sensitive to more factors.

Note that there is one problem one might suspect to arise that can easily be discarded: When a previously unmarked nominal N_C receives a case value of its own, making it ineligible to serve as case-competitor, its inaccessibility to the probe on N_P can be modeled straight-forwardly by specifying the probe to be “discriminating” in the sense that it only considers nodes with particular properties (here: with unvalued [case_{self}: ____]) as possible goals (Bobaljik 2008; Deal 2017; Preminger 2014).

3.6.5 Summary

In this section I have argued that the data from §3.4 on Sakha favor Preminger’s (2014) implementation of DCT in Minimalism over Poole’s (2024) Agree-based implementation. In the relevant data from Sakha, Fig. 3.16, Baker & Vinokurova (2010) have carefully shown that dependent-down ACC case is assigned to two independent NPs based on case-competition with a single competing/assigning NP. Since one of these ACC-marked NPs originates from an adjunct clause, and a sentence can in principle contain arbitrarily many adjunct clauses, we can assume that a competing NP can license not only two instances of ACC, but arbitrarily many, as indicated in Fig. 3.17 (Laszakovits 2018). In this section I have argued that this poses a problem for Agree-based implementations of DCT where dependent case is assigned by a probe on a functional head, because we can hardly add arbitrarily many such probes to the derivation. On the contrary, the novel operation ‘Case Competition’, being an operation, can apply freely whenever its requirements are met, thus accounting for the Sakha data.

3.7 Conclusions

In this chapter I have presented an important property of dependent-up case: the possibility of $1 : n$ assignment/licensing, i.e., the fact that the same competing NP can serve as case-competitor for more than one dependent-down (ACC) case-marked nominal. We have seen the data from Turkic illustrating this pattern in section 3.4, and I have discussed the theoretical implications for this in section 3.6, arguing that the implementation of DCT by Baker (2015b) and Marantz (1991) and many others, which assumes a syntactic operation ‘Case Competition’, can handle $1 : n$ assignment, while the Agree-based implementation by Clem & Deal (2024) and Poole (2024) cannot: in a nutshell, because Agree requires probes on lexical items, and the lexicon needs to be bounded, so anything in the grammar that is unbounded cannot be lexical.

Sections 3.2 and 3.3 have laid the groundwork for this argument, illustrating (§3.2) that ACC in Turkic is a structural case in that it is sensitive to properties of the syntactic structure and can disappear or change with the syntactic structure. In §3.3 I have illustrated the domain sensitivity of DCT’s rule of ACC case assignment: ACC can only be assigned in the clausal case-assignment domain (CAD), and cannot be assigned in the verbal CAD. We see this on direct objects that can only receive ACC-marking if the direct object has topicalized from the verbal into the clausal domain. A similar argument was made in §3.4 where subjects of embedded clauses can topicalize to the edge of the embedded clause, which puts them into the matrix CAD. I have carefully summarized data from Turkish, Sakha, and Uyghur that show that the ACC-marking on topicalized subjects of embedded clauses is sensitive to the presence of a matrix subject, thus clearly illustrating that ACC is a dependent-down case.

Having shown that dependent-down cases can take $1 : n$ patterns, I will now turn in chapter 4 to dependent-up cases, which famously are not able to take an $n : 1$ pattern (e.g. Nie 2017) and discuss reasons for this difference.

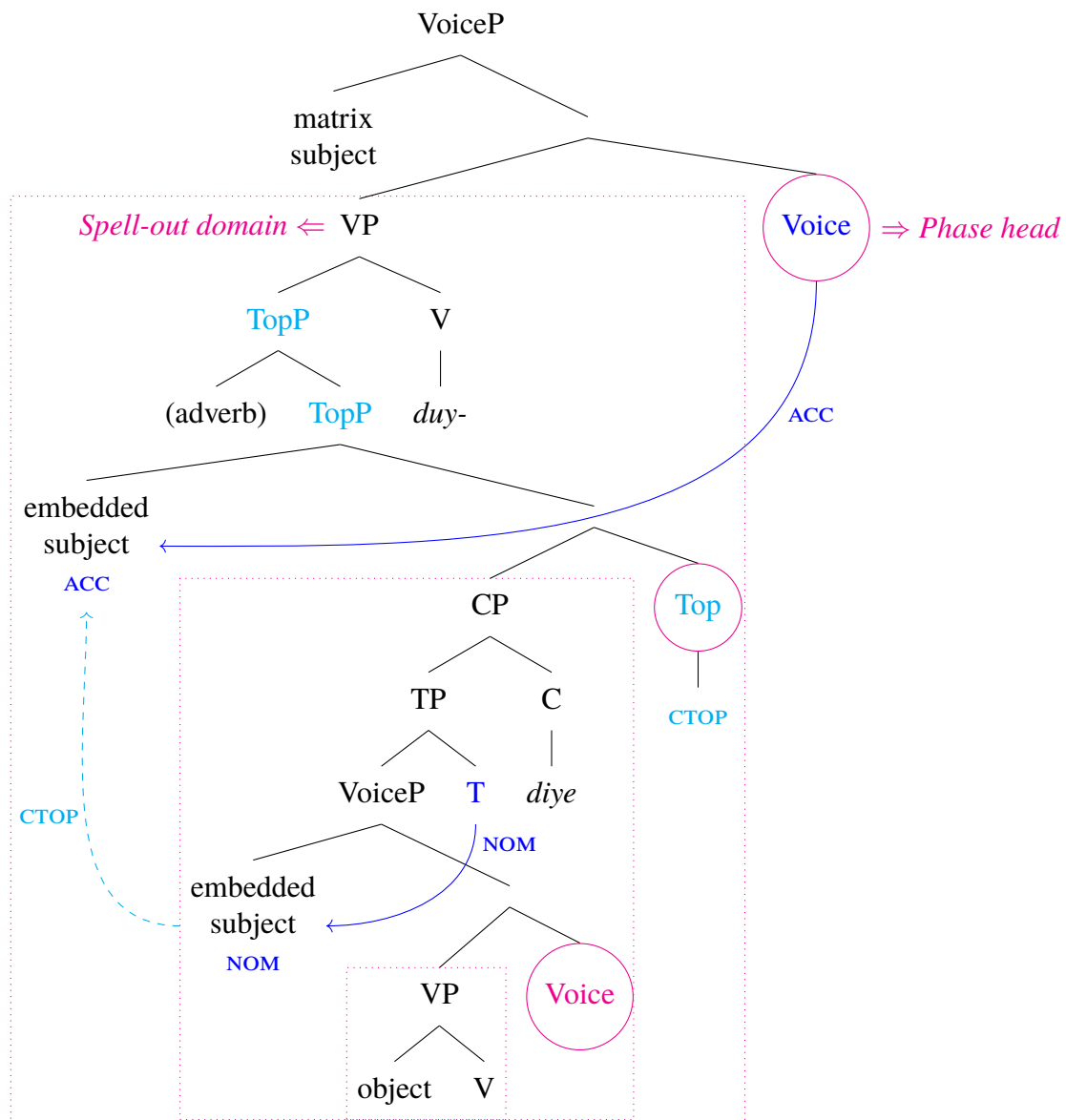
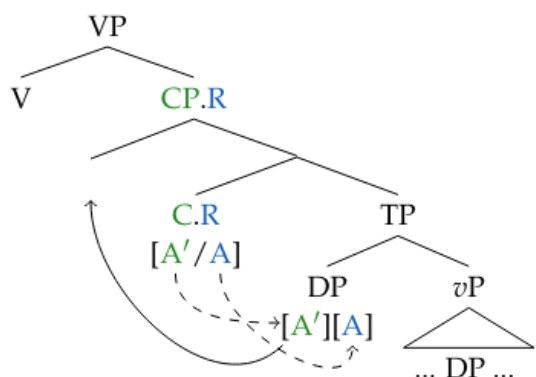
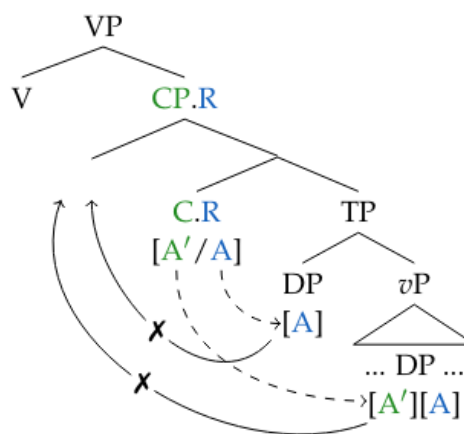
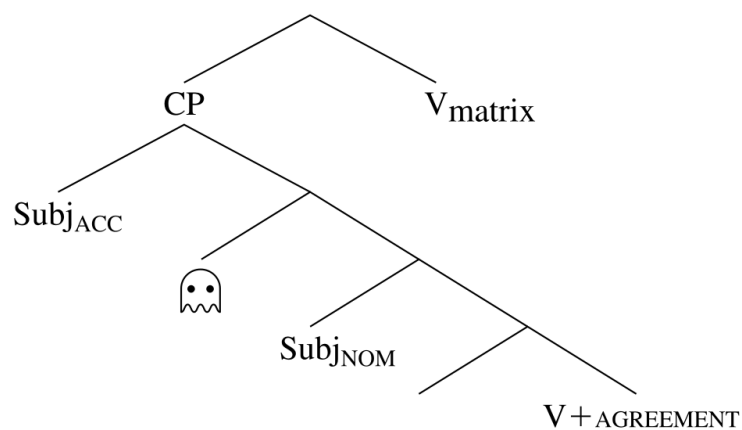


Figure 3.5: Şener's (2008) analysis of finite complement clauses with an ACC-marked subject under FHCT

a. Dependent satisfaction



b. A-Minimality

Figure 3.6: Lohninger, Kovač & Wurmbrand's (2022) analysis of type-4 CCAs⁶⁰Figure 3.7: Shklovsky & Sudo's (2014) analysis of the interaction between accusative and indexical shifting in Uyghur⁷¹

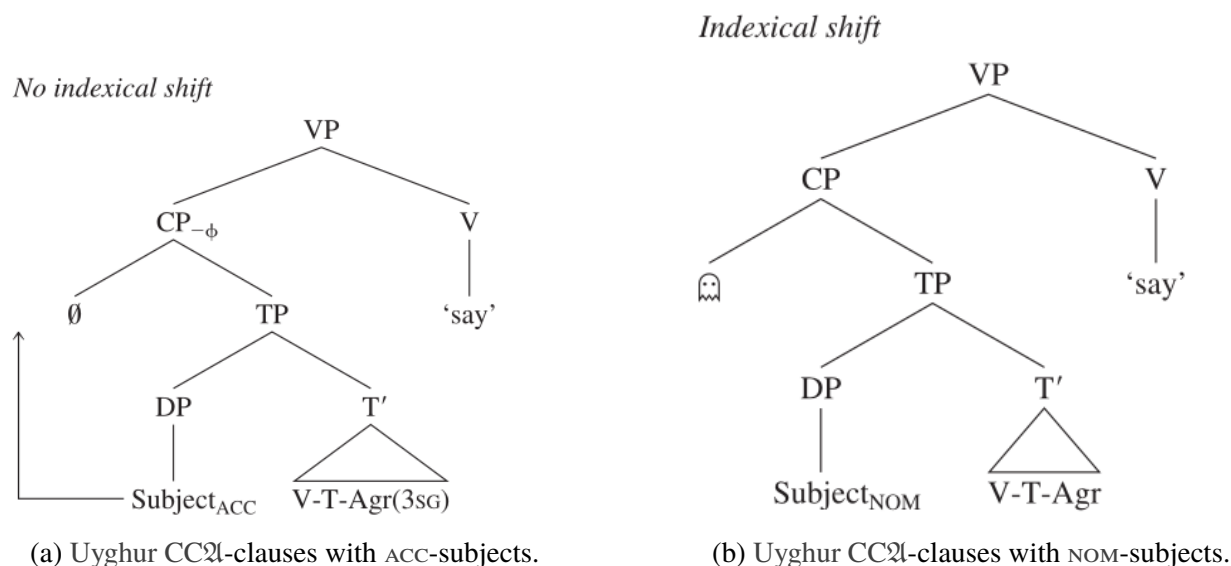


Figure 3.8: Major's (2022) analysis of Uyghur CCʌ-clauses⁷²

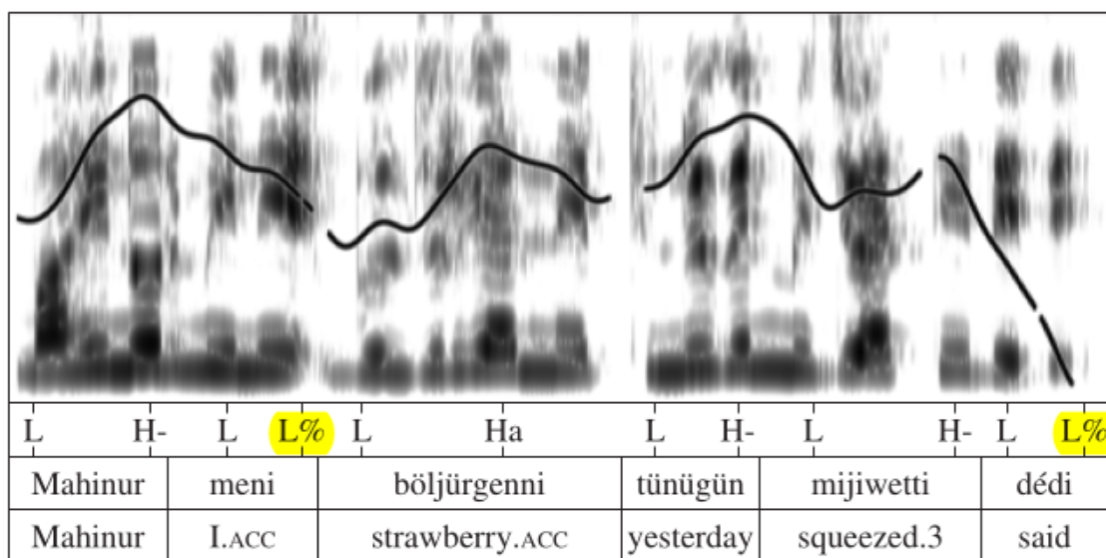


Figure 3.9: Spectrogram for non-shifted (48) (from Major & Mayer 2019: p. 252, ex. (12); Major 2022: p. 787, Fig. 1).

$$\begin{array}{c}
 [\text{TP} \quad [\text{Subj}_{\text{matrix}}] \quad [[\text{Subj}_{\text{ACC}i} \quad] \quad [\text{CP Op.}[\text{TP } t_i \text{ V}]] \quad \text{V}_{\text{matrix}}]] \\
 (\text{IP} \quad (\text{IP} \quad \quad \quad \text{L\%}) \quad \quad \quad \text{L\%})
 \end{array}$$

Figure 3.10: The prosodic schema of non-shifted declaratives in Uyghur (from Major & Mayer 2019: p. 250, ex. (9))



Figure 3.11: Spectrogram for indexically shifted (49) (from Major & Mayer 2019: p. 251, ex. (11)).

$$\begin{array}{c}
 [\text{TP} \quad [\text{Subj}_{\text{matrix}} \quad] \quad [[\text{CP Op.} \quad [\text{TP}(\text{Subj}_{\text{NOM}}) \text{ V}]] \quad \text{V}_{\text{matrix}}]] \\
 (\text{IP} \quad (\text{ip} \quad \text{H-}) \quad (\text{ip} \quad \text{L/H\%}))
 \end{array}$$

Figure 3.12: The prosodic schema of indexically shifted declaratives (right edge L%) and questions (H%) in Uyghur (from Major & Mayer 2019: p. 250, ex. (9))

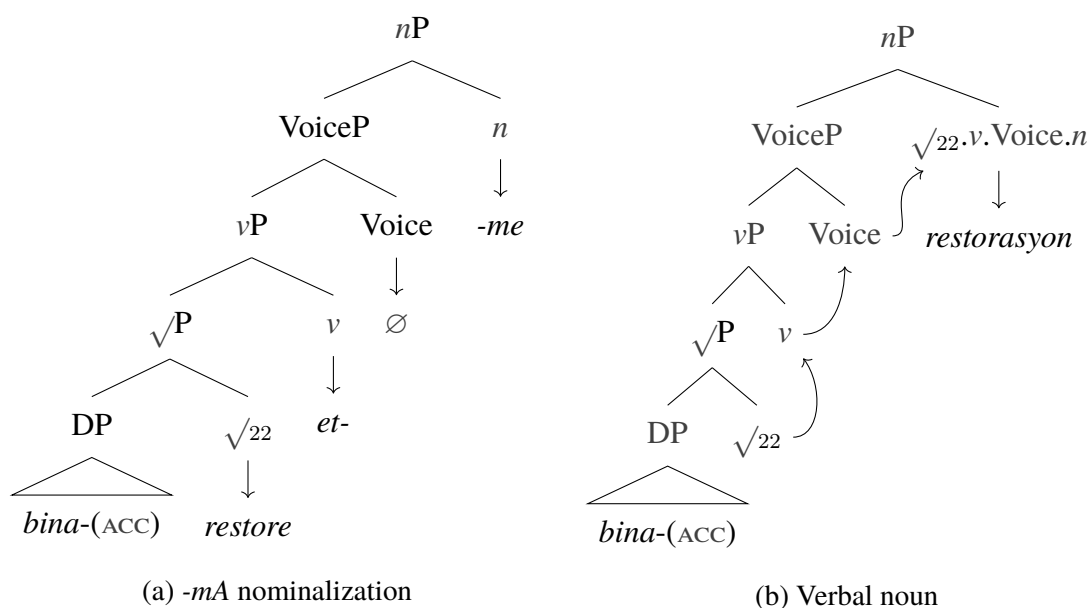


Figure 3.13: The assumed functional projections making up a -mA nominalization and a verbal noun

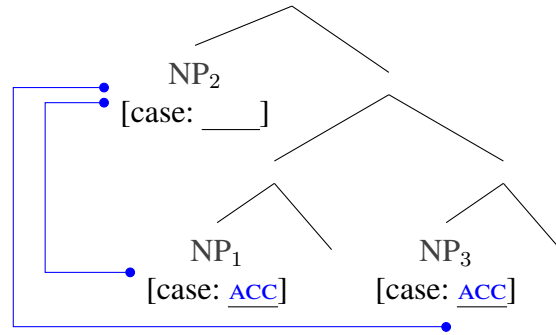


Figure 3.14: 1 : n licensing of ACC (1 licenser licenses ACC on n licensees)

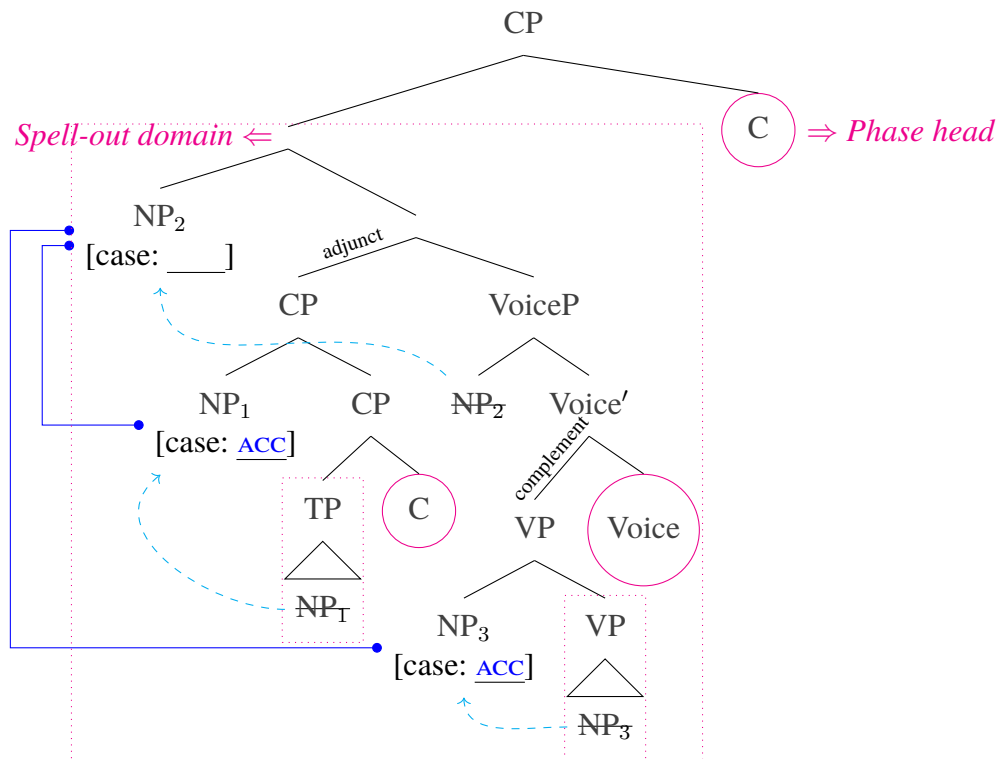


Figure 3.15: 1 : n dependent-down case licensing according to Baker (2015b) and Preminger (2014)

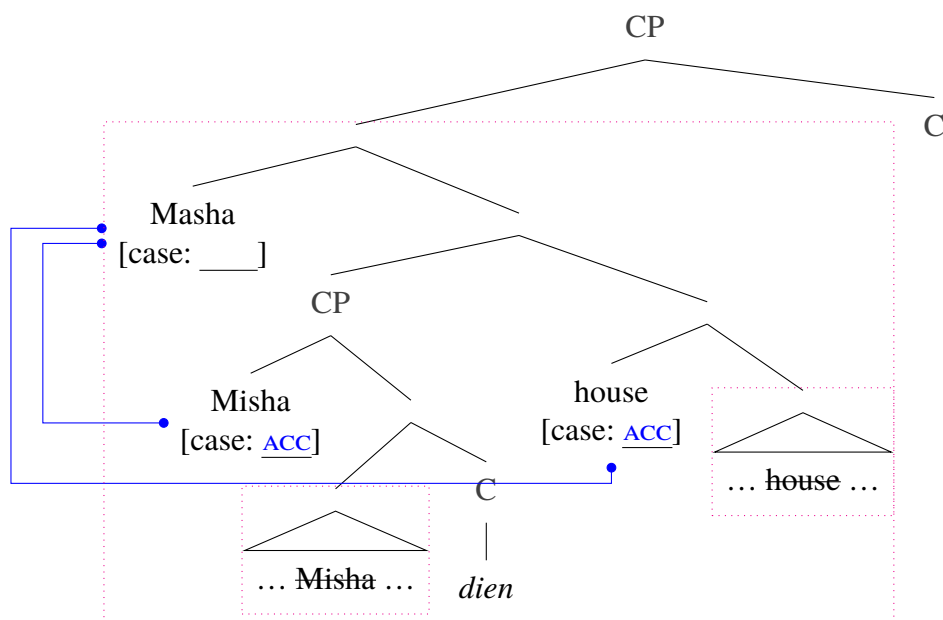


Figure 3.16: One licensing NP licenses ACC on two other NPs (attested in Sakha)

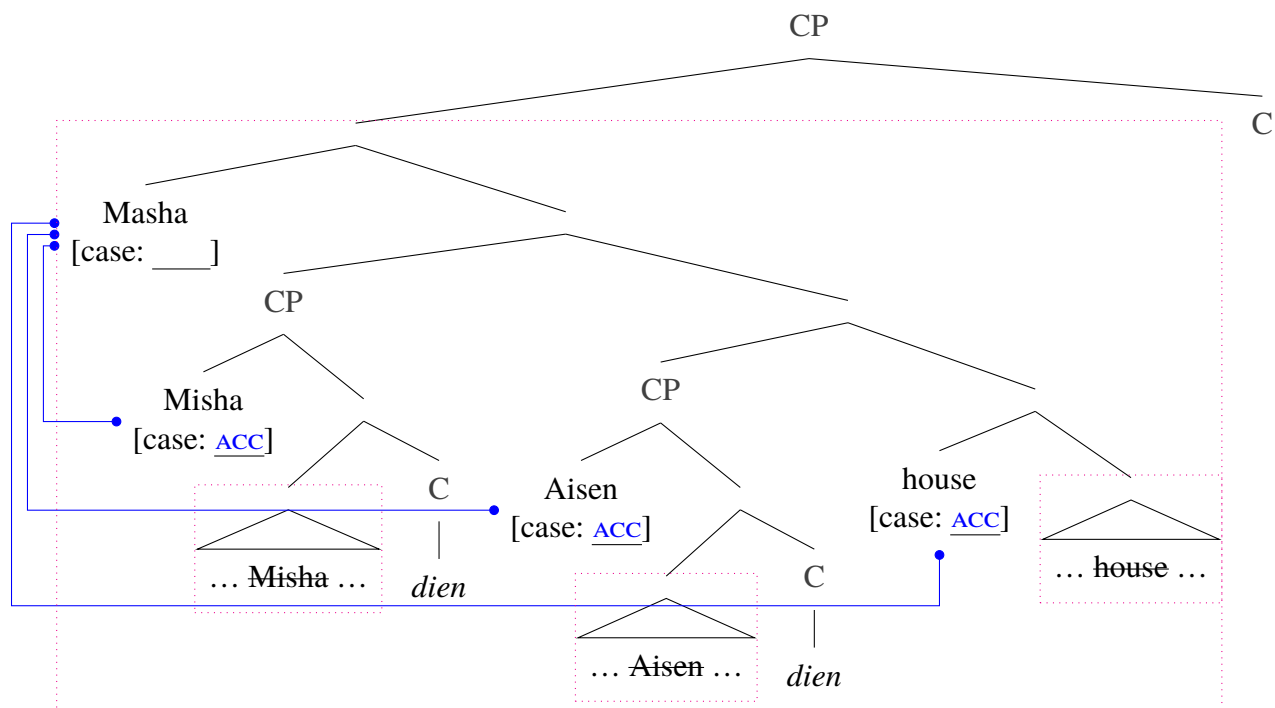


Figure 3.17: One licensing NP licenses ACC to three other NPs (predicted)

Chapter 4

Formal differences between dependent-up and dependent-down case assignment

4.1 Introduction

In this chapter I will discuss formal properties of the derivational mechanism that assigns resp. licenses dependent-up case. First (§4.2) I will review proposals that the case labeled “ergative (ERG)” in (non-Turkic) languages is a dependent-up case, i.e., a dependent case assigned in a case-competition configuration to the *higher* of two nominals. The case-competitor of ERG-assignment often appears with “absolutive (ABS)” marking, which I take to correspond to the absence of a case value [case: ____], and is thus identical to what is in other languages referred to as “nominative (NOM)”. (See my discussion in §2.5.3 on an alternative proposal where NOM/ABS is its own case value.)

Then, in section 4.3 I present the puzzle about the distribution of ERG: ERG seems to always appear on only one argument per CAD, and always appears on the highest argument in this CAD. Following Nie (2017) I label this pattern the “Single Ergative Generalization”. I discuss why we would expect ERG to pattern differently, namely parallel to the dependent-down case accusative (ACC), which we

have seen in §§3.4–3.6 can appear on multiple nominals in the same CAD, and even through case-competition against the same competitor.

I then (§4.4) turn to a review of previous accounts for this distribution of ERG. There are two main camps: One camp takes ERG to be an inherent case assigned by Voice, but only if the complement of Voice is transitive. Since inherent cases come with a thematic interpretation, and typically every thematic role is only assigned once per predicate, this derives why there can only be a single ERG-marked nominal per CAD. The other camp takes ERG to be a structural (dependent) case, as I present in §4.2, and argues that in principle, dependent-up cases can appear multiple times just like dependent-down cases (see Baker (2015b) and Baker & Vinokurova (2010) for DAT as the dependent-up case in the verbal domain, which I argue against, at least for Turkic, in §5), and it is essentially accidental that we only ever see one ERG-marked nominal, namely because there is only one argument that gets first-Merged outside the verbal phase, and this argument is necessarily the structurally highest argument (Baker 2014b, 2015b).

I propose an alternative theory of dependent-up case licensing. In §4.5, I introduce my hypothesis that it is not accidental that we only ever see one ERG, and that in fact this is a hallmark-characteristic of dependent-up case. In §5 and §6 I will even use the fact that there can be multiple DAT-marked arguments as evidence that DAT cannot be a dependent-up case (the missing “Single Dative Generalization”), and the fact that Turkic languages do not allow multiple GEN-markings in the nominal CAD as supporting evidence for my claim that GEN is a dependent-up case. To support this hypothesis, I provide arguments from formal language theory resp. mathematical syntax, showing that the assignment/licensing of dependent-up case differs substantially from that of dependent-down case.

Section 4.6 summarizes this chapter.



(a) “Ergative alignment”: S and O pattern together

(b) “Accusative alignment”: S and A pattern together

Figure 4.1: Ergative alignment versus accusative alignment²

4.2 Ergative is a dependent-up case

ERG is the name typically given to a case that appears on subjects of transitive predicates (A), but not on subjects of intransitives (neither unergatives nor unaccusatives¹) (S), nor on objects of transitives (O). This is depicted in the common diagram in Fig. 4.1, where in Fig. 4.1a, A-arguments, which receive ERG-marking and are colored in red, stand in contrast to S- and O-arguments, which, on the other hand, pattern together and receive a case typically referred to as ABS. This is known as an “ergative alignment” with respect to case marking. Meanwhile, famously, in languages following an “accusative alignment”, Fig. 4.1b, A- and S-arguments pattern together and both receive a case known as NOM, while O-arguments are marked with ACC. Examples illustrating the ergative case pattern are given in (1) for Koryak and in (2) for Shipibo.

- (1) a. $\gamma\text{əm} \text{ -nan } t\text{-} \quad \text{ə-} \quad \text{nu} \text{ -ne} \text{ -w} \quad \text{ʔəvənʔ} \text{ -u} \quad .$
 1SG -ERG 1SG.S/A- EP- eat -3.O -3PL berry -ABS.PL
 A -ERG O -ABS
 ‘I ate berries.’ (transitive) *Koryak*³
- b. $\gamma\text{ojat}\zeta\text{ek} \quad \emptyset\text{-} \quad \text{aŋaŋja} \text{ -j} \quad .$
 man .ABS 2/3.S/A.IND- sing -AOR
 S -ABS
 ‘The man sang.’ (unergative) *Koryak*⁴

¹ I will not be considering “active” or “split-S” languages here, where the subject of an unergative, carrying the thematic role AGENT, patterns with the subject of a transitive, also carrying the thematic role AGENT. See Baker & Bobaljik (2017: §5.5) for arguments that there are no known languages with this case pattern.

² Images from Wikimedia Commons, https://en.wikipedia.org/wiki/File:Ergative_alignment.svg and https://commons.wikimedia.org/wiki/File:Accusative_alignment.svg, downloaded on 2025-05-15.

³ Abramovitz 2020: p. 7, ex.(9a)

⁴ Abramovitz 2020: p. 8, ex. (10a)

- c. $\text{?et?} \text{ -}\partial \text{ -}n \text{ } \emptyset\text{-} \text{ } vi? \text{ -}i \text{ } .$
 dog -EP -ABS 2/3.S/A.IND- die -AOR
 S -ABS
 ‘The dog died.’ (unaccusative) *Koryak*⁵
- (2) a. *Maria -nin -ra ochiti noko -ke .*
 Maria -ERG -PRT dog .ABS find -PRF
 A -ERG O -ABS
 ‘Maria found the dog.’ (transitive) *Shipibo*⁶
- b. *Rosa -ra bewa -ke .*
 Rosa.ABS -PRT sing -PRF
 S -ABS
 ‘Rosa sang.’ (unergative) *Shipibo*⁷
- c. *Kokoti -ra joshin -ke .*
 fruit .ABS -PRT ripen -PRF
 S -ABS
 ‘The fruit ripened.’ (unaccusative) *Shipibo*⁸

In recent decades, there have been two main views on the nature of ergative case: that it is an inherent case assigned by Voice (Legate 2008 and many others), or, that it is a dependent case assigned by case-competition to the higher of two nominals in the clausal CAD (Baker 2015b; Marantz 1991). As Baker & Bobaljik (2017) discuss in detail, the main argument for the inherent-ERG view used to be that the available evidence did not include examples of ERG being assigned to a nominal that did not also have the thematic role AGENT (Marantz’s (1991) “Ergative Case Generalization”) — but this missing evidence has since then been presented by Baker (2014b): In Shipibo, applicatives of unergatives and unaccusatives can lead to the S-argument receiving ERG in the presence of an applicative argument. Compare (3a) to (2b), and (3b) to (2c).

- (3) a. *Papashoko -n -ra Rosa bewa -xon -ai .*
 grandfather -ERG -PRT Rosa.ABS sing -APPL -IPFV
 S -ERG
 ‘The grandfather is singing for Rosa.’ (applicative of unergative) *Shipibo*⁹
- b. *Bimi -n -ra Rosa joshin -xon -ke .*
 fruit -ERG -PRT Rosa.ABS ripen -APPL -PRF
 S -ERG

⁵ Abramovitz 2020: p. 8, ex. (10b)

⁶ Baker & Bobaljik 2017: p. 111, ex. (1a)

⁷ Baker & Bobaljik 2017: p. 116, ex. (5a)

⁸ Baker 2015b: p. 55, ex. (40b); Baker & Bobaljik 2017: p. 116, ex. (5b)

⁹ Baker & Bobaljik 2017: p. 116, ex. (6b)

‘The fruit ripened for Rosa.’ (applicative of unaccusative)

*Shipibo*¹⁰

Baker & Bobaljik (2017) present further data from Kalaallisut, Chukchi, and Yup’ik, and Deal (2019) from Nez Perce, that go in the same direction as the above data from Shipibo, providing further arguments that *ERG* can appear on arguments that do not carry the θ -role *AGENT*, so that this *ERG* cannot be inherent case assigned by Voice.

Meanwhile, the view that *ERG* is a dependent-up case accounts for the data in (2) and (3) perfectly: *ERG* is assigned to an argument if and only if there is a second, lower argument in the same CAD. Baker & Bobaljik (2017) show that this captures further predictions, namely that *ERG* can disappear when the lower argument is removed by detransitivizing processes such as incorporation or antipassive, as evidenced by Chukchi, Kalaallisut, and other languages. I will not repeat the relevant data points here and instead refer the interested reader to that paper.

Let us take away from this section, among others, the generalization in (4):

- (4) *ERG* can appear only if there is a lower/c-commanded case-competitor in the same domain.

4.3 Two puzzles about the distribution of ergative case

4.3.1 Introduction

This section describes the distribution of *ERG*, introducing two puzzling properties: that there seems to always only be one *ERG*-marked nominal per CAD (§4.3.2), and that the single *ERG*-marking is always on the highest argument (§4.3.3). I then proceed to discuss why we might expect *ERG* to be distributed differently, especially given the close similarity to dependent-down cases like *ACC*, in §4.3.4. Finally, in §4.3.5 I summarize the distribution of *ERG* in more succinct terms, which will be of relevance in the next two sections.

¹⁰ Valenzuela 2003: pp. 691, 694; Baker 2015b: p. 55, ex. (41b); Baker & Bobaljik 2017: p. 117, ex. (7b)

4.3.2 The Single Ergative Generalization

Ergative case is notorious for one particular property: It can appear on only one nominal per case-assignment domain (CAD). Baker (2015b) formulates this in the following way:

[t]here is a striking asymmetry between ergative languages and accusative languages [in regard to double object constructions]. In ergative languages with no VP-internal cases, the subject is ergative, while both internal arguments are absolutive. This is true across a broad range of ergative languages (as long as they do not have structural dative [to be picked up below, SL]), including Shipibo (Valenzuela 2003: pp. 346–348; Torres-Bustamante 2011; Baker 2014b) Burushaski (Willson 1996: p. 61), and Kewa (Franklin 1971: p. 71), see also Merlan (1994: pp. 66, 68) on Wardaman. [...] Meanwhile, in accusative languages the subject is nominative and both internal arguments are accusative [...]. [fn. 1:] Languages that are not relevant here because they have dative case are [...] Greenlandic, Chukchi, and Ingush on the ergative side. (Baker 2015b: p. 231)

This property of ERG, although often observed, has not been much investigated (to be discussed in more detail in §4.4), and has been termed the “Single Ergative Generalization” by Nie (2017), a terminology that I adopt here. If ERG is a structural case, it is unique among the structural cases in this respect: NOM can appear on two arguments in some stative transitive constructions in Japanese; ACC can appear on two arguments in the raising-to-ACC CC₂ constructions in Turkic, (Baker & Vinokurova 2010: pp. 615–620; discussed in §3.4 above); and ABS can appear on multiple nominals in Shipibo EXPERIENCER predicates (Baker 2015b: pp. 240–242). Regarding multiple ERG-marked arguments, Nie (2017: p. 2) writes,

I have not found a single instance of ERG-ERG in any language. This result is confirmed by Jessica Coon (p.c.), Julie Anne Legate (p.c.), and Diane Massam (p.c.), among others. No language marks ergative case on more than one argument of a single verb.

Let us define this puzzle as (5):

(5) The Single Ergative Generalization:

A syntactic clause may contain at most one ergative argument. (Nie 2017: p. 2)

Note that the wording only makes claims about “arguments”. Baker (2015b) presents data from Warlpiri and Diyari, where in addition to the AGENT, high adverbs can also receive ERG-marking. I have already presented these data in §2.5.2 above. It would be interesting to see if multiple adverbs will all receive ERG-marking or if there are further restrictions. I do not have access to the required data and thus have to leave the issue with this.

4.3.3 Ergative is always on the highest argument

Relatedly, there is a second puzzle regarding ERG: While in predicates with two arguments, it is well-known that ERG appears on the structurally higher argument (§4.2), in predicates with three or more arguments, ERG seems to always appear on the highest of all arguments, yielding an ERG-ABS-ABS pattern. This is illustrated for ditransitives in (6):

- (6) a. *Maria -nin -ra Jose piti meni -ke.*
 Maria -ERG -PRT Jose.ABS fish.ABS give -PRF
 ‘Maria gave Jose fish.’ Shipibo¹¹
- b. *Hilés -e dasín taswúir mó- o- ltir -imi .*
 boy -ERG girl .ABS picture(Y).ABS 3SG.F- CAUS- show -3SG.M.PST
 ‘The boy showed the girl the picture.’ Burushaski¹²
- c. *Né -mé mená sápi kála -wa .*
 me -ERG pig .ABS sweet potato.ABS give -PST.1SG
 ‘I gave sweet potato to the pig.’ Kewa¹³

Similarly, the tripartite language Diyari shows that ditransitives employ an ERG-ACC-ACC pattern, with the middle (GOAL) argument also not marked with dependent-up ERG:

- (7) *ñu -lu pula -ña [ñi -ña putu] yĩŋki -ña wara -yi .*
 he -ERG them -ACC 3SG.NF -ACC thing give -PTCP AUX -PRS
 ‘He gave them that thing.’ Diyari¹⁴

¹¹ Baker 2015b: p. 231, ex. (2a)

¹² Willson 1996: p. 61; Baker 2015b: p. 231, ex. (2b)

¹³ Franklin 1969: p. 121, ex. (31); Franklin 1971: p. 71; Baker 2015b: p. 231, ex. (2c)

¹⁴ Austin 1981: p. 115; Baker 2015b: p. 232, ex. (5)

Even when the ditransitive argument structure is derived such as through causativization or applicatives of a transitive predicate, the case pattern adapts such that, again, the highest argument will be the one (and the only one) to receive ERG-marking. Consider the Niuean causative of a transitive in (8). In the transitive version, (8a), it is the AGENT of ‘hold’ that is marked ERG. In the causative, (8b), however, the AGENT of ‘hold’ has changed its case marking from ERG to ABS, and it is the causer that receives the single ERG-marking.

- (8) a. *Ne totō [e ia] [e kapiniu].*
PST hold [ERG].PROPER 3SG ABS.COMMON cup
 ‘[He] held the cup [in his hand].’ Niuean¹⁵
- b. *Kua faka- totō aki [e ia] [e kato] [e tama haaku].*
PRF CAUS- hold INS ERG.PROPER 3SG ABS.COMMON basket [ABS].COMMON
child 1SG.GEN
 ‘She made [my child] hold the basket.’ Niuean¹⁶

Let us take away from this the generalization in (9), which is a re-formulation of (5), and the second generalization about the distribution of ERG besides (4).

- (9) A nominal marked ERG must not be c-commanded by a case-competitor.

4.3.4 Why *ERG-ERG-ABS is surprising

The generalization in (9) is unexpected because, while (4) (the need for a case-competitor) parallels the distribution of the dependent-down case ACC, there is no parallel to (9) for ACC, which, if it existed, might look like (10).

- (10) A nominal marked ACC must not c-command a case-competitor. (unattested)

If ERG were parallel to ACC in that it depended only on the presence of a case-competitor but did not have a negative requirement as well, we would expect to see more than a single ERG-marking in constructions with three or more arguments — because when computing a structural case for the middle argument, the lowest argument should be able to serve as case-competitor for the middle

¹⁵ Nie 2017: p. 4, ex. (7a); Gould, Massam & Patchin 2009: p. 7, ex. (16b)

¹⁶ Nie 2017: p. 4, ex. (7b); Gould, Massam & Patchin 2009: p. 6, ex. (16a)

argument and assign it ERG, as illustrated in Fig. 4.2 for ditransitives, compare (6), and causatives, compare (8b). In Fig. 4.2, the THEME serves as case-competitor for both the GOAL/causee and the

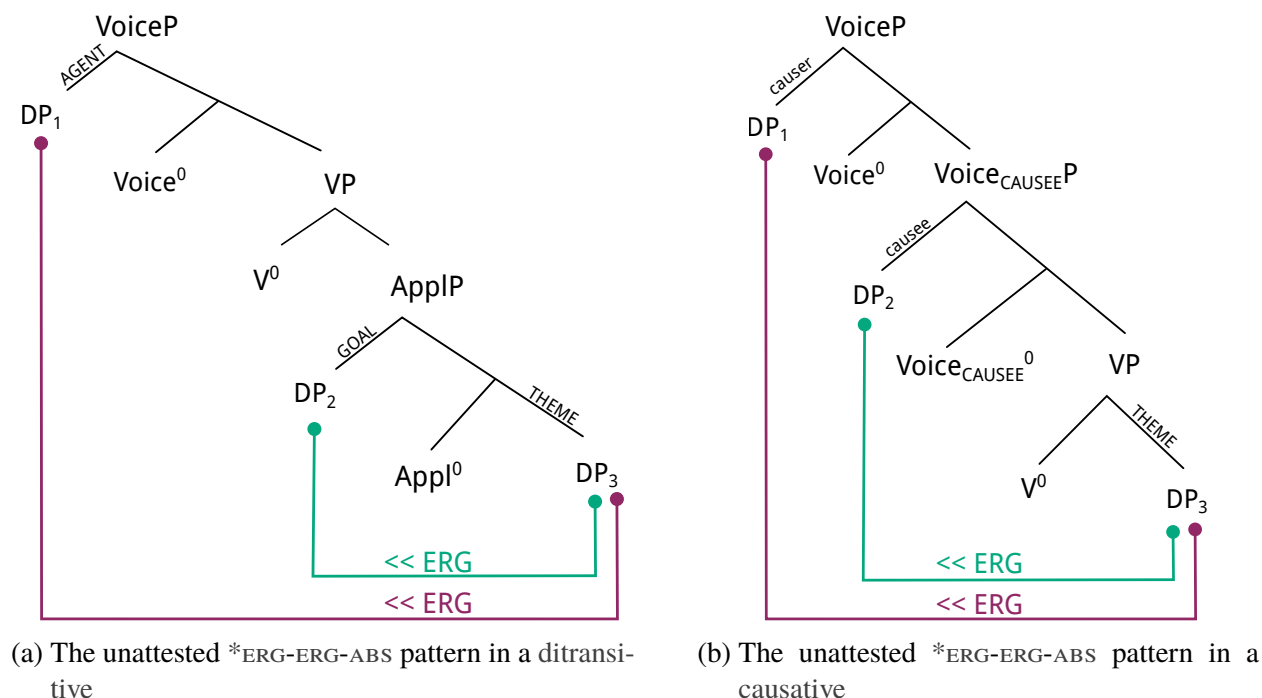


Figure 4.2: The unattested *ERG-ERG-ABS pattern where the lowest argument competes against the top and middle arguments

AGENT.

This unattested pattern is what Yip, Maling & Jackendoff (1987: pp. 219 f., p. 224) explicitly state as a prediction that their proposal (cf. §2.4.4) makes. In ergative languages, the association of structural cases on the “case tier” to nominal arguments proceeds from right to left: First, the right-most nominal (the lowest argument) is associated with ABS. Then, the second-lowest (if present) nominal is associated with ERG. ERG being the “last” (left-most) case on the tier will have the ability to “spread”, i.e., any other nominals further to the left (higher in the structure) will also be associated with ERG. This is illustrated in Fig. 4.3.

One might ask whether the constructions in Fig. 4.2 are perhaps ruled out because both ERG-markings originate through case-competition against *the same* competitor (the THEME). Perhaps there is a restriction on case-competitors such that they can only participate in one case-competition. This is not so, as I have shown in §3.6 above. There, we have seen that it is possible in Sakha for the

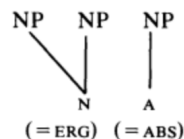


Figure 4.3: The proposal by Yip, Maling & Jackendoff (1987: p. 224, Fig. 10) about “spreading” of ERG

same NOM-marked nominal to serve as case-competitor for two ACC-markings. The two ACC-marked nominals did not stand in a c-command relation to each other, thus one could not have served as case-competitor for the other, and it was necessarily the NOM-marked nominal, c-commanding both ACC-markings, that was the case-competitor for both. We would thus expect that the ABS-marked THEME in Fig. 4.2 should also in principle be able to serve as case-competitor more than once.

Second, as already discussed in §2.5.3, Abramovitz (2020) has presented evidence that this is exactly what happens in Koryak when an ABS-marked nominal undergoes successive cyclic long-distance movement: The Koryak verb *valom-* ‘hear’ takes a finite clausal complement (surrounded by squared brackets) and an ABS-marked subject (boxed):

- (11) *ɣəmmo t- ə- valom-ə -k*
 me.ABS 1SG.S/A-EP-hear -EP-1SG.S
 [*əno ʔewŋəto -na -k Ø- j- ə- t̤im -aw -nin*
 that Hewngyto-OBL.SG-ERG 2/3.S/A.IND-CAUS-EP-break-VBLZ-3SG.A>3.O
kojŋ-ə -t].
 cup -EP-ABS.DU
 ‘I heard [that Hewngyto broke two cups].’ Koryak¹⁷

Koryak wh-words undergo movement all the way to the left of the matrix clause. When replacing the embedded THEME ‘two cups’ in (11) with the wh-word ‘what.DU’, we can observe that the matrix subject (‘me’ in (11), ‘you’ in (12), both boxed) changes its case from ABS to ERG, Abramovitz (2020) writes. The embedded subject ‘Hewngyto’ keeps its ERG-marking. Thus, ‘what.ABS’ must have served as case-competitor for both ‘Hewngyto.ERG’ and for the matrix subject ‘you.ERG’. This differs from the ACC-pattern from §3.6, where the same NOM-marked nominal serves as competitor for two ACC-marked nominals, only in that ‘what.ABS’ competes from two distinct positions.

¹⁷ Abramovitz 2020: p. 17. ex. (27b)

- (12) *jej -i* {[✓]*ʏə -nan* / **ʏət̪t̪t̪i*} *∅-* *valom-na -t*
 what-ABS.DU you-ERG you.ABS 2/3.S/A.IND-hear -3.O-3DU
 [*əno ʔewŋəto -na -k* *∅-* *j- ə- t̪im -aw -nin*
 that Hewngyto-OBL.SG-ERG 2/3.S/A.IND-CAUS-EP-break-VBLZ-3SG.A>3.O
jej -i]?
 what-ABS.DU
 ‘What two things did you hear [that Hewngyto broke ~~what two things~~]?’ *Koryak*¹⁸

The derivation of (12) proposed by Abramovitz (2020) is given in Fig. 4.4.

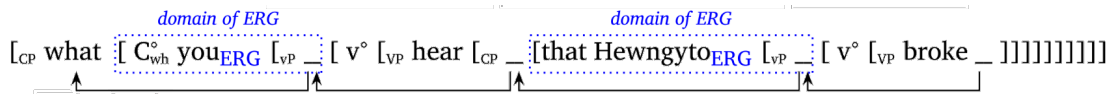


Figure 4.4: The analysis of (12) by Abramovitz (2020: p. 23, ex. (44e))

4.3.5 Summary: the distribution of ergative case

Given the above presentation of the distribution of *ERG*, we may characterize it more succinctly as in (13), whereby (13a) is a reformulation of (4), and (13b) is a reformulation of (9).

- (13) An NP N_1 carries [case: ERG] iff
- it locally c-commands another NP N_2 that qualifies as case-competitor, *and*
 - there is no other NP N_0 that qualifies as case-competitor and locally c-commands N_1 .

Given that (13) is the intersection of two constraints, we would expect the Language Faculty to also utilize both constraints independently. Indeed, (13b) is reminiscent of Principle C of Binding Theory, where a proper name must not be c-commanded by a nominal, pronoun, or anaphor with the same referential index. However, I cannot think of a phenomenon that utilizes (13a) to the exclusion of (13b).¹⁹

The difference between dependent-up and dependent-down case can then be restated as (14):

¹⁸ Abramovitz 2020: p. 17, ex. (27a)

¹⁹ Susi Wurmbrand has suggested to me that the following phenomena might be candidates for (13a): (i) inflected infinitives in Brazilian Portuguese (papers in Santos & Gonçalves 2018); (ii) hyperraising in Greek embedded clauses (Alexiadou & Anagnostopoulou 2002); (iii) determiner-noun agreement in Russian (Glushan 2013; Smith 2015). I leave a detailed investigation into these phenomena for future research.

- (14) Why does dependent-up case assignment in, as far as we know, all languages with ergative case alignment follow the pattern (13a+b) and never follows the simpler pattern (13a)?

4.4 Previous accounts of the Single Ergative

Generalization

4.4.1 Introduction

In this section I will discuss some existing accounts of the distribution of ERG as presented in §4.2 and §4.3. First, in §4.4.2, I revisit the proposal that ERG is not a structural but an inherent case assigned by Voice if and only if the complement of Voice is transitive, with explicit mention of Nie's (2017) version. Then, in §4.4.3, I discuss the proposal by Baker (2014b, 2015b), who argues that the distribution of ERG is actually completely parallel to that of ACC, with the Single Ergative Generalization being essentially accidental because of independent properties. Then, in §4.4.4, I return to the proposal by Clem & Deal (2024) and Poole (2024), which states that Case Competition can be implemented by Agree. I make explicit a footnote in Poole's (2024) paper, where he mentions that his implementation cannot derive ERG-ABS-ABS and instead makes the prediction *ABS-ABS-ERG. §4.4.5 concludes.

4.4.2 Ergative is an inherent case iff the complement of Voice is transitive

On the background of Case Theory as developed by Chomsky & Lasnik (1977), Vergnaud (1977), and Chomsky (1980, 1981) (cf. §2.3) as well as regarding data from split-S languages, where the subjects of unergatives pattern with the subjects of transitives in carrying ERG, while the subjects of unaccusatives pattern with the objects of transitives in carrying ABS, and also lacking the data from Shipibo and other languages mentioned above, many authors have argued that ERG is best analyzed as an inherent case assigned by Voice to its specifier, i.e., ERG will always and only appear

on the nominal carrying the thematic role AGENT (Aldridge 2004, 2008, 2012; Johns, Massam & Ndayiragije 2006; Coon 2013; Laka 2006; Legate 2006, 2008, 2012; Mahajan 1990, 2012; Massam 2006; Nash 1996; Woolford 1997, 2006 and many of the chapters in Coon, Massam & deMena Travis (2017)).

In particular, Nie (2017), after observing the Single Ergative Generalization, accounts for the distribution of ERG by making it a head case (in Preminger’s (2024) terminology) assigned by Voice to its specifier. She assumes that a nominal first-Merged in Spec,VoiceP is interpreted as AGENT, but crucially Nie also allows movement into Spec,VoiceP such as by the THEME in Shipibo applicatives of unaccusatives. To account for the absence of ERG on the subjects of unergatives, which are by common assumptions first-Merged in Spec,VoiceP, Nie (2017) postulates that the rule for ERG-assignment includes a transitivity requirement on the complement of Voice. Thus, in her system, ERG cannot recurse because Voice cannot recurse and can only have a single specifier.

This proposal is thus an attempt at salvation of the inherent-ERG theories by deriving the pattern from Shipibo through movement into Spec,VoiceP. I refer the reader to the discussion in Baker & Bobaljik (2017: §5.4.2) for arguments why supplementing the complement of Voice with a transitivity requirement renders the inherent-ERG theory superfluous because DCT can capture the transitivity requirement much more directly.

4.4.3 The Single Ergative Generalization is accidental

Baker (2014b, 2015b) analyzes dependent-up case in the following way: In principle, it is completely equivalent to dependent-down case. Many instances of what is called “DAT” are dependent-up cases in the verbal domain, and we see that it can appear on multiple arguments. Thus, there is nothing inherent to dependent-up case that would lead to the Single Ergative Generalization. However, it is true that the Single Ergative Generalization seems to hold; that is, for ERG, the dependent-up case in the clausal domain, but for DAT. There must thus be something about the clausal domain that prohibits the occurrences of multiple ERG-marked nominals. Baker (2014b, 2015b) proposes to capitalize on the fact that all the lower arguments of a verb form a constituent to the

exclusion of the highest argument, which is the one receiving ERG. Thus, Baker (2014b, 2015b: §6.1.1) builds an elaborate system that derives *ERG-ERG-ABS in ditransitives in the following way.

The clause contains two CADs: the verbal domain, and the clausal domain. Both domains are independently specified for dependent-up case, and/or dependent-down case, (and for unmarked case). In languages with DOM effects, such as Turkic languages, only moved DOs (that have entered the clausal domain) receive ACC resp. trigger ERG on the subject. In languages without DOM, such as Shipibo, nominals inside the verbal domain are inaccessible to the rules of the clausal domain — because Voice is a “hard phase head”, Baker proposes. Meanwhile, in languages without DOM effects, the verbal domain remains accessible to the rules of the clausal domain (Voice is a “soft phase head”). Thus, nominals inside the verbal domain are essentially evaluated twice: First on the verbal domain cycle, and then again on the clausal cycle. The asymmetry between dependent-down ACC, which can appear on multiple nominals, and dependent-up ERG, which cannot, is derived as follows:

First, the THEME and GOAL are Merged, followed by Voice, which triggers Spell-out of its complement, and thereby the application of Case Competition on the pair $\langle \text{GOAL}, \text{THEME} \rangle$. No cases are assigned because no applicable rules are defined; i.e., Shipibo has neither a dependent-down nor a dependent-up case in the verbal CAD. This concludes case assignment in the verbal domain. Then the AGENT is Merged. This triggers Case Competition between the pairs $\langle \text{AGENT}, \text{GOAL} \rangle$, $\langle \text{AGENT}, \text{THEME} \rangle$, and $\langle \text{GOAL}, \text{THEME} \rangle$. By the cases defined for the clausal domain (ERG as dependent-up, no dependent-down, ABS as unmarked case), the first two pairs both yield ERG-marking on the AGENT, while the third pair would yield ERG-marking on the GOAL. However, Baker (2014b, 2015b) proposes, the third pair is not considered in the clausal domain because it has already been considered in the verbal domain, where it was subject to the case definitions of the verbal domain. Thus, the derivation keeps tracks of the pairs of nominals that have been subjected to Case Competition and ensures that the same pair is not considered more than once. Baker (2015b: §6.1.2) shows how these assumptions correctly derive the ABS-ABS patterns in Shipibo EXPERIENCER predicates (dyadic unaccusatives), and also the ERG-ABS patterns of Shipibo applicatives of intransitives,

shown in (3b) above, where, likewise, both arguments are first-Merged inside the verbal domain, yielding the pair $\langle \text{BENEFACTIVE}, \text{THEME} \rangle$, but crucially the THEME moves above the applied argument, creating a new pair $\langle \text{THEME}, \text{BENEFACTIVE} \rangle$ and triggering Case Competition again, which leads to ERG -marking on the THEME . Thus, Baker (2015b) proposes that a combination of independent factors lead to the phenomena of only-one- ERG and ERG -always-highest.

To create a derivation with more than one ERG -marked nominal in Baker's (2015b) system, we'd need to observe one of the following configurations:

- A. First-Merge another nominal argument inside the clausal domain. Then this argument should also receive ERG -marking by Case Competition against any case-less argument inside the accessible verbal domain.

If two arguments are first-Merged in the clausal domain while there are no arguments in the verbal domain, the higher argument should appear with ERG , despite the complement of Voice not being transitive.

- B. Move a nominal argument from inside the verbal domain into the clausal domain, such that it then c-commands an argument that it did not c-command from its base position. This is similar to Shipibo applicatives of unaccusatives, except we would want the base predicate to be transitive. Then, the AGENT will be ERG -marked by Case Competition against the lower arguments, and after movement of the lowest argument above the AGENT , the lowest argument should also receive ERG -marking by Case Competition against the middle argument.²⁰

As far as I know, by current understanding and assumptions, both options are not feasible.

Regarding option A, the functional head that closes off the thematic domain, here labelled Voice, is the same head that Merges the AGENT , thus there can be no further arguments first-Merged above the AGENT . Similarly, because Voice closes off the thematic domain, we call it a phase head, and, if the assumption holds that phasal Spell-out domains are CADs, this makes it impossible for any

²⁰ Case Competition against the already- ERG -marked AGENT might be ruled out for independent reasons, namely because an ERG -marked nominal might not be able to serve as case-competitor; cf. §2.5.2.

argument Merged below Voice to still count as first-Merged in the clausal domain. The possibility of multiple specifiers of Voice is also commonly excluded.

Option B does not seem feasible either because any movement that precedes further applications of Case Competition, such as EPP-driven movement to SpecTP but not *wh*-movement, will target the closest accessible nominal. While some arguments can be inaccessible to such movement, such as applied arguments in Shipibo, which Baker (2015b) argues are contained inside a PP, it seems that *AGENTS* are never inaccessible to movement to SpecTP. Thus, in the presence of an *ERG*-marked *AGENT*, it would not happen that, say, the *THEME* moves to SpecTP and receives *ERG*-marking. Furthermore, for the *THEME* of a ditransitive predicate to undergo such movement, it would not only be the *AGENT* that would need to be inaccessible to move but also the middle argument.

But there is another possibility how we might expect to be able to derive more than one *ERG*-marking: if *ERG* is not only defined as the dependent-up case in the clausal domain, but also as the dependent-up case in the verbal domain. Then we would expect *ERG*-marking on the *GOAL*, assigned on the verbal cycle, and again *ERG*-marking on the *AGENT*, assigned on the clausal cycle. Baker (2015b) calls the dependent-up case in the verbal domain “*DAT*” (but see my §5 on why in Turkic, *DAT* is not a dependent-up, but an inherent case assigned by Appl) — thus essentially this amounts to *ERG-DAT* syncretism, crucially with structural (dependent-up) *DAT*.

This would not (necessarily) always be an instance of accidental syncretism, because it is a well-known typological phenomenon that *ERG* may extend to the nominal domain and appear on possessors, thus creating *ERG-GEN* syncretism, which is not usually analyzed as accidental.

The question can then be rephrased as:

(15) Why do we see so little *ERG-DAT* syncretism, when we see so much *ERG-GEN* syncretism?

There are only two languages reported in Baker (2015b) that may have *ERG-DAT* syncretism, but the data are rather fragile.

In Ika (Chibchan, also known as Arhuaco), the *AGENT* of a transitive verb, the *GOAL* of a ditransitive verb, and some *SOURCE* arguments are marked with the same suffix *-se*? — however, there is

no available example with this marking appearing on an AGENT and a GOAL in the same sentence.

The other potential language with ERG-DAT syncretism is the extinct Ubykh (Caucasian), where ERG, DAT, and also GEN are marked with the suffix *-n* for singular and *-n3* for plural. In a ditransitive, (16), the AGENT is ERG-marked and the GOAL is DAT-marked.

- (16) *Nq̣ə̄m beyə-n sy°a a-sə-n-q'adaž'-q'a.*
Nâzim bey-ERG I.DAT 3SGA-1SGD-3SGE-tell-PST
 'Nâzim bey told it to me.' (Dumézil 1965: 43) *Ubykh*^{21,22}

Fell (2012b: p. 65, fn. 1) notes: “Traditional descriptions of Ubykh grammar note only two primary case markers, the zero-marked absolutive and the oblique. The oblique is used to mark ergative, dative, adverbial and genitive cases, without discrimination.”

We can also find examples with one ERG-marked and two DAT-marked arguments: Example (17) shows an applicative of a ditransitive, where the GOAL (a low applicative in the analysis by Fell 2012b) is DAT-marked, and the BENEFACTIVE (a high applicative, Fell 2012b, but possibly a lexical DAT assigned by the incorporated P glossed as ‘BEN’) was originally DAT-marked and has been relativized (on relativization of DAT-marked nominals cf. Fell 2012a: §4.5), in addition to the ERG-marked AGENT ‘I’.

- (17) *za-má-Ø Ø-t-x'a-wə-s-t°ə-n tət-Ø.*
one-apple-ABS 3SGA-REL-BEN-2SGD-1SGE-give-PRS man-ABS
 ‘The man for whom I give you an apple.’ (Charachidzé 1989: 441) *Ubykh*²³

Similarly, (18) illustrates a sentence with an ERG-marked AGENT and two DAT-marked nominals, (all of them pro-dropped), although, again, both DAT-markings might be lexical cases assigned by the incorporated Ps glossed as ‘BEN’ and ‘ABL’, respectively.

- (18) *a-s-x'á-w-ya-nə-wt°'-ay-awt.*
3SGA-1SGD-BEN-2SGD-ABL-3SGE-take-ITER-FUT2
 ‘He will take it back from you for me.’ (Dumézil 1975: 102) *Ubykh*²⁴

²¹ Fell 2012b: p. 67, ex. (19)

²² In the glosses in (16), and also in the following two examples from Ubykh, the letters ‘A’, ‘D’, and ‘E’ on agreement affixes indicate the case of the agreement target: ABS, DAT, and ERG, respectively. For instance, the gloss ‘3sgA’ indicates 3sg-agreement with the ABS-marked argument (pro-dropped ‘it’ in (16)).

²³ Fell 2012b: p. 79, ex. (58)

²⁴ Fell 2012b: p. 79, ex. (59)

That this is an instance of syncretism, and not of markedness reversal²⁵, follows from the fact that Ubykh verbal agreement prefixes (or clitics, as Fell 2012a argues) track case-marking, and the ERG- and DAT-tracking prefixes display allomorphy in the 3rd persons, showing that the grammar must in some way distinguish between the ERG- and DAT-marked target nominals.

4.4.4 Agree-based implementations of DCT

Recall the Agree-based implementation of DCT by Clem & Deal (2024) and Poole (2024), introduced in §2.4.5, which rejects the extra operation Case Competition. Instead, nominals are valued as $[\text{case: DEP}]$ by a functional head as reflex of φ -agreement, but only if this head has first Agreed with another nominal with unmarked case, thereby implementing the transitivity condition of dependent case. Fig. 4.5 illustrates the derivation of ERG-ABS in a transitive clause. In §3.6.3 I

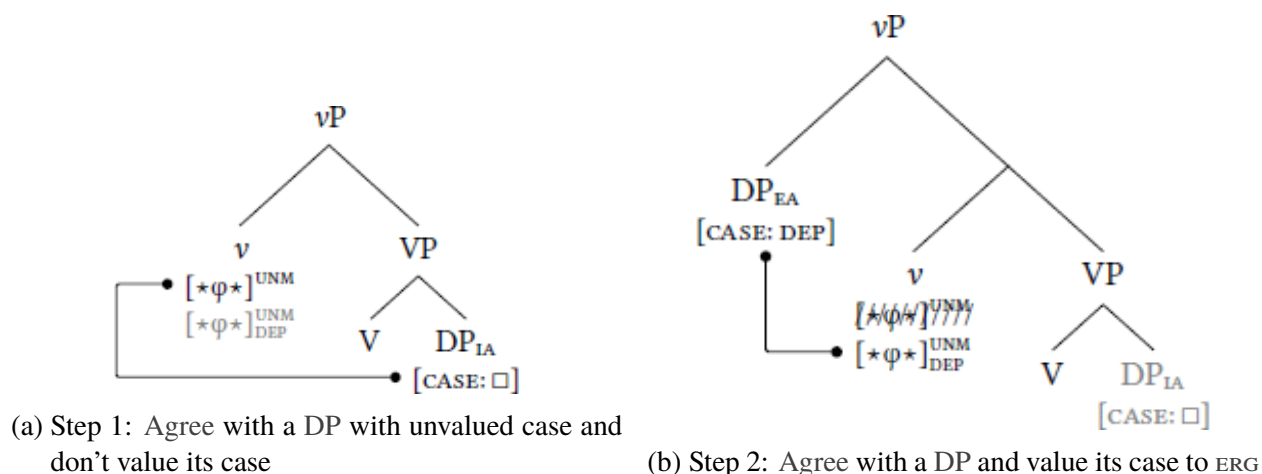


Figure 4.5: Assignment of dependent-up case (here: ERG) via Agree (figures from Poole 2024: exs. (11a, b))

have argued that this approach cannot capture the $1 : n$ pattern of multiple ACCS, where the same NOM-marked nominal serves as case-competitor against more than a single ACC-marked nominal: the Agree-based approach would need to allow this functional head to carry potentially arbitrarily many ACC-valuing probes — a configuration that I have formally proven in Laszakovits (2018)

²⁵ As Preminger (2024: §4.3.1.2, pp. 84 f.) has proposed for English: English ACC, which appears in many disparate contexts is the unmarked case (the elsewhere form), and English NOM, which appears in restricted contexts, is a “head case” assigned by a functional head like T.

would blow up the derivable syntactic patterns way beyond what is attested in natural languages.²⁶

However, perhaps this shortcoming of the Agree-based approaches to ACC-assignment DCT is a virtue with respect to the assignment of *ERG*, inherently capturing the Single Ergative Generalization. Poole (2024: p. 9, fn. 6) discusses the predictions that his proposal makes for configurations containing three DPs, noting that — without further assumptions, such as that the lowest DP is actually in a different CAD, thereby reverting to a two-DP configuration — he actually predicts the pattern **ABS-ABS-ERG*. Or, more generally, the single *ERG*-marking will always appear on the third-highest nominal (the second-highest under the assigning head). This prediction is given in Fig. 4.6.

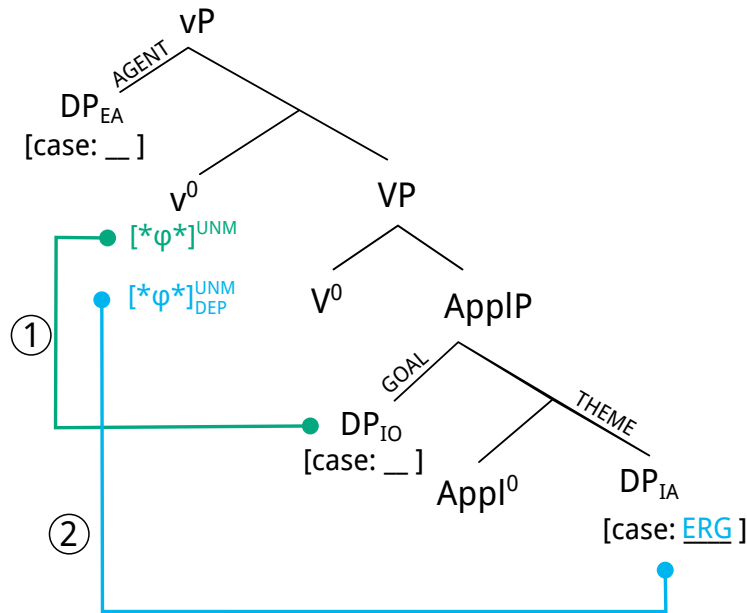


Figure 4.6: Poole's (2024) prediction **ABS-ABS-ERG*

4.4.5 Summary

In this section I have shown three alternative proposals to my theory which I will present in the next section 4.5. First, §4.4.2, a short reiteration of the inherent-*ERG* view, which derives the Single Ergative Generalization, but is not a satisfying theory for other reasons.

²⁶ As Whit Tabor points out to me, the available sample size may well not be adequate to draw final conclusions. My argument is based on trying to find the smallest complexity class that can still derive all attested data.

Then, in §4.4.3, I have discussed the proposal by Baker (2014b, 2015b), where the Single Ergative Generalization is accidental and not a meaningful generalization at all. Baker (2014b, 2015b) predicts that dependent-up cases are completely parallel to dependent-down cases and there is nothing principled that stops multiple assignments of ERG — as can be seen from the distribution of DAT, Baker argues, which, on his account, is a dependent-up case as well.

I have discussed some possibilities to elicit these predicted multiple ERG-markings, but I cannot at this time come up with good testing grounds for them.

I have argued that Baker's theory only shifts the problem to the question: Why can't ERG be generalized from the clausal to the verbal domain, such that ERG-morphology appears not only on AGENTS, but also on GOALS etc., which would yield the actual ERG-ERG-ABS pattern? This absence of ERG-DAT syncretism is especially puzzling because typologically we see a lot of ERG-GEN syncretism.

I thus maintain that there must be something more going on; the Single Ergative Generalization cannot be discarded as accidental and essentially due to the facts that only the AGENT is first-Merged outside the verbal domain, and the AGENT is rarely, if ever, ineligible for movement targeting the closest nominal (such as movement satisfying the EPP).

Finally, in §4.4.4, I have discussed the proposal by Poole (2024), where dependent cases are assigned by functional heads with a transitivity requirement built in. Poole (2024) himself notes that his proposal cannot account for the distribution of ERG, incorrectly predicting the pattern ABS-ABS-ERG-ABS* for predicates with more than 2 arguments.

I now turn to my proposed theory.

4.5 Possibilities to account for the Single Ergative Generalization by computational means

4.5.1 Introduction

In this section I will discuss implications by considerations from formal language theory resp. mathematical syntax to the unattested pattern of multiple `ERG`-markings per `CAD`, and to the attested pattern of the Single Ergative Generalization. First, in §4.5.2, I begin by making explicit my proposed theory: that the Single Ergative Generalization is real and not accidental. Then I show how two formalisms that have received a lot of attention in recent publications might account for the patterns we observe regarding dependent-up case. Subsection 4.5.3 shows that the unattested pattern of `ERG`*-`ABS` is computationally easy and can be expressed by the complexity class known as Tier-based Strictly 2-Local (TSL-2). In §4.5.4 I show that another frequently explored formalism for syntactic dependencies, namely string languages over sequences of c-commanders, known as c-strings, cannot handle the Single Ergative Generalization — a result that I expand to Sensing Tree Automata (STAs) in §4.5.5. In short, because the case-competitor (which, so to speak, licenses the `ERG`-marking) is further down in the tree, and because the tree branches downwards, it is formally speaking *hard* to know whether the case-competitor has been found in *some* sub-tree.

4.5.2 Hypothesis

Recall that the previous proposals for the Single Ergative Generalization, discussed in §4.4, suggest that the Single Ergative Generalization essentially falls out as accidental because of properties independent of the dependent-up nature of `ERG`. Here, I would like to propose an alternative, which I split into two parts:

- (19) dependent-up case assignment-licensing is fundamentally different from dependent-down case licensing. The change in directionality comes with consequences.
- (20) One of these consequences is the Single Ergative Generalization: The distribution of `ERG`-

marked nominals is restricted in the observed way because this helps keep the complexity class of the grammar within acceptable bounds.

In this section I will present my arguments for (19). At this point, I do not have arguments for (20), so I leave it as hypothesis to be evaluated at a later point.

4.5.3 Multiple instances of ergative are computationally easy

It is a well-known fact in mathematical syntax that common models in formal language theory implementing Minimalism, most notably MGs (Stabler 1997, 2011), can in principle express any constraint that can be stated as over regular (Type 3) tree languages (Graf 2011; Kobele 2011).

This holds for dependent cases as well. In Laszakovits (2018) I have shown that dependent case can be modelled by covert Move and by c-selection (Merge) in Minimalist Grammars, illustrating that its complexity falls into the range of things that are natural in syntax, and allowing us to formalize it as a regular constraint on tree languages. From this it follows that the rules for the distribution of both dependent-up and dependent-down case can be computed by a deterministic bottom-up tree automaton. This is a clear upper bound for the complexity class of dependent case.²⁷

However, while regular languages are the most restrained class in the Chomsky (1956, 1959) hierarchy, Fig. 4.7, they are still fairly powerful in terms of the patterns that they can express, and they overgenerate compared to the constraints attested in natural languages. For instance, regular languages allow modulo counting, which allows the specification of grammars with constraints like “the number of NPIs in a sentence must be even”. They also allow XOR conjunctives, which might state a constraint like “either the constraint on NPIs must hold, or the constraint on anaphor binding must hold, but the sentence is ungrammatical if they are both fulfilled”. Such constraints have yet to be observed in natural languages, and until they do, it is preferable to write grammars of a schematic

²⁷ Thomas Graf points out to me (p.c.) that in combination with his results in Graf (2012), this means that dependent-up case is LR-deterministic, i.e., top-down deterministic if, at every node, we can test for exactly one of its two daughters whether the subtree rooted in that daughter belongs to a particular regular tree language. Intuitively, we can ask for one, and only one, of the daughter nodes whether it locally dominates at least one suitable licenser. The question then becomes whether other locality conditions, such as my musings about Phases in §4.5.5, are enough to reduce that kind of limited regular look-ahead to finite look-ahead.

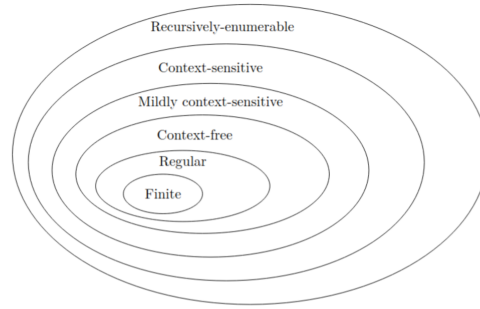


Figure 4.7: The Chomsky hierarchy of formal languages

that do not allow the formulation of such constraints.

In recent decades, work has been undertaken to define and investigate some schematics that are less powerful than that of regular languages, collectively called the subregular languages (Chandlee 2014; Chandlee & Heinz 2017; Heinz 2011a,b). As the name suggests, subregular grammars can do a subset of the things that regular grammars can do. Since assuming the class of regular grammars for human language overgenerates the actual grammars we see for individual natural languages, it is necessary to find a weaker complexity class.

A subregular class of languages that has received a lot of attention recently is the class of tier-based strictly local (TSL) languages. A strictly k -local (SL- k) language (also known as k -grams or Markov chain) is a formal language (i.e., a set of well-formed expressions) whose well-formedness is determined by passing a window of finite size k over every expression and checking the contents of the window against a list of allowed (resp. disallowed) combinations. Thus, SL- k grammars can check local dependencies up to a distance of k . For long-distance dependencies, it is necessary to first delete all irrelevant symbols (by projecting the relevant ones onto a “tier”) so that the dependency becomes local enough to be checked by an SL- k grammar. Graf (2018: p. 131 f.), Vu, Shafiei & Graf (2019), and Graf & Kostyszyn (2021) describe this process in the following way:

- The projection function T determines which symbols/nodes are ignored and which are copied onto the tier. A simple projection function may make reference only to the label of the node as it decides. A more complex projection function may also take the local context of the node into account.

- The licensing function S maps each node n on the tier to a string language L_n . The tier is illicit if the string of n 's daughters is not a member of L_n . Again, a simple licensing function might consider only the label of n to determine its L_n , while a more complex licensing function might also take n 's structural context into account.
- Finally, the complexity of the string languages L_i influences the overall complexity of the TSL- k tree grammar. Graf & Kostyszyn (2021) propose the notation TSL- $k[X]$, with X the complexity class of L_i .

This mechanism is employed by Vu, Shafiei & Graf (2019) and Hanson (2023) for English and Japanese, respectively; both accusative languages. For instance, to rule out (21), Vu, Shafiei & Graf (2019) propose the tier-projection in Fig. 4.8, which is ruled out by the constraint in (22).

(21) *He saw she.

English

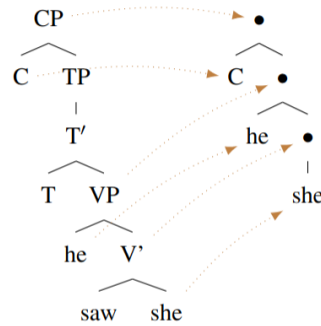


Figure 4.8: A tier-projection of “all C-heads and their mothers as well as all D-heads with nominative case and their mothers” (Vu, Shafiei & Graf 2019: p. 269).

- (22) “If a node on the tier is labeled $\bullet$ and has a sister carrying nominative case, then it must not have a daughter carrying nominative case. Formally, a node n that is labeled $\bullet$ and appears in the context $[\square[\text{NOM}, n]]$ is mapped to the string language $(\Sigma \setminus \{\text{NOM}\})^*$ (where $\square$ is a placeholder matching every node label and Σ is a fixed set of node labels).” (Vu, Shafiei & Graf 2019: p. 269)[

Similarly, Hanson (2023) proposes a TSL analysis for the case patterns of Japanese. A sentence

like (23) is mapped onto a tier as in Fig. 4.9, on which an SL-2 grammar is applied, part of which is given in (24) for the permissible daughter strings of v .

- (23) *Taroo -ga piano -o hiita.*
Taroo -NOM piano -ACC played.
‘Taroo played the piano.’

*Japanese*²⁸

- (24) $\{\text{NOM, GEN, lexical DAT}\} \cdot (\text{DAT}^* \cdot \{\text{ACC, lexical DAT}\})$

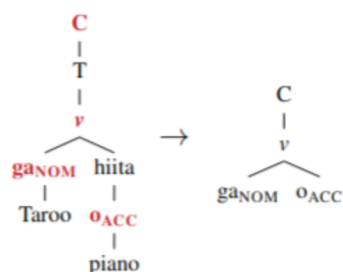


Figure 4.9: A dependency tree and the projection of “nodes of category K, C, and v ” (Hanson 2023: p. 16, Fig. 1)

(24) allows strings such as $\checkmark \text{NOM}$, $\checkmark \text{NOM} \cdot \text{ACC}$, $\checkmark \text{NOM} \cdot \text{DAT} \cdot \text{ACC}$, etc., but doesn’t allow strings like $*\text{ACC}$, $*\text{NOM} \cdot \text{DAT}$, or $*\text{NOM} \cdot \text{ACC} \cdot \text{ACC}$. From this it can be seen that Hanson (2023) implicitly analyzes Japanese structural DAT as an “intermediate dependent” case in the sense of Podobryaev (2013), where the pattern $\text{NOM} \cdot \text{DAT}^* \cdot \text{ACC}$ is stated over one large CAD without making a distinction between verbal and clausal domain. In an accusative-aligning language without DAT, such as Korean or Finnish, we might project the tiers in a similar way as Hanson (2023), but modify the constraints on the daughters of v as in (25).

- (25) $\text{NOM} \cdot \text{ACC}^*$

This yields the combinations $\checkmark \text{NOM}$, $\checkmark \text{NOM} \cdot \text{ACC}$, $\checkmark \text{NOM} \cdot \text{ACC} \cdot \text{ACC}$, etc., but does not yield $*\text{ACC}$, $*\text{ACC} \cdot \text{ACC}$, etc..

Then what prohibits an ergative-aligning language from imposing a constraint on the daughter

²⁸ Hanson 2023: p. 16, ex. (1)

string of v that corresponds to the reversal of (25) in (26)?

$$(26) \quad \text{ERG}^* \cdot \text{ABS}$$

In this framework of TSL-2, there is nothing that can account for the conjunction of constraints in (13) because the constraint (13a) on its own should be as easy for the Language Faculty to implement as its reversal for dependent-down accusative.

Let us now look at other formalisms.

4.5.4 The Single Ergative Generalization cannot be stated over c-strings

Graf & Shafiei (2019) have observed that two things go hand-in-hand: the ubiquity of syntactic constraints stated over c-command, and prominence with respect to the timing of feature checking: if x c-commands y , this means that x has been Merged (or Moved) more recently (in a bottom-up derivation) than y .²⁹ They note that syntactic constraints involving c-command do not in fact need to be stated over trees, but can be stated over strings, which they term “c-strings”. The transformation is done by a recursive mapping procedure where given a phrase-structure tree, Fig. 4.10a, and the corresponding dependency tree, Fig. 4.10b, we can define the c-string of a node as (27).

$$(27) \quad \begin{array}{ll} \text{a.} & \text{cstring}(d_i) = \text{leftsibblings}(d_i) \uparrow \text{cstring}(m) \quad \text{with } m \text{ the mother node of } d_i \\ & \text{in Fig. 4.10b,} \\ \text{b.} & \text{cstring}(r) = \text{leftsibblings}(r) = r \quad \text{if } r \text{ is the root node.} \end{array}$$

$$(28) \quad \text{leftsibblings}(d_i) = d_i \cdot d_{i-1} \cdots d_1 \quad \text{given the configuration in Fig. 4.10b}$$

Then, constraints involving c-command, such as NPI-licensing or Binding Theory, Graf & Shafiei (2019) argue, become constraints no longer over trees, but over strings: over sequences of c-com-

²⁹ To be precise, Graf & Shafiei (2019) use a formulation of c-command where heads c-command their specifiers, corresponding to a mixture of c-command and m-command Aoun & Sportiche (1983), which they term “d-command” for “derivational command”. This matches their recency observation more closely: A head H d-commands its specifier S because the highest projection of H (HP) will be Merged after S has been Merged.

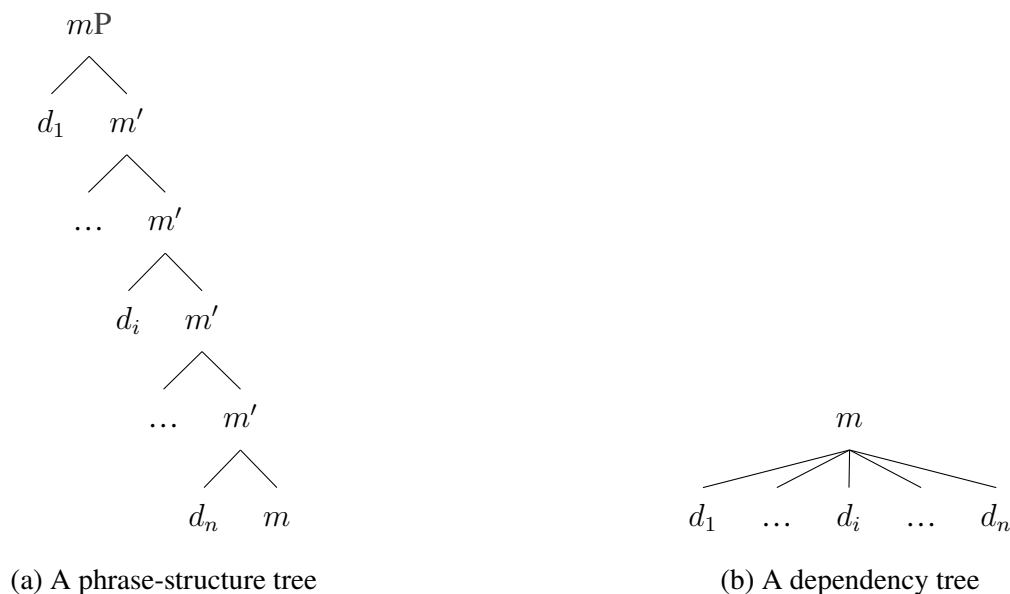


Figure 4.10: A phrase-structure tree and its corresponding dependency tree

manders. For instance, from the sentence in (29a), the c-string of the anaphor becomes (29b), which can be recognized by a TSL string grammar, informally stating something like (29c) (for the formal implementation, see Graf & Shafiei 2019).

- (29) a. *John said that Mary likes { ✓ herself / *himself }.* English
- b. $\text{cstring}(\text{herself}) = \text{herself} \cdot \text{Mary} \uparrow \text{likes} \uparrow \text{T} \uparrow \text{that} \cdot \text{John} \uparrow \text{said} \uparrow \text{T} \uparrow \text{C}$
- c. Project onto the tier the initial anaphor, all finite T heads, and all D heads whose φ -features match that of the initial anaphor. Then, pass a window of size 2 over this new sequence. You may not encounter the sequence $\text{herself} \uparrow \text{T}$, which would mean that the anaphor was not c-commanded by an appropriate DP in its binding domain.

Thus, given the definition of c-strings, which is simple enough, a local c-command dependency between first-Merge positions can be stated with a subregular grammar over strings. For a fuller analysis of English anaphor licensing (Principle A) I refer the reader to the extensive discussion in Shafiei (2022: §3.1).

Given that c-command features prominently in case-competition (cf. §2.6), we might try to capture the distribution of structural cases via c-strings. Consider again the dependency tree in Fig. 4.9

on page 127. To validate both case markings we can consider their c-strings, (30).

- (30) a. $\text{cstring}(\text{ACC}) = \text{ACC} \uparrow \text{'played'} \cdot \text{NOM} \uparrow v \uparrow T \uparrow C$
 b. $\text{cstring}(\text{NOM}) = \text{NOM} \uparrow v \uparrow T \uparrow C$

Then, we expect that the c-string of ACC should contain NOM (or any other nominal that qualifies as case-competitor), meaning that the ACC-marked nominal is c-commanded by a NOM-marked nominal, and the c-string of NOM should not contain any other nominals (that qualify as case-competitors). In (30), this is fulfilled for both c-strings.

To check the distribution of ERG, consider again Fig. 4.2 on page 111, showing the ergative-absolutive case patterns of a ditransitive (on an applicative structure) and of a causative of a transitive, respectively. The corresponding dependency tree for the ditransitive is given in Fig. 4.11 below. From Fig. 4.11 we project a tier containing all nominals that qualify as case-competitors as well as

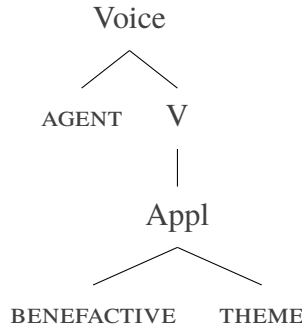


Figure 4.11: The dependency tree of a ditransitive (applicative of a transitive)

the edges of the CADs (let's assume Voice), and from this tier we then read off the c-strings for all three arguments, (31).

- (31) a. $\text{cstring}(\text{THEME}) = \text{THEME} \cdot \text{BENEFACTIVE} \uparrow \text{AGENT} \uparrow \text{Voice}$
 b. $\text{cstring}(\text{BENEFACTIVE}) = \text{BENEFACTIVE} \uparrow \text{AGENT} \uparrow \text{Voice}$
 c. $\text{cstring}(\text{AGENT}) = \text{AGENT} \uparrow \text{Voice}$

Then, we can enforce an SL-2 grammar over these c-strings with constraints such as those in (32).

4.5 Possibilities to account for the Single Ergative Generalization by computational means

- (32) a. *_{ERG} _{ERG} (only a single _{ERG}; nothing above an _{ERG})
 b. *_{ERG} _{ABS} (nothing above an _{ERG})
 c. ✓ _{ABS} _{ERG}

(32a) will rule out a case pattern where the _{GOAL/BENEFACTIVE} is _{ERG}-marked, which is the desired result. However, with an SL-2 grammar over c-strings, we cannot capture the fact that _{ERG} is only allowed if there is a lower nominal. We cannot make this constraint part of the c-string of the _{ERG}-marked nominal, because c-strings look up. We also cannot make this constraint part of the c-string of the lower (_{ABS}-marked) nominal, such as via (33a), because in intransitives, there is a single _{ABS}-marked nominal without the requirement that there be a c-commanding _{ERG}. (I gloss over the fact that unaccusatives might not contain Voice at all; my argument stands for unergatives in any case.) We also cannot use the constraint in (33b) because that would incorrectly rule out the attested case-marking in ditransitives where the _{THEME} and the _{GOAL} are both _{ABS}-marked.

- (33) a. *_{ABS} Voice
 b. *_{ABS} _{ABS}

If there is an upper limit k to the number of _{ABS}-marked nominals below the single _{ERG}, k any natural number (perhaps 3, or 5), we could expand our grammar to an SL- $(k + 1)$ grammar with the rules in (34), for $k = 3$.

- (34) a. *_{ERG} _{ERG} (no two _{ERGS})
 b. *_{ERG} _{ABS} (nothing above an _{ERG})
 c. *_× _{ERG} (_{ERG} is not the lowest)
 d. *_{ABS} _{ABS} _{ABS} _{ABS} (no four _{ABS}, maximally $k = 3$)

Considering the problem from the perspective of c-strings shows us that there is a fundamental difference between dependent-up and dependent-down cases: Dependent-down case (_{ACC}) can be licensed by considering one clearly defined string of potential licensors, namely the _{ACC}-marked nominal's c-string. On the other hand, licensing dependent-up (_{ERG}) case is not so simple. This is

so because we cannot read off an analogous sequence of c-commandees and check that for the occurrence of a case-competitor (ABS-marked nominal). Since trees branch downwards, there would need to be several such sequences. Furthermore, the constraint that there needs to be a case-competitor in the sequence of c-commandees of an ERG-marked nominal only needs to hold for *at least one* of these sequences, and not for all of them. The computational device would thus need to perform an OR-combination of the results, such that ERG is licensed if at least one sequence fulfills the constraints, and is not licensed if none of the sequences fulfills the constraints.

Thomas Graf (p.c.) has mentioned to me an alternative way of phrasing this issue: While for dependent-down case we can look at c-strings, for dependent-up case, where c-strings are not sufficient, we could define a new string representation c' -strings such that a c' -string of node n contains all of n 's accessible local c-commandees. (A left-corner parser and similar devices gets these c' -strings for free as a side effect of how it builds the trees.) Then, we can employ the same string constraint for both dependent-down and dependent-up case, namely “have at least one licenser”, which is TSL-2. The difference lies not in the complexity of the constraint, but in the representation: c-strings are a lot easier to compute from the tree representation than c' -strings.

4.5.5 Too many search paths for the case-competitor

Graf & De Santo (2019) describe a “sensing tree automaton” (STA), which is a formal automaton over tree-structures (here, again, dependency trees) whose rules can have a limited amount of look-ahead. The automaton reads the input trees top-down and assigns a state to each node, starting with a defined start state. When assigning a state to a node n , the automaton’s “sensing rules” make reference to (i) the state assigned to the parent node of n , (ii) the label of n , and (iii) the labels of all of n 's siblings. As any automaton, an STA can encompass information about the input tree through the states it assigns. For instance, it might utilize the state q_1 to denote that a particular element (such as an anaphor) is licensed. Then, the STA would assign the state q_1 to a node n if one of n 's left siblings is a relevant licenser (recall that n 's left siblings are c-commanders of n 's children), and also if n 's parent already had state q_1 , but not if n 's label denotes a barrier for this licensing

relationship. Then, when encountering a node a labeled with an anaphor, the STA will reject the input tree if the state of a 's parent is not q_1 , and will continue the decision procedure otherwise.

This is a highly promising concept for dependency checking. We have seen how STAs can be used to capture dependent-down licensing relationships, such as anaphor binding, with ACC-assignment working analogously. What about dependent-up case assignment?

We might employ the STA's states to encode information such as in (35):

- (35) q_{ERG} = "I have encountered an ERG, so now I need to be on the look-out for a case-competitor. Once I have found one, I will remove this notice that I'm still looking. If I reach a leaf without having found one, or if I encounter a second ERG, I will reject the input tree."

But this raises the same problems as with c-strings above: The STA can pass information down the tree (from parent node to child node), and to some extent sideways (from sibling to sibling), but not back up. So when the automaton is in state q_{ERG} and the tree branches, and q_{ERG} is assigned to multiple sub-trees, the automaton will reject this input unless *all* of these sub-trees remove the state q_{ERG} by finding a case-competitor. This is not the desired result: We would want the input to be accepted as long as *at least one* sub-tree has managed to remove this state, and to be rejected only if *none* of the sub-trees have removed it. Thus, the sub-trees need to pass the information whether they have found a case-competitor back up the tree to the node where the branching happened, so that the information from the sub-trees might be combined. This is not possible with an STA.

Graf & De Santo (2019) observe that STAs are limited to encoding such constraints where at the point of branching, the search for a licenser (in the above example, the case-competitor; in their example below, a mover satisfying the equivalent of a Probe) need only be continued in a single sub-branch. They write,

STA-recognizability would [...] hold [A] if movers could only escape from the left-most argument, or [B] if the label of the selecting head decides which one of its arguments can be extracted from. Extraction from arguments could also be parameterized for each feature so that [C] wh-movers may only leave complements whereas [D] topicalization

is only allowed from the last but one argument, if it exists. Or [E] the STA could switch between these four constraints depending on the number of ancestors of the current node modulo 4. STA-recognizability holds as long as distributing the head's [unsatisfied Probes] across its arguments is fully deterministic based on the information available to a sensing rule (current state, label of selecting head, labels of selected heads). (Graf & De Santo 2019: p. 19)

For their implementation, they propose a constraint where licensors/movers may only be searched for in complements (the first argument of a head), not in specifiers (the second and higher arguments as well as landing sites of movement), which they misleadingly label the “Specifier-Island Constraint”.³⁰ This proposed constraint is not of relevance to natural languages, but it does provide the formal grounds of exploring the properties of an STA with deterministic search, which they argue is highly desirable from a computational perspective.

Can we find similar restrictions on the search paths of licensors in dependent-down licensing configurations that are well-founded in natural language data?

I want to propose that Phases, and in particular the Phase Impenetrability Condition, take care of a large part of the problem. I adopt a theory of dynamic phasehood (Bobaljik & Wurmbrand 2005; Bošković 2013, 2014; den Dikken 2007; Gallego 2010; Wurmbrand 2013, 2014), where in an extended spine the phase head is the highest head on the spine. For instance, the verbal spine includes the functional heads Appl_{low} , v , $\text{Appl}_{\text{high}}$, $\text{Voice}_{\text{CAUSEE}}$, $\text{Voice}_{\text{PASS}}$, Voice , and others, which are ordered along a functional sequence. The highest projected head along this sequence will close off the verbal phase. Similar considerations hold for the nominal and the clausal domain, and possibly others.

From this, it follows that any sub-trees that get Merged as arguments into, say, the verbal phase, will not be able to extend the verbal phase, and will not be able to use the selecting head in the verbal phase to extend their own (nominal or clausal) phase. Thus, if the nominal and clausal

³⁰ Specifiers are not per se islands, as Stepanov (2007) has shown in detail, because movement out of subjects that remain in Spec,VoiceP is possible. Rather, the generalization seems to be that there is no movement out of moved elements (Bošković 2018). In their paper, Graf & De Santo (2019) omit this distinction and apply the term “Specifier-Island Constraint” to both kinds of specifiers.

arguments are impenetrable Phases, we can exclude them for the search of the licensor. Indeed, for the search for a case-competitor, a nominal phase head will itself carry the relevant identification, thus it suffices to look at the phase head, and is not necessary to look inside the phase. However, as is well-known, Phases have escape hatches for movers. For instance, as we have seen in §3.4 above, a clausal phase may topicalize an argument to its edge as final landing site from where the argument participates in the matrix CAD, as in the raising-to-object constructions in Sakha; and a clausal phase can contain a moving *wh*-element which undergoes successive cyclic movement and participates in the CADs of every intermediate landing site, as discussed in §4.3.4 for Koryak. Thus, when looking for a licensor in the verbal phase, it does not suffice to look at the features of the phase heads of the arguments in the verbal phase, but it is also necessary to look at the content of the phase edges of the arguments (SpecCP, etc.). But, importantly, it is not necessary to look further down than that.

However, I have been assuming that the PIC holds and renders the complement of phase heads inaccessible to the next-higher phase. There is much debate on the PIC in current syntactic literature, with much evidence given that it can be violable in some languages for some phase heads. For instance, the data from Shipibo presented by Baker (2014b, 2015b) and discussed in §4.2, where the *THEME* moves past the *ABS*-marked applicative and thus is able to receive *ERG*, on Baker's analysis are derived such that the applicative, while being the structurally higher argument, is inaccessible for EPP-movement to SpecTP because it is contained inside a PP, but crucially, *is* accessible for case-competition against the *THEME*, which means that the PP is a transparent phase. I do not know at this point whether we might argue for similar restrictions on transparent Phases as I have proposed for oblique Phases. Until that time comes I do not see how an STA could be employed for natural language phenomena where the licensor is to be found further down the tree than the licensee. Perhaps a prediction is to be made here: In a language where all Phases obey the PIC, the grammar would be STA-recognizable, which might make further predictions, pending further research into STAs and repercussions of the proposed complexity of grammar.

4.5.6 Summary

In this section I have proposed my theory about dependent-up case: that it is fundamentally different from dependent-down case because the licensing relationship (between the dependently marked nominal and its licensor, the case-competitor) is computationally *easier* if the licensor c-commands the ACC-marked nominal, and computationally *harder* if the licensor is c-commanded by the ERG-marked nominal. This is because searching for the licensor towards the root leaves only a single (deterministic) path to search, while searching for the licensor towards the leaves substantially increases the complexity of such a computational device.

This argument supports the first part of my hypothesis, repeated below in (36):

- (36) Dependent-up case assignment-licensing is fundamentally different from dependent-down case licensing. The change in directionality comes with consequences.

I have not provided arguments for the second part, repeated in (37):

- (37) One of these consequences is the Single Ergative Generalization: The distribution of ERG-marked nominals is restricted in the observed way because this helps keep the complexity class of the grammar within acceptable bounds.

I must leave this second part for future research.

4.6 Conclusions

In this chapter I have presented and discussed a phenomenon that has received different proposals in the existing literature, termed the Single Ergative Generalization. In short, while for dependent-down cases like ACC, we see that within a CAD, this dependent case can appear multiple times assigned by multiple calls of Case Competition, this seems to be different for the dependent-up case ERG. Some proposals (Baker 2014b, 2015b; Nie 2017) claim that this is accidental, and other dependent-up cases (DAT has been proposed) can in fact appear multiple times, in full parallel to dependent-down cases. However, these proposals are then faced with further questions, such

as: A language with ERG-DAT syncretism would violate the Single Ergative Generalization. Is it also accidental that this syncretism does not exist? Note that ERG-GEN syncretism is typologically widespread.

I have presented my alternative hypothesis that dependent-up and dependent-down cases are not the same modulo the direction of case assignment during Case Competition. Rather, I have argued, dependent-up cases must be computed differently from dependent-down cases because searching for the licenser (the case-competitor) towards the leaves is computationally hard because the tree also branches in this direction. On the other hand, searching for the licenser of dependent-down case proceeds towards the tree's root, which is a single path of c-commanders, and thus deterministic.

Thus, I claim that any dependent-up case will pattern like ERG and should obey a “Single X Generalization”. I will pick up this claim in §5, where I argue that DAT in Turkic cannot be a dependent-up case precisely because it can occur multiple times per CAD, and in §6, where I argue that GEN in Turkic is the dependent-up case in the nominal domain, and that this is supported by the fact that GEN is not repeatable, like ERG. However, I do not yet know how exactly the Single Ergative Generalization follows from the fact that dependent-up cases are harder to compute than dependent-down cases. This is something I must leave for future research.

I finish by repeating that there might be data points available to test the different predictions of my theory compared to Baker's (2015b) theory that the Single Ergative Generalization is accidental. Baker (2015b: p. 219) notes that in Warlpiri and Diyari, in addition to one single ERG-marking on the AGENT, a high adverb can also receive ERG-marking (by competition against the THEME, Baker proposes. Baker (2015b: p. 219) analyzes these data by saying that the case-marking on adjuncts such as adverbs is determined on a later cycle than that of arguments.) Since adverbs should easily be repeatable, the accidental condition of a single ERG on arguments would disappear, and in this way these two languages might provide an ideal testing ground to see whether the ERG-marking on adverbs is in fact repeatable. I do not at this point have the necessary data to validate or disprove my theory.

Chapter 5

Turkic dative is a lexical/inherent case assigned by lexical/functional heads

5.1 Introduction

In this section, I discuss the evidence for structural DAT in Turkic languages under DCT's analysis of dative as the dependent-up case inside the verbal case-assignment domain (CAD). Recall the distinction between lexical, inherent, and structural cases that I have discussed in section 2.2 above: lexical cases are assigned by particular lexical items such as adpositions, adjectives, or verbs, inherent cases are assigned by particular functional heads that provide thematic interpretation, such as the head Appl introducing the applied argument, while structural cases in the sense of DCT are dependent and unmarked cases. In this chapter, I will argue that we do not have sufficient evidence to say that dative in Turkic is a structural (dependent-up) case, and that rather all alleged instances of “structural” dative can be analyzed as inherent case assigned by Appl (§5.3.2) or causative Voice (§5.4). I proceed as follows:

In §5.2 I discuss the different kinds of datives found in Turkic, some of which are clearly lexical cases, and others may be inherent or structural cases, the topic of this chapter.

In §5.3–§5.5 I review constructions and phenomena that have been claimed in the literature to be

instances of structural dative. I review their syntax in detail, and I will draw the conclusion that none of these constructions warrant the conclusion that the DAT-marking observed there must necessarily be structural DAT. Rather, in all constructions, a functional head appears that may be responsible for the DAT-marking, making this DAT an inherent case. The constructions under discussion are low applicatives (§5.3.2), dyadic unaccusatives (§5.3.3), RECIPIENTS/high GOALS of ditransitives (§5.3.4), causatives (§5.4), possessor-raising constructions (§5.5.2), and raising-to-object in THEMES influencing the case on the RECIPIENT (§5.5.3).

The strongest evidence for structural DAT have been claimed to be causatives (§5.4) where causees that are AGENTS of the embedded predicate can take ACC or DAT depending on whether the embedded predicate contains a THEME or not. However, ACC/DAT causees differ not only in their case, but also in animacy restrictions and in their interpretation if omitted (pronominal vs. existential), from which I conclude that there must be more going on than simply the presence or absence of a THEME inside the VP, namely a specific functional head that provides this interpretation and may then also be responsible for the DAT case marking.

Another strong argument for structural DAT is the case alternation ACC→DAT on RECIPIENTS/high GOALS (§5.5.3) that seems to be fed by the subject of the clausal THEME raising-to-ACC as discussed in §3.4. However, the only predicate for which this ACC→DAT fed by raising-to-object alternation has been documented is ‘promise’ in Sakha, which has some puzzling properties that may influence the conclusion that this DAT must be structural.

In §5.6 I discuss similarities and differences between DAT as dependent-up case in comparison with ERG (see chapter 4) and GEN (chapter 6) as dependent-up cases.

§5.7 concludes that unless more evidence comes along, or the analyses in §5.3 turn out to be wrong, or either the conclusions from chapter 4 are incorrect or the data presented in §5.6 can’t be reproduced, we have no evidence for DAT as dependent-up case, but we do see the likely possibility that DAT is assigned by the functional heads present in all constructions, making it an inherent case.

5.2 Lexical dative case in Turkic

The distinction between structural, inherent, and lexical case is particularly difficult for dative case, as Baker (2015b: p. 12–14) notes, and as I will show throughout this chapter. The most basic distinction between them is that structural and inherent case depend on syntactic structure¹ — thus, variations in structure triggering variations in case are a strong argument for the structural-ness or inherent-ness of this case —, and lexical case depends on particular lexical items (V, P, A, N) — thus it should vary with exchanging the lexical item, and should remain constant under changes to the structure. However, sometimes we see case that doesn't vary with syntactic structure, but there also is no obvious assigning lexical item. We might then postulate a phonologically null lexical item such as a P as case assigner, as proposed by Baker (2012a,b, 2015b) and McFadden (2004), a.o. This is more likely if we can identify a uniform semantics that this P would assign (such as 'from' for ABL and 'at' for LOC in Turkic), and less likely if the meaning varies.

In general, the 'meanings' of dative case in Turkic range from likely-adpositional 'towards', via RECIPIENTS of transitive 'help' and RECIPIENTS of ditransitive 'give', to BENEFACTIVES 'for', and finally to causees, which can be EXPERIENCERS or AGENTS of the embedded predicate. Göksel & Kerslake (2005: p. 142, pp. 177 f.) list the following contexts for Turkish dative:

1. The complement of some postpositions: *doğru* 'towards', *göre* 'according to', *kadar* 'until'.
2. The complement of some adjectives: *yakın* 'close to'/'near', *uygun* 'suitable for'.
3. The complement of many² verbs of emotion: *acı-* 'feel sorry for', *bayıl-* 'adore', *gıcık ol-* 'be irritated by' (slang), *güven-* 'trust', *inan-* 'believe', *kıl ol-* 'be irritated by' (slang), *kız-* 'be angry with/about', *kıy-* 'be merciless towards', *sevin-* 'be pleased about', *sıkıl-* 'be upset by', *canı sıkıl-* 'be annoyed about', *tap-* 'worship', *üzül-* 'be sorry about'.

¹ Cf. chapter 2 for a discussion of how inherent cases depend on syntactic structure. In short, because thematic roles are implemented in Minimalism by corresponding to particular structural positions, such as AGENT in Spec, VoiceP, and THEME in the complement of V, following the UTAH (Baker 1988, 1997).

² Many other verbs of emotion assign ABL case to the THEME.

4. The complement of some other verbs: *benze-* ‘resemble’, *boyun eǵ-* ‘submit to’, *razı ol-* ‘agree to’, *uy-* ‘conform to’/ ‘comply with’/ ‘match’.
5. The destination of an action: ‘to’, ‘into’, etc..
6. The price at which something is sold or offered for sale.
7. To mark an infinitive as purpose clause.
8. The RECIPIENT or beneficiary of an action: *bak-* ‘look at/after’, *ver-* ‘give’.
9. The causee of a causative construction based on a transitive verb.

The first four items are clearly dependent on the lexical item that introduces the DAT-marked NP as its complement, and can thus be seen as lexical case. Items 5 and 6 can be seen as involving a silent postposition that assigns lexical DAT, since the semantics are so very narrow and correspond to adpositions in other languages (such as English *to* and *per*, resp.). The same can possibly be said about item 7.

Items 8 and 9 are more likely candidates for non-lexical case since these datives are assigned in a predictable manner in certain configurations, and not exceptionally because of certain lexical items. In the next section 5.3 I will take up these constructions (§5.3.4, §5.4) and others and investigate whether these datives are instances of inherent or of structural case.

5.3 Instances of “structural dative” in Turkic and their compatibility with inherent case

5.3.1 Introduction

In this section I discuss some constructions that have been claimed in the literature to be instances of structural dative case in Turkic. In particular, Baker (2015b) and Baker & Vinokurova (2010) have analyzed DAT in Sakha as the dependent-up case in the verbal domain: as structural case

assigned to a case-less NP iff it c-commands another case-less NP (typically the THEME). This is in addition to lexical DAT assigned by particular lexical items as discussed in §5.2. In Preminger’s (2014) terminology, both lexical dative [case: DAT] and the dependent-up case in the verbal domain [case: DEP-UP] are realized morphologically as ‘dative paradigm’.

This section goes over three of the constructions that Baker (2015b) and Baker & Vinokurova (2010) have used to argue that DAT in Sakha is the dependent-up case in the verbal case-assignment domain. I will draw the conclusion that none of these constructions show this convincingly because they are compatible with inherent DAT assigned by the applicative head Appl: Appl is present in the structure of low applicatives (§5.3.2), of ditransitives on an applicative-analysis (§5.3.4), and of dyadic unaccusatives such as ‘need’ (§5.3.3). In the following two sections, §5.4 and §5.5, I will discuss the remaining constructions that Baker (2015b) and Baker & Vinokurova (2010) have argued contain dependent-up DAT.

In the absence of convincing evidence for the dependency of structural DAT on the presence of a lower c-commanded NP with structural case, and not on lexical items or particular functional heads, I conclude (§5.3.5) that we cannot at this point say for sure whether these datives are structural (dependent-up) or inherent datives.

5.3.2 Applicatives

Pylkkänen (2008) famously distinguishes two structural positions for applicatives: “high” applicatives establish a thematic relation between an applied argument and the event described by the VP and are introduced by a syntactic head Appl that selects VP and is selected by Voice (Fig. 5.1); while “low” applicatives establish transfer-of-possession of the direct object from or to the applied argument, and are represented by Appl selecting the internal argument and being selected by V (Fig. 5.2).

Turkish has “non-core” (i.e., optional) dative arguments that have often been analyzed as high applicatives (Akkuş 2019; Tonyalı 2015) because the DAT-marked argument does not always stand in a possession-relation to the THEME, thus they can’t be low applicatives. However, Key (2022) has

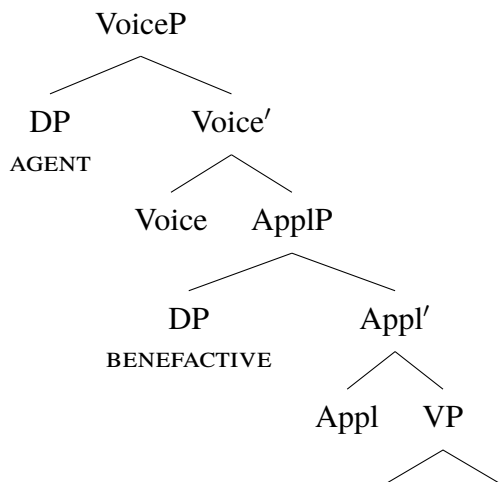


Figure 5.1: High applicative (Pylkkänen 2008: p. 14, ex. (6a))

argued that these datives are actually low applicatives, but that the definition of “low applicative” needs to be extended to “affected applicatives” in the sense of Cuervo (2003). This, he argues, allows to keep Knecht’s (1986: §5.4.1) generalization on “BEN-to-3 advancement”³ in Turkish:

BEN-3 may apply just when an agent[’s] activities make it possible for the entity denoted by the BEN to use or enjoy something or [to] further [their] ability to use or enjoy it. Disposition of whatever the agent acts upon should be understood to pass to the BEN.
Knecht 1986: p. 163 f., cited after Key 2022: p. 48

The affected argument is thus not necessarily in possession of the internal argument, but instead needs to stand in a relation to the state of the internal argument, unlike high applicatives, which have no such requirement. Some of these cases where a non-core dative can have an affected (low) applicative reading without possession entailment, but crucially not a high applicative reading, are given below.

- (1) *Ahmet kadın -a palto -sun -u tut -tu.*
 Ahmet woman -DAT coat -POSS.3SG -ACC hold -PST
 ‘Ahmet held the woman’s coat for her.’

Turkish

³ This is Relational Grammar terminology for the alternation between *için* ‘for’ (‘BEN’, BENEFACTIVES) and indirect objects (‘3’; i.e., DAT-marked constituents).

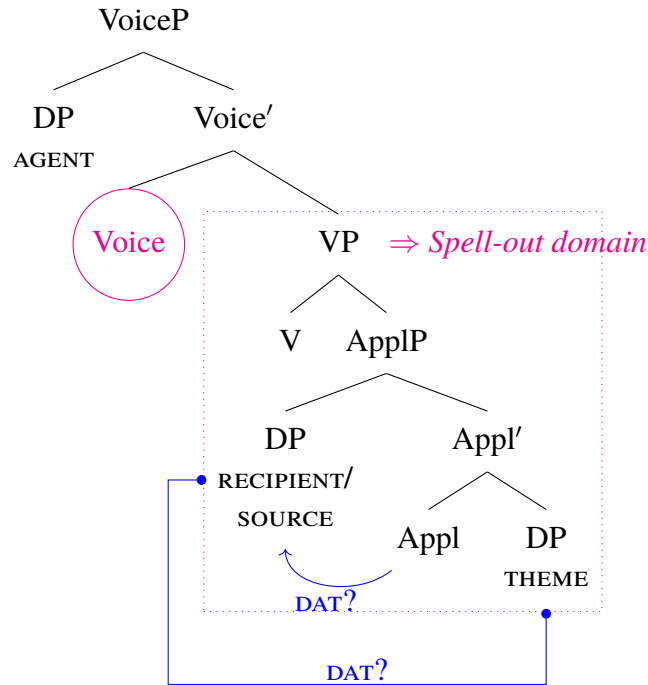


Figure 5.2: Low applicative (Pylkkänen 2008: p. 14, ex. (6b))

- a. ^{ok}... to help her put it on. (affected applicative)⁴
- b. #... while she goes into the restroom. (high applicative)⁵
- (2) *Bana şu kapı -yı bir tut -ar mı -sin?*
 me.DAT this door -ACC one hold -AOR Q -COP.2SG
 ‘Could you just hold this door for me?’ *Turkish*
- a. ^{ok}... so I can walk through. (affected applicative)⁶
- b. #A handyman is fixing a door. He wants to go outside for a smoke, but the door is
 partway off its hinges and he needs someone to hold it. (high applicative)⁷
- (3) *Ali bana Cengiz -in sigara -sin -ı yak -tı.*
 Ali me.DAT Cengiz -GEN cigarette -POSS.3SG -ACC put.on.fire -PST
 ‘Ali lit Cengiz’s cigarette for me.’ *Turkish*
- a. ^{ok}Ali lit one of Cengiz’s cigarettes for the speaker to smoke. (affected applicative)⁸

⁴ Key 2022: p. 53, ex. (34)⁵ Key 2022: p. 53, ex. (35) and the text above.⁶ Key 2022: p. 53, ex. (36) and the text above.⁷ Key 2022: p. 53 f., ex. (37)⁸ Knecht 1986: p. 162, ex. (31c) and the text below.

- b. #Ali lit Cengiz's cigarette, on behalf of the speaker, for Cengiz to smoke.

(high applicative)⁹

What is interesting for structural DAT is Pylkkänen's (2008) prediction that since low Appl selects the internal argument, it cannot appear in structures that lack an internal argument, such as with unergative verbs. This is borne out, as (4) illustrates.¹⁰

- (4) #*Burcu Elif -e yürü -dü.*

Burcu Elif -DAT walk -PST

- a. #‘Burcu walked for Elif.’

(high applicative)

- b. ^{ok}‘Burcu walked to Elif.’

*Turkish*¹¹

Under an analysis of DAT case as dependent-up case, it follows directly why Turkish low applicative arguments would appear in DAT case: they are Merged inside the verbal case-assignment domain, and there is another argument in a position that they c-command, and so they receive dependent-up case, which in this CAD is DAT. These datives are thus compatible with structural DAT case. However, due to the obligatory presence of an internal argument, we cannot test whether they *must* be structural, leaving the possibility that they might be inherent case assigned by low Appl to its specifier.

5.3.3 Dyadic unaccusatives

Baker & Vinokurova (2010: §3.2) describe dyadic predicates in Sakha where neither argument is external to the VP (an AGENT). In these constructions, the higher argument receives DAT and the lower argument NOM, as illustrated in (5).

⁹ Knecht 1986: p. 162, ex. (31c) and the text below.

¹⁰ Regarding some seemingly unergative verbs that do allow dative-applicatives, Key (2022: p. 54 f.) argues that they are either actually transitives (such as *dans et-* ‘to dance (a dance)’ and *şarkı söyle-* ‘to sing (a song)’), or are actually lexically dative-marked arguments (such as *X-DAT çalış-* ‘to work for X’ in the sense of ‘to be in the employ of X’).

¹¹ Tonyalı 2015: p. 113, ex. (17)

- (5) a. *Ejiexe massyyna tiij -bet.*
 you.DAT car.NOM reach -NEG.AOR.3SG
 ‘You lack a car.’ Sakha¹²
- b. *Ejiexe massyyna baar.*
 you.DAT car.NOM exist
 ‘You have a car.’ Sakha¹³
- c. *Ejiexe massyyna naada.*
 you.DAT car.NOM need
 ‘You need a car.’ Sakha¹⁴

The same construction can be found in Turkish¹⁵, (6), and other Turkic languages.

- (6) a. *Bana bir araba gerek -iyor.*
 me.DAT a car.NOM need -PROG
 ‘I need a car.’ Turkish
- b. *Bana bir araba lazım.*
 me.DAT a car.NOM necessary
 ‘I need a car.’ Turkish

Baker & Vinokurova (2010: p. 605 f.) apply a diagnostic to test the c-command relationships between the two arguments: In the construction with *naada* ‘need’ in (5c), the DAT-marked argument can bind the anaphor *beje-* inside the NOM-marked argument, (7a), but the reverse does not hold: the NOM-marked argument cannot bind the anaphor inside the DAT-marked argument, (7b).

^{12, 13, 14} Baker & Vinokurova 2010: p. 605, ex. (19)

¹⁵ (5b) with *baar* does not replicate to Turkish *var*, but we see (i) with lexical LOC:

- (i) *Ben -de bir araba var.*
 me -LOC a car.NOM exist
 ‘I have a car at my disposal.’ Turkish

Note that in Sakha, LOC has become syncretic with DAT (Menz & Monastirev 2021: pp. 447, 454, 456), so (5b) may be an instance of lexical LOC=DAT. If the DAT on the higher internal argument is structural case, as Baker & Vinokurova (2010) claim, and if lexical case is always assigned by a (possibly null) adposition, as Baker (2015b) proposes, this makes a prediction: there should be a difference between (5a) and (5c) on the one hand, which contain structural DAT, and (5b) and (i) on the other hand, which contain lexical LOC. Anaphor binding in a construction like (ii) should be possible with (5a), (5c), (6a), and (6b), which contain structural case, but not with (5c) and (i), which contain lexical case: because the higher internal argument, being contained inside a PP, cannot c-command out of the PP, and thus cannot bind the lower anaphor.

- (ii) The children_i have/need/lack each other_i’s toys.

- (7) a. [*Beje -tin* *oqo -to*] *kim -iexe naada?*
 self -POSS.3SG_i.GEN child -POSS.3SG.NOM who_i -DAT need
 ‘Who_i needs their_i own child?’ Sakha¹⁶
- b. *[*Beje -tin* *ije -ti*]-ger *kim naada?*
 self -POSS.3SG_i.GEN mother -POSS.3SG -DAT who_i need
 intended: *‘Who_i does their_i own mother need?’, or
 ‘Who_i is needed by their_i own mother?’ Sakha¹⁷

This shows that the DAT-argument is structurally higher than the NOM-argument because by the acceptability of (7a), the NOM-marked anaphor-containing NP must be able to reconstruct below the DAT-marked argument, and we expect the same reconstruction to be available in principle for the DAT-marked anaphor-containing NP in (7b), but since (7b) is unacceptable, its inability to reconstruct must be due to there not being a landing site for the DAT-marked NP beneath the NOM-marked NP. If this binding relation between a wh-word *kim* ‘who’ and the variable anaphor *beje*-POSS requires c-command, (7a) would be evidence for structural DAT on the higher internal argument of *naada* ‘need’ because if it were lexically DAT-marked, we would expect there to be a PP that blocks c-command by the lexically DAT-marked NP.

The dyadic unaccusative *naada* ‘need’ contrasts with an AGENTIVE NOM-DAT predicate like *kömölöh* ‘help’, where we see the opposite pattern: the NOM-argument can bind into the DAT-argument, (8a), but not vice versa, (8b). By the same argumentation as above, this indicates that the NOM-marked argument is structurally higher than the DAT-marked argument. For this verb, the DAT-marked argument must be lexical/inherent case.

- (8) a. [*Beje -tin* *oqo -tu*]-gar *kim kölömös -tö?*
 self -POSS.3SG_i.GEN child -POSS.3SG -DAT who_i help -PST.3SG
 ‘Who_i helped their_i own child?’ Sakha¹⁸
- b. *[*Beje -tin* *ije -te*] *kim -iexe kölömös -tö?*
 self -POSS.3SG_i.GEN mother -POSS.3SG.NOM who_i -DAT help -PST.3SG
 ‘Who_i did their_i own mother help?’ in the sense of ‘Who_i was helped by their_i own

¹⁶ Baker & Vinokurova 2010: p. 606, ex. (21b)

¹⁷ Baker & Vinokurova 2010: p. 606, ex. (21a)

¹⁸ Baker & Vinokurova 2010: p. 606, ex. (22a)

mother?’

Sakha¹⁹

Dyadic unaccusative predicates might thus have a structure like Fig. 5.3, inspired by Pylkkänen’s (2008) low applicatives (cf. §5.3.2). Other structures where the higher DAT-argument c-commands the lower NOM-argument inside the verbal CAD are also possible. Note that the possessor-like relationship between the applied argument and the THEME described in §5.3.2 matches the semantics of the predicates discussed here, potentially indicating that an applicative structure is on the right track. As discussed in §5.3.2, in an applicative structure, we cannot tell whether the DAT originates

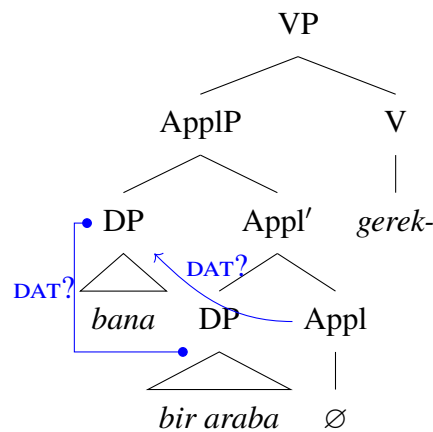


Figure 5.3: A dyadic unaccusative verb with an applicative structure

as inherent case from the Appl head, or as dependent-up case in case competition with the obligatory THEME.

5.3.4 Ditransitives

5.3.4.1 Introduction

In this subsection, I discuss the possibility that DAT case on RECIPIENTS resp. on high GOALS of ditransitives is structural case, assigned because the RECIPIENT c-commands the THEME inside the verbal CAD. This subsection is split into two parts. First (§5.3.4.2) I discuss the evidence for saying that RECIPIENTS are higher than THEMES. Then (§5.3.4.3) I discuss passives of ditransitives, which

¹⁹ Baker & Vinokurova 2010: p. 606, ex. (22b)

pattern differently from English passives of ditransitives in that, while in English it is the RECIPIENT that promotes to subject (being the higher of the two structurally case-marked arguments), in Turkic, the RECIPIENT keeps its DAT-marking and the THEME changes its case from ACC to NOM. We will see that under a dependent-up case analysis of this DAT, the Turkic facts are exactly what is expected — however, under an inherent case analysis of DAT, they are, too. §5.3.4.4 summarizes the discussion of ditransitives. We will see many of the arguments from this section resurface in §5.4 on causatives.

5.3.4.2 Low versus high GOALS

Öztürk (2005: §3.2.3, 2007) argues that double-object constructions resp. ditransitives (i.e., verbs that take one ACC-marked and one DAT-marked argument) fall into two categories: those where the DAT-marked argument is a “high GOAL” (base-generated above the THEME), standing in a “possession relation” to the THEME; and those where the DAT is a “low GOAL” with a base-ordering THEME > GOAL and a “locative interpretation” for the GOAL. (9) illustrates examples of high GOALS and (10) of low GOALS. Öztürk (2005, 2007) presents evidence from binding in combination with reconstruction for the two different base orders, (9a) and (10a), based on Kural’s (1992) observations about the interaction of reconstruction, scrambling, and contrastive focus in Turkish (not repeated here).

- (9) a. *Her_i adam -a resm -in_i -i ver -di -m.*
 every man -DAT picture -POSS.3SG -ACC give -PST -1SG
 ‘I gave every_i man his_i picture.’ *Turkish*²⁰
- b. *Ayşe Ali -ye bu kitab -ı yolla -dı.*
 Ayşe.NOM Ali -DAT this book -ACC send -PST
 ‘Ayşe sent this book to Ali.’ *Turkish*²¹
- c. *Masha aqa -ty -gar surug -u yyt -ta.*
 Masha.NOM father -POSS.3SG -DAT letter -ACC send -PST.3SG
 ‘Masha sent her father a letter.’ *Sakha*^{22,23}

²⁰ Öztürk 2005: p. 154, ex. (99a), 2007: p. 307, ex. (9a)

²¹ Öztürk 2005: p. 160, ex. (114a), 2007: p. 313, ex. (28a)

²² Baker & Vinokurova 2010: p. 599, ex. (7c)

²³ Baker & Vinokurova (2010) give the translation as ‘a letter’, but given the ACC-marking, I suspect it may more correctly translate as ‘the letter’. Note the word order DAT-ACC, though, which, as I have discussed in §3.3, is described by Baker & Vinokurova (2010) as typically ACC-DAT at least in predicates where the THEME can undergo DOM, which I strongly suspect is the case here.

- (10) a. *Resm_i -i çerçeve -sin_i -e koy -du -m.*
 picture -ACC frame -POSS.3SG -DAT put -PST -1SG
 ‘I put the picture_i in its_i frame.’ *Turkish*²⁴
- b. *Misha kumaaqy -ny xoruopka -qa uk -ta.*
 Misha.NOM paper -ACC case -DAT put -PST.3SG
 ‘Misha put the paper in the case.’ *Sakha*²⁵

Baker & Vinokurova (2009) and Baker & Vinokurova (2010) come to the same conclusion and observe that in Sakha, high and low GOALS pattern differently with respect to relativization possibilities of the DAT-marked argument: low GOALS can be relativized, (11), while high GOALS cannot, (12).

- (11) a. *Suruk ostuol -ga uur -ulun -na.*
 letter.NOM table -DAT put -PASS -PST.3SG
 ‘The letter was put on the table.’ *Sakha*²⁶
- b. [*suruk_i østuol -ga uur -ullu -but*] *ostuol -a_i*
 letter.NOM table -DAT put -PASS -PTCP table -POSS.3SG.NOM
 ‘the table that the letter was put on’ *Sakha*²⁷
- (12) a. *Misha -qa suruk yyt -ylyn -na.*
 Misha -DAT letter.NOM send -PASS -PST.3SG
 ‘The letter was sent to Misha.’ *Sakha*²⁸
- b. ?*[*kihi -exe suruk_i yyt -ylly -byt*] *kihi -te_i*
 person -DAT letter.NOM send -PASS -PTCP person -POSS.3SG.NOM
 ‘the person that the letter was sent to’ *Sakha*^{29,30}

Object-relativizing clauses in Sakha and other Turkic languages involve a possessive agreement suffix on the head noun that tracks the φ -features of the subject inside the relative clause. In the examples above, this agreement relation is indicated by subscripts. Baker & Vinokurova (2010: p. 628) argue that this agreement is the reason for the difference in acceptability: In (12b), the trace of the relativized NP (marked as strike-through) intervenes between the poss-agreement on the

²⁴ Öztürk 2005: p. 154, ex. (100a), 2007: p. 308, ex. (10a)

²⁵ Baker & Vinokurova 2010: p. 627, ex. (61a)

^{26,27} Baker & Vinokurova 2010: p. 628, ex. (63a)

^{28,29} Baker & Vinokurova 2010: p. 628, ex. (63b)

³⁰ I have constructed the deleted element after Landmann (2016: p. 13).

head noun and the subject *suruk* ‘letter’ inside the relative clause because the trace is higher than the subject. Contrary to this, in (11b), the trace is lower than the subject *suruk*, thus not creating such an intervention effect.

It is also possible, though slightly degraded, to have both a high GOAL and a low GOAL in the same clause, (13). Crucially, Tonyalı (2015: p. 31, fn. 10) notes that “39 out of 40 subjects in the survey interpreted [(13)] as ‘my being in Ankara is not required’”, nicely illustrating the non-locative but possessive interpretation of the high GOAL.

- (13) ?*Ali bana kitab -ı Ankara -ya yolla -dı.*
 Ali.NOM me.DAT book -ACC Ankara -DAT send -PST
 ‘Ali sent me the book to Ankara.’ *Turkish*³¹

Given this possessive interpretation, I will follow Key (2022: §5.2) among many others in assuming a low applicative structure for “high GOALS”. This immediately opens up the possibility that the source of the DAT on “high GOALS” (RECIPIENTS) could be inherent case assigned by Appl, as we have seen in the previous subsections.

On the other hand, low GOALS are uncontroversially accepted to be lexically case-marked, i.e., assigned DAT by V resp. by a null P with locative (directional) semantics. Since low GOALS do not c-command the THEME, their DAT-marking could not possibly be dependent-up case.

There is another contrast between DAT-nominals reported by Baker & Vinokurova (2009, 2010): In Sakha, the agentive nominalizer -AAccY can only nominalize some VPs containing DAT-marked nominals. In (14a), nominalizing a ditransitive VP is acceptable, while in (14b), nominalizing a transitive VP with a lexically DAT-marked argument is marginal, and in (14c), nominalizing a VP containing a BENEFACTIVE adjunct is unacceptable.

- (14) a. [*Oqo -lor -go emp bier*]-ecci ol tur -ar.
 child -PL -DAT medicine give -er.NOM there stand -AOR
 ‘A giver of medicine to children is over there.’ *Sakha*³²

³¹ Öztürk 2005: p. 156, ex. (103a), 2007: p. 308, ex. (14); Tonyalı 2015: p. 30, ex. (10a)

³² Baker & Vinokurova 2009: p. 524, ex. (14), 2010: p. 614, ex. (37)

- b. ??[*presidien -ŋe kömölöh*]-ööccü
 president -DAT help -er.NOM
 ‘the helper of the president’ Sakha³³
- c. *[*djoŋ -ŋo son tig*]-ecci *manna olor -or*.
 people -DAT coat sew -er.NOM here live -AOR
 ‘The sewer of coats for people lives over here.’ Sakha³⁴

Baker & Vinokurova (2009) argue at length that the complement of the AGENTIVE nominalization suffix -AAccY is the VP: the internal argument can undergo DOM, which by their and my (§3.3) analysis of DOM necessitates a VP, but higher functional projections (low adverbs, negation, aspect, voice) cannot be included in the complement of -AAccY. Baker & Vinokurova (2009: p. 524, 2010: p. 614 f.) argue that DAT-marked high GOALS are allowed inside AGENTIVE nominalizations, (15a), while lexically DAT-marked arguments are not, (15b), and BENEFACTIVE datives aren’t either, (15c). In Baker & Vinokurova (2010: p. 615) they argue that (14a) is acceptable because the DAT-marked nominal is a high GOAL, which is inside the VP and can thus be inside the complement of -AAccY, while in (14b), the DAT-nominal is marked with lexical DAT, and lexical DAT is assigned by a null P, and PPs can’t modify NPs in Sakha, as in most languages. Finally, in (14c), Baker & Vinokurova (2009) argue that *djoŋ-ŋo* ‘for people’ is a BENEFACTIVE adjunct, and adjuncts are not part of the VP proper.

I don’t agree with this analysis, but I do not have a better one to offer at this point. Accepting their conclusion that the complement of -AAccY is the VP, I do not see how the unacceptability of (14b) follows, because we would expect the PP to be inside the VP, which should not constitute a problem for the rest of Baker & Vinokurova’s (2009) analysis.

I also do not know how to derive the unacceptability of (14c). Given the data from Turkish that I have discussed in §5.3.2, this construction looks more like a low applicative to me because of the possessive interpretation: the ‘people’ will own the ‘coats’. As we have seen, a low applicative is part of the VP proper. It is, however, possible that the ‘*’ judgment of (14c) might be intended as a high applicative reading in the sense of ‘the sewer of coats on behalf of people’ (i.e., a subcontractor

³³ Baker & Vinokurova 2010: p. 614, ex. (38)

³⁴ Baker & Vinokurova 2009: p. 524, ex. (15b)

of a tailor, and not someone who interacts with end-customers themselves), which by Pytkänen (2008) is indeed Merged outside of VP (Fig. 5.1). Note however that the high Appl in Fig. 5.1 can be selected for by Voice, and Baker & Vinokurova (2009) claim that the AGENTIVE nominalizer replaces Voice, so we would still expect high ApplP to be able to be part of an AGENTIVE nominalization. I remain agnostic as to why (14c) is unacceptable.

5.3.4.3 Passives of ditransitives

Famously (Barss & Lasnik 1986; Lacerda 2020), in English double-object constructions, the indirect object can passivize by virtue of being structurally higher than the THEME:

- (15) a. John sent Mary a letter. *English*³⁵
 b. Mary was sent a letter. *English*³⁶
 c. ?*A letter was sent Mary. *English*³⁷

Given the lengthy arguments above that in Turkish and Sakha as well, high GOALS are higher than THEMES, and especially if they are structural datives, we might expect them to be able to passivize as well. However, this is not so. Only the structurally lower ACC-marked THEME can lose its case and appear with NOM.

- (16) a. *Bu kitap Ali -ye ver -il -di.*
 this book.NOM Ali -DAT give -PASS -PST
 ‘This book was given to Ali.’ *Turkish*³⁸
 b. **Ali bu kitap -(ı) ver -il -di.*
 Ali.NOM this book -ACC give -PASS -PST
 ‘Ali was given this book.’ *Turkish*³⁹

³⁵ Larson 1988: p. 335, ex. (1a)

³⁶ Larson 1988: p. 362, ex. (39a)

³⁷ Larson 1988: p. 363, ex. (41)

³⁸ Öztürk 2007: p. 305, ex. (2b)

³⁹ Öztürk 2007: p. 306, ex. (2c)

- (17) a. *Suruk Masha -ga yyt -ylyn -na.*
 letter.NOM Masha -DAT send -PASS -PST.3SG
 ‘The letter was sent to Masha.’

Sakha⁴⁰

- b. **?Masha suruk yyt -ylyn -na.*
 Masha.NOM letter send -PASS -PST.3SG
 intended: ‘Masha was sent a letter.’

Sakha⁴¹

If the RECIPIENT is marked with lexical DAT, the data are just what we would expect. If the RECIPIENT is marked with structural DAT, however, we might think that Turkish and Sakha should pattern with English above. Baker & Vinokurova (2010: p. 610 f.) show that DCT can account for (16) and (17) assuming structural (dependent-up) DAT. The only difference between an active and a passive clause is that in a passive, Voice does not project an AGENT in its Spec; the complement of Voice, i.e., the extended vP, resp. the Spell-out domain of the verbal CAD, remains unchanged. Thus, the fact that the RECIPIENT c-commands the THEME inside the verbal CAD does not change between a passive and an active clause, thus the DAT-marking of the RECIPIENT does not change either. This is illustrated in Fig. 5.4.

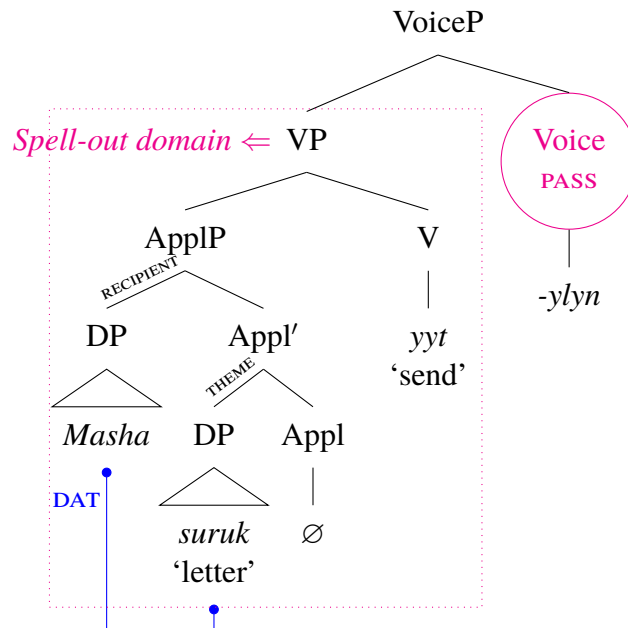


Figure 5.4: Passive of ditransitive (after Baker & Vinokurova 2010: p. 610 f.)

⁴⁰ Baker & Vinokurova 2010: p. 610, ex. (29)

⁴¹ Baker & Vinokurova 2010: p. 610

That these DAT-retaining passives do not cause a locality violation (in that it is not the highest structural NP that gets promoted to subject) is further argued for by Öztürk (2005, 2007): Turkish passives do not involve movement of an argument to Spec,TP. Rather, Öztürk assumes DCT for ACC-assignment, arguing that the case change that we see on the THEME is simply due to the AGENT not being projected, and not because the THEME would move to Spec,TP to receive NOM under agreement with T etc. The THEME remains inside the VP, which makes it scope beneath negation, (18a,b), and makes T not agree with animate subjects, (18a), and we can observe the THEME appearing to the left of RECIPIENTS, (18b).

- (18) a. *Bütün çocuk -lar okul -a yolla -n -ma -dı.*
all child -PL.NOM school -DAT send -PASS -NEG -PST
‘All children were not sent to school.’ *Turkish*⁴²
- (i) *all > not
(ii) ^{ok}not > all
- b. *Ali -ye bütün paket -ler yolla -n -ma -dı.*
Ali -DAT all package -PL.NOM send -PASS -NEG -PST
‘All the packages were not sent to Ali.’ *Turkish*⁴³
- (i) *all > not
(ii) ^{ok}not > all

When the THEME does move out of the VP (for reasons other than passivization), it scopes above negation, (19a,b), T agrees with animate subjects, (19a), and the THEME appears to the right of the RECIPIENT, (19b).

- (19) a. *Bütün çocuk -lar okul -a yolla -n -ma -dı -lar.*
all child -PL.NOM school -DAT send -PASS -NEG -PST -3PL
‘All children were such that they were not sent to school.’ *Turkish*⁴⁴
- (i) ^{ok}all > not
(ii) *not > all

⁴² Öztürk 2005: p. 161, ex. (115a)

⁴³ Öztürk 2005: p. 161, ex. (116a)

⁴⁴ Öztürk 2005: p. 161, ex. (115b)

- b. *Bütün paket -ler Ali -ye yolla -n -ma -di.*
 all package -PL.NOM Ali -DAT send -PASS -NEG -PST
 ‘All the packages were not sent to Ali.’

Turkish⁴⁵(i) ^{ok}all > not

(ii) *not > all

5.3.4.4 Summary

In this subsection I have illustrated how ditransitives of the form RECIPIENT-THEME (but not of the form THEME-GOAL) are compatible with the assumption that the RECIPIENT is marked with structural (dependent-up) DAT case. In Sakha AGENTIVE nominalizations, RECIPIENTS behave differently from GOALS, which may indicate that, while GOALS clearly receive lexical DAT, RECIPIENTS do not — although it is not clear to me how this difference arises. Assuming a low applicative structure for RECIPIENTS (“high GOALS”), the same issue arises as discussed in §5.3.2 above: the DAT-marking on the RECIPIENT *can* be dependent-up case assigned in competition against the obligatory THEME, but it can also be inherent case assigned by Appl to its specifier. I do not know how to tell the difference.

5.3.5 Summary

In this section I have presented and discussed three constructions that I have found discussed in the literature that claim to provide evidence for structural DAT case in Turkic. I have argued for each construction that it does not actually show beyond reasonable doubt that it contains structural DAT. Rather, all DAT-marked nominals discussed in this section appear in configurations where their position is closely related to the functional head Appl that may likely be responsible for assigning this DAT to the nominal in its specifier, making these DAT inherent case.

⁴⁵ Öztürk 2005: p. 162, ex. (116b)

5.4 Not even the dative case in causatives is structural

5.4.1 Introduction

In this section I discuss what Öztürk (2007: p. 309) calls “definitely structural” case, and Baker & Vinokurova (2010: p. 608) have called “perhaps the single strongest reason for saying that dative case can be a dependent case in Sakha”: that the case on the causee (undergoer of causativization) varies between ACC and DAT depending on the presence of other NPs inside the verbal CAD. Baker (2015b: p. 132) writes:

The dative in [(23b)] seems like a structural case, in that it does not have a set theta role — not one typically associated with dative case, anyway. It is an agent, not a goal, and there is little reason to think that the causee’s thematic role is different in [(22b)] and [(23b)], such that difference in case can be explained thematically. (Baker 2015b: p. 132)

In §5.4.2 I discuss the case patterns that serve as the foundation of the claim that DAT case on causees is structural case. In §5.4.3 I illustrate Öztürk’s (2007) and Baker’s (2015b) argument that passives of causatives support the theory of DAT on causees as structural case. Section 5.4.4 introduces data from Akkuş (2021) showing that DAT- and ACC-marked causees differ in more than just their case marking, which draws doubt to the analysis proposed by Öztürk (2007), Baker & Vinokurova (2010), and Baker (2015b) and makes an analysis of DAT on causees being inherent case assigned by certain Voice more likely. §5.4.5 summarizes the argument for causatives.

5.4.2 Variations on the case of the causee

As has been observed for Japanese in e.g. Miyagawa (2001) and Italian in e.g. Folli & Harley (2007), and as I have already begun to illustrate in §3.2.5 above, in Turkic languages as well, the case on the causee depends on the argument structure of the embedded event: in causatives of intransitives (unaccusatives, (20), unergatives, (21), and intransitive EXPERIENCER predicates, (22)), the causee

is marked with ACC, while in causatives of transitives, (23) and ditransitives, (24), the causee is marked with DAT. Verbs that only take lexically-case-marked objects pattern with intransitives, (25). In all examples, replacing ACC with DAT and vice versa on the causee is ungrammatical.

(20) Unaccusatives (causee is a THEME)

- a. *Çocuğ -u uyu -t -tu -m.*
child -ACC sleep -CAUS -PST -1SG
'I made the child sleep.' *Turkish*⁴⁶
- b. *Murat stul -dí sín -dír -dí.*
Murat.NOM chair -ACC break -CAUS -PST
'Murat broke the chair.' *Karakalpak*⁴⁷
- c. *Apa -m sorpa qayna -t -íp atír.*
mother -POSS.1SG.NOM soup boil -CAUS -CVB PROG
'My mother is cooking soup.' *Karakalpak*⁴⁸

(21) Unergatives (causee is an AGENT)

- a. *Aygúl Murat -tí jur -t -íp atír.*
Aygül.NOM Murat -ACC run -CAUS -CVB PROG
'Aygül is making Murat run.' *Karakalpak*⁴⁹

(22) Intransitive EXPERIENCER predicates (causee is an EXPERIENCER)

- a. *Adam çocuğ -u gül -dür -dü.*
man.NOM child -ACC laugh -CAUS -PST
'The man made the child laugh.' *Turkish*⁵⁰
- b. *Sardaana Aisen -y yta(a) -t -ta.*
Sardaana.NOM Aisen -ACC cry -CAUS -PST.3SG
'Sardaana made Aisen cry.' *Sakha*⁵¹

(23) Transitives (causee is an AGENT)

- a. *Adam çocuğ -a süit (-ü) iç -ir -di.*
man.NOM child -DAT milk -ACC drink -CAUS -PST
'The man made the child drink (the) milk.' *Turkish*⁵²

⁴⁶ Erguvanlı 1979: p. 92, ex. (2)

^{47,48,49} Data from fieldwork in Spring 2019 in Asia Pietraszko's field methods class.

⁵⁰ Dede 1981: p. 41, ex. (4b)

⁵¹ Baker & Vinokurova 2010: p. 607, ex. (23a)

⁵² Dede 1981: p. 41 f., exs. (5b) and (5d)

- b. *Misha Masha -qa miin (-i) sie -t -te.*
Misha.NOM Masha -DAT soup -ACC eat -CAUS -PST.3SG
‘Misha made Masha eat (the) soup.’ *Sakha*⁵³
- c. *Aygúl Gúlnara -ğa sorpa ísh -tír -dí.*
Aygül.NOM Gulnara -DAT soup drink -CAUS -PST
‘Aygül made Gulnara eat soup.’ *Karakalpak*⁵⁴
- (24) Ditransitives (causee is an AGENT)
- a. *Baba -m -a çocuk -lar -a masal anlat -tır -dı -m.*
father -POSS.1SG -DAT child -PL -DAT story tell -CAUS -PST -1SG
‘I had my father tell stories to the children.’ *Turkish*⁵⁵
- b. *Mama -m mağan kítap -tı Murat -qa ber -dír -dí.*
mom -POSS.1SG.NOM me.DAT book -ACC Murat -DAT give -CAUS -PST
‘My mom made me give the book to Murat.’ *Karakalpak*⁵⁶
- (25) Intransitives with a lexically case-marked argument (causee is an AGENT)
- a. *Ali -yi at -a bin -dir -di -m.*
Ali -ACC horse -DAT mount -CAUS -PST -1SG
‘I made Ali get on the horse.’ *Turkish*⁵⁷
- b. *Aygúl Murat -tı qala -ğa bara -dír -dí.*
Aygül.NOM Murat -ACC market -DAT go -CAUS -PST
‘Aygül made Murat go to the market.’ *Karakalpak*⁵⁸

This is exactly what we expect to find under the theory that DAT on causees is structural case: DAT is assigned inside the verbal CAD iff there is another lower structurally case-marked argumental NP inside the verbal domain. This holds for internal arguments such as direct objects, which receive structural case, but does not hold for lexically case-marked arguments such as in (25). Under this theory of DAT-marked causees, we can thus use the case on the causee to test whether another NP has lexical or structural case if that NP is the sole argument of the embedded verb. I have made this argument in §3.2.5 to argue that ACC is structural.

⁵³ Baker & Vinokurova 2010: p. 607, ex. (23b)

^{54,56,58} Data from fieldwork in Spring 2019 in Asia Pietraszko’s field methods class.

⁵⁵ Göksel 1993: p. 216, ex. (40)

⁵⁷ Erguvanlı 1979: p. 93, ex. (12)

5.4.3 Passives of causatives

As I have discussed in §5.3.4.3 above for passives of ditransitives, under a standard Indo-European-centric subject-demoting object-promoting theory of passives, if the DAT on the causee is structural case, one might expect that it is the causee that promotes to subject, given that it is structurally higher than the internal argument, as the RECIPIENT does in English passives of double-object constructions. However, in Turkish and Sakha, we find that the exact opposite is true: like in ditransitives, the ACC-marked THEME can promote, but the allegedly structurally DAT-marked causee, like the RECIPIENT, cannot, (26).

- (26) a. *Ali Ayşe -ye bu kitab -ı oku -t -tu.*
 Ali.NOM Ayşe -DAT this book -ACC read -CAUS -PST
 ‘Ali made Ayşe read this book.’ *Turkish*⁵⁹
- b. *Bu kitap Ayşe -ye oku -t -ul -du.*
 this book.NOM Ayşe -DAT read -CAUS -PASS -PST
 ‘This book was caused to be read by Ayşe.’ *Turkish*⁶⁰
- c. **Ayşe bu kitab -ı oku -t -ul -du.*
 Ayşe.NOM this book -ACC read -CAUS -PASS -PST
 intended: ‘Ayşe was made to read this book.’ *Turkish*⁶¹

In contrast to this, an ACC-marked causee can passivize just fine. In (27), *bak-* ‘to look at’ selects for a lexical-DAT argument, which, unsurprisingly, cannot lose its lexical DAT, (27c).

- (27) a. *Ayşe Ali -yi resm -e bak -tır -dı.*
 Ayşe.NOM Ali -ACC picture -DAT look -CAUS -PST
 ‘Ayşe made Ali look at the picture.’ *Turkish*⁶²
- b. *Ali resm -e bak -tır -ıl -dı.*
 Ali.NOM picture -DAT look -CAUS -PASS -PST
 ‘Ali was made to look at the picture.’ *Turkish*⁶³

⁵⁹ Öztürk 2007: p. 309, ex. (17b)

⁶⁰ Öztürk 2007: p. 309, ex. (18b)

⁶¹ Öztürk 2007: p. 309, ex. (18a)

⁶² Öztürk 2007: p. 310, ex. (19b)

⁶³ Öztürk 2007: p. 310, ex. (19c)

- c. **Resim Ali -yi bak -tır -ıl -dı.*
 picture.NOM Ali -ACC look -CAUS -PASS -PST

intended: ‘The picture was made to be looked at by Ali.’

*Turkish*⁶⁴

Does this mean, then, that either DAT on the causee is lexical case after all, like the DAT in (27), or that we need to rethink locality, because it is not the highest structurally case-marked argument that promotes to subject? No, as Öztürk (2005, 2007) argues, because just like with passives of ditransitives above (§5.3.4.3), under the theory of DAT as dependent-up case in the verbal domain, the pattern in (26) is precisely what is expected. Following Akkuş (2021) and Nie (2020, 2022) on the structure of Turkish causatives and passives, (26) is represented by the structure in Fig. 5.5.

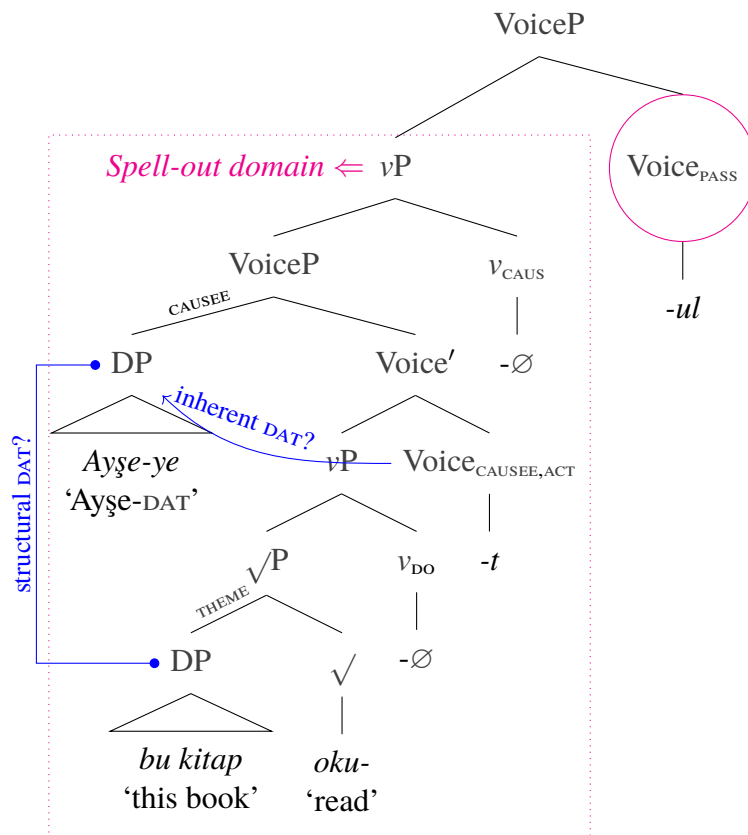


Figure 5.5: The passive of the causative of the transitive in (26)

⁶⁴ Öztürk 2007: p. 310, ex. (19d)

5.4.4 Differences between DAT- and ACC-marked causees

The discussion of causatives so far has made it seem as if the only difference between DAT- and ACC-marked causees was the argument structure of the embedded predicate, which would explain the difference in case-marking if the DAT is a structural dependent-up case. There are, however, two other important differences between the two kinds of causees:

1. DAT-marked causees must be animate, (28a–b,e), while ACC-marked causees can be animate or inanimate, (28c–d), and
2. ACC-marked causees, when omitted, must be interpreted as pro-dropped, i.e., discourse-given, (29a–b), while null DAT-marked causees can receive an interpretation as pro or as indefinite ‘someone’ via existential closure, (29c–d).

(28) DAT-marked causees must be animate. ACC-marked causees can be animate or inanimate.

- a. *Anahtar kapı -yı aç -tı.*
key.NOM door -ACC open -PST
‘The key opened the door.’ Turkish⁶⁵
- b. **Müdür anahtar -a kapı -yı aç -tır -dı.*
director.NOM key -DAT door -ACC open -CAUS -PST
intended: ‘The director made the key open the door.’ Turkish⁶⁶
- c. *Bekçi kapı -yı aç -tı.*
watchman.NOM door -ACC open -PST
‘The watchman opened the door.’ Turkish⁶⁷
- d. *Müdür bekçi -ye kapı -yı aç -tır -dı.*
director.NOM watchman -DAT door -ACC open -CAUS -PST
‘The director made the watchman open the door.’ Turkish⁶⁸
- e. *Müdür kapı -yı aç -tır -dı.*
director.NOM door -ACC open -CAUS -PST
(i) ^{ok} ‘The director made someone open the door.’
(ii) # ‘The director made something open the door.’ Turkish

⁶⁵ Key 2013: p. 187, ex. (32b)

⁶⁶ Modified after Key 2013: p. 187, ex. (31b)

⁶⁷ Key 2013: p. 187, ex. (32a)

⁶⁸ Key 2013: p. 187, ex. (31a)

(29) Absent ACC-marked causees must be interpreted as pro-drop.

Absent DAT-marked causees can be interpreted as as pro-drop or as indefinites.

- a. *Şoför yolcu -lar -ı otobüs -e bin -dir -di.*
 driver.NOM passenger -PL -ACC bus -DAT board -CAUS -PST
 ‘The driver made the passengers board the bus.’ *Turkish*⁶⁹
- b. *Şoför otobüs -e bin -dir -di.*
 driver.NOM bus -DAT board -CAUS -PST
 (i) ‘The driver made him/her/them/... board the bus.’
 (ii) *‘The driver had the bus boarded.’ *Turkish*^{70,71}
- c. *Bütün misafir -ler -e araba -yı temizle -t -ti.*
 all guest -PL -DAT car -ACC clean -CAUS -PST
 ‘S/he made all the guests clean the car.’ *Turkish*⁷²
- d. *Araba -yı temizle -t -ti.*
 car -ACC clean -CAUS -PST
 (i) ‘S/he made him/her/them/... clean the car.’
 (ii) ‘S/he had the car cleaned.’ or ‘S/he had someone clean the car.’ *Turkish*⁷³

This immediately raises the question whether the connection between the case-marking on the causee is accidentally or directly related to its interpretive properties. Akkuş (2021) provides the following overview (Table 5.1) that clearly shows the correlation between case-marking and interpretation of causees.

I will not discuss the animacy requirement on causees of transitive, but not on causees of intransitive predicates, as this might be a more general cross-linguistic phenomenon, also described

⁶⁹ Akkuş 2021: p. 278, ex. (592a)

⁷⁰ Akkuş 2021: p. 278, ex. (592b)

⁷¹ Özkarağöz 1986: p. 120, ex. (12b)

⁷² Akkuş 2021: p. 277, ex. (588a)

⁷³ Akkuş 2021: p. 277, ex. (588b)

⁷⁴ Abbreviations used in this table:

“Impersonal” “Can this predicate appear in an impersonal construction (that looks a lot like passive, but isn’t, Akkuş 2021; Legate et al. 2020)?”

“Passive” “Is this predicate passivizable?”

“Causee ∃” “Does a null causee have an existential interpretation?”

“Trans ACC” transitive predicate with ACC-marked internal argument

“Trans P-INC” transitive predicate with pseudo-incorporated internal argument

“Trans FIN CLAUSE” transitive predicate with a finite clause as internal argument

“Trans OBL” ‘transitive’ predicate with an oblique (lexically case-marked) argument

	Impersonal	Passive	Causee $\exists$	Case of Causee
Trans ACC	*	✓	✓	DAT
Trans P-INC	✓	%	✓	DAT
Trans FIN CLAUSE	✓	*	✓	DAT
Trans OBL	✓	%	*	ACC
Unerg	✓	*	*	ACC
Refl	✓	*	*	ACC
Recip	✓	*	*	ACC
Unacc	✓	*	N/A	N/A

Table 5.1: The correlation between predicate types, interpretation of causees, and case marking of causees (from Akkuş 2021: p. 280, Table 4.5)⁷⁴

for Japanese by Miyagawa (2001), for Italian by Folli & Harley (2007), and for Hungarian by Horvath & Sioni (2011) and Key (2013), among others. In the following I will discuss the proposals by Akkuş (2021) and Key (2013) about causatives in Turkish, who both conclude that DAT- and ACC-marked causees are projected in different syntactic positions, i.e., differ in more than just their case-marking. In particular, Akkuş (2021) argues that the DAT-marked causee is projected in the specifier of a special VoiceP, which he calls Voice_{CAUSEE}P, and from this proposal it follows that the causee's DAT-marking may be inherent case assigned by this Voice_{CAUSEE}.

After arguing in detail that Turkish causatives (except causatives of unaccusatives) contain a second causing VoiceP (Voice_{CAUSEE}P), Akkuş (2021) argues that this Voice_{CAUSEE}P can undergo an active/passive alternation that does not show up morphologically — i.e. both Voice_{CAUSEE,ACT} and Voice_{CAUSEE,PASS} spell out as $\emptyset$, identically to Voice_{ACT}, but contrasting with Voice_{PASS}, which has an overt exponent. The active/passive alternation corresponds only to whether an animate DAT-causee is merged in Spec,VoiceP (active) or not, in which case the causee is interpreted via existential closure (passive). Akkuş (2021) carefully runs tests showing that the null DAT-causee is not syntactically projected, like the implicit argument of a regular passive. He shows that the null causee cannot be picked up by sluicing⁷⁵, cannot bind anaphors (reflexives⁷⁶, reciprocals⁷⁷), cannot bind

⁷⁵ Akkuş 2021: §4.4.2, ex. (540)

⁷⁶ Akkuş 2021: p. 261, ex. (555)

⁷⁷ Akkuş 2021: p. 259, ex. (550)

pronouns⁷⁸, cannot be picked up by *pro*⁷⁹, and cannot license depictives⁸⁰. That passive is the correct feature to capture the causatives with a missing DAT-causee is evidenced by the fact that some idioms which resist passivization in general, (30a–b), cannot appear in causatives when the causee is null, (30d), but *can* appear in causatives with an overt DAT-causee, (30c). This contrasts with passivizable idioms, (31a–b), which can be causativized with overt, (31c), or covert, (31d), DAT-causee.

(30) Non-passivizable idioms can be causativized, but a DAT-marked causee cannot be silent.

- a. *Kemal ayva -yı ye -di.*
Kemal.NOM quince -ACC eat -PST
(i) ^{ok} ‘Kemal ate the quince.’ (literal)
(ii) ^{ok} ‘Kemal got into trouble.’ (idiomatic) *Turkish*⁸¹
- b. *Ayva (Kemal tarafından) ye -n -di.*
quince.NOM Kemal by eat -PASS -PST
(i) ^{ok} ‘The quince was eaten (by Kemal).’ (literal)
(ii) *‘Kemal got into trouble.’ (idiomatic) *Turkish*⁸²
- c. *Valla bu oğlan bana şimdiden ayva -yı ye -dir -di.*
honestly this boy.NOM me.DAT already quince -ACC eat -CAUS -PST
(i) ^{ok} ‘Honestly this boy already made me eat the quince.’ (literal)
(ii) ^{ok} ‘Honestly this boy already got me into trouble.’ (idiomatic) *Turkish*⁸³
- d. *Valla bu oğlan şimdiden ayva -yı ye -dir -di.*
honestly this boy.NOM already quince -ACC eat -CAUS -PST
(i) ^{ok} ‘Honestly this boy already had someone eat the quince.’ (literal)
(ii) *‘Honestly this boy already had someone get into trouble.’ (idiomatic) *Turkish*⁸⁴

(31) Passivizable idioms can be causativized with overt or silent DAT-marked causee.

- a. *Sonunda bakla -yı ağız -dan çıkar -dı.*
finally bean -ACC mouth -ABL take.out -PST

⁷⁸ Akkuş 2021: p. 260, ex. (553b)

⁷⁹ Akkuş 2021: p. 261, ex. (556b)

⁸⁰ Akkuş 2021: §4.4.4, ex. (563a, b)

⁸¹ Akkuş 2021: p. 252, ex. (533a)

⁸² Akkuş 2021: p. 253, ex. (533b)

⁸³ Akkuş 2021: p. 253, ex. (534a)

⁸⁴ Akkuş 2021: p. 253, ex. (534b)

- (i) ^{ok} ‘S/he finally took the bean out of his/her mouth.’ (literal)
- (ii) ^{ok} ‘S/he finally disclosed the secret.’ (idiomatic) *Turkish*⁸⁵
- b. *B20 toplantı -sın -da bakla ağız -dan çıkar -ıl -dı.*
 B20 meeting -CMPD -LOC bean.NOM mouth -ABL take.out -PASS -PST
 (i) ^{ok} ‘At the B20 meeting the bean was taken out of the mouth.’ (literal)
- (ii) ^{ok} ‘The secret was disclosed at the B20 meeting.’ (idiomatic) *Turkish*⁸⁶
- c. *En sonunda bakla -yı çıkar -t-tır -dı bana.*
 very finally bean -ACC take.out -CAUS-CAUS -PST me.DAT
 (i) ^{ok} ‘Eventually s/he made me take the bean out of my mouth.’ (literal)
- (ii) ^{ok} ‘Eventually s/he made me disclose the secret.’ (idiomatic) *Turkish*^{87,88}
- d. *Anne -ler eninde sonunda bakla -yı ağız -dan çıkar -t-(tır)*
 mother -PL.NOM eventually bean -ACC mouth -ABL take.out -CAUS-(CAUS)
-ir -lar.
 -AOR -3PL
 (i) ^{ok} ‘Mothers always make someone take the bean out of their mouth one way or another.’ (literal)
- (ii) ^{ok} ‘Mothers always make someone disclose the secret one way or another.’ (idiomatic) *Turkish*⁸⁹

Akkuş (2021) discusses DAT-marked causees in great detail, but does not say much about ACC-marked causees. Key (2013: p. 190 f.) suggests that ACC-marked causees might be arguments projected in the specifier of v_{DO} (i.e., the verbalizing v with the ‘flavor’ of ‘to do’, as opposed to the flavors ‘to become’ and ‘to cause’; Folli & Harley 2005, 2007; Hale & Keyser 1993a). Causatives of unaccusatives, (32), are mono-eventive, i.e., have a single vP -layer. Key (2013) proposes the structure in Fig. 5.6.

- (32) a. *Gemi bat -tı.*
 boat.NOM sink -PST
 ‘The boat sank.’ *Turkish*

⁸⁵ Akkuş 2021: p. 253, ex. (535a)

⁸⁶ Akkuş 2021: p. 253, ex. (535b)

⁸⁷ Akkuş 2021: p. 253, ex. (536a)

⁸⁸ Turkish allows doubling of the causative suffix for emphatic purposes. Akkuş (2021: p. 253, fn. 41) writes that this is the case for this example and the next, i.e., that there are not two causing events understood here.

⁸⁹ Akkuş 2021: p. 254, ex. (536b)

- b. *Çocuk gemi -yi bat -ır -dı.*
 child.NOM boat -ACC sink -CAUS -PST
 ‘The child sank the boat.’

*Turkish*⁹⁰

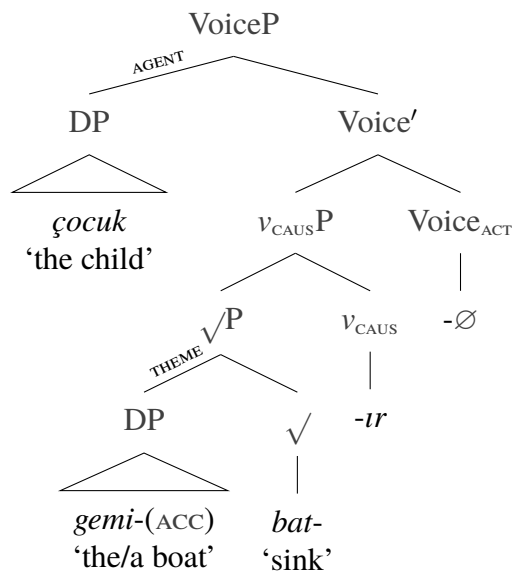


Figure 5.6: The structure of the causative of an unaccusative proposed by Key (2013: p. 88, ex. (1b), p. 192, ex. (40a))

Causatives of unergatives, (33), might have a structure as in Fig. 5.7, Key (2013: p. 190 f.) speculates.

- (33) a. *Araba çalış -tı.*
 car.NOM work -PST
 ‘The car started.’

*Turkish*⁹¹

- b. *Mehmet araba -yı çalış -tır -dı.*
 Mehmet.NOM car -ACC work -CAUS -PST
 ‘Mehmet started the car.’

*Turkish*⁹²

Accepting Akkuş’s (2021) arguments that causees that receive an indefinite interpretation when omitted receive this interpretation through existential closure like implicit arguments of passives, it must be that these causees, when present, are projected in the specifier position of this VoiceP, and that causees that cannot be interpreted as indefinites when absent, are not projected as specifiers of

⁹⁰ Key 2013: p. 88, ex. (1a), p. 192, ex. (40b)

⁹¹ Key 2013: p. 190, ex. (37a)

⁹² Key 2013: p. 190, ex. (37b)

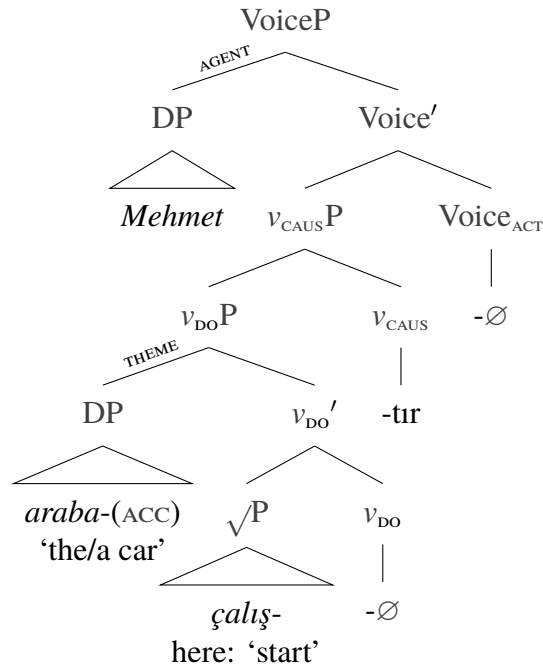


Figure 5.7: The structure of the causative of an unergative proposed by Key (2013: p. 191, ex. (39), fn. 50)

this VoiceP. No matter the exact positions of ACC-marked causees (resp. pro, when absent), it must be a different position than that of DAT-marked causees. It follows that the DAT-marking is associated with a particular structural position, and the presence of the causativizing Voice head, and thus it becomes possible that this DAT-marking is not structural dependent-up case, assigned through case-competition against the structurally ACC-marked theme inside the causativized predicate, but that it is inherent case, assigned by Voice to its specifier.

5.4.5 Summary

In this section I have presented the case patterns on the causees of Turkic causatives and shown that upon close examination, they are compatible with both a theory of DAT-marking on causees as structural case marking, and a theory of DAT-marking on causees as inherent case marking. Baker (2015b) and Baker & Vinokurova (2010) have argued for the first option, namely that this DAT is the dependent-up case in the verbal domain assigned in competition against the ACC-marked

THEME, which accounts for the fact that the DAT-marking on causees correlates with the presence of a structurally case-marked THEME inside the embedded predicate. The second option receives support from Akkuş's (2021) investigation into the voice domain, where he argues that causatives of transitives contain a projection that he labels $\text{Voice}_{\text{CAUSEE}}$, which, when active, projects the causee in its specifier, and possibly assigns it inherent DAT case, and when passive, accounts for the indefinite interpretation of implicit DAT-marked causees.

The correlation between case-marking, interpretation, and valency of the embedded predicate (Table 5.1) remains a puzzle. What is important to the present discussion is that given this correlation, an inherent analysis of the DAT-marking on some causees is possible, and the case-alternation on causees depending on the embedded predicates argument structure is not an argument that the DAT-marking on causees must be structural case.

5.5 Inconclusive constructions with “structural dative” in Sakha

5.5.1 Introduction

For completeness, I discuss two more constructions that have been proposed by Baker & Vinokurova (2010) as evidence for structural dative in Sakha: In §5.5.2 I discuss possessor-raising constructions, where a GEN-marked possessor that moves into the verbal CAD appears with DAT-marking, but when it moves further up into the clausal CAD, it appears with no marking (which corresponds to NOM or GEN; we don't know which case it is). I will argue that the data as presented by Baker & Vinokurova (2010) are not clear enough to warrant their conclusion that this DAT is dependent-up case in the verbal domain. In §5.5.3 I discuss evidence of a CC21 dependency (the subject of an embedded clause topicalizes to its clause's edge, becomes part of the matrix CAD, and thereby may receive dependent-down ACC; cf. §3.4) feeding the assignment of dependent-up dative to the RECIPIENT of the verb *erenner*- ‘promise’: when the subject of the THEME clause remains low, the

Baker & Vinokurova (2010: §3.6) present evidence from Sakha for what they argue is a possessor-raising construction. “[...] under certain conditions, [...] in particular [...] when the possessed NP is the sole argument of an unaccusative verb” (p. 621), the (usually GEN-marked) possessor, (34a), can optionally leave its DP and raise to the left where it can appear with either no case marking (whether this is the exponent of NOM or GEN is an empirical question, but the data are not known to me at this time), (34b), or DAT, (34c):

- Baker & Vinokurova (2010) assume that the constructions in (34) are related by movement of the possessor to the left because this separation of possessor and possessum is not possible if the φ -features of the separated possessor and the possessive suffix don't match, (35a), nor if the pos-

^{94, 95} Baker & Vinokurova 2010: p. 621, ex. (51b)

sessum does not carry any possessive φ -features, (35b). This is not conclusive evidence that this construction involves movement, as Baker & Vinokurova (2010) themselves note; however, if it were something like an ethical dative, the restrictions in (35) would be unexpected.

- (35) a. **Masha* -(qa) *beqehee* *massyyna* -m *aldjan* -na.
Masha -(DAT) yesterday car -POSS.1SG.NOM break -PST.3SG
intended: ‘My car broke down on Masha yesterday.’ *Sakha*⁹⁶
- b. **Misha* -(qa) *beqehee* *at* *öl* -lö.
Misha -(DAT) yesterday horse.NOM die -PST.3SG
intended: ‘The/A horse died on Misha yesterday.’ *Sakha*⁹⁷

Baker & Vinokurova (2010) propose an analysis where this alternation between DAT and NOM resp. GEN is due to the possessor’s landing site being at different structural positions: DAT is assigned as dependent-up case to the higher of two nominals when the possessor lands within the verbal CAD where it locally c-commands the THEME, and where it must appear to the right of TP-level adjuncts such as ‘yesterday’, (36)–(37), and ‘probably’, (38), which we can assume don’t scramble. On the other hand, NOM/GEN appears when the possessor raises into the clausal CAD, (37a), (38a). Thus, *Masha* and *Masha-qa* in (36) appear in the same linear position, but are actually in two different landing sites. This can be seen more clearly when the possessor raises to the left of TP-level adjuncts, which results in DAT-marking being judged as ‘??’, (37b) and (38b).

- (36) a. *Beqehee* *Masha* *emiske* [~~*Masha*~~ *massyyna*-ta] *aldjan* -na.
yesterday Masha.NOM suddenly car-POSS.3SG.NOM break -PST.3SG
‘Yesterday Masha’s car suddenly broke.’ *Sakha*⁹⁸
- b. *Beqehee* *Masha* -qa *emiske* [~~*Masha*~~ *massyyna*-ta] *aldjan* -na.
yesterday Masha -DAT suddenly car-POSS.3SG.NOM break -PST.3SG
‘Yesterday Masha’s car suddenly broke.’ *Sakha*⁹⁹
- (37) a. *Masha* *beqehee* [~~*Masha*~~ *massyyna*-ta] *aldjan* -na.
Masha.NOM yesterday car -POSS.3SG.NOM break -PST.3SG
‘Yesterday Masha’s car broke.’ *Sakha*¹⁰⁰

⁹⁶ Baker & Vinokurova 2010: p. 621, ex. (52a)

⁹⁷ Baker & Vinokurova 2010: p. 621, ex. (52b)

^{98,99} Baker & Vinokurova 2010: p. 622, ex. (54a)

¹⁰⁰ Baker & Vinokurova 2010: p. 622, ex. (54b)

- b. ??*Masha -qa beqehee* [*Masha massyyna -ta*] *aldjan -na*.
 Masha -DAT yesterday car -POSS.3SG.NOM break -PST.3SG
 ‘Yesterday Masha’s car broke.’ *Sakha*¹⁰¹
- (38) a. *Masha baqar* [*Masha massyyna -ta*] *aldjan -na*.
 Masha.NOM probably car -POSS.3SG.NOM break -PST.3SG
 ‘Probably Masha’s car broke.’ *Sakha*¹⁰²
- b. ??*Masha -qa baqar* [*Masha massyyna -ta*] *aldjan -na*.
 Masha -DAT probably car -POSS.3SG.NOM break -PST.3SG
 ‘Probably Masha’s car broke.’ *Sakha*¹⁰³

I propose here that this construction does not present evidence for DAT in Sakha being a dependent/up case. My first two arguments stem from irregularities in the data that make this paradigm much less clear-cut than Baker & Vinokurova (2010) present it to be. My third argument comes from the derivation of the alleged case alteration by possessor-raising: Given present assumptions, and especially the assumptions that Baker & Vinokurova (2010) would presumably make, it is very unclear to me how to derive the NOM or GEN marking on possessors that raise all the way up into the clausal domain.

First, there seems to be an inconsistency between (37b) above and (39), which is also from Baker & Vinokurova (2010), regarding whether a DAT-marked raised possessor can or cannot appear to the left of the adverb *beqehee* ‘yesterday’. Assuming that this adverb always adjoins at the same position (such as to TP), this is a puzzling contrast.

- (39) *Misha -qa beqehee* [*Misha at -a*] *öl -lö*.
 Misha -DAT yesterday horse -POSS.3SG.NOM die -PST.3SG
 ‘Misha’s horse died (on him) yesterday.’ *Sakha*¹⁰⁴

Second, Vinokurova (2005), while discussing GEN- and DAT-marking alterations on possessors in clausal possessives (the Sakha equivalent of English ‘have’ and ‘there is’ sentences), notes the following data point, where unexpectedly, moving a bare (NOM/GEN-marked) possessor into the clausal domain is not possible despite the matrix predicate being unaccusative, as Baker & Vino-

^{101, 102, 103} Baker & Vinokurova 2010: p. 622, ex. (54b)

¹⁰⁴ Baker & Vinokurova 2010: p. 621, ex. (51a)

kurova (2010: p. 621) note is relevant.

- (40) $\langle *en \rangle$ *toqo bügün uruok -ka* [$\langle^{ok}en \rangle$ *kinige -ŋ*] *kirtij*
 $\langle *you.NOM/GEN \rangle$ why today lesson -DAT $\langle^{ok}you.GEN \rangle$ book -POSS.2SG become.dirty
-de -j?
 -PST.3SG -Q
 ‘Why did your book become dirty today at the lesson?’ Sakha¹⁰⁵

I argue that we cannot count this construction as evidence for DAT as dependent-up case in Sakha until a more thorough paradigm of this construction becomes available.

Third, even if the paradigm as presented by Baker & Vinokurova (2010) were stable and accurate, it still raises questions as to the exact steps in the derivation of separated possessors with NOM/GEN-marking. I discuss two possibilities in turn: (A) that DAT-marked possessors, appearing in the verbal CAD, are an intermediate step towards deriving NOM/GEN-marked separated possessors; (B) that there are two independent possessor-raising operations: one targeting a landing site in the verbal CAD, and one targeting a landing site in the clausal CAD directly. Regarding (A), assume that the derivation of a separated NOM/GEN-marked possessor proceeds in two steps: First, the possessor moves out of its base position inside the possessed nominal phrase to a position in the verbal CAD, where it is assigned DAT-marking. Then, in a second step, the now-DAT-marked possessor moves again into the clausal CAD. This creates a problem: Given the assumptions about DCT made in Baker & Vinokurova (2010) and Baker (2015b: pp. 272–281, §6.3.3) (cf. my discussion in §2.5) it should not be possible for a nominal once assigned a dependent case (DAT) to lose its dependent case and receive default unmarked case (NOM or GEN), resp. to overwrite its dependent case with an unmarked case. In §6 I will argue that Turkic GEN is actually a dependent-up case itself assigned inside the nominal CAD by case-competition against the possessum, which would make it theoretically possible to overwrite dependent-up DAT with dependent-up GEN (if the bare marking on the separated possessor *is* GEN-marking at all, rather than NOM-marking) — however, the high landing site of the possessor is not in a nominal CAD such that GEN could be assigned there. I conclude that given current assumptions the derivation of possessor-raising cannot proceed

¹⁰⁵ Vinokurova 2005: p. 138, ex. (18a, b)

in two steps with an intermediate landing site where dependent-up DAT is assigned; i.e., option (A) is not feasible.

Now let us explore option (B) and assume that there are two independent operations of possessor-raising: one where the possessor raises to a position inside the verbal CAD as discussed above, and an independent second one where the possessor raises directly in a one-fell-swoop manner from its base position into the clausal CAD. Then, if it turns out that the separated possessor carries GEN-marking, this GEN could simply be preserved from the base position (thus, GEN is a case value [case: GEN], not the realization of an empty case feature [case: ____], supporting my argument that Turkic GEN is not an unmarked case); or, if it turns out that the separated possessor carries NOM-marking, this NOM could simply be the unmarked case [case: ____] spelled out in the clausal CAD, which would mean that the possessor did not receive a case value [case: GEN] in its base position, possibly going against my argument from §6. Having seen that there are no issues with case-value-inheritance in chains as in option (A) above, there remains the issue of how to implement the one-fell-swoop movement from the possessor’s base position inside the possessed nominal all the way into the clausal domain. The difficulty lies in locality: If the DP containing the possessor’s base position as well as the ν P containing this DP are both intransparent Phases, upon Merge of the possessor’s landing site inside the CP, the possessor’s lower copy will no longer be accessible for Move. One could try to find ways around this, such as arguing that the ν P is not phasal or at least not an intransparent phase (Baker ‘soft phases’) (perhaps because this kind of possessor-raising is only possible with unaccusatives, Baker & Vinokurova 2010); or perhaps the argument can be made that the possessor’s base position is at the edge of the DP, thus not part of the DP’s Spell-out domain, and the DP is at the edge of the ν P, and in this way the possessor’s base position remains accessible to the clausal Phase. All of these considerations depend on understanding the phenomenon of Sakha possessor-raising in greater detail, so until this empirical matter is researched more, I leave the discussion at this point, concluding that this construction cannot at present be used to argue that the DAT-marking on raised possessors must be dependent-up case.

5.5.3 CCAs feed dative case assignment

Baker & Vinokurova (2010: p. 620) make a strong argument — one that I consider stronger than the case alternation in causatives (§5.4), at least if it extends to more verbs — for structural DAT in raising-to-object resp. cross-clausal A/A'-dependency (CCAs) constructions (introduced in §3.4). They argue that when a ditransitive verb like ‘to promise’ takes as THEME a finite *dien*-clause and as RECIPIENT an NP (with the RECIPIENT c-commanding the THEME as discussed in §5.3.4), the RECIPIENT will appear with ACC, as expected, (41).

- (41) *Sargy Keskil -i [[kim daqany] kel -im -ie dien] erenner -de.*
 Sargy.NOM Keskil -ACC who.NOM PRT come -NEG -FUT that promise -PST.3SG
 ‘Sargy promised Keskil that nobody will come.’ Sakha¹⁰⁶

The subject of the *dien*-clause may undergo raising-to-object to the *dien*-clause’s edge and participate in the matrix CAD, receiving ACC if there is a matrix subject or other case-competitor, as discussed in §3.4. Interestingly now, the RECIPIENT-NP is affected by this raising of the embedded subject and can no longer appear with ACC, (42a), taking DAT instead, (42b).

- (42) a. **Sargy Keskil -i [kim -i daqany] [~~kim-daqany~~ kel -im -ie dien]*
 Sargy.NOM Keskil -ACC who -ACC PRT come -NEG -FUT that
erenner -de.
 promise -PST.3SG Sakha¹⁰⁷
- b. *Sargy Keskil -ge [kim -i daqany] [~~kim-daqany~~ kel -im -ie dien]*
 Sargy.NOM Keskil -DAT who -ACC PRT come -NEG -FUT that
erenner -de.
 promise -PST.3SG
 ‘Sargy promised Keskil that nobody will come.’ Sakha¹⁰⁸

Baker & Vinokurova (2010) argue that this case alternation arises because when the embedded subject *kim daqany* has left the embedded CP and moved to its edge, the matrix RECIPIENT *Keskil* and the embedded subject are now both in the same CAD (the matrix verbal CAD), with the RECIPIENT c-commanding the embedded subject, which triggers the rule for dependent-up dative case

¹⁰⁶ Baker & Vinokurova 2010: p. 620, ex. (48)

^{107, 108} Baker & Vinokurova 2010: p. 620, ex. (49)

assignment to the RECIPIENT. This configuration is illustrated in Fig. 5.8, assuming a low applicative structure for ‘promise’. Furthermore, this ACC→DAT alternation on the RECIPIENT only happens

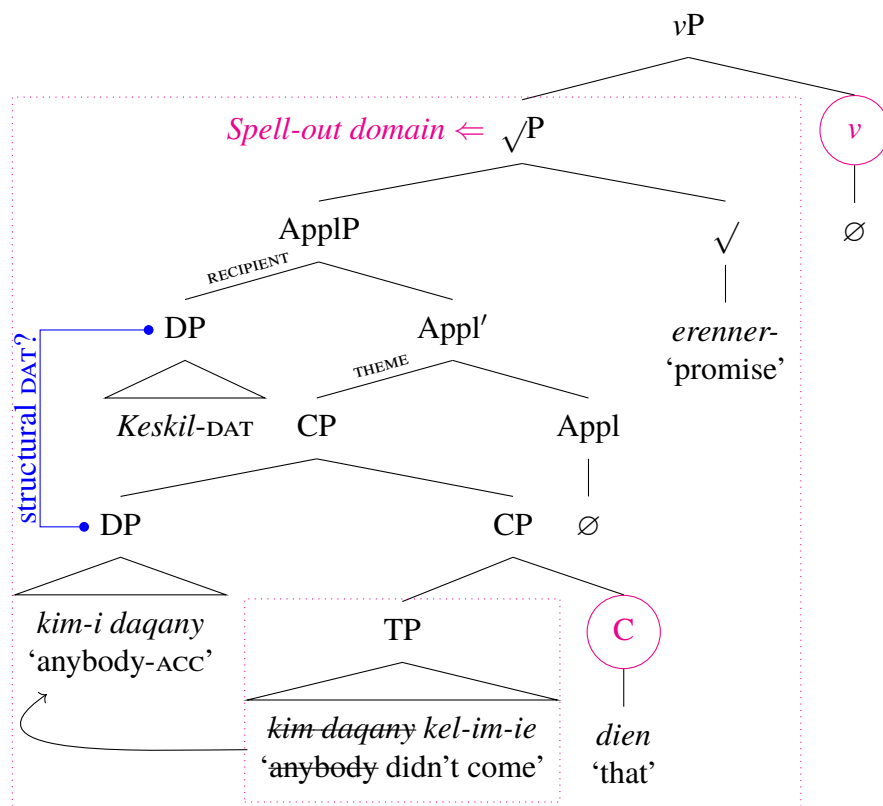


Figure 5.8: CCʼl feeding DAT assignment

if the embedded clause is argumental and not if it is an adjunct (which it was in the data in §3.4). In Baker & Vinokurova’s (2010) theory, this follows directly because adjuncts are never inside the verbal CAD, and thus raising to the edge of an adjunct clause will not put the embedded subject into the matrix verbal domain where it can compete against the matrix RECIPIENT.

What Baker & Vinokurova (2010) don’t discuss but what is briefly mentioned in Baker (2015b)¹⁰⁹

¹⁰⁹ Baker (2015b: p. 198, fn. 14) writes: “[A variant of (41), given in (i)] is also grammatical with *Keskil* marked dative (Vinokurova 2005: p. 367) [(ii)]. I assume that in this situation dative is an inherent case, assigned by a null adposition (probably allowed because the verb is a lexicalized causative) of *eren-* ‘rely upon’ (Vinokurova 2005: p. 449).

- (i) *Sargy Keskil -i [Aisen kel -ie dien] erenner -de.*
 Sargy.NOM Keskil -ACC Aisen.NOM come -FUT that promise -PST.3SG
 ‘Sargy promised Keskil that Aisen would come.’ *Sakha* (Vinokurova 2005: p. 367, ex. (116a))
- (ii) *Sargy Keskil -ge [Aisen kel -ie dien] erenner -de.*
 Sargy.NOM Keskil -DAT Aisen.NOM come -FUT that promise -PST.3SG
 ‘Sargy promised Keskil that Aisen would come.’ *Sakha* (Vinokurova 2005: p. 367, ex. (116b))

and shown in detail in Vinokurova (2005), is the following evidence that the DAT on *Keskil-ge* in (42b) above is not necessarily a structural case: In addition to the data given above, *erenner-* ‘promise’ can also appear with a DAT-marked RECIPIENT when the embedded subject stays inside its clause, (43).

- (43) *Sargy Keskil -ge [Aisen kel -ie dien] erenner -de.*
 Sargy.NOM Keskil -DAT Aisen.NOM come -FUT that promise -PST.3SG
 ‘Sargy promised Keskil that Aisen will come.’ Sakha¹¹⁰

This means that when the embedded subject remains low, we have variation: the matrix RECIPIENT can be either ACC- or DAT-marked. However, when the embedded subject raises to ACC, variation disappears and the matrix RECIPIENT must appear with DAT-marking. This is summarized in Table 5.2. How to account for this variation? Multiple options come to mind. (A) It could be lexical

	<i>Keskil</i> -ACC	<i>Keskil</i> -DAT
<i>Aisen</i> -NOM	✓(41)	✓(43)
<i>Aisen</i> -ACC	*(42a)	✓(42b)

Table 5.2: Case marking options for the RECIPIENT and the embedded subject of ‘promise’ in Sakha finite *dien*-clauses

variation on *erenner-* ‘promise’ where the RECIPIENT argument does or does not receive lexical (or, on an applicative analysis of ‘promise’, cf. §5.3: inherent) dative: if the RECIPIENT does not receive non-structural case, it will later receive structural case, which might be dependent-up dative, as Baker & Vinokurova (2010) propose. (B) The observed variation could also be optionality on whether or not the clausal THEME argument counts as case-competitor for the assignment of dependent-up DAT to the RECIPIENT: (B.1) Perhaps there is variation in the argument structure of *erenner-* ‘promise’ such that the THEME clause and the RECIPIENT can or cannot stand in the local c-command configuration necessary for Case Competition. Susi Wurmbrand has suggested to me (p.c.)

He goes on to say, “The crucial point is that the other internal argument can be accusative if the theme argument is a true CP, but not if it is a participial clause.” I cannot tell whether by “the other internal argument” Baker means the RECIPIENT *Keskil* or the embedded subject *Aisen* or some third element. If he means the RECIPIENT, his statement does not make sense because ACC on the RECIPIENT is possible when the THEME clause is a participial clause, (44a). If he should mean the embedded subject *Aisen* (which I would not consider an internal argument of ‘promise’), again, ACC-marking is possible in participial clauses, (45b).

¹¹⁰ Vinokurova 2005: p. 367, ex. (116b)

that we might be dealing with a raising and a control version of ‘promise’ here. I currently lack the necessary data to test this. (B.2) Or, perhaps there is more detailed variation in the operation Case Competition that requires further attention. (I do not explore this option further.)

In any case, what remains of great interest is the absence of this variation when the embedded subject undergoes raising-to-ACC, (42a). Could it be that Baker & Vinokurova (2010) are correct in proposing that Sakha has dependent-up dative in the verbal domain, and when the subject has raised-to-ACC, the two variants of *erenner*- ‘promise’ (whatever their nature) simply collapse, and we simply cannot, with the currently available data, differentiate whether the DAT-marking on the RECIPIENT *Keskil* is lexical/inherent case or structural dependent-up case?

I will now argue that this is not possible. The relevant considerations come from *erenner*- ‘promise’ when its THEME argument is not a finite clause headed by *dien*, but a nominalized clause, which will itself, by virtue of being nominalized, receive ACC-marking. In Sakha the embedded NOM-marked subject of a participial nominalization can still undergo raising-to-ACC.¹¹¹ The pattern between NOM/ACC-marking on the embedded subject and ACC/DAT-marking on the matrix RECIPIENT is identical to that in *dien*-clauses above. This is illustrated in (44)–(45) and summarized in Table 5.3.

- (44) a. *Sargy Keskil -i* [*Aisen kel -er -in*]
 Sargy.NOM Keskil -ACC Aisen.NOM come -AOR -POSS.3SG, ACC
erenner -de.
 promise -PST.3SG
 ‘Sargy promised to Keskil that Aisen will come.’ Sakha ¹¹²
- b. **Sargy Keskil -i Aisen -y* [*Aisen kel -er -in*]
 Sargy.NOM Keskil -ACC Aisen -ACC come -AOR -POSS.3SG, ACC
erenner -de.
 promise -PST.3SG
 ‘Sargy promised to Keskil that Aisen will come.’ Sakha ¹¹³

¹¹¹ This is not possible in Turkish and other Turkic languages. I suspect that it may be related to the case marking of the embedded subject, which in Sakha is NOM-marked, while in (as far as I know: all) other Turkic languages is GEN-marked by virtue of the nominalization being a possessive construction (compare English *his arriving late*).

^{112, 113} Vinokurova 2005: p. 367, ex. (117a)

- (45) a. *Sargy Keskil -ge* [*Aisen kel -er -in*]
 Sargy.NOM Keskil -DAT Aisen.NOM come -AOR -POSS.3SG. ACC
erenner -de.
 promise -PST.3SG Sakha ¹¹⁴
- b. *Sargy Keskil -ge Aisen -y* [*Aisen kel -er -in*]
 Sargy.NOM Keskil -DAT Aisen -ACC come -AOR -POSS.3SG. ACC
erenner -de.
 promise -PST.3SG Sakha ¹¹⁵

	<i>Keskil</i> -ACC	<i>Keskil</i> -DAT
<i>Aisen</i> -NOM	✓(44a)	✓(45a)
<i>Aisen</i> -ACC	*(44b)	✓(45b)

Table 5.3: Case marking options for the RECIPIENT and the embedded subject of ‘promise’ in Sakha ACC-marked participial clauses

Let us assume that Baker & Vinokurova (2010) are correct and Sakha has a dependent-up case assignment rule that assigns DAT to a nominal under Case Competition with a lower not-yet-case-marked nominal in the same verbal domain, and let us also assume that this rule is not optional. Now, consider (44a). Let us further assume that since the THEME-clause receives case-marking, it can in principle also serve as case-competitor (cf. §2.5.4 for a discussion of which elements can serve as competitors). Since the RECIPIENT is not DAT-marked, we can then conclude that Case Competition did not apply. Thus, the RECIPIENT and the nominalized THEME in (44a) must not stand in the relevant local c-command configuration. Now, consider (44b). If Baker & Vinokurova (2010) are correct that (44b) is unacceptable because the operation Case Competition with the rule for dependent-up DAT could have applied but did not, this would mean that the RECIPIENT must locally c-command the raised subject in the verbal domain. Now, what could the structure be such that the RECIPIENT c-commands the raised subject but does not c-command the nominalized THEME containing the subject’s base position? Such a structure would require something like sideward movement or very peculiar assumptions about locality. I consider it far more likely that this construction simply does not involve dependent-up dative.

^{114, 115} Vinokurova 2005: p. 367, ex. (117b)

Susi Wurmbrand (p.c.) has suggested that the contrast in acceptability between (44a) and (44b) could be independent of case reasons and could instead be due to the raised subject *Aisen-y* and the ACC-marked matrix RECIPIENT *Keskil-i* competing for the same position. Together with an argument structure where the RECIPIENT and the nominalized THEME do not stand in a local c-command relation, this would indeed make it possible to account for all the present data under Baker & Vinokurova’s (2010) model with dependent-up dative case. However, I have no ideas what this unique position might be. Furthermore, this competition for the same position would need to only apply when the RECIPIENT *Keskil-i* is ACC-marked, while a separate position would need to be available for the DAT-marked RECIPIENT *Keskil-ge*, which can cooccur with the ACC-marked embedded subject *Aisen-y*. This consequence, then, seems to me to introduce another difficulty for Baker & Vinokurova’s (2010) proposal that the DAT-marking on the RECIPIENT be structural case: In DCT, why should a structurally-ACC-marked RECIPIENT and a structurally-DAT-marked RECIPIENT need to appear in different syntactic positions?

In conclusion, the argument for DAT as dependent-up case as illustrated in Baker & Vinokurova (2010), where raising of the embedded subject into the matrix verbal CAD has effects on the case-marking of the matrix RECIPIENT, would be a beautiful argument for dependent-up case, but upon closer inspection of the available data for the one proposed matrix predicate (*erenner*- ‘promise’), I find that there seems to be variation or optionality or further factors involved that do not allow us to make the argument that the DAT→ACC alternation on the RECIPIENT *must* be dependent-up case. In the absence of this argument, and especially if ‘promise’ has a low applicative structure, the DAT-marking on the RECIPIENT *may* also be inherent dative assigned by the same mechanism as shown in §5.3.2 for low applicatives in general.

5.5.4 Summary

In this section I have discussed two phenomena that have been proposed to constitute examples of structural dependent-up dative in Sakha: 1. raising of a possessor out of the THEME into the verbal domain, where it receives DAT instead of GEN, allegedly by case-competition against the THEME,

and 2. raising of an embedded subject to the edge of the embedded clause, such that this subject becomes part of the matrix clause's verbal CAD and allegedly triggers DAT-marking on a higher argument in the matrix clause (the RECIPIENT of 'promise'). I have discussed the available evidence for both constructions and concluded that the data available in the literature are not sufficiently detailed to warrant the conclusion that this dative must be structural dependent-up case. Thus, at this point, we do not have *any* conclusive evidence that Sakha (or any other Turkic language) contains dependent-up dative case.

5.6 The missing Single Dative Generalization

I have argued above in chapter 4 that dependent-up cases on arguments are always singular within their case-assignment domain (CAD). If DAT case were the dependent-up case in the verbal domain, as Baker (2015b) and Baker & Vinokurova (2010) claim, and if the argument in chapter 4 is correct, this makes a prediction:

- (46) It is impossible for two or more structurally DAT-marked nominals to receive their structural DAT case in the same CAD.
- a. If we get two or more DAT-marked nominals in the same CAD, some or all of these DAT must be non-structural cases.
 - b. If DAT case is a dependent-up case, we expect to see patterns like we do with ERG, namely that a pattern $\langle \text{ERG}, \text{ABS} \rangle$ upon Merge of another nominal on top of the structure should turn into $\langle \text{ERG}, \text{ABS}, \text{ABS} \rangle$, i.e., a case change dependent-up→unmarked on the top→middle nominal.

Let us try to combine a ditransitive (represented as low applicative) with a causative. Zimmer (1976: p. 411, ftn. 12) gives the following example and describes one consultant's judgment as "unwilling to accept" and those of "a number of informants" as "quite awkward":

- (47) *?Dişçi müdür -e mektub -u Hasan -a göster -t -ti.
 dentist.NOM director -DAT letter -ACC Hasan -DAT show -CAUS -PST
 ‘The dentist made the director show the letter to Hasan.’ Turkish¹¹⁶

Is the unacceptability of (47) an indication that both these DAT are structural cases? No, as Zimmer (1976) notes: It seems to be an effect of processing overload. If we replace the subject NP with *pro*, reducing the sequence of four NPs in (47) to three in (48), the sentence becomes acceptable with both DAT-marked arguments still there.

- (48) Biz -e mektub -u Hasan -a göster -t -ti -ler.
 us -DAT letter -ACC Hasan -DAT show -CAUS -PST -3PL
 ‘They made us show the letter to Hasan.’ Turkish¹¹⁷

This structure is illustrated in Fig. 5.9 Thus, we can conclude that either the DAT-marking on the RECIPIENT of ditransitives, or the DAT-marking on the causee of ditransitives, or both, must be instances of non-structural case, else (48) should be unacceptable.

Similarly, the following examples (repeated from above) show that there can be multiple DAT-marked arguments in the same clause. In (49), repeated from (24a), there is a DAT-marked causee and a DAT-marked RECIPIENT — or, alternatively, *anlat-* ‘to tell’ can be analyzed as its own causative *anla-t-* ‘to make understand’, then we (49) would contain a causative of a causative with two DAT-marked causees. Two DAT-marked causees would, by (46a) be proof that this DAT is not dependent-up case.

- (49) Baba -m -a çocuk -lar -a masal anlat -tır -dı -m.
 father -POSS.1SG -DAT child -PL -DAT story tell -CAUS -PST -1SG
 ‘I had my father tell stories to the children.’ Turkish¹¹⁸

Example (50), repeated from (24b) shows a causative of a ditransitive. Recall from my discussion in §5.3.4.2 that the DAT-marked argument of a ditransitive can in principle be a low or high GOAL. As illustrated there, when the GOAL is a person, it is usually interpreted as RECIPIENT, i.e., a high GOAL with inherent DAT, while when it is a location, it is a low GOAL with lexical DAT. Thus, again,

¹¹⁶ Zimmer 1976: p. 411, fn. 12, ex. (i), after Comrie 1976.

¹¹⁷ Zimmer 1976: p. 411, fn. 12, ex. (iii)

¹¹⁸ Göksel 1993: p. 216, ex. (40)

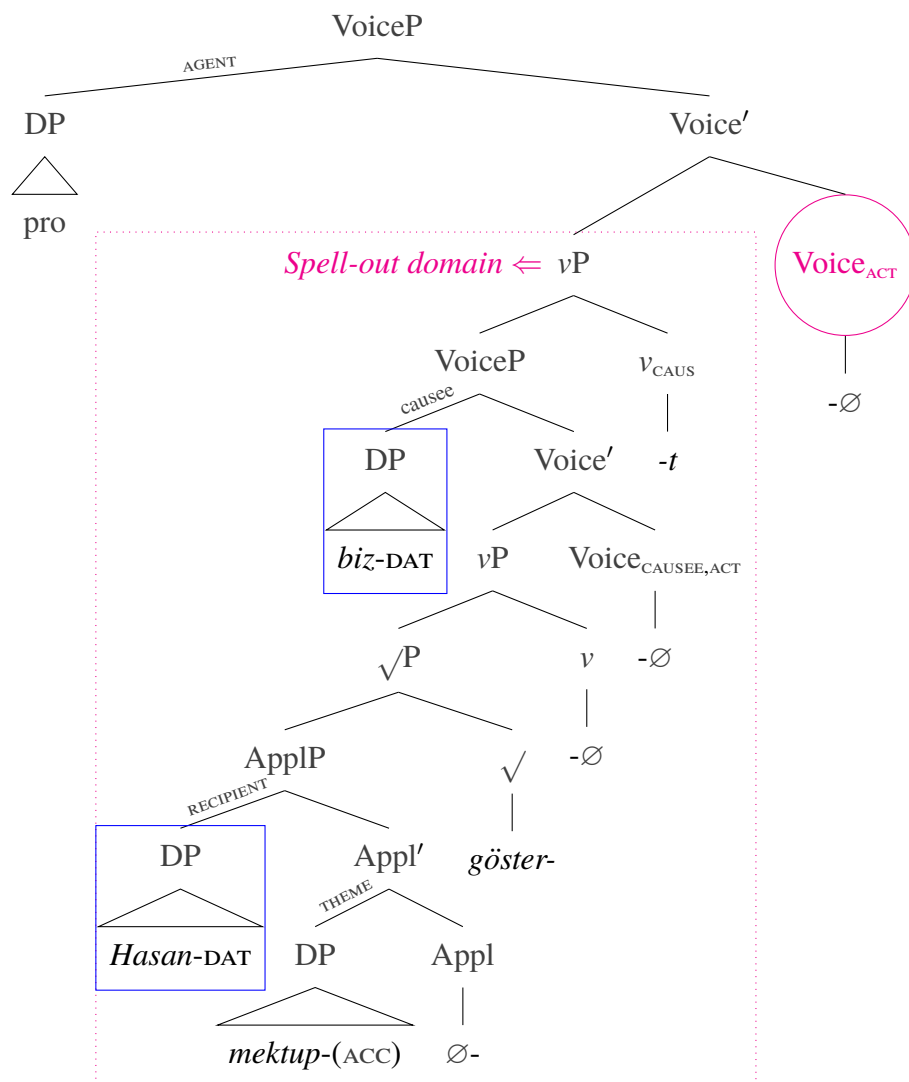


Figure 5.9: A causative of a low applicative: two DAT-marked arguments

here we see that either the DAT-marking on the causee, or the DAT-marking on the RECIPIENT, or both must be non-structural DAT.

- (50) *Mama -m mağan kitap -tı Murat -qa ber -dír -dí.*
 mom -POSS.1SG.NOM me.DAT book -ACC Murat -DAT give -CAUS -PST
 ‘My mom made me give the book to Murat.’

Karakalpak ¹¹⁹

¹¹⁹ Data from fieldwork in Spring 2019 in Asia Pietraszko’s field methods class.

5.7 Conclusions

In this chapter, I have reviewed and discussed the evidence that has been submitted under the theory that in Turkic languages, some datives can be instances of structural case (dependent-up case), as proposed by Baker (2015b), Baker & Vinokurova (2010), and Marantz (1991), among others. After careful consideration of all the phenomena involved (§5.3–§5.5) I have come to the conclusion that not one of the discussed phenomena warrants the conclusion that in this phenomenon, the observed DAT-marking must be structural case. Rather, in all discussed phenomena, there is also a functional head present that may be responsible for the respective DAT-marking, making this DAT an inherent case. In particular, the DAT-marking in causatives (§5.4), which has often been called the clearest evidence for structural DAT in Turkish because the DAT-marking appears in correlation with a lower c-commanded structurally case-marked nominal, has been shown by Akkuş (2021) to also correlate with the presence of a functional head ($\text{Voice}_{\text{CAUSEE}}$), which may well be responsible for the appearance of DAT case.

What I consider the strongest possible evidence for structural DAT is the case alternation on RECIPIENTS of ditransitives depending on movement of another nominal into a lower c-commanded position in the same case-assignment domain (§5.5.3). However, the only data that I have found discussed for this construction are complex, not clearly understood, and may be due to independent factors, such as optional lexical DAT assigned by the one verb where this case alternation has been attested. Unless this pattern turns out to be productive, I cannot conclude that it must contain structural dative case.

In §5.6 I have argued that the results of chapter 4 make a prediction about structural dative: that it is impossible for more than one structural DAT-marking to appear on arguments in the same case-assignment domain. It has often been observed that multiple DAT-markings in Turkic languages are not acceptable, but in this section I have cited a data point that shows that this restriction against multiple datives is due to processing and not due to the restriction against multiple dependent-up cases from chapter 4.

Taken together, I draw the conclusion that Turkic languages may not contain dependent-up dative

case at all, and that all DAT-markings are either lexical cases or inherent cases.

Chapter 6

Turkic genitive is the dependent-up case in the nominal domain

6.1 Introduction

In this chapter, I present a novel proposal on how to analyze *GEN* in Turkic in the framework of DCT: that *GEN* is the dependent-up case in the nominal case-assignment domain. First, in §6.2, I will show how to differentiate between structural *GEN*, which I argue is always a dependent-up case, from lexical *GEN*, which is assigned by adpositions and which Kuzgun & Atlamaz (2023) have shown differs morphologically. Then I proceed to the main argument about structural *GEN*. My argument proceeds in two steps. First, I show that it is impossible for *GEN* to be an unmarked case (§6.3) or inherent case (§6.4), because if it were, it would behave differently than it does: If *GEN* were an unmarked case, and assuming with Preminger (2014) that unmarked cases are unvalued cases, a nominal should be able to lose its *GEN*-marking when it moves into a different CAD, in the same way that a *NOM*-marked nominal can become *ACC*, as discussed previously throughout this thesis. If *GEN* were an inherent case, we would expect it to only express a particular thematic role, which Öztürk & Taylan (2016) have shown it does not, and to depend on an assigning functional head, which it does not either. Given the framework of DCT, being a dependent case is the only

remaining option for GEN.

In §6.5 I show how it works out that GEN is the dependent-up case in the nominal case-assignment domain. Building on Baker (2015a), who has shown how some languages assign dependent-up ERG to the possessor by case-competition against the possessum, I argue in §6.5.2 that the same happens in Turkish. In §6.5.3 I argue that another indication that GEN is a dependent case comes from the prediction made by Bobaljik's (2008) Revised Moravcsik (1974, 1978) Hierarchy: If an agreement probe can target dependent case, it must also be able to target unmarked case. I show that this holds for the possessive agreement probe on Poss in Turkish and Uzbek.

Finally, in §6.6 I turn to a much-debated construction in Turkic: Differential Subject Marking (DSM), i.e., the phenomenon that subjects of nominalized clauses can appear with NOM- or GEN-marking. I show that, given the reduced functional projections in nominalized clauses that Kelepir (2015, 2021) has identified, GEN as dependent-up case makes the correct predictions for DSM in nominalized argument clauses while GEN as unmarked case does not. For nominalized adjunct clauses, where the subject is always NOM-marked, I follow many others in adopting a domain-extension approach, claiming that the element that links the nominalized clause to the matrix clause extends the case-assignment domain from a nominal to a clausal domain, thus voiding GEN-assignment.

6.2 Lexical genitive

Kuzgun & Atlamaz (2023) make the point that GEN in Turkish can be a structural or a lexical case, and the two sometimes differ morphologically. They argue that GEN assigned by adpositions is a lexical case because the adpositions that take a GEN-marked complement form neither a semantic nor a syntactic class, and the translations of these adpositions into German and Hungarian assign different cases from GEN and from each other, thus this property is idiosyncratic to the respective lexical items. The GEN-assigning adpositions in Turkish are listed in (1).

- (1) a. *için* 'for'

- b. *gibi* ‘like’
- c. *kadar* ‘as much as’
- d. *ile*, often cliticized to *-(y)la* ‘with’
- e. *hakkında* ‘about’, ‘by’ in passives

Kuzgun & Atlamaz observe that the GEN-marking on the complement of these adpositions is different from the GEN-marking on possessors, which they call structural GEN: on non-inflected pronouns, (2a), and indexical demonstratives¹, lexical GEN looks identical to structural GEN, while on inflected pronouns, (2b)², non-indexical demonstratives³, common nouns, (2c), and proper names, (2d), lexical GEN looks like NOM, i.e., there is no morphological marking. This contrasts with the GEN-marking on possessors, where the suffix is always present (§6.4.2, cf. also Kuzgun & Atlamaz 2023: p. 177).

- (2)
- | | | |
|----|---|-------------------------------|
| a. | <i>kim -in için</i>
who -GEN for
‘for who’ | <i>Turkish</i> ^{4,5} |
| b. | <i>kim -ler { *-in /^{ok}-∅ } için</i>
who -PL -GEN for
‘for who all’ | <i>Turkish</i> ⁶ |
| c. | <i>kitab { *-in /^{ok}-∅ } için</i>
book -GEN for
‘for the book’ | <i>Turkish</i> ⁷ |
| d. | <i>Ayten { *-in /^{ok}-∅ } için</i>
Ayten -GEN for
‘for Ayten’ | <i>Turkish</i> ⁸ |

¹ Kuzgun & Atlamaz 2023: p. 176 f., ex. (10)

² Kuzgun & Atlamaz (2023: p. 175 f.) write that this inflection applies to pronouns marked with PL *-lar*, which can appear on *on-lar* s/he-PL ‘they’, *kim-ler* who-PL ‘who all’ (2b), *biz-ler* us-PL ‘we.HON’, *siz-ler* y’all-PL ‘you.HON’, but also the POSS.3SG- resp. CMPD-suffix *-(s)I(n)* forming *kim-i* ‘someone’ and the NPI *-(y)sA* forming *kim-se* ‘anyone’.

³ Kuzgun & Atlamaz 2023: p. 177, ex. (11)

⁴ Kuzgun & Atlamaz 2023: p. 175, ex. (5c)

⁵ It is not impossible for non-inflected pronouns to appear without the GEN-suffix (e.g., *ben gibi* vs. *ben-im gibi* ‘like me’), as Kuzgun & Atlamaz (2023: p. 177, fn. 2) note, but they say that at this point, they “cannot precisely characterize” the difference between these forms.

⁶ Kuzgun & Atlamaz 2023: p. 176, ex. (7b)

⁷ Kuzgun & Atlamaz 2023: p. 175, ex. (6a)

⁸ Kuzgun & Atlamaz 2023: p. 175, ex. (6b)

Kuzgun & Atlamaz (2023) proceed to make an argument about the structure of pronouns that derives the different morphological realizations of lexical GEN and accounts for the syncretism between lexical GEN and structural GEN. The details of their proposal are not relevant here; I take away from this that Turkish has a contrast between lexical and structural genitives and crucially that they differ morphologically in all contexts except on minimal pronouns as described above. I will use this as a diagnostic that genitives with morphological realizations such as possessors are not lexical genitives.

6.3 Structural genitive is not an unmarked case

6.3.1 Introduction

In this section I will argue that GEN case in Turkic languages is not the Spell-out of an unmarked case, for instance in the nominal case-assignment domain (CAD). As discussed above in §2.5.3, unmarked cases have a particular property: they can be replaced resp. overwritten with dependent cases. In §2.5.3 I have compared two proposals about the nature of unmarked case as unvalued [case: ____] (Preminger 2014) resp. as its own value [case: NOM] (Baker 2015b). This distinction does not matter for the present section; I merely use the observation that unmarked cases can be overwritten, which Preminger and Baker share, as diagnostic for Turkic: In §6.3.2, I show that NOM behaves like an unmarked case in that it can be overwritten after movement, and in §6.3.3, I show that GEN cannot be overwritten after movement of the GEN-marked nominal. From this difference I conclude that Turkic GEN cannot be an unmarked case.

6.3.2 Turkic nominative is overwritable

6.3.2.1 Introduction

In all the Turkic languages discussed in this thesis, cases are expressed as suffix on the head of the case-marked NP, with the exception to what has been called “NOM case”, which is expressed by the

absence of a case suffix. We have (at least) three ways of analyzing this:

- (i) “*NOM*-marked” NPs are valued as [case: *NOM*], and the morphological exponent of this value is zero/null ($\emptyset$); or
- (ii) what we call *NOM* corresponds to the absence of a value [case: ____], which spells out context-sensitively in the clausal CAD as $\emptyset$, and perhaps in other domains with a different exponent; or
- (iii) *NOM* corresponds not only to the absence of a value, but to the absence of the feature [case] itself, resp. if case is not a feature on some nominal projection but its own projection *K*, then *K* may not be projected.

Many papers and grammars choose approach (i), while famously Jakobson (1936), Andrews (1982), and Bittner & Hale (1996) have opposed the existence of a value “*NOM*”, similar to more recent literature in the framework of Minimalism, where Preminger (2014: §9, 2024) and Kornfilt & Preminger (2015) have argued for (ii)⁹, while Paparounas & Akkuş (2019, 2023) and Satık (2020a,b) have proposed (iii), presenting data where Turkish *GEN* behaves like a phase head and *NOM* doesn’t.

The only marking that I have seen claimed to distinguish suffix-less NPs that are subjects (and by assumption *NOM*-marked) from, for instance, those that are “bare” objects (which are presumably case-less; see §3.3 on *DOM*) is prosody: “*NOM*-marked” subjects are spoken with rising prosody followed by a prosodic break (cf. e.g. Christopher 2018: p. 116 f. on Kazakh), while bare objects are prosodically (pseudo-) incorporated into the immediately following verb. However, the “*NOM*”-prosody is more likely a “topic”-prosody as it occurs on all topicalized (non-focalized) NPs no matter their case-marking.

I will in the rest of this section 6.3.2 present the argument by Kornfilt & Preminger (2015) that *NOM* case in Sakha corresponds to an empty feature [case: ____] and not to a “*NOM*”-value

⁹ Marantz’s (1991) “default case” would then be the elsewhere-form of unvalued case by not being context-sensitive to the surrounding CAD. Cf. also Preminger (2024) on an account of English “*ACC*” as default case that does not make reference to Marantz’s (1991) “default case”: English “*ACC*” is the unmarked case, and English “*NOM*” is an inherent case assigned by finite *T*. Cp. also Poole (2024: §6.1) on an account of English “*NOM*” using Baker’s (2015b) negative dependent-up cases (that he ends up arguing against).

[case: NOM]: We see that a “NOM-marked” nominal can undergo movement into a different CAD and receive a different case-marking there. Baker (2015b) proposed that this overwritability is a property of unmarked cases, while Kornfilt & Preminger (2015) show that it follows directly if unmarked case is unvalued case. In section 6.3.3 I will apply the same test to genitive, showing that it is not overwritable after movement into a different CAD and thus not an unmarked case.

6.3.2.2 The timing of agreement and case assignment in Sakha raising-to-object constructions

Recall from §3.4.2 the Sakha raising-to-object construction where the subject of a finite or nominalized embedded clause moves to the edge of its clause and enters a CC_Q-dependency with an element in the matrix clause, receiving ACC by case-competition against a matrix subject; repeated below.

- (3) a. *Min ehigi -ni [ehigi bügün kyaj -yax -xyt dien] erem -mit -im.*
 me.NOM y'all -ACC y'all.NOM today win -FUT -2PL that hope -PRF -1SG
 ‘I hoped that y’all would win today.’ Sakha¹⁰
- b. *Ehigi bihigi -ni [bihigi kyajtar -dy -byt dien] xomoj -du -gut.*
 y'all.NOM we -ACC we.NOM lose -PST -1PL that become.sad -PST -2PL
 ‘Y’all were disappointed that we lost.’ Sakha^{11,12}

Kornfilt & Preminger (2015) notice that we see agreement on the embedded T (boxed in (3)) with the ACC-marked embedded subject (and also on matrix T with the matrix subject), but crucially T never ever agrees with an ACC-marked NP anywhere else in the language. It thus seems reasonable to say that in general, T tracks “NOM-marked” NPs only (in the sense of Bobaljik 2008). The timing of these examples must thus be such that 1. the embedded subject receives the value NOM (to be revised), 2. T finds it and copies its φ -features (Agree), and then 3. the embedded subject raises to

¹⁰ Baker & Vinokurova 2010: p. 615, ex. (39a); Kornfilt & Preminger 2015: p. 110, ex. (1)

¹¹ Kornfilt & Preminger 2015: p. 110, ex. (2)

¹² The astute reader may notice that Sakha *kyajtar*- ‘lose’ in (3b) looks like *kyaj*- ‘win’ in (3a) followed by a causative suffix *-tar*. According to Wiktionary (<https://en.wiktionary.org/wiki/%D0%BA%D1%8B%D0%B0%D0%B9%D1%82%D0%B0%D1%80#Yakut>, accessed on 2024-05-19), this is indeed the case: losing means letting/making the other person win. However, I do not know at this point whether *kyaj-tar*- could take the causee as additional argument and what its syntax would be, or whether *kyajtar*- is perhaps lexicalized.

Spec,CP and 4. overwrites its case value with ACC in a CC_Q-dependency with the matrix clause.

This poses a problem, Kornfilt & Preminger (2015) argue, because once the embedded subject has received its first case (NOM), why and how would it enter into further case-competitions? Usually ACC is only assigned to NPs that have not yet received any case. Noting this problem too, Baker (2015b: p. 278–281) writes,

[T]here is some empirical reason to think that unmarked case assigned to a lower copy in a chain is not automatically inherited by a higher copy in the same chain. (p. 278)

and,

[T]he lower member of a multi-membered chain does not receive unmarked case because there is a chance of a higher member of the chain getting a more distinctive case, as it does in [(3)] (accusative). (p. 279)

and goes on to suggest that this property may be used as a diagnostic for unmarked case. This, Kornfilt & Preminger (2015) argue, calls for one of two possibilities: Either NOM is special in that it can be overwritten by ACC¹³; or NOM is simply the absence of a value, and NPs can be valued after movement into a different CAD.

¹³ This is reminiscent of Case Attraction as described, among others, for relative clauses by Georgi & Salzmann (2014, 2017). These authors, following Grosu (1994) and Pittner (1995), assume a case-obliqueness hierarchy such as (i), derived by subset relation from the decomposition of cases into privative features, as proposed by Caha (2009) and McFadden (2004), a.o.

- | | | | |
|-----|----|--|-----------------------------------|
| (i) | a. | P-Kase $\succ \dots \succ$ GEN $\succ$ DAT $\succ$ ACC $\succ$ NOM | (Grosu 1994: p. 122, ex. (1.125)) |
| | b. | other $\succ$ ACC $\succ$ NOM | (Pittner 1995: p. 201, ex. (5)) |

Then, in relative clauses in some languages it can be observed that a case c_i on the relative pronoun can be “overwritten” by a case c_j on the head noun only if c_j is more marked than c_i , i.e., to the right of c_i in the hierarchy in (i).

If this “overwriting” in Case Attraction is similar to the “overwriting” discussed here, the two proposals might be united by saying that languages differ in their cut-off point on the hierarchy in (i), such that perhaps Turkic languages allow only the overwriting of NOM by ACC, but not the overwriting of ACC by any other case.

For my purposes here, this would not be a problem: Let us assume that the hierarchy in (i) is generalized to something like (ii).

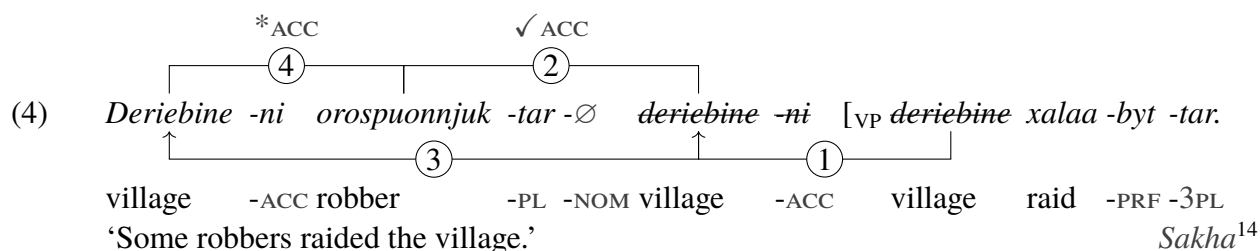
- (ii) lexical case $\succ$ dependent case $\succ$ unmarked case

This would then make the prediction that if Turkic GEN were an unmarked case, it should be overwritable by dependent ACC, just like unmarked NOM is. The data that I present here show clearly that GEN is not overwritable by ACC. Thus, GEN and NOM cannot be the same kind of case.

6.3.2.3 Scrambling of the object past the subject

To differentiate between these two possibilities, Kornfilt & Preminger (2015) discuss scrambling or wh-movement of the ACC-marked object past the NOM-marked subject: After this movement by the object, the subject is now c-commanded by the object, and since NOM can be overwritten by ACC, we might expect that the subject will appear with ACC-marking. However, this does not happen.

In (4), in step 1 the direct object moves from the verbal into the clausal CAD, where it (step 2) receives ACC (cf. §3.3). Subsequently (step 3), the object undergoes topicalization movement (or the like) past the subject. Then, Kornfilt & Preminger (2015) argue, we might expect that the object can re-value the subject's case from NOM to ACC (step 4); however, this is clearly unacceptable.



Baker & Vinokurova (2010) exclude this scenario (i.e., step 4) based on the fact that the moved object could not be assigning ACC to the subject, because even if NPs with unmarked case could enter further case-competitions, this would only apply to the subject, but not to the object, because at the time of step 4, it has already been valued as ACC (cf. §2.5.2 on when case-competition may apply to already-case-marked nominals). Thus, while in (3a), the c-commanding pair $\langle \text{min} [\text{case: } \underline{\quad}], \text{ehigi} [\text{case: } \underline{\text{NOM}}] \rangle$ succeeds in overwriting the case value of *ehigi* with $[\text{case: } \underline{\text{ACC}}]$, this is not possible in (4): the pair $\langle \text{deriebine} [\text{case: } \underline{\text{ACC}}], \text{orospuonnjuk-tar} [\text{case: } \underline{\text{NOM}}] \rangle$ is not being considered by the grammar because of the existing ACC-value. It is unclear to me why Kornfilt & Preminger (2015) think step 4 would be applicable here.

Baker (2015b) introduces two further mechanisms for the application of case-competition that are of relevance for this construction: One mechanism (Baker 2015b: §6.1, pp. 230–252) is that the derivation remembers prior applications of case-competition and ensures that it applies only to new

¹⁴ Baker & Vinokurova 2010: p. 604, ex. (16); Kornfilt & Preminger 2015: p. 112, exs. (5) and (6)

“pairs” (sets of size 2; i.e., unordered pairs) of c-commanding nominals: First (step 2) the subject c-commands the object, which produces the set {subject, object} and marks the object as ACC, and then (step 3) the object moves past the subject, which yields the set {object, subject}, but as this is the same set as in step 2, case-competition cannot apply again (step 4). The second mechanism (Baker 2015b: §6.2.2–6.3.2, pp. 256–272) is that adjuncts cannot serve as case-competitors against arguments. Depending on one’s assumptions about the landing site of the moved object in (4), this mechanism may thus preclude the object serving as case-competitor against the subject on principled grounds. Kornfilt & Preminger (2015) seem unaware of these two additions, perhaps due to the close publication dates of these papers.

I believe that the only reason Kornfilt & Preminger (2015) take up this issue at all is because they allow case stacking/re-valuation more freely than is necessary, namely as (5a), when, I believe, (5b) would suffice for their purposes and would make the discussion of (4) unnecessary, applying to (3) but not to (4), as desired.

- (5) a. “Any nominative noun phrase locally c-commanded by another noun phrase will surface as accusative, irrespective of whether the noun phrases in question have previously been assigned nominative.” (Kornfilt & Preminger 2015: p. 111, ex. (4))
- b. Any nominative-marked or not-(yet)-case-marked noun phrase locally c-commanded by another not-(yet)-case-marked noun phrase will surface as accusative.

In any case, Kornfilt & Preminger (2015) conclude that a “case-stacking” analysis of Sakha is not tenable and that thus “NOM” must be the absence of a value for case, so that the embedded subject in (3) doesn’t need to stack or re-value its case, but can simply value it for the first time in the matrix CAD.

I, however, conclude that treating NOM as case-less-ness is theory-internally more elegant than Baker’s (2015b) stipulation that unmarked cases, but not dependent cases, can be overwritten later on in the derivation, but that it does not change the empirical coverage. Note that my goal in this section 6.3 is to use the overwritability of unmarked cases to diagnose GEN as not-unmarked, so it

is crucial that Baker's (2015b) observation concern unmarked case in general (as he proposes), and not single out NOM.

6.3.2.4 Summary

I have illustrated here that while some of the arguments given in Kornfilt & Preminger (2015) are answered by Baker (2015b) and Baker & Vinokurova (2010), their proposal of treating morphologically null NOM as case-less-ness is appealing on conceptual grounds in that it explains more easily Baker's (2015b) observation that unmarked cases can be overwritten if the thusly marked NPs move into new case-competition configurations: In Kornfilt & Preminger's (2015) proposal there is no necessity for overwriting because all that is happening is valuation. I conclude that for the present purpose it does not matter whether NOM is overwritable or "no case [value] at all": We can treat it like an unmarked case in Marantz's (1991) original terminology, whatever that may be, noting that unmarked cases seem to be overwritable with dependent cases.

6.3.3 Turkic genitive is not overwritable

6.3.3.1 Introduction

In §6.3.2 we have seen that unmarked case is unvalued case [case: ____] and can thus be valued later in the derivation, which we can observe as the overwriting of unmarked case by dependent case. Turkic nominative behaves like an unmarked case in this way.

In this section I will show that genitive case in Turkic is not overwritable, i.e., cannot be replaced with a different case after optional movement of the GEN-marked nominal. From this I draw the conclusion that the GEN-marking must originate from a GEN-value [case: GEN], and that GEN case is not unmarked case.

6.3.3.2 Raising of possessors in Sakha

Recall the Sakha possessor-raising data from §5.5.2, repeated in (6). We have a chain of an NP (the possessor) with its case feature valued as [case: GEN] in its base position prior to movement, (6a), and re-valued as [case: DAT] after movement, (6b). This is reminiscent of [case: NOM] getting (re)-valued to [case: ACC] in §6.3.2.2 above. Does this mean, then, that GEN can be re-valued resp. valued after movement, as we expect all unmarked cases, and only¹⁵ unmarked cases to do?

- (6) a. $\emptyset$ Emiske [Masha massyyna -ta] aldjan -na.
suddenly Masha.GEN car -POSS.3SG.NOM break -PST.3SG
'Masha's car suddenly broke down.' Sakha¹⁶
- b. Beqehee Masha -qa emiske [~~Masha~~ massyyna -ta] aldjan -na.
yesterday Masha -DAT suddenly car-POSS.3SG.NOM break -PST.3SG
'Yesterday Masha's car suddenly broke.' Sakha¹⁷
- c. Masha beqehee [~~Masha~~ massyyna -ta] aldjan -na.
Masha.NOM/GEN? yesterday car-POSS.3SG.NOM break -PST.3SG
'Yesterday Masha's car broke.' Sakha¹⁸

Baker & Vinokurova (2010: §3.6, pp. 620–622) use these data to argue that the first step of this raising, (6a–b), shows that this DAT is the dependent-up case in the verbal domain. I have discussed in §5.5.2 that I find this argument problematic and that I do not agree that it constitutes evidence for DAT as dependent-up case.

I will now put aside the DAT-marking and discuss the bare marking on the raised possessor in (6c), assuming that the possessor can move from (6a) to (6c) directly, i.e., without receiving DAT-marking in an intermediate position in (6b). Baker & Vinokurova (2010: p. 621) do not explore whether this possessor is marked with NOM or GEN, which should be a straight-forward empirical matter, testable by replacing the proper name *Masha* with a possessed NP such as 'Masha's father'. Since I do not have these data available at this time, I will discuss both possibilities in turn.

Possibility 1: The raised possessor in (6c) is NOM-marked. If the GEN on the possessor in (6a)

¹⁵ But see fn. 13 for a sketch of an proposal on how this may be subject to variation.

¹⁶ Example constructed after Baker & Vinokurova 2010: p. 621, exs. (50), (51), and (53). Judgment unknown to me.

¹⁷ Baker & Vinokurova 2010: p. 622, ex. (54a)

¹⁸ Baker & Vinokurova 2010: p. 622, ex. (54b)

is unmarked case, then, under Preminger's (2014) view that unmarked case is unvalued case, the possessor was simply not valued in its base position and still is not valued in the clausal domain, realizing its [case: ____] as NOM.¹⁹ Assuming that the landing site of the possessor is not so high up that it is outside the search domain of the agreement probe on T, this makes a predication that is not fulfilled: The high possessor, by virtue of being a NOM-marked nominal in the clausal domain higher up than the NOM-marked possessed NP it has moved out of, should become the goal for the agreement probe on T. However, it doesn't: Baker & Vinokurova (2010: p. 622) write explicitly that it is the THEME that agrees with T even in a construction analogous to (6c).

Possibility 2: The raised possessor in (6c) is GEN-marked.²⁰ Let us first assume with Preminger (2014) that unmarked cases are unvalued cases. Then, if GEN is an unmarked case, it is the realization of an unvalued case feature [case: ____], perhaps in the nominal domain, but crucially not in the clausal domain, where the Spell-out of [case: ____] is NOM. Since the possessor in (6c) has left the nominal domain and moved into the clausal domain, it could not possibly realize its [case: ____] as GEN. Thus, if it *is* GEN-marked, this GEN must be its own value [case: GEN], valued in base position, and thus cannot be unmarked case. Now let us assume with Baker (2015b) that unmarked cases are valued cases with special overwritability properties as discussed in §2.5.3. Then perhaps this construction can be seen as an argument that the possessor gets valued in base position, (6a), as unmarked [case: GEN], then moves into the clausal domain, (6c), but unmarked GEN cannot be overwritten by unmarked NOM (unlike in Finnish, cf. fn. 19), thus preserving the value [case: GEN] and also accounting for the absence of agreement. To explore this option further, one should test whether this GEN on raised possessors (if it is GEN at all) *can* be overwritten by dependent ACC, perhaps in a raising-to-object (CC2l) construction.

¹⁹ Note that Baker's (2015b) contrasting view on unmarked case, where unmarked cases correspond to their own feature values with certain overwritability properties, as discussed in §2.5.3, would need to allow that unmarked [case: GEN] can be overwritten by unmarked [case: NOM]. This pattern is also observed in Finnish, he argues (p. 278), where the unmarked case in the verbal domain is partitive and can be overwritten by the unmarked case in the clausal domain, which is nominative.

²⁰ This is the possibility that Baker & Vinokurova (2010: p. 622) mention briefly: that movement of the possessor goes directly from its base position, (6a), into the clausal domain, (6c), and simply retains its GEN case. Crucially, Baker & Vinokurova (2010: pp. 632 f.) do not assume that GEN in Sakha is an unmarked case, but that it is inherent case assigned by POSS, which values the nominal as [case: GEN], thus voiding the problem described here. See §6.4 on why I think that GEN is not an inherent case assigned by POSS.

There are many open questions regarding this construction, some empirical, some theoretical. I propose that at this time, given the available data, we should not draw a conclusion about the nature of this GEN, thus not counting it as evidence that the GEN on possessors is an unmarked case.

6.3.3.3 Scrambling of possessors in Turkish

Consider possessors in Turkish, which are also marked with GEN (to be discussed further in §6.4.2). Unlike Sakha, Turkish possessors cannot raise out of their NP, but they can scramble out, (7a,b).²¹ In (7c) we see that it is not possible for a GEN-marked possessor to overwrite its case value with ACC after scrambling into the clausal CAD. If GEN on possessors were unmarked case, we would expect (7c) to be acceptable and (7b) to be unacceptable, contrary to fact.

- (7) a. *Ali dün [Can -ın araba -sın] -ı kır -mış.*
 Ali.NOM yesterday John -GEN car -POSS.3SG -ACC break -PRF
 ‘Yesterday Ali broke John’s car.’ *Turkish*
- b. *Ali Can -ın dün [Can -ın araba -sın] -ı kır -mış.*
 Ali.NOM John -GEN yesterday car -POSS.3SG -ACC break -PRF
 ‘Of John’s things, Ali yesterday broke the car.’ *Turkish*
- c. **Ali Can -ı dün [Can -ın araba -sın] -ı kır -mış.*
 Ali.NOM John -ACC yesterday car -POSS.3SG -ACC break -PRF
Turkish

6.3.3.4 Raising-to-object in Sakha and Turkish

In addition to possessors, Turkic languages use GEN case to mark subjects of nominalized clauses (argument clauses §6.6.2, sometimes adjunct clauses §6.6.3, and object relative clauses). However, I am not aware of any Turkic language in which a GEN-marked subject can undergo raising-to-object (a CC᳚) and receive ACC, unlike NOM-marked subjects of finite clauses as discussed in §3.4.

Data from Sakha are especially instructive here because subjects of nominalized clauses *can* undergo CC᳚, but crucially Baker (2011) has shown that these subjects are NOM-marked despite occurring in *nominalized* clauses.

²¹ The raising in §6.3.3.2 may differ from the scrambling here in that the former might be an instance of A-movement while the latter is an instance of A'-movement.

Compare (8) from Sakha to (9) from Turkish. (8a,b) show that the subject of a nominalized clause in Sakha must bear NOM and cannot bear GEN. In (8c) we see that the subject of a nominalized clause can move into the matrix clause and receive ACC-marking in a CC₂ configuration.

- (8) a. [[*Masha aqa -ta*]-Ø *bu miin -i söbül(ee) -üür -e*
Masha.GEN father -POSS.3SG -[NOM] this soup -ACC like -AOR -POSS.3SG
]-Ø *ücügej.*
-NOM good
‘It is good that Masha’s father likes this soup.’ Sakha²²
- b. ?*[[*Masha aqa -ty*]-n *bu miin -i söbül(ee) -üür -e*
Masha.GEN father -POSS.3SG -[GEN] this soup -ACC like -AOR -POSS.3SG
]-Ø *ücügej.*
-NOM good Sakha²³
- c. *Min ehigi -ni* [*ehigi бүгүн kyaj -byk -kyt*]-yn *ihit -ti -m.*
me.NOM y’all -[ACC] y’all.NOM today win -PTCP -2PL.POSS -ACC hear -PST -1SG
‘I heard y’all won today.’ Sakha²⁴

Turkish (9a,b) shows that the subject of a nominalized argument clause must bear GEN and cannot bear NOM. As discussed in §3.4, subjects of nominalizations cannot move into the matrix clause in Turkish to receive ACC, (9c).

- (9) a. [[*Maşa -nın baba -sın*]-ın *bu çorba -yı sev -diğ -i*]-Ø
Masha -GEN father -POSS.3SG -[GEN] this soup -ACC love -PTCP -POSS.3SG -NOM
iyi.
good
‘It is good that Masha’s father loves this soup.’ Turkish
- b. *[[*Maşa -nın baba -sı*]-Ø *bu çorba -yı sev -diğ -i*]-Ø
Masha -GEN father -POSS.3SG -[NOM] this soup -ACC love -PTCP -POSS.3SG -NOM
iyi.
good Turkish
- c. *Ben [*Maşa -nın baba -sın*]₁-ı [*t₁ bu çorba -yı sev -diğ -in*
me Masha -GEN father -POSS.3SG -[ACC] this soup -ACC love -PTCP -POSS.3SG

^{22,23} Baker 2011: p. 1175, ex. (23b)

²⁴ Baker & Vinokurova 2010: p. 615, ex. (39b)

] -i duy -du -m.

-ACC hear -PST -1SG

intended: ‘I heard that Masha’s father loved this soup.’

Turkish

These data show that NOM can be overwritten by ACC, but GEN cannot, indicating that GEN is not the realization of unvalued case.

6.3.3.5 Summary

In this section I have provided evidence that neither the GEN-marking on possessors in Sakha (§6.3.3.2) or Turkish (§6.3.3.3), nor the GEN-marking on subjects of nominalizations in Turkish (§6.3.3.4) can be overwritten by a different case marking after movement into the clausal case-assignment domain. We have seen previously that NOM-marked nominals can change their case marking, and this is illustrated again in §6.3.3.4 for NOM-marked subjects of nominalizations in Sakha, which differ minimally from Turkish subjects of nominalizations both in their case marking and in their ability to undergo raising-to-object and receive ACC-marking in this CC₂-configuration.

6.3.4 Summary

In this section I have argued that, following Preminger’s (2014) proposal that unmarked case is unvalued case and Baker’s (2015b) matching observations that unmarked case can be overwritten with dependent case, we see that in Turkic, NOM case can be overwritten, i.e., valued after movement into a different case-assignment domain or case-competition configuration (§6.3.2), and GEN case cannot be overwritten after such movement (§6.3.3). Having tested this overwritability with some of the most structural instances of GEN (possessors, §6.3.3.2 and §6.3.3.3, and subjects of nominalizations with possessive agreement tracking the subject, §6.3.3.4), I propose to conclude that not even these genitives are morphological realizations of unvalued case [case: ____] — thus, GEN is never an unmarked case in Turkic.

6.4 Structural genitive is not an inherent case

6.4.1 Introduction

In the literature on GEN in Turkic it is often proposed that GEN is assigned by the functional head POSS that attaches to the possessum and agrees with the possessor, spelling out the possessor's φ -features. This is motivated in part by theoretical considerations where case is a reflex of Agree (§2.3), and in part by the fact that GEN almost always occurs in the vicinity of the POSS head. In this section, I discuss two constructions where GEN appears without a POSS-head. From this it follows that in these constructions, GEN cannot be assigned as a reflex of agreement with POSS, and thus economy considerations on our case theory suggest that it may not be assigned in this way in other constructions either. Furthermore, in §6.5.3 I will present data from Gribanova (2017) on Uzbek where POSS agrees with one NP while GEN appears on another NP.

In §6.4.2 I present regular possessive constructions in Turkic, where we see the cooccurrence of GEN and POSS. In §6.4.3 I discuss the first construction where GEN appears without POSS: contextually salient possessors that do not fill a semantic argument slot of the head noun can appear as adjuncts to the head noun. The dependent-up analysis for GEN that I will present in §6.5 accounts for this GEN-marking. §6.4.4 presents the second construction where GEN appears without POSS: in predicative constructions with a strict ownership reading ('belongs to'), the possessor is marked with GEN in the absence of a possessum and of POSS. I leave open the nature of the assignment of these genitives.

6.4.2 Possessive constructions in Turkic

Turkic possessive constructions are formed by GEN-marking the possessor and adding an agreement suffix of the possessive paradigm (glossed as POSS) to the possessum, which copies the possessor's person and number features.

- (10) a. *biz -im kitab -ımız*
 we -GEN book -POSS.1PL
 ‘our book’ Turkish
- b. *pro kitab -ımız*
 we -GEN book -POSS.1PL
 ‘our book’ Turkish
- c. *Ali -nin kitab -ı*
 Ali -GEN book -POSS.3SG
 ‘Ali’s book’ Turkish

As in many other languages, these constructions can be used for a wide variety of thematic relationships between possessor and possessum, such as the following given in Öztürk & Taylan (2016) (after Vikner & Jensen 2002):

- (i) kinship terms (*the child’s sister*),
- (ii) verb-related nouns (*the building’s destruction*),
- (iii) other inherently relational nouns (*this country’s queen, Oliver’s birthday*),
- (iv) dependent part-whole relations (*the table’s surface*),
- (v) autonomous part-whole relations (*the car’s wheel*),
- (vi) body parts (*my nose*),
- (vii) telicity (*Susi’s festschrift*),
- (viii) location (*Istanbul’s museums*),
- (ix) temporal relations (*yesterday’s news*),
- (x) agentivity (*the author’s poem, the photographer’s picture, the murderer’s victim*),
- (xi) THEME (*the model’s picture, the victim’s murderer*),
- (xii) control (*this customer’s rental car, my copy of this book*),

and surely others. If the GEN-marking on possessors were inherent case, we would expect all possessors to have the same θ -role, but clearly they don't. This is the first argument against GEN being an inherent case.

There are two similar constructions that are often discussed in comparison to these possessives and that many linguists assume are derivationally linked to possessives: 1. compounds, which lack the GEN-marking, but keep what looks like a 3SG possessive agreement suffix on the head noun (I will return to compounds in §6.5.3); and 2. alternate possessive constructions with GEN-marked possessors but without the agreement suffix (§6.4.3). Öztürk & Taylan (2016) have shown that the latter can only express the thematic relations (v) through (xii), i.e. not (i) to (iv), which are semantically transitive (Vikner & Jensen 2002). Given that in these possessive constructions, the GEN always appears together with a possessive agreement suffix, it has often been proposed that it is the agreement suffix that assigns GEN to the possessor, much like T has been argued to assign NOM to the subject, and *v* to assign ACC to the internal argument. If the possessive agreement suffix were the source of the possessor's GEN-marking, this GEN might be an inherent case, if one can argue around the variety of θ -roles assigned to possessors. However, in the next two subsections (§6.4.3, §6.4.4) I will present two constructions where GEN appears without the possessive agreement suffix. Thus, we need to account for these genitives differently, and then we might as well use that approach for all genitives, making the inherent case analysis superfluous because it cannot cover all assignments of structural GEN.

6.4.3 Adjoined possessors

In addition to the regular possessive construction introduced in §6.4.2 above, repeated in (11a), there are also possessive constructions without an agreement suffix, while the GEN-marked possessor remains, (11b).

- (11) a. *sen -in kitab -ın*
 you -GEN book -POSS.2SG
 ‘your book’

Turkish

- b. *sen -in kitap*
 you -GEN book
 ‘your book’

Turkish

This construction is often discarded in the literature on possessives as a colloquial abbreviation (cf. Öztürk, Taylan & Zimmer 2015), but Öztürk & Taylan (2016) have shown that there are interesting constraints on the absence of the agreement suffix: The construction in (12b), labeled “possessive-free genitives” by Öztürk & Taylan (2016), cannot appear with head nouns that require an argument; i.e., it cannot appear with inherently semantically transitive nouns that require a relation to their argument of types (i) kinship, (12a), (ii) verb-related nouns, (12b), (iii) relational nouns, (12c), and (iv) dependent part-whole relations, (12d), as discussed above. All these examples would be grammatical with the agreement suffix *-(s)I(n)*.

- (12) a. **öğretmen -in hala*
 teacher -GEN aunt
 intended: ‘the teacher’s paternal aunt’ (kinship) *Turkish*²⁵
- b. **bina -nın yıkım*
 building -GEN destruction
 intended: ‘the demolition of the building’ (verb-related) *Turkish*²⁶
- c. **makale -nin başlık*
 article -GEN title
 intended: ‘the title of the article’ (relational noun) *Turkish*²⁷
- d. **masa -nın kenar*
 table -GEN edge
 intended: ‘the edge of the table’ (dependent part-whole) *Turkish*²⁸

This contrasts with the other types of relations between nouns and possessors listed in §6.4.2 above, some of which are illustrated in (13):

²⁵ Öztürk & Taylan 2016: p. 92, ex. (11a)

²⁶ Öztürk & Taylan 2016: p. 92, ex. (11c)

²⁷ Öztürk & Taylan 2016: p. 92, ex. (11b)

²⁸ Öztürk & Taylan 2016: p. 92, ex. (12c)

- (13) a. *araba -nın lastik*
 car -GEN tire
 ‘the car’s tire’ (autonomous part-whole, (v)) *Turkish*²⁹
- b. *çocuğ -un burun*
 child -GEN nose
 ‘the child’s nose’ (body part, (vi)) *Turkish*³⁰
- c. *çocuğ -un şiir*
 child -GEN poem
 ‘the child’s poem’ (agentive, (x)) *Turkish*³¹
- d. *kadın -ın araba*
 woman -GEN car
 ‘the woman’s car’ (control, (xii)) *Turkish*³²

Thus, possessive-free genitives are only possible when the head noun does not have a transitive argument structure (and when certain presuppositions are fulfilled, Öztürk, Taylan & Zimmer 2015). Öztürk & Taylan (2016) argue that because nouns cannot project their own argument structure, a semantically transitive head noun requires the functional head $-(s)I(n)$ to project the syntactic position of the argument. The nouns in (12) are such that their semantics require an argument, thus $-(s)I(n)$ is required — the unacceptability of the sentences in (12) is due to the same reason as the unacceptability of sentences like (14).

- (14) **John devoured.* *English*

Öztürk & Taylan (2016) take this to indicate that GEN-marked NPs in (14) are not arguments but modifiers, and are syntactically adjuncts (Fig. 6.1). This accounts for the following distributional difference between these genitives and the genitives in possessives: Adjunct-genitives can precede, (15a), or follow, (15b), demonstratives (which are typically analyzed as adjuncts in Turkish, e.g. Bošković & Şener 2014), while possessors must precede demonstratives, (15c,d):

²⁹ Öztürk & Taylan 2016: p. 92, ex. (12b)

³⁰ Öztürk & Taylan 2016: p. 92, ex. (12a)

³¹ Öztürk & Taylan 2016: p. 92, ex. (13)

³² Öztürk & Taylan 2016: p. 92, ex. (14)

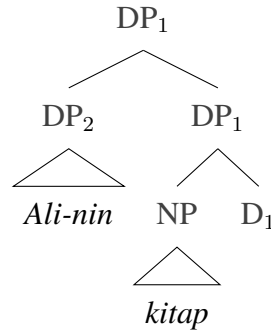


Figure 6.1: The structure of a ‘possessive-free genitive’ proposed by Öztürk & Taylan (2016: p. 105, ex. (55a))

- (15) a. *sen -in o araba*
 you -GEN that car
 ‘that car of yours’ *Turkish*³³
- b. *o sen -in araba*
 that you -GEN car
 ‘that car of yours’ *Turkish*³⁴
- c. *sen -in o araba -n*
 you -GEN that car -POSS.2SG
 ‘that car of yours’ *Turkish*³⁵
- d. **o sen -in araba -n*
 that you -GEN car -POSS.2SG
 intended: ‘that car of yours’ *Turkish*^{36,37}

The GEN-marking on the adjoined possessor cannot be assigned by the possessive agreement head POSS, because that head is not present in possessive-free genitives. I conclude that this GEN cannot be inherent case.

6.4.4 Predicative possessors

The second construction that I’m aware of to contain GEN-marked nominals, but no functional head like POSS that might be responsible for the GEN-marking, are predicative possessors, (16).

³³ Öztürk, Taylan & Zimmer 2015: p. 194, ex. (17b)

³⁴ Öztürk, Taylan & Zimmer 2015: p. 194, ex. (17a)

³⁵ Öztürk, Taylan & Zimmer 2015: p. 194, ex. (18b)

³⁶ Öztürk, Taylan & Zimmer 2015: p. 194, ex. (18a)

³⁷ Note that this contrasts with the judgment given in Bošković & Şener (2014: p. 112, ex. (27a)).

- (16) a. *Bu kitap Ali -nin.*
 this book.NOM Ali -GEN
 ‘This book is Ali’s.’ resp. ‘This book belongs to Ali.’ *Turkish*
- b. *Bu kalem ben -im.*
 this pencil.NOM me -GEN
 ‘This pencil is mine.’ resp. ‘This pencil belongs to me.’ *Turkish*³⁸

One might ask whether we can be sure that there is no silent POSS-head present that has perhaps been elided after assigning GEN under a deletion-at-LF analysis of ellipsis. I will proceed to show that this is not the case for the genitives in (16).

It can be observed cross-linguistically that predicative possessives fall into two classes: those that also project the possessum in some form, and those that do not. These two classes also differ in their possible interpretations, and many languages differentiate between them morphologically, although crucially English doesn’t.

Class 1, involving some projection of the possessum, can receive an interpretation of the relation between possessor and silent possessum that matches the context-dependent flexibility of this relation in regular possessives. That is, in the examples below, ‘This country is mine’ and ‘These children are mine’ can be uttered by a newly naturalized citizen resp. a nanny on a playground, which constitutes context-dependent relations between possessor and possessum, and not only a strict ownership reading. Class 2 does not contain the projection of a regular possessive, but contains only a predicate of type $\langle e, t \rangle$ with strict possessive semantics in the sense of ownership, paraphrasable as ‘belongs to’.

In English, both classes are realized morphologically by strong pronouns such as *mine* or possessives such as *John’s*. I refer the reader to Zribi-Hertz (1997) for detailed discussions of predicative possessives English and on using the distribution of *own* as a diagnostic between them.

In some other languages these two classes *are* differentiated morphologically. Polish differentiates between equative NP-*be*-NP constructions, where the predicate NP requires³⁹ INS case, and

³⁸ Öztürk & Taylan 2016: p. 93, ex. (16)

³⁹ Bondaruk (2013: §5) notes that NP-*be*-NP.NOM constructions with the predicate NP marked NOM do exist but are “heavily constrained” and have (perhaps thusly) not been explored much in previous literature. She writes, “[t]hey typically appear with 1st and 2nd person subjects and the nominative predicates found in this type of clause represent

predicational NP-*be*-A constructions, where the adjective requires⁴⁰ NOM case. If ellipsis has occurred in an equative NP-*be*-NP construction, the stranded adjective will appear in INS case, (17a). If the adjective appears in NOM, (17b), no elided NP can have been present. The interpretation of these possessives lines up with this diagnostic for ellipsis: (17a) might be uttered by a person who changed their citizenship away from the country under discussion, thus in a relatively free context-dependent relation between speaker (possessor) and country (possessum). Aneta Gąsiorek (p.c.) reports that empress Sisi of Austria-Hungary might utter (17a) about Bavaria, where she was born. This contrasts with the non-ellipsis construction in (17b), which only has a strict ownership-reading: Empress Sisi could not utter (17b) about Bavaria, but crucially Austrian-Hungarian emperor Franz Joseph I might utter (17b) about parts of Italy, which he lost the rule of in 1866.

- (17) a. *Ten kraj był kiedyś moim (krajem).*
 that.M.SG.NOM country(M).SG.NOM was.M.SG once my.M.SG.INS (country(M).SG.INS)
 ‘That country was once mine/my country.’ *Polish*⁴¹
- b. *Ten kraj był kiedyś mój.*
 that.M.SG.NOM country(M).SG.NOM was.M.SG once my.M.SG.NOM
 ‘That country was once mine.’ *Polish*^{42,43}

a limited class, covering professions, nationalities, titles and emotive expressions. [...] Taking into account some syntactic tests, such as lifetime effects, aspect [including stage-level predicates] and the small clause complement of a lexical verb, proposed for other languages by Roy (2006), [it has been][I have] concluded that *być* [‘to be’] clauses with the instrumental DP predicate represent *characterising* predicational clauses, while the ones with the nominative DP predicate belong to the class of *defining* predicational structures.” (Bondaruk 2013: p. 210) [emphasis mine, SL]

⁴⁰ Bondaruk (2013: p. 202, fn. 25) notes that Witkoś (2010) and Kallas (1993) have reported constructions where a raising or control matrix predicate contains a NP-*be*-A structure with the A marked INS, but that both authors have suggested that these adjectives are attributive adjectives with an elided noun.

⁴¹ Partee & Borschev 2001: p. 108, ex. (34a, b)

⁴² Partee & Borschev 2001: p. 108, ex. (35a)

⁴³ For the fourth sentence in the paradigm, i.e., with a full NP in nominative case, consider the following scenario and the minimally different sentence (i): Franz Joseph I time-travels and visits modern Slovenia and utters about this now-independent country that it was once a region in his empire.

- (i) *#Ten kraj był kiedyś mój region.*
 that.M.SG.NOM country(M).SG.NOM was.M.SG once my.M.SG.NOM region
 ‘That country was once my region.’ *Polish*

Aneta Gąsiorek (p.c.) reports that the inappropriateness of uttering (i) in this context comes from the fact that it presupposes the uniqueness of a region possessed by Franz Joseph I, contrary to fact. I understand this to mean that my consultant is trying to construct an equative reading — perhaps because the “heavy constraints” for a predicational

The following Polish examples are constructed after a comment in Partee & Borschev (2001: p. 112, endnote 12), where Janusz Bien is credited with the observation that Polish *dom* can denote either the common noun ‘house’ or the inherently relational (semantically transitive) noun ‘home’. Aneta Gąsiorek (p.c.) reports that in a context where the speaker inherited a house, never lived in it, i.e., never considered it their home, and then sold it, (18a–c) meaning ‘my home’ are not acceptable — perhaps because by adding a possessor to *dom*, the relational reading ‘home’ becomes so very salient that the interpretation ‘my house’ is not available —, while (18d) indicating pure possession of a ‘house’ is felicitous.

- (18) a. #*Ten dom był kiedyś moim domem.*
 that.M.SG.NOM house(M).SG.NOM was.M.SG once my.M.SG.INS home(M).SG.INS
 ‘That house was once my home.’ Polish

Speaker comment: “This is literally false.”

- b. ??*Ten dom był kiedyś moim.*
 that.M.SG.NOM house(M).SG.NOM was.M.SG once my.M.SG.INS
 (i) #‘That house was once my home.’ Polish

Speaker comment: “This is literally false in this context.”

- (ii) ??‘That house was once my house.’

Speaker comment: “This is weird because you should have used (18d).”

- c. #*Ten budynek był kiedyś mój dom.*
 that.M.SG.NOM building(M).SG.NOM was.M.SG once my.M.SG.NOM home(M).SG.NOM
 (i) #‘That building was once my home.’ Polish

Speaker comment: “grammatically fine, but doesn’t work in any context I can think of”⁴⁴

- (ii) ^{ok}‘[Pointing to a ruin] That building was once the house that I owned.’

- d. *Ten dom był kiedyś mój.*
 that.M.SG.NOM house(M).SG.NOM was.M.SG once my.M.SG.NOM
 ‘That house was once mine.’ Polish

Speaker comment: “This is literally true because my name was on the deed.”

NP-*be*-NP.NOM construction reported by Bondaruk (2013: §5) are not met, and/or to make sense of the fact that the speaker chose to not use the predication NP-*be*-NP.INS construction — and fails in this context.

⁴⁴ This is expected under Bondaruk’s (2013) proposal because a *być*-NP.NOM construction is an equative and cannot

In German as well, NP-ellipsis can be seen because it leaves a remnant on the stranded adjectives, which surfaces as inflection morphology. Thus, (19a), having undergone NP-ellipsis can have the same context-flexible interpretations as (19b) with the overt NP, while (19c), whose predicate is a pure adjective of type $\langle e, t \rangle$, which is not inflected for agreement as are all adjectives in predicative position in German, has only the narrow ownership reading.

- (19) a. *Das Land ist jetzt mein -es.*
 this.N.SG.NOM country(N).SG is now mine -N.SG.NOM
 ‘This country is now mine.’ German⁴⁵
- (i) ^{ok} ... because I now have its citizenship.
 (ii) ^{ok} ... because I just conquered it.
- b. *Das Land ist jetzt mein Land.*
 this.N.SG.NOM country(N).SG is now my.N.SG.NOM country(N).SG
 ‘This country is now my country.’ German
- (i) ^{ok} ... because I now have its citizenship.
 (ii) ^{ok} ... because I just conquered it.
- c. *Das Land ist jetzt mein.*
 this.N.SG.NOM country(N).SG is now mine
 ‘This country is now mine.’ German⁴⁶
- (i) # ... because I now have its citizenship.
 (ii) ^{ok} ... because I just conquered it.

Partee & Borschev (2001: p. 105) report that (19c) is of high register and has an archaic flavor. However, it surfaces naturally in the lyrics of a song about the colonization of North America by Europeans when the invaders proclaim possession of the land, (20).

- (20) a. *Mein, Jungs, mein sind die Berge.*
 mine boys mine are the mountains
 ‘Mine, boys, mine ev’ry mountain’⁴⁷ German⁴⁸

co-occur with temporal restrictions (Bondaruk 2013: pp. 155, 215).

⁴⁵ After Partee & Borschev 2001: p. 106, ex. (29a)

⁴⁶ Partee & Borschev 2001: p. 106, ex. (29b)

⁴⁷ The German translation cannot reproduce the English play-upon-words that builds on the homophony between pronominal *mine* and the verb *to mine*.

⁴⁸ Lyrics from the song “Mine, mine, mine” (lyrics by Stephen Schwartz) in the Walt Disney movie “Pocahontas”

- b. *Gold, es ist mein, mein, mein.*
gold it is mine mine mine
'It's gold and it's mine, mine, mine' *German*⁴⁹
- c. *Mein, mein, mein wird das alles.*
mine mine mine will.become this all
'It's mine, mine, mine for the taking' *German*⁵⁰
- d. *All dieses Gold ist mein.*
all this gold is mine
'When all of the gold is mine' *German*⁵¹
- e. *Das größte Erlebnis wär mein.*
the greatest adventure would.be mine
'The greatest adventure is mine' *German*⁵²

Similar data are available for Russian (Babby 1973; Partee & Borschev 2001; Siegel 1976), French (Tremblay 1989, 1991), Dutch (Partee & Borschev 2001: p. 111, endnote 6), and Latin (Zribi-Hertz 1997: p. 529).

Returning to Turkish, Hankamer (2004) and Bošković & Şener (2014: §3.2.1, pp. 131–134) have observed that Turkish predicative possessives, as in (16) above, do not involve projection of the head noun, because such a projection, when silent, would be replaced morphologically with the suffix *-ki(n)*, comparable to English *one*:

- (21) *Pamuk -un kitab -ın -ı oku -du -m, ama Auster -ın -kin -i oku -ma*
Pamuk -GEN book -POSS.3SG -ACC read -PST -1SG but Auster -GEN -one -ACC read -NEG
-dı -m.
-PST -1SG
'I read the book by Pamuk, but I didn't read the one by Auster.' *Turkish*⁵³

Like the first possessive in (21), 'Pamuk's book', the second possessive with *-ki(n)*, literally 'Auster's one', can have a context-relative interpretation, such as authorship, proximity, control, or any other

(1995) following the translation into German ("Mein, mein, mein") by Frank Lenart und Lutz Riedel, according to the Wikipedia page [https://de.wikipedia.org/wiki/Pocahontas_\(Film\)](https://de.wikipedia.org/wiki/Pocahontas_(Film)) accessed on 2024-10-06. The translations in (20) follow the respective verse in the English lyrics.

^{49,50,51,52} Lyrics from the song "Mine, mine, mine" (lyrics by Stephen Schwartz) in the Walt Disney movie "Pocahontas" (1995) following the translation into German ("Mein, mein, mein") by Frank Lenart und Lutz Riedel, according to the Wikipedia page [https://de.wikipedia.org/wiki/Pocahontas_\(Film\)](https://de.wikipedia.org/wiki/Pocahontas_(Film)) accessed on 2024-10-06. The translations in (20) follow the respective verse in the English lyrics.

⁵³ After Bošković & Şener 2014: p. 131, ex. (70)

context-defined relation, including, but crucially not limited to, ownership. In (22), *-ki(n)* is even obligatory because the context is such that the owner is the speaker, and the predicative possessor thus must have a non-ownership relation to the possessum.

(22) Context: I know my friends are sending me presents for my birthday. I come home and find a number of presents on the table. Having opened one of them, I say:

a. *#Bu (hediye) Pelin -in ol -mal.*
 this present.NOM Pelin -GEN be -NECESS
 ‘This (present) must belong to Pelin.’ *Turkish*⁵⁴

b. *Bu (hediye) Pelin -in -ki ol -mal.*
 this present.NOM Pelin -GEN -one be -NECESS
 ‘This (present) must be Pelin’s.’ *Turkish*⁵⁵

(23) and (24) express a strict ownership reading and as such, *-ki(n)* is not possible. Bošković & Şener (2014: p. 133) draw the conclusion that stranded possessors without *-ki(n)* have only a possessive reading, not a relational reading.

(23) *Pelin -in bu kitab -ı aslında Mete -nin -(*ki).*
 Pelin -GEN this book -POSS.3SG.NOM in.fact Mete -GEN -(*one)
 ‘This book of Pelin’s actually belongs to Mete.’ *Turkish*⁵⁶

(24) *Borç -lar -ı bit -tik -ten sonra ev ben -im -(*ki) ol -acak.*
 debt -PL -ACC end -PTCP -ABL after house.NOM me -GEN -(*one) be -FUT
 ‘After I pay off the mortgage, the house will be mine.’ *Turkish*⁵⁷

This illustrates that the two classes of predicative possessives also exist in Turkish and that, like in many other languages, if the possessum is not projected in the syntax, as evidenced by the absence of *-ki(n)*, the interpretation of the possession-relation is limited to strict ownership.

From this alone, it does not straight-forwardly follow that the possessive agreement suffix that attaches to the possessum in regular possessives (§6.4.2) is also absent. If present but silent, it might still account for the GEN-marking as inherent case. The argument that the POSS head, too,

^{54,55} Bošković & Şener 2014: p. 133, ex. (75), and the text on the bottom of p. 133.

⁵⁶ Bošković & Şener 2014: p. 133, ex. (77)

⁵⁷ Bošković & Şener 2014: p. 133, ex. (78b)

is absent in *-ki(n)*-less predicative possessives, comes from the fact that these possessives are not only restricted to a pure-ownership reading, but also obey the same restrictions as possessive-free genitives as discussed in §6.4.3 above: They cannot appear with a possessum that is inherently semantically transitive in selecting a possessor, (25).⁵⁸

- (25) a. **On-un baba -sı Ali -nin de.*
 he-GEN father -POSS.3SG.NOM Ali -GEN too
 ‘His father is also Ali’s.’ (kinship)
 cp. English *#His father belongs to Ali.* *Turkish*⁵⁹
- b. *#Bu katil kadın -ın.*
 this murderer.NOM woman -GEN
 ‘This murderer is the woman’s.’ *Turkish*⁶⁰
- (i) ... is the murderer associated with the woman, e.g., who the woman hired
 (ii) *... is the murderer who killed the woman (verb-related)
- c. *ØŞu başkent Türkiye -nin -dir.*
 this capital.NOM Türkiye -GEN -COP
 intended: ‘This capital is Türkiye’s.’ (relational noun) *Turkish*
- d. *ØBu kenar masa -nın.*
 this edge.NOM table -GEN
 intended: ‘This edge is the table’s.’ (dependent part-whole) *Turkish*⁶¹

From (25) it follows that there is no POSS head present in these constructions that could be responsible for assigning the GEN-marking on the owner. I do not know where this GEN comes from,

⁵⁸ If this is a semantic restriction, this should hold for the same construction in the other languages discussed above as well. Testing this prediction is outside the scope of this thesis.

⁵⁹ Öztürk & Taylan 2016: p. 94, ex. (18b)

⁶⁰ Öztürk & Taylan 2016: p. 93, ex. (17a)

⁶¹ Constructed after Öztürk & Taylan 2016: p. 92, ex. (12c) and the discussion on (p. 94) including (p. 94, fn. 3). Predicted to be unacceptable.

but crucially for my purposes here, it is not inherent case assigned by POSS.^{62,63,64,65}

6.4.5 Summary

In this section I have argued that GEN is not an inherent case assigned by the functional head POSS to its specifier together with a thematic interpretation as possessor. While most genitives occur together with POSS, crucially not all do, as I've discussed in §6.4.3 and §6.4.4. An analysis of GEN-assignment that accounts for all appearances of GEN is on philosophical grounds preferable to a collection of different analyses for different constructions. I now proceed to lay out my proposed analysis of GEN as the dependent-up case in the nominal domain.⁶⁶

⁶² Given the narrow meaning of this predicate, Freeze (1992) and Larson & Cho (1999, 2003), a.o., have proposed that this possessive relation is introduced by a silent adposition P with a meaning like 'to' or 'of'. Perhaps it is this P that assigns the GEN case to the owner. Then we would be dealing with lexical GEN that, crucially, would be different from the lexical GEN assigned by other adpositions (§6.2), because the GEN-marking on predicative possessors does not follow the allomorphy pattern described in §6.2 for overt adpositions. To repeat, (16a) above, repeated as (ia), contrasts with the allomorphy of lexical GEN assigned by overt adpositions in (ib).

- | | | | |
|-----|----|--|----------------|
| (i) | a. | <i>Bu kitap Ali</i> { ^{ok} - <i>nin</i> / *-Ø }.
this book Ali -GEN
'This book is Ali's.' | <i>Turkish</i> |
| | b. | <i>Ali</i> { *- <i>nin</i> / ^{ok} -Ø } <i>için</i>
Ali -GEN for
'for Ali' | <i>Turkish</i> |

I conclude from this that the GEN-marking on predicative possessors is not lexical GEN.

⁶³ It also cannot be dependent-up case, as I claim for other structural genitives in this thesis, because I have just shown in detail that there is no competing NP present in these constructions.

⁶⁴ Tremblay (1989, 1991) argues that the corresponding DAT in French is a default case. Perhaps this GEN in Turkish could be argued to be an unmarked case, but at this point I do not see how this might work out.

⁶⁵ Zribi-Hertz (1997) suggests that the English suffix 's when denoting ownership-only predicative possessors is a derivational suffix added in the lexicon (in contrast to the functional head 's that appears in context-flexible possessives). She presents some arguments that the functional 's attaches to phrases while the derivational 's attaches to heads. I doubt that this approach would work for Turkish -(n)In, but I list it here for completeness.

⁶⁶ I do not see how my proposed analysis can cover the data on predicative possessors in §6.4.4 and so I leave this problem open. Crucially, however, I don't know of any another analysis that covers predicative possessors and the remaining instances of structural GEN. My analysis does cover the data in §6.4.3, which would require an independent explanation if structural GEN were assigned by POSS.

6.5 Structural genitive is a dependent-up case

6.5.1 Introduction

In this section, I present my arguments that structural GEN in Turkic is a dependent case, more precisely the dependent-up case in the nominal CAD. The argument proceeds in two steps: First, in §6.5.2, I argue that it is possible for the GEN-marking on possessors to be dependent case, by presenting Baker's (2015a) proposal, which argues exactly that (although not for Turkic languages). In a nutshell: dependent-up GEN is assigned in a case-competition configuration where the case-competitor is the possessum.

Then, in §6.5.3, I argue that it is necessary for structural GEN to be a dependent case. If we assume Bobaljik's (2008) Revised Moravcsik (1974, 1978) Hierarchy about which case-marking nominals are accessible to an agreement probe, it follows from the agreement patterns of possessive agreement in Turkic that structural GEN *must* be a dependent case.

§6.5.4 summarizes this discussion.

6.5.2 Structural genitive is assigned to the possessor by competition against the possessum

Baker (2015a) discusses the assignment of dependent cases inside of nominal domains. He argues that the case visible on possessors, typically labeled 'genitive', can vary between languages as (i) unmarked case (e.g. in Japanese and Tamil), (ii) dependent-up case (e.g. in Shipibo, Burushaski, Kalaallisut, Nez Perce), or (iii) inherent case (such as in Sakha, Baker & Vinokurova argue). Baker's (2015a) main reason to propose that GEN can be a dependent-up case, comes from ergative or tripartite languages that use ERG not only to mark subjects of transitive clauses, but also to mark possessors; i.e. these languages have systematic syncretism between ERG and GEN⁶⁷, (26).

⁶⁷ With a few exceptions: In Shipibo, ERG and GEN are distinct for the 1SG and 3SG pronouns (Baker 2015a: p. 24, fn. 6), and the same suffix can additionally also mark INS (cf. (26)), intercessive, locative/allative, and temporal (Valenzuela 2010: p. 68–70). In Burushaski, ERG and GEN are distinct on feminine nominals (Baker 2015a: p. 25, comment in ex. (17b)).

- (26) *Cesar* -nin -*ra* [*Maria* -nin *wai*] *rera* -*ke machito* -*nin*.
 Cesar -ERG -PRT Maria -GEN field.ABS fell.tree -PRF machete -INS
 ‘Cesar cleared Maria’s field with a machete.’ *Shipibo*⁶⁸

Baker, having argued at length that ERG is a dependent-up case (Baker 2014b; Baker & Bobaljik 2017), takes this syncretism to mean that in these languages, the case on possessors is also a dependent-up case. He observes that similarly to ERG, this GEN cannot appear more than once in the same nominal phrase, (27), unlike the GEN in Japanese and Tamil, (28), which Baker takes to mean that GEN is an unmarked case there, assigned in the absence of other case-assignment rules.

- (27) a. *Cesar* -*nin foto*
 Cesar -GEN picture
 ‘Cesar’s picture’ *Shipibo*⁶⁹
- b. **Cesar* -*nin Maria* -*nin foto*
 Cesar -GEN Maria -GEN foto
 intended: ‘Cesar’s picture of Maria’ *Shipibo*⁷⁰
- (28) a. *yuubokumiN* -*no toshi* -*no hakai*
 nomad -GEN city -GEN destruction
 ‘the nomad’s destruction of the city’ *Japanese*⁷¹
- b. *John* -*oota Mary* -*oota padam*
 John -GEN Mary -GEN picture
 ‘John’s picture of Mary’ *Tamil*⁷²

Baker (2015a) argues that this dependent-up GEN is assigned by case-competition of the possessor against the possessum. This opens the immediate question whether the same case-competition shouldn’t also yield dependent-down (ACC) case on the possessum in the presence of a possessor in languages with a dependent-down case defined for the nominal domain. Baker (2015a) investigates the possibility that it might but never gets realized morphologically because of the following case-marking of the possessum’s maximal projection in the surrounding clause, but discards this possibility because then we might expect *some* languages to show it, such as Korean, which has

⁶⁸ Baker 2015a: p. 24, ex. (15)

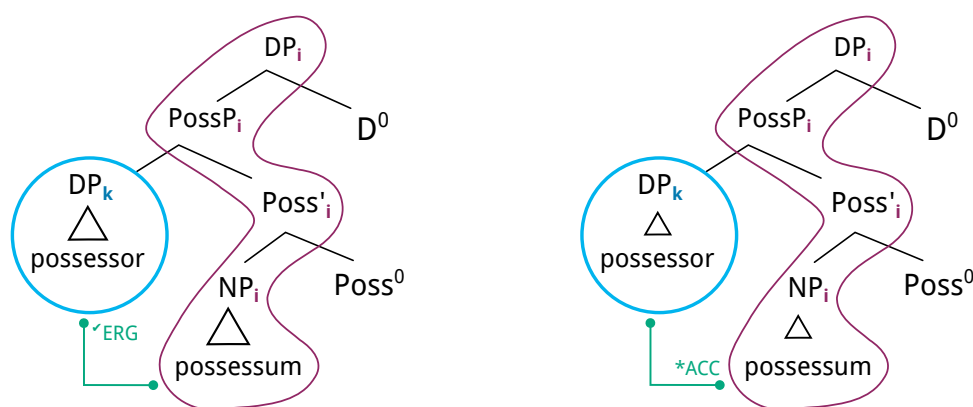
^{69,70} After Baker 2015a: p. 29, ex. (23)

⁷¹ Baker 2015a: p. 22, ex. (12b)

⁷² Baker 2015a: p. 23, ex. (13)

case-stacking, or languages like Turkic where NOM is morphologically null, such that the underlying ACC might surface.

Instead, he argues, the possessor and possessum are differentiated by a crucial property: at the time of the case-competition, the possessor is already a maximal nominal, in particular a maximal projection with its assigned referential index, while the possessum is not yet maximal. He proposes that the rules for dependent case should be modified such that the phrase receiving dependent case needs to be the maximal projection of its referential index, while the case competitor does not need to be maximal. This is illustrated in Fig. 6.2. This proposal is compatible with different



(a) ERG can be assigned because the receiving nominal is maximal. (b) ACC cannot be assigned because the receiving nominal is not maximal.

Figure 6.2: The case-competition between possessor and possessum according to Baker (2015a: p. 33, ex. (29))

assumptions about the exact timing of case-competition: It might occur when the possessor is merged into the larger nominal, i.e., at the node labeled $PossP$. It might also occur at the Spell-out of the domain, i.e., at Merge of the next highest phase head. This might be D , yielding the same result. Additionally, $Poss$ might also be a phase head, as Baker (2015a) proposes. This would also work out: In the Spell-out domain of $Poss$, labeled NP , no case assignment takes place because there are no maximal nominals. Subsequently, in the Spell-out domain of D , case assignment proceeds as described above, with the NP (the entire lower Spell-out domain) acting as case-competitor for ERG.

Baker (2015a) argues for the phasal status of $Poss$ because of the case patterns of nouns like ‘pic-

ture' that take an internal argument: this argument never appears with dependent-down ACC, even though it might be in case-competition with the head noun, and with a possessor if present. Fig. 6.3 illustrates Baker's (2015a) proposal why here, again, dependent-down case cannot be assigned: case-competition between the IA and the possessor is not possible because they are in different Spell-out domains, assuming that Poss is a phase head. Case-competition between the IA and the head noun 'picture' is ruled out, I assume Baker intends, because the c-command relation is not asymmetric.

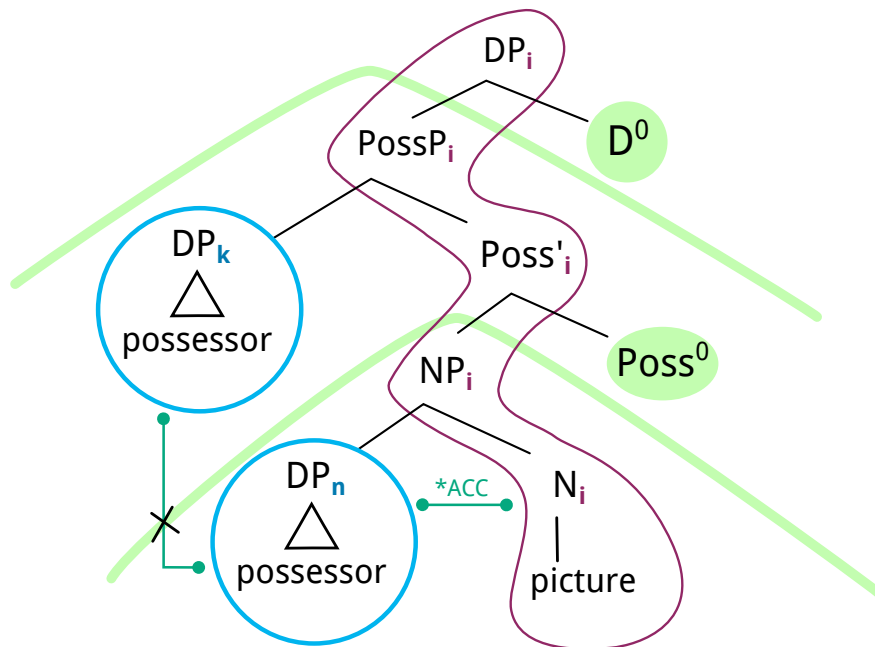


Figure 6.3: The structure of a possessed content noun and why it cannot receive ACC (after Baker 2015a: p. 35, ex. (31))

Thus, Baker (2015a) has shown how dependent-up case assignment in the nominal domain, to the possessor in case-competition against the possessum, would work. He proposes two points of evidence for this: (i) some languages have systematic syncretism between dependent-up *ERG* and this *GEN*, and (ii) the same languages, but also others, do not allow multiple occurrences of this *GEN* in the same nominal domain. In the following section I will use Bobaljik's (2008) insight that the accessibility of cases to agreement follows the (revised) Moravcsik (1974, 1978) hierarchy, to argue that Turkic *GEN* *must* be a dependent-up case.

6.5.3 The accessibility of genitive case to possessive agreement in Turkic

6.5.3.1 Introduction

In this section I will discuss a prediction made by Bobaljik (2008): Since possessive agreement tracks GEN-marked nominals (§6.4.2), if GEN is a dependent case, this agreement should also track nominals marked with unmarked case, i.e., NOM. In §6.5.3.2 I walk through this prediction in detail. Then I present two sets of data where we can see that it is fulfilled: in §6.5.3.3 I discuss three kinds of nominalized clauses where a NOM-marked subject or object is the target for possessive agreement. In §6.5.3.4 I discuss compounds, which at first glance seem like a counter-example since possessive agreement does not track a NOM-marked modifier, but following the detailed analysis by Öztürk & Taylan (2016) we see that the agreement head is absent altogether. Section 6.5.3.5 concludes.

6.5.3.2 The prediction

Bobaljik (2008) has shown that agreement on T can only track ERG-marked NPs (subjects of transitives) if it also tracks ABS-marked NPs (subjects of intransitives, and objects). He argues that it is a parametric setting of languages which NPs are accessible to agreement, and that the possible values of this parameter are along the hierarchy in (29) (after Moravcsik 1974, 1978), and that for whatever value on this hierarchy T is set to agree with, T can also agree with all cases to the left of this value on the hierarchy.

(29) Unmarked case > Dependent case > Lexical case (Bobaljik 2008: p. 303, ex. (13))

Thus, if Turkic GEN is an unmarked case, possessive agreement with a GEN-marked NP would not be surprising. See the discussion in §6.3 on arguments why GEN is not an unmarked case.

If GEN is an inherent or lexical case, there are two possibilities: (A) The agreeing head (Poss) is also the head that assigns GEN: then we expect that the agreement on Poss always copies the φ -features of the GEN-marked nominal. I will show below that this does not hold: In some constructions,

a nominal N_1 receives GEN while POSS agrees with another nominal N_2 . (B) One head (say, P) assigns this inherent/lexical GEN, and another head (say, POSS) can agree with the φ -features of the nominal marked with inherent/lexical GEN. Then (29) predicts that this same head (POSS) can *also* agree with the φ -features of all nominals marked with dependent or unmarked cases. This possibility is tenable on theoretical grounds, but weakly so. If the nominal domain is not defined for any dependent case, and if no additional arguments can be made distinguishing this inherent/lexical GEN from structural GEN, the present argument from the accessibility of case-marking to agreement would make no contribution regarding possibility (B). Claims that the GEN on possessors and subjects of nominalizations is assigned by a functional head abound in the literature — crucially, however, the head that is claimed to assign GEN is POSS, which also displays the agreement, bringing us back to possibility (A). Claiming (B) proper would thus also run counter to common proposals for inherent GEN, making it a hard stance to take, in addition to the problems that come with assuming such a GEN-assigning head ‘P’, which I have never seen any evidence of.

I now return to the evidence against possibility (A). If GEN is a dependent case, its being accessible to the probe on POSS together with the hierarchy in (29) necessarily implies that NPs marked with unmarked case should also be accessible to the probe on POSS. Let us assume that the unmarked case in the nominal domain is NOM.

Two⁷³ constructions come to mind that we can look at to see whether POSS can also agree with NOM-marked NPs. The first construction is nominalized copula clauses, which beautifully illustrate that POSS can in principle agree with the GEN-marked subject or the NOM-marked equative object, and will choose its target based on a person hierarchy, irrespective of the nominals’ case marking. The second construction is compounds which seem to be a counter-example at first glance, but I will argue below that they aren’t. I now turn to both constructions in turn.

⁷³ A potential third construction might be pseudo-incorporated subjects in nominalized clauses, which lose their GEN-marking, to be discussed in §6.6.2. However, I do not at this time know how to test whether there is agreement with the bare subject there or perhaps default 3SG agreement. Subjects can only be pseudo-incorporated when they are highly indefinite and non-specific, and 1 and 2 person pronouns tend to be the opposite and thus resist Differential Subject Marking.

6.5.3.3 Nominalized clauses

The first construction are certain nominalized clauses, namely (i) nominalized copula clauses in Turkish and Uzbek; (ii) nominalized adjunct clauses in Kazakh, Sakha, Turkish⁷⁴, and Uzbek; and (iii) nominalized argument clauses in Sakha⁷⁵. In nominalized copula clauses, (i), we see possessive agreement (Poss) tracking NOM-marked “objects” of the copula, obligatorily so along a person hierarchy; see (30) for Uzbek and (31) for Turkish.

- (30) a. [*Zamira* -(ning) *siz* *e* -kan -(lig) -ingiz -ni] ayt -d -im.
Zamira -(GEN) *you*.NOM *be* -PTCP -(NMLZ) -POSS.2PL -ACC *say* -PST -1SG
 ‘I said that Zamira was you (polite).’ Uzbek^{76,77}
- b. *[*Zamira* -(ning) *siz* *e* -kan -(lig) -i -ni] ayt -d -im.
Zamira -(GEN) *you*.NOM *be* -PTCP -(NMLZ) -POSS.3SG -ACC *say* -PST -1SG
Uzbek^{78,79}
- c. [*Siz* -(ning) *men* *e* -kan -(lig) -im -ni] ayt -di.
you -(GEN) *me*.NOM *be* -PTCP -(NMLZ) -POSS.1SG -ACC *say* -PST.3SG
 ‘S/he said you (polite) were me.’ Uzbek^{80,81,82}
- d. *[*Siz* -(ning) *men* *e* -kan -(lig) -ingiz -ni] ayt -di.
you -(GEN) *me*.NOM *be* -PTCP -(NMLZ) -POSS.2PL -ACC *say* -PST.3SG
Uzbek^{83,84}
- (31) a. *Ben* [*Deniz* -in *siz* *ol* -duğ -unuz -u] bil -iyor -um.
me.NOM *Deniz* -GEN *you*.NOM *be* -PTCP -POSS.2PL -ACC *know* -PROG -1SG
 ‘I know that Deniz is you (polite).’ Turkish⁸⁵
- b. **Ben* [*Deniz* -in *siz* *ol* -duğ -un -u] bil -iyor -um.
me.NOM *Deniz* -GEN *you*.NOM *be* -PTCP -POSS.3SG -ACC *know* -PROG -1SG
Turkish⁸⁶

In nominalized adjunct clauses (ii), we see Poss tracking the NOM-marked subjects of the nominalized clauses, (32). I will return to this construction in §6.6.3.

⁷⁴ To be taken up in §6.6.3.

⁷⁵ Cf. §6.6.2 below. This construction has already appeared in ex. (8a) on page 200 above.

^{76,78} Gribanova 2017: p. 33, ex. (59b)

^{77,79,81,84} I have corrected the φ -features of *siz* from 2SG to 2PL, following Landmann (2010: p. 20).

^{80,83} Gribanova 2017: p. 33, ex. (60b)

⁸² I have amended the translation to include an interpretation with a discourse-given feminine (F) matrix subject.

^{85,86} Satik 2020a: p. 6, ex. (12a)

- (32) [*Sen konser -e git -tiğ -in -de*] *ben ev -e dön -üyor*
 you.NOM concert -DAT go -PTCP -POSS.2SG -LOC me.NOM home -DAT return -PROG
 -du -m.
 -PST -1SG
 ‘When you were going to the concert, I returned home.’ *Turkish*⁸⁷

The same holds for nominalized argument clauses in Sakha, (iii), which contain a NOM-marked subject, (33) and (34).

- (33) [[*Masha aqa -ta*]-Ø *bu miin -i söbül(ee) -üür -e*]-Ø
 Masha.GEN father -POSS.3SG -NOM this soup -ACC like -AOR -POSS.3SG -NOM
üçügej.
 good
 ‘It is good that Masha’s father likes this soup.’ *Sakha*⁸⁸

- (34) *Min [ehigi бүгүн кыј -byk -kyt -yn] ihit -ti -m.*
 me.NOM y’all.NOM today win -PTCP -POSS.2PL -ACC hear -PST -1SG
 ‘I head that y’all won today.’ *Sakha*⁸⁹

From these examples, it becomes clear that POSS can target not only GEN-marked nominals, but also NOM-marked nominals. This is consistent with, and expected under, an analysis of GEN as dependent case, given Bobaljik’s (2008) Revised Moravcsik (1974, 1978) Hierarchy in (29).

6.5.3.4 Compounds

In compounds, the modifying part receives no case suffix, consistent with NOM, and the head noun receives a suffix that looks like the possessive agreement suffix valued as [φ : 3SG], in Turkish realized as *-(s)I(n)*⁹⁰. Since the modifier in compounds is typically a common noun, it is pretheoretically unclear whether the agreement suffix is valued as 3SG by agreement with the common noun or by default. I have found one example in the literature where the modifier has inherent φ -features other than 3SG. This example is given in (35), containing the compound ‘you-complaint’.

⁸⁷ Kornfilt 2008: p. 97, ex. (35)

⁸⁸ Baker 2011: p. 1175, ex. (23b)

⁸⁹ Baker 2011: p. 1165, ex. (1b)

⁹⁰ This notation indicates phonologically conditioned allomorphy: initial *s* is present iff the preceding sound is a vowel, capital *I* indicates 4-way vowel harmony, yielding *i*, *ι*, *u*, or *ü*, and final *n* appears iff the next morpheme is a case suffix.

The compounding suffix cannot agree with the NOM-marked modifier ‘you.SG’, (35b), and must instead remain valued as 3SG, (35a).

- (35) a. *Bu tipik bir sen şikayet -i.*
 this typical a you.NOM complaint -POSS.3SG
 ‘This is a typical complaint of yours.’ (lit. ‘a typical you-complaint’;
 cp. English *This is a typical you-problem.*) *Turkish*⁹¹
- b. **Bu tipik bir sen şikayet -in.*
 this typical a you.NOM complaint -POSS.2SG
 ‘This is a typical complaint of yours.’ *Turkish*⁹²

This might be taken as a counter-example to the proposal that GEN is a dependent case: If POSS can agree with a nominal marked with dependent GEN, it should also be able to agree with a nominal marked with unmarked NOM, as in (35b), contrary to fact.

However, Öztürk & Taylan (2016) have argued at length that the suffix *-(s)I(n)* is not in fact a possessive agreement suffix, but rather a functional head that serves to introduce a syntactic argument to the head noun (cp. §6.4.3). Evidence comes from the fact that semantically transitive nouns, which are not possible in possessive-free genitives without *-(s)I(n)*, as discussed in §6.4.3, are allowed in compounds: Like regular possessives (§6.4.2), semantically transitive nouns ((i) kinship terms, (iii) relational nouns like ‘capital’, (ii) verb-related nouns like ‘destruction’, and (iv) dependent part-whole nouns like ‘edge’) will provide the interpretation of the relation between head noun and modifier, (36), while other nouns will receive a more flexible interpretation based on the noun’s lexical semantics and/or pragmatic context, (37).

- (36) *kadın katil -i*
 woman murderer *-(s)I(n)*
 ‘a murderer of women’ (who kills women; verb-related, (ii)),
 but not: ‘a murderer for women’ (hired by women; telic, (vii)) *Turkish*

⁹¹ Öztürk & Taylan 2016: p. 100, ex. (38a)

⁹² Öztürk & Taylan 2016: p. 100, ex. (38b)

- (37) *çocuk şiir -i*
 child poem -(s)I(n)
 ‘a poem by a child’ (agentive, (x)) or ‘a poem for children’ (telic, (vii)) *Turkish*⁹³

From the above data as well as some distributional differences, including morpheme ordering, between -(s)I(n) and the other possessive agreement suffixes, Öztürk & Taylan (2016) conclude that -(s)I(n) is not an agreement marker but a derivational suffix. They argue that -(s)I(n) has often been misanalyzed as the possessive agreement marker for 3SG because it is also present in possessives and because it has weird allomorphy, yielding the somewhat unintuitive possessive agreement paradigm in Table 6.1. I will not present the evidence for this here in detail, but refer the reader

	F	Poss[φ : $\underline{x}$]
$x = 1\text{SG}$	$-\emptyset$	-(I)m
$x = 2\text{SG}$	$-\emptyset$	-(I)n
$x = 3\text{SG}$	-(s)I(n)	$-\emptyset$
$x = 1\text{PL}$	$-\emptyset$	-(I)mIz
$x = 2\text{PL}$	$-\emptyset$	-(I)nIz
$x = 3\text{PL}$	-(lAr)-(s)I(n)	$-\emptyset$

Table 6.1: Analysis of the possessive agreement paradigm in Turkish by Öztürk & Taylan (2016)

to the existing literature on this topic, such as Dede (1978), Göksel (2009), Hayasi (1996), Kharytonava (2011), and Kunduracı (2013). What matters for the present purposes is that there are good reasons to not view the lack of agreement with the NOM-marked modifier of the compound in (35) as evidence against GEN being a dependent case.

6.5.3.5 Summary

In this section I have shown that if GEN is a dependent case in Turkic, a certain prediction arises from Bobaljik’s (2008) Revised Moravcsik (1974, 1978) Hierarchy: The possessive agreement head Poss, which can agree with GEN-marked nominals such as possessors (§6.4.2), should also be able to agree with nominals marked with an unmarked case. This section has shown that this prediction is borne out: In three kinds of nominalized clauses (§6.5.3.3), the possessive agreement on the nominalized

⁹³ Öztürk & Taylan 2016: p. 97, ex. (28) and p. 98

verb can track the φ -features of a NOM-marked nominal. Data from compounds (§6.5.3.4) seems to violate this prediction, but on a closer look we can see that it does not: in fact, agreement is absent altogether, the Turkish suffix *-(s)I(n)* is not the exponent of $\text{Poss}[\varphi: \underline{3}_{\text{SG}}]$ but a derivational suffix (Öztürk & Taylan 2016).

6.5.4 Summary

In this section, I have argued that structural GEN case in Turkic is a dependent-up case in the nominal CAD. It is assigned by case-competition against the possessum, following Baker's (2015a) proposal, presented in §6.5.2. The agreeing head Poss, which spells out as possessive agreement (see also table 6.1 on page 225), can find and agree with nominals that are marked with structural GEN or unmarked NOM, obeying Bobaljik's (2008) Revised Moravcsik (1974, 1978) Hierarchy in (29), §6.5.3. Thus, together with the arguments in §6.3 that GEN is not an unmarked case, and §6.4 that GEN is not an inherent case, and distinguishing it from lexical GEN as presented in §6.2, it follows that GEN *must* be a dependent case.

6.6 Differential Subject Marking in Turkish

6.6.1 Introduction

In this section, I turn to the application of the present theory of GEN-assignment to a much-debated construction: the case marking of subjects of nominalized embedded clauses. These nominalizations look a lot like possessives, introduced above in §6.4.2, with the subject corresponding to the possessor and the nominalized predicate corresponding to the possessum. In §6.6.2 I will argue that this is exactly how the GEN-marking on the subject of nominalizations arises. Because these nominalized clauses have a verbal core, the subject can also stay lower than the possessor-position, much like the direct object in DOM patterns, and does not receive GEN-marking. I will argue that this difference in GEN-marking is only explicable if GEN is a dependent-up case in the nominal domain, and not if it is an unmarked case.

In §6.6.3 I discuss some nominalized adjunct clauses, which, despite also looking like possessors, do not allow GEN-marking on their subjects. I follow many existing proposals in saying that this is because in adjunct clauses, the case-assignment domain is expanded from a nominal domain to a clausal domain, thus voiding the availability of GEN case as the dependent-up case in the nominal domain.

6.6.2 DSM in nominalized argument clauses

Factive nominalized argument clauses in Turkish are marked with the suffixes *-DIG* (present or perfect) or *-(y)AcAG* (future) creating a participle. The subject of this nominalization carries GEN case like a possessor, and the participle carries possessive agreement that agrees with the GEN-marked subject. This entire nominalization can then receive case marking in the superordinate clause. Some examples are given in (38).

- (38) a. [*Sen -in dün sabah ev -de yemek pişir -diğ -in*]-i duy
 you -GEN yesterday morning home -LOC food cook -DIG -POSS.2SG -ACC hear
 -du -m.
 -PST -1SG
 ‘I heard that you had cooked food at home yesterday morning.’ *Turkish*⁹⁴
- b. [*Sen -in yemek pişir -diğ -in*]-den kork -tu -m.
 you -GEN food cook -DIG -POSS.2SG -ABL fear -PST -1SG
 ‘I feared that you had cooked food.’ *Turkish*⁹⁵
- c. [*Sen -in yemek pişir -diğ -in*]-Ø bana sürpriz yap -tı.
 you -GEN food cook -DIG -POSS.2SG -NOM me.DAT surprise do -PST
 ‘That you cooked food surprised me.’ *Turkish*⁹⁶

Very similarly to non-specific indefinite non-topicalized direct objects (as discussed in detail in §3.3 above), non-specific indefinite subjects can pseudo-incorporate into the verb. Such subjects remain in the linearly immediately preverbal position and do not carry GEN-marking.⁹⁷ (39a) and (40a)

⁹⁴ Kornfilt 2003: p. 139, ex. (6a)

⁹⁵ Laszakovits 2017a: p. 269, ex. (1b)

⁹⁶ Laszakovits 2017a: p. 269, ex. (1c)

⁹⁷ Pseudo-incorporation of subjects is also possible in finite clauses, but since regular subjects in finite clauses carry NOM-marking, we cannot observe a difference in case marking there. The interpretive properties of pseudo-incorpora-

show a nominalized argument clause with a definite and topicalized subject. In (39b) and (40b), the subject is non-specific, non-referential, and even underspecified between a singular and plural reading (much like pseudo-incorporated objects in DOM), as the translations indicate. If the subject carries GEN-marking in the immediately preverbal position, (39c) and (40c), it is interpreted as definite and focalized.

- (39) a. [*Arı -nın çocuğ -u sok -tuğ -un*]-u duy -du -m.
bee -GEN child -ACC sting -DIG -POSS.3SG -ACC hear -PST -1SG
'I heard that the bee stung the child.' *Turkish*
- b. [*Çocuğ -u arı sok -tuğ -un*]-u duy -du -m.
child -ACC bee sting -DIG -POSS.3SG -ACC hear -PST -1SG
'I heard that bees stung the child.' or 'I heard that a bee stung the child.' *Turkish*⁹⁸
- c. [*Çocuğ -u arı -nın sok -tuğ -un*]-u duy -du -m.
child -ACC bee -GEN sting -DIG -POSS.3SG -ACC hear -PST -1SG
'I heard that (it was) the BEE (that) stung the child.' *Turkish*⁹⁹
- (40) a. [*Haydut -un köy -ü bas -tığ -ın*]-ı duy -du -m.
robber -GEN village -ACC raid -DIG -POSS.3SG -ACC hear -PST -1SG
'I heard that the robber raided the village.' *Turkish*¹⁰⁰
- b. [*Köy -ü haydut bas -tığ -ın*]-ı duy -du -m.
village -ACC robber raid -DIG -POSS.3SG -ACC hear -PST -1SG
'I heard that robbers raided the village.' *Turkish*¹⁰¹
- c. [*Köy -ü haydut -un bas -tığ -ın*]-ı duy -du -m.
village -ACC robber -GEN raid -DIG -POSS.3SG -ACC hear -PST -1SG
'I heard that (it was) the ROBBER (who) raided the village.' *Turkish*¹⁰²

Where in the structure is this bare resp. NOM-marked pseudo-incorporated subject and how does its [case: ____] feature remain unvalued, while the case feature of a topicalized subject gets valued as [case: GEN]?

Many investigations into the syntactic structure of Turkic nominalized argument clauses have

tion as described here still hold.

⁹⁸ Kornfilt 1997: p. 398, ex. (1377), 1997: p. 218, ex. (769)

⁹⁹ Kornfilt 1997: p. 398, ex. (1378)

¹⁰⁰ Von Heusinger & Kornfilt 2005: p. 15, ex. (14); Kornfilt 2008: p. 84, ex. (2): After

¹⁰¹ Von Heusinger & Kornfilt 2005: p. 15, ex. (15b); Kornfilt 2008: p. 84, ex. (3b)

¹⁰² After Kornfilt 2008: p. 84, ex. (3a)

come to the conclusion that they are reduced (truncated) structures compared to finite matrix clauses. Kelepir (2015, 2021) argues that the Turkish suffix *-DIG* forms a participle that selects a verb stem that can optionally be amended by stem-extending suffixes such as negation *-mA-* or ability *-(y)Abil-*, but no other TAM morphemes can appear together with *-DIG*.¹⁰³ I understand this to mean that nominalized argument clauses like those above do not contain a clausal CAD. Instead, we are looking at a verbal CAD dominated by a nominal CAD directly.

Consider now the position of pseudo-incorporated subjects of unergative and transitive predicates. In Fig. 3.3 on page 63 above I have assumed that such subjects are base-generated in Spec,VoiceP. We know from the discussion of DOM above in §3.3 that Spec,VoiceP is outside the verbal domain because there needs to be a landing site for topicalized objects where they are c-commanded by a subject in base position (this subject may remain pseudo-incorporated, which I have not discussed in §3.3) such that these objects can be valued as [case: ACC] by Case Competition against the subject, and ACC is the dependent-down case in the clausal domain but not in the verbal domain (else we could not derive DOM effects).

In truncated nominalized clauses, then, which do not contain a clausal CAD, the subject's base-position must be in the nominal domain. If GEN were an unmarked case, we would thus expect even pseudo-incorporated subjects of unergative and transitive predicates that remain in Spec,VoiceP to always appear with GEN-marking, contrary to fact. We would also expect them to contrast with subjects of unaccusative predicates, which are base-generated as internal arguments inside the verbal domain and *should* be able to remain case-less when pseudo-incorporated; again, this contrast does not exist.

Instead, analyzing GEN as dependent-up case in the nominal domain derives the DSM pattern nicely. Consider Fig. 6.4. In Spec,VoiceP, the subject may license ACC case on a moved direct object, as discussed in §3.3. Recall that, as discussed previously in §3.5 above, I take ACC to be the dependent-down case not only in the clausal, but also in the nominal domain. In §3.5 I have argued that the ACC-rule applies in verbal-noun constructions (VNCs) and *-mA* gerunds, and here I

¹⁰³ This contrasts, Kelepir argues, with other Turkic languages such as Turkmen and Noghay, where the equivalents of Turkish *-DIG* occupy higher functional heads and can thus embed more TAM morphemes.

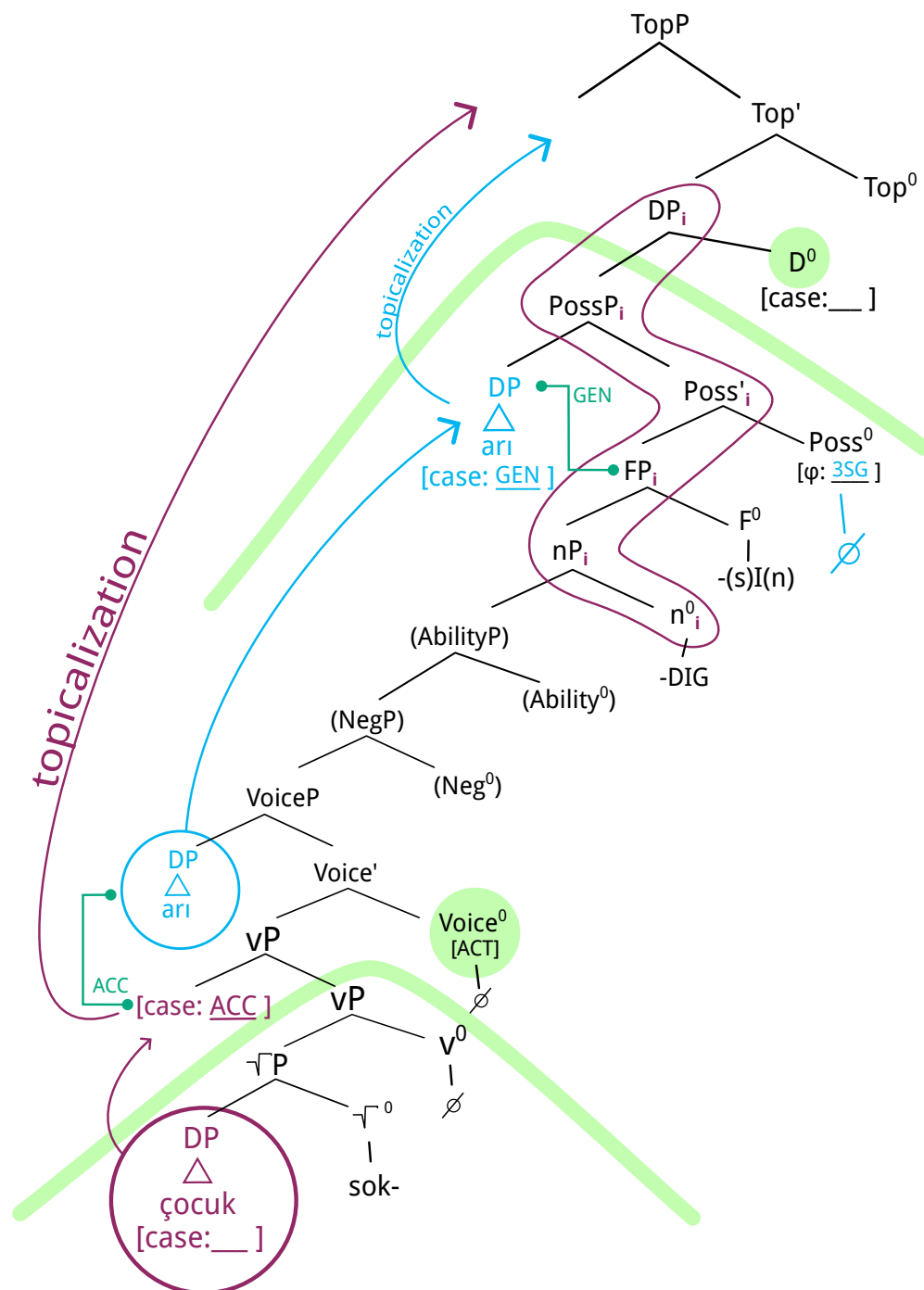


Figure 6.4: Differential Subject Marking in Turkish

add nominalized subordinate clauses to this collection.

In its base position, the subject cannot receive dependent-up GEN. I stipulate that this is so because the ACC-rule takes precedence over the GEN-rule, and once the direct object has been valued as [case: ACC], it can no longer participate in the case-competition that would assign GEN to the

subject. No other nominal is available to value the subject's case. The subject itself cannot value any other nominal's case as either ACC or GEN because the only other nominal in the structure is the nominalization *n* resp. the *-DIG* participle, and it does not c-command the subject but dominates it.

The subject may undergo movement out of its base position, which allows it to be interpreted referentially. This movement may target the landing site in Spec,PossP. There, the subject will then be in a configuration to undergo case-competition against the nominalization/participle, which will value the subject as [case: GEN], parallel to possessors, as discussed above in §6.4.2. The same holds for subjects of unaccusatives, which differ only in their base position.

6.6.3 The absence of DSM in nominalized adjunct clauses

Nominalized adjunct clauses come in many different shapes in Turkic. Here I will focus on those that differ minimally from nominalized argument clauses: focusing on Turkish, they contain the participial marker *-DIG* or its future version *-(y)AcAG*, followed by possessive agreement, and then optionally a case-marking suffix and/or other linking element to the matrix clause. The contrast to argument clauses is that the subject can never be GEN-marked and must always appear as NOM. The linking element often looks like a postposition, (41a–c), or a noun, (41d), but may also simply be case-marking, (41e), which, given the stable semantics of these constructions, I assume corresponds to a silent postposition that assigns this case.

- (41) a. [*Sen yemek pişir -diğ -in için*] *ben konser -e gid -ebil -di*
 you.NOM food cook *-DIG* -POSS.2SG for me.NOM concert -DAT go -can -PST
 -m.
 -1SG
 ‘Because you cooked, I was able to go to the concert.’ *Turkish* ^{104,105}
- b. [*Sen yemek pişir -diğ -in -e göre*] *hepiniz ev -de*
 you.NOM food cook *-DIG* -POSS.2SG -DAT according.to all of you home -LOC

¹⁰⁴ The postposition *için* ‘for’ assigns GEN, which in its lexical allomorph is $\emptyset$ and omitted in this example. For details, refer to the discussion in §6.2.

¹⁰⁵ Kornfilt 2003: p. 151, ex. (18), 2008: p. 97, ex. (31)

kal -acak -sınız.

stay -FUT -2PL

‘Given that you cooked, all of you will stay at home.’

*Turkish*¹⁰⁶

- c. [*Ben yemek pişir -diğ -im -den dolayı*] *konser -e gid -e -me*
me.NOM food cook -DIG -POSS.1SG -ABL because.of concert -DAT go -can -NEG
-di -m.

-PST -1SG

‘Because I cooked, I was unable to go to the concert.’

*Turkish*¹⁰⁷

- d. *Ben [Ali cam -ı kır -diğ -ı zaman] gerçeğ -i bil*
me.NOM Ali.NOM glass -ACC break -DIG -POSS.3SG time truth -ACC know
-iyor -du -m.

-PROG -PST -1SG

‘I knew the truth when Ali broke the glass.’

*Turkish*¹⁰⁸

- e. [*Sen opera -ya git -tiğ -in -de*] *ben konser -e git -miş ol*
you.NOM opera -DAT go -DIG -POSS.2SG -LOC me.NOM concert -DAT go -PRF be
-acağ -ım.

-FUT -1SG

‘I will have gone to the concert when you’ll go/will have gone to the opera.’ *Turk-*

*ish*¹⁰⁹

Proposals on how to capture the difference in case-marking of subjects between adjunct and argument clauses abound in the literature. Theories of case where GEN is licensed by the possessive agreement head are hard-pressed to say why GEN is impossible in (41), since possessive agreement is present and agrees with the subject just like in argument clauses. For instance, Kornfilt (2003, 2006) proposes that the agreement head itself needs to be ‘licensed’ before it can license GEN on the subject, and that the agreement head is ‘licensed’ iff it is θ -marked by the matrix verb, i.e., iff the nominalized clause is an argument clause and not an adjunct clause. This, of course, amounts to a restatement of the facts.

Other proposals argue for different case-assignment domains: that argument clauses are in some sense more nominal, drawing a connection between GEN-marked subjects and GEN-marked possessors, and adjunct clauses are more clausal, drawing a connection between NOM-marked subjects in

¹⁰⁶ Kornfilt 2003: p. 151, ex. (19), 2008: p. 97, ex. (32)

¹⁰⁷ Kornfilt 2003: p. 151, ex. (20), 2008: p. 97, ex. (33)

¹⁰⁸ Aygen 2007: p. 2, ex. (3)

¹⁰⁹ Kornfilt 2006: p. 150, ex. (12)

nominalized adjunct clauses and in finite clauses. Asarina (2011), Aygen (2002), Aygen (2007), Laszakovits (2017a), and Lees (1965) have argued for Uyghur resp. Turkish that argument clauses are larger than adjunct clauses: both are CPs, and argument clauses contain an additional nominal layer on top of the CP. Thus, argument clauses are essentially always complex NPs, i.e., CPs selected by a content noun like ‘fact’, possibly semantically bleached, and most of the time the content noun remains silent. On the other hand, Colley & Davis (2017) and Ulutaş (2010) have argued that adjunct clauses are larger than argument clauses: both structures are CPs and adjunct clauses contain an additional PP on top. This PP is immediately visible in (41a–c,e) above.

I will not go into a detailed analysis of the functional projections of these adjunct clauses here. Instead, I will follow the proposals above that claim that through phase extension or some similar mechanism, the case-assignment domain of adjunct clauses is clausal, even though they contain nominalization morphology. That is, the CAD is closed off by the phasal head C, not by D. This is illustrated in Fig. 6.5. Thus, the rule for GEN as dependent-up case assigned in case-competition against the nominalization participle *-DIG* can no longer apply, prohibiting GEN-marking on the subject. I assume that the highest projection in these adjunct clauses is C and not P, following Aygen (2002) and Aygen (2007), who provides diagnostics showing that the postpositions in (41a–c) and the noun in (41d) have been reanalyzed as complementizers: the postpositions have changed their meaning (such as *göre* from ‘according to’ to causal ‘since’ or ‘given that’) and the noun *zaman* ‘time’ is exchangeable with its synonym *vakit* when it is a noun, but not when it serves as complementizer *when*; it also cannot take adjectival modification or plural marking when it serves as complementizer.

6.6.4 Summary

In this section I have illustrated how the present proposal that GEN is the dependent-up case in the nominal case-assignment domain works out for the phenomenon of Differential Subject Marking in Turkish, whereby only topicalized subjects of nominalized clauses are marked with GEN. In §6.6.2 I have argued that given the truncated projections in nominalized clauses, the absence of GEN on

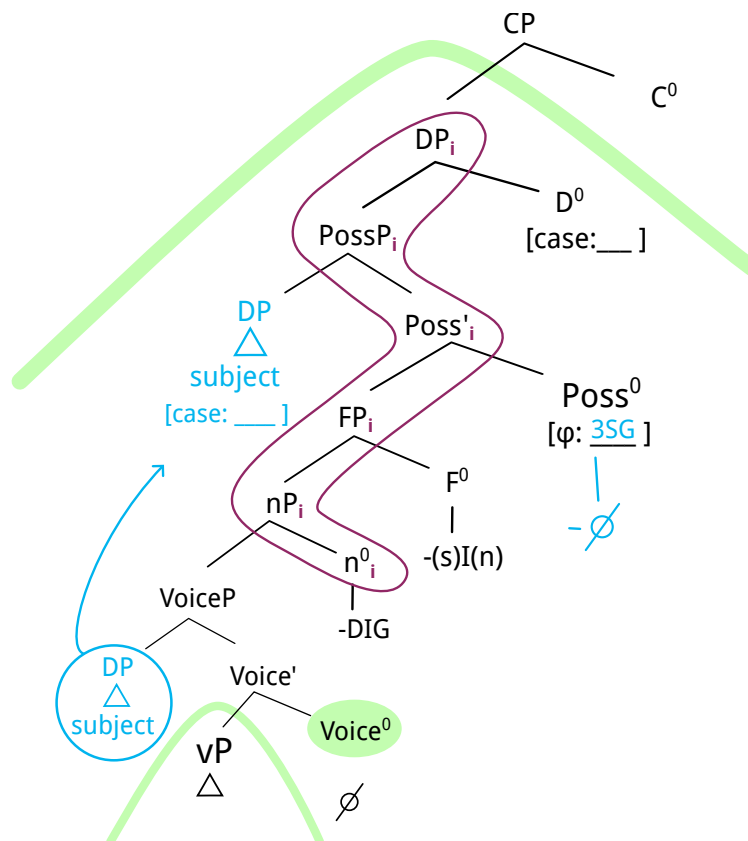


Figure 6.5: Turkish adjunct clauses contain a clausal case-assignment domain

subjects that remain in base position can only be explained if GEN is a dependent-up case, and not if GEN is the unmarked case in the nominal domain. In §6.6.3 I have discussed the absence of DSM effects in nominalized adjunct clauses — a much-discussed and puzzling phenomenon. I follow many existing proposals that argue that in adjunct clauses, the case-assignment domain is extended (such as through phase-extension) from the nominal to the clausal domain, which voids the applicability of GEN as dependent-up case in the nominal domain.

6.7 Conclusions

In this chapter, I have argued that Turkic structural GEN is the dependent-up case in the nominal case-assignment domain. I have shown (§6.3) that if we assume with Preminger (2014) that unmarked case is unvalued case, structural GEN cannot be an unmarked case because then it should be overwritable like NOM, i.e., a nominal should be able to value its case feature after moving into a

different case-assignment domain, contrary to fact. I have also shown (§6.4) that structural GEN is not an inherent case because on the one hand, it does not correspond to a single θ -role, and on the other hand, there is no functional head that could assign this GEN that is present in all constructions where we see structural GEN.

In §6.5 I have presented my main argument: First, given the framework of DCT, if GEN is neither unmarked nor inherent case, it must be dependent case. Second, it is possible for GEN to be the dependent-up case in the nominal domain because Baker (2015a) has shown us how case-competition of the possessor against the possessum works out technically. Third, GEN being a dependent case makes predictions for Bobaljik's (2008) Revised Moravcsik (1974, 1978) Hierarchy, which work out: possessive agreement can target not only GEN-marked nominals, but also NOM-marked nominals.

Then (§6.6) I have turned to a much-debated construction in Turkic, namely Differential Subject Marking (DSM): subjects of nominalized clauses sometimes appear with GEN-marking and sometimes appear with NOM-marking. I have shown that, given the analysis by Kelepir (2015, 2021) of the functional projections being present in these nominalizations, GEN as dependent-up case makes the correct predictions for DSM in nominalized argument clauses, while GEN as unmarked case does not.

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Lists

List of abbreviations

Notation	Description
1	1 st person
2	2 nd person
3	3 rd person
a.m.o.	among many others
a.o.	among others; and others
ℳ	A/A'
A	adjective
A	agent
ABIL	ability
ABL	ablative
ABS	absolute
ACC	accusative
ACT	active
ADJ	adjective
ADV	adverb(ial)
AOA	agent-oriented adverb
AGR	agreement
ALL	allative
ANTIP	antipassive
AOR	aorist
Appl	applicative
APPL	applicative
*	Kleene star, indicating 0 or more repetitions of its argument
*	ungrammatical, unacceptable
AUX	auxiliary
BEN	benefactive
BG	Burzio's Generalization
C	complementizer
CAD	case-assignment domain
CAUS	causative

Notation	Description
CC ₂	cross-clausal A/A'-dependency
cf.	Latin <i>confer</i> 'compare'
✓	acceptable
CMPD	compound
COMP	complementizer
COMPL	completive
COP	copula
cp.	compare
CVB	converb
D	determiner
DAT	dative
DCT	Dependent Case Theory
DEF	definite
DEM	demonstrative
DM	Distributed Morphology
DO	direct object
DOM	Differential Object Marking
DSM	Differential Subject Marking
DU	dual
DUR	durative
e.g.	Latin <i>exemplum gratiae</i> 'for example'
EA	external argument
ECM	Exceptional Case Marking
ECP	Empty Category Principle
EP	epenthesis
EPP	Extended Projection Principle
ERG	ergative
et seq.	Latin <i>et sequentes</i> 'and following [works]'
etc.	Latin <i>et cetera</i> 'and so on'
EVID	evidential
EXCL	exclusive
F	feminine
FHCT	functional-head case theory
FOC	focus
CFOC	contrastive focus
PFOC	presentational focus
FUT	future
GB	Government and Binding theory
GEN	genitive

Notation	Description
GF	grammatical function
#	nonsensical; in general, or in the given context
HON	honorific
i.a.	Latin <i>inter alia</i> ‘among others’
i.e.	Latin <i>id est</i> ‘this is’, ‘this means’
IA	internal argument
iff	if and only if
ILL	illative
IMP	imperative
IMPS	impersonal
INCL	inclusive
IND	indicative
INDF	indefinite
INE	inessive
INF	infinitive
INS	instrumental
INTR	intransitive
IO	indirect object
IPFV	imperfective
LDA	long-distance agreement
LF	Logical Form
LFG	Lexical-functional grammar
LI	lexical item
lit.	literally
LOC	locative
LV	light verb
LVC	light-verb construction
M	masculine
MG	Minimalist Grammar
N	noun, nominal
<i>n</i>	nominalizer
N	neuter
NECESS	necessitative
NEG	negation
NF	non-feminine
NMLZ	nominalizer
NOM	nominative
∅	Judgment unknown to the author; elicitation left for future work.

Notation	Description
NPI	negative polarity item
NPST	non-past
∅ (null)	null morpheme, zero morpheme
O	object
OBL	oblique
ok	acceptable
p.c.	personal communication
P	preposition, postposition, adposition
P	phrase
PART	partitive
PASS	passive
PCC	Person-Case Constraint
PF	Phonological Form
PFV	perfective
φ (phi)	the feature bundle comprising person, number, and gender (if applicable)
PIC	Phase Impenetrability Condition
PL	plural
POSS	possessive
Pred	predicate
PRF	perfect
PROG	progressive
PRS	present
PRT	particle
PST	past tense
PTCP	participle
PPP	perfect passive participle
Q	question particle/marker
??	Negative judgment, but not completely unacceptable.
RECP	reciprocal
REFL	reflexive
REL	relative
resp.	respectively
RG	Relational Grammar
√ (root)	the lexical specification of the root of a word
S	subject
SG	singular
SOA	subject-oriented anaphor
Spec	specifier

Notation	Description
T	tense, tempus
<i>t</i>	trace
TAG	Tree-Adjoining Grammar
TAM	Tense-Aspect-Mood
θ -role	thematic role
Top	topic
TOP	topic
CTOP	contrastive topic
TR	transitive
UG	Universal Grammar
UTAH	Uniformity of Theta-Assignment Hypothesis
V	verb
<i>v</i>	verbalizer
VLZ	verbalizer
VI	Vocabulary Insertion
VN	verbal noun
VNC	verbal-noun construction
Voice	
vs.	versus
WCO	weak cross-over
Wh	constituent question

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